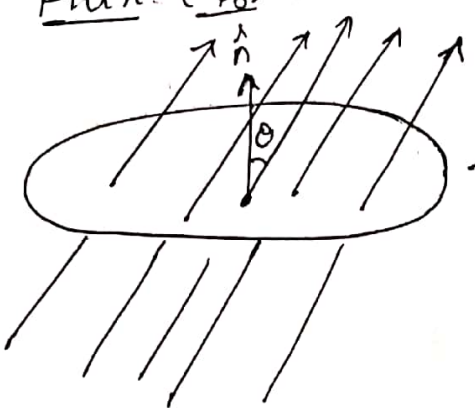
Electro Magnetic Induction

2/06/25/Monday

* Oersted ⟶ current wire can produce magnetic field.

* Faraday ⟶ converse is also true .i.e, magnetic field can produce the current. (with some conditions).

Magnetic Flux (ϕ_B) :



B = magnetic field / induction.

— circular coil

The total no. of lines of force passing through an area gives magnetic flux.

Mathematically, $\boxed{\phi_B = \vec{B} \cdot A\,\hat{n}}$ *

$\boxed{\phi_B = BA\cos\theta}$ *

'θ' is angle b/w $\vec{B}$ & $\vec{A}$. ($\theta \in [0, 180^\circ]$)

* If coil contains 'n' turns.

$\boxed{\phi_B = nBA\cos\theta}$

* ϕ is a scalar quantity and is +ve, -ve, zero

units- C.G.S unit – maxwell

S.I unit – Weber (wb)

$\boxed{1\,wb = 10^8\,maxwell}$

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* $\phi = BA \quad (\theta = 0^\circ)$

$$\boxed{B = \phi/A}$$

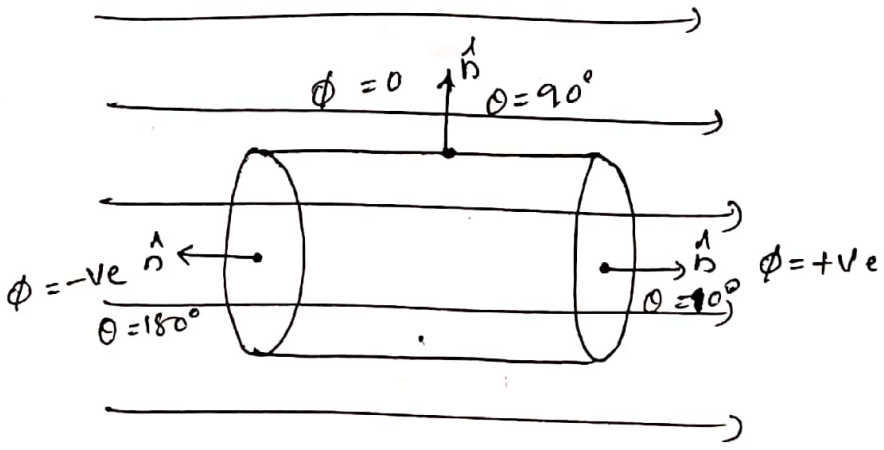
B — unit : wb/m^2 (or) Tesla(T)

$$1\,T = 1\,wb/m^2$$

$$\boxed{1\,T = 10^4 \text{ guass}}$$

* The total no. of lines of force passing through unit area when the plane is considered normal to the magnetic field is called magnetic ~~flux~~ field.

Ex:



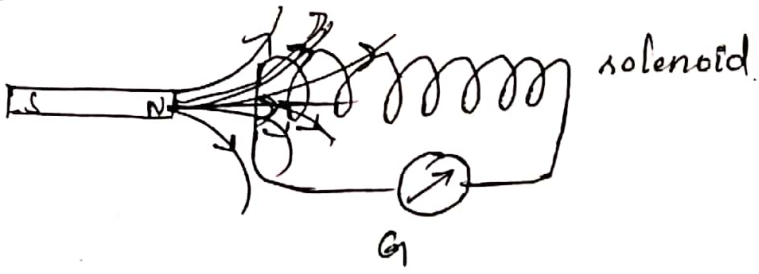
we have,

$$\phi = BA\,n\cos\theta$$

→ ϕ can be changed when B (or) A (or) n (or) θ are changed.

Faraday's Experiments

Experiment-1:



A, n, θ — const

* B is changed by moving barmagnet towards

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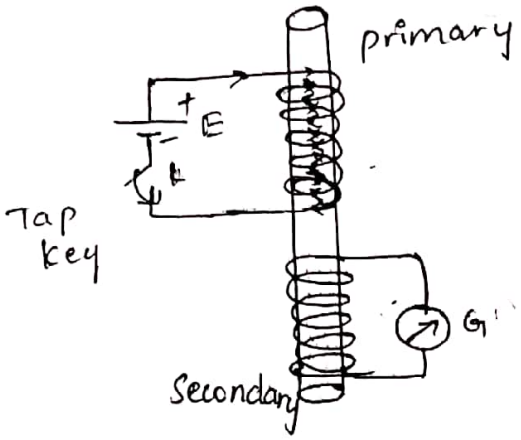
the coil we can observe deflection in galvanometer during motion of magnet.

* No deflection when it is at rest.
* with change in flux, emf is induced in the conductor.
* when ever there is relative motion blw coil & barmagnet then there will be deflection.
* And induced emf is produced inside the coil & (either closed/open circuit) induced current is produced inside (only when circuit is ~~closed~~ closed)

Experiment-2:



* There is deflection in galvanometer when tapkey is closed (may be for along time or not) and opened. (just for a moment and then becomes zero).

Faraday's Laws of Electromagnetic Induction: 3|06|25|Tuesday

L-I & * whenever there is a magnetic flux linked with the coil is changed, an emf will be induced in it.

L-II & * The induced emf in a coil is always directly proportional to the rate of change of magnetic flux linked with the coil. i.e, $e \propto \frac{-d\phi}{dt}$. $\Rightarrow$ $\boxed{e = -\frac{d\phi}{dt}}$

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where $\phi = NBA\cos\theta$.

-ve sign indicates, emf opposes the change in flux, which is given by Lenz Law.

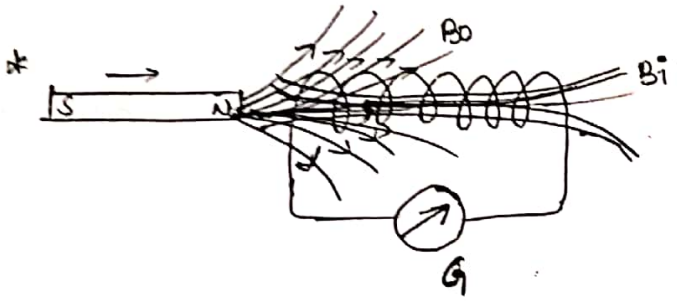
Lenz Law:

It gives us the direction of induced emf or induced current in the coil.

It is based on "law of conservation of energy".

Statement -

The direction of induced emf is always such that it opposes the cause that has produced it.

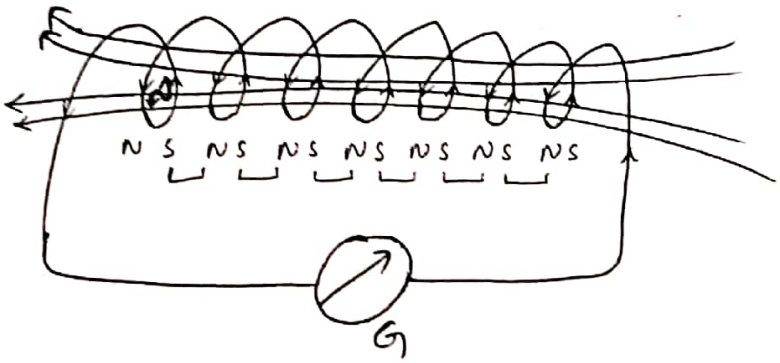


* due to changing flux => induced emf => induced current

(circuit must be closed) — induced current

(circuit is opened / closed) — induced emf

induced current => induced magnetic field.



here coil will compress due to attractions b/w S, N.

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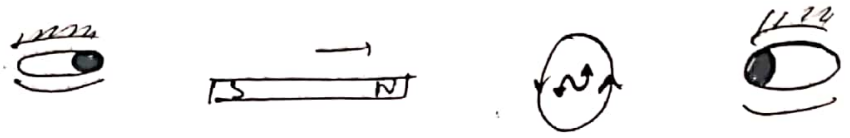
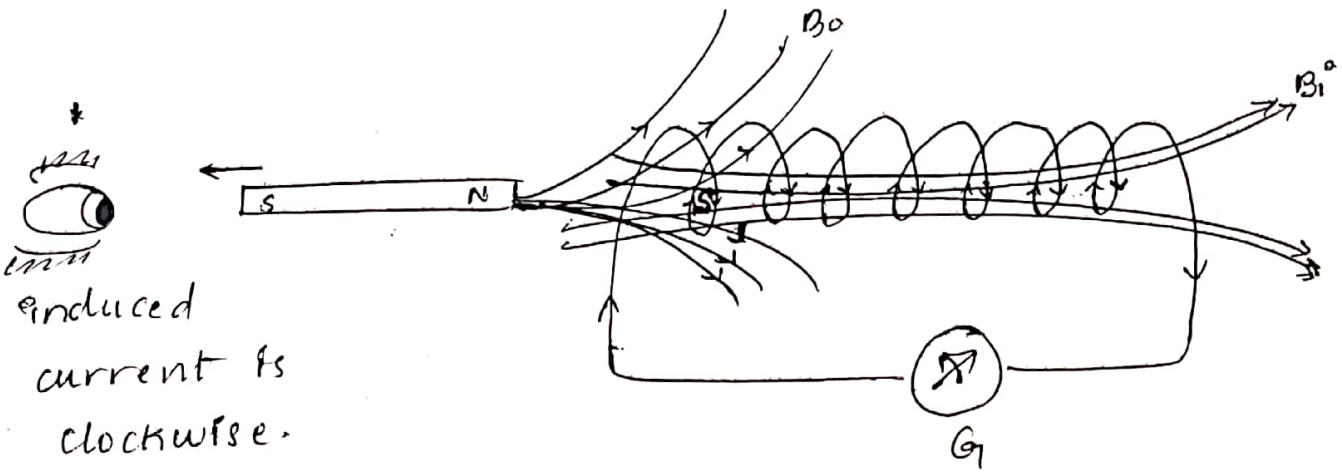


Induced current is clockwise.

Induced current is anticlockwise.



Induced current is clockwise.

* Barmagnet has to do some work in order to move towards / away from the coil.

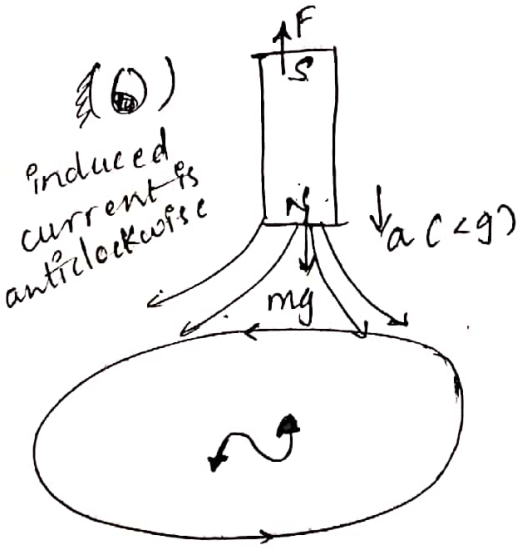
[mechanical energy ⇒ electric energy ⇒ Heat energy]

principle of electromagnetic induction.

{* There is an inductor which stores energy.}

Points:

1)



(i) induced current is anticlockwise

Metal ring

$a = \frac{mg - F}{m}$

$= g - \frac{F}{m}$

$a < g$

(ii) induced current is in clockwise.

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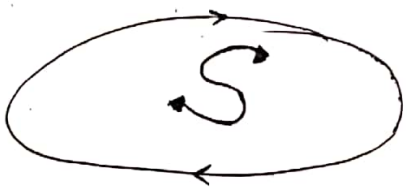
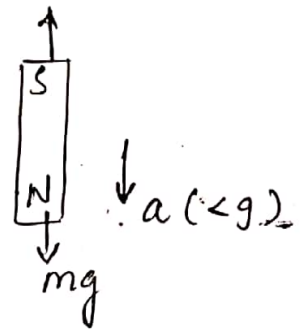
v)



induced current is clockwise



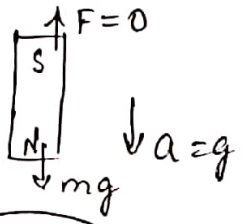
$$a = \frac{mg - F}{m}$$

$$= g - \frac{F}{m}$$

$\boxed{a < g}$

induced current is anti clockwise

3)



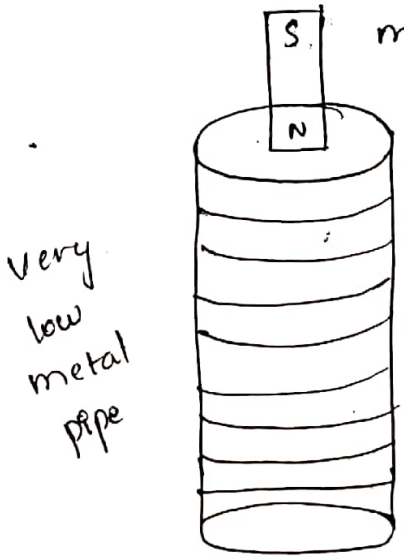
cut ring

$\frac{d\phi}{dt} \neq 0$

$e \neq 0$

$i = 0$ (open circuit)

4)



magnet is from the mouth of pipe.

$$a = \frac{mg - F}{m} = g - \frac{F}{m}$$

As $v(\uparrow) \Rightarrow e(\uparrow) \Rightarrow i(\downarrow)$

$\Rightarrow F(\uparrow)$

$a(\downarrow)$

when speed 'v' becomes max, $\frac{dv}{dt} = 0 \Rightarrow \boxed{a = 0}$

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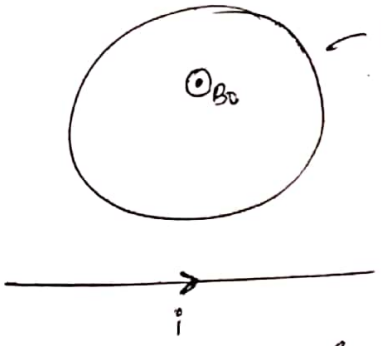
Then that v_{max} is called terminal velocity

5)

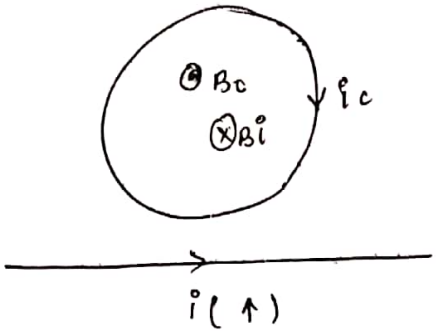


$i = const$

$B_0 = const$

$\phi = const$

$e = 0 , i_c = 0$

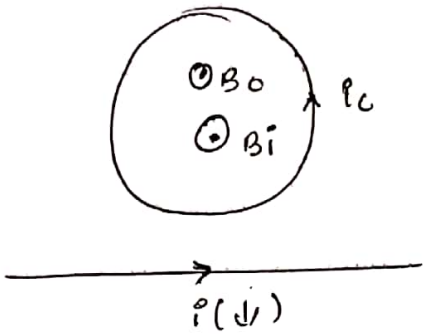


$B_0(\uparrow)$

$\phi(\uparrow)$

$e \neq 0, i_c \neq 0$

clockwise



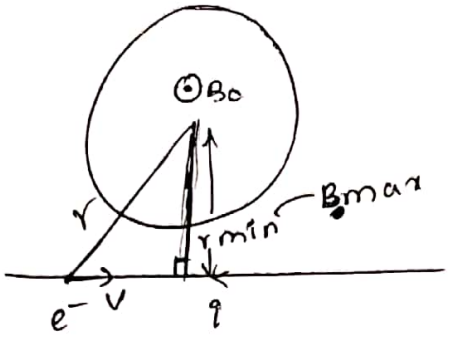
$B_0(\downarrow)$

$\phi(\downarrow)$

$e \neq 0, i_c \neq 0$

anticlockwise

6)



$$B_0 = \frac{\mu_0}{4\pi} \frac{q(\vec{v} \times \hat{r})}{r^2}$$

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Important Formulae:

1) Flux linked with the coil is,

$$\phi = NBA\cos\theta$$

2) induced emf in the coil is,

$$e = -\frac{d\phi}{dt}$$

$$e_{avg} = \frac{\int_{t_1}^{t_2} e_{inst}\,dt}{\int_{t_1}^{t_2} dt}$$

3) induced current in the coil is,

$$i = \frac{e}{R} = -\frac{1}{R}\frac{d\phi}{dt}$$

4) induced charge flowing in the circuit,

$$dq = i\,dt = -\frac{1}{R}\frac{d\phi}{dt} \times dt$$

$$dq = -\frac{d\phi}{R}$$

$$\Delta q = -\frac{\Delta\phi}{R} = -\frac{(\phi_f - \phi_i)}{R}$$

↓

total induced charge

$$\Rightarrow \Delta\phi = \Delta q(R) = R \times \text{area of } i\text{-}t \text{ graph.}$$

points:

1) Clearly induced current & induced charge depends on resistance of the circuit whereas induced emf doesnot depends on resistance of the circuit.

2) Induced emf and induced current depends on rate of change of flux but induced charge depends on change in flux only.

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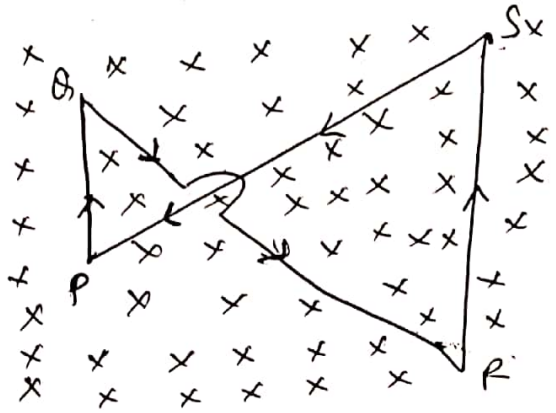
Again,

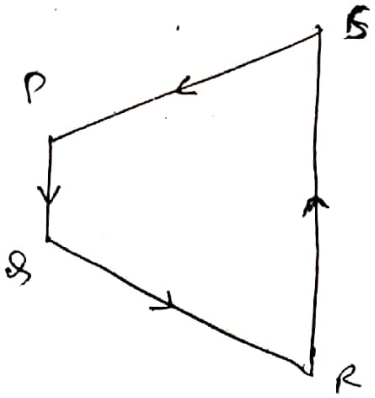
a) If only B is changed,

$$e_{inst} = NB\cos\theta \cdot \frac{dB}{dt}$$

$$e_{avg} = NA\cos\theta \left(\frac{B_2 - B_1}{t_2 - t_1}\right)$$

Eg!





b) If only N is changed,

$$e_{inst} = BA\cos\theta \cdot \frac{dN}{dt} \qquad \& \; i_{inst} = \frac{e_{inst}}{R}$$

$$e_{avg} = BA\cos\theta \left(\frac{N_2 - N_1}{t_2 - t_1}\right) \qquad \& \; i_{avg} = \frac{e_{avg}}{R}$$

c) If only θ is changed,

$$e_{inst} = BAN \frac{d}{dt}(\cos\theta)$$

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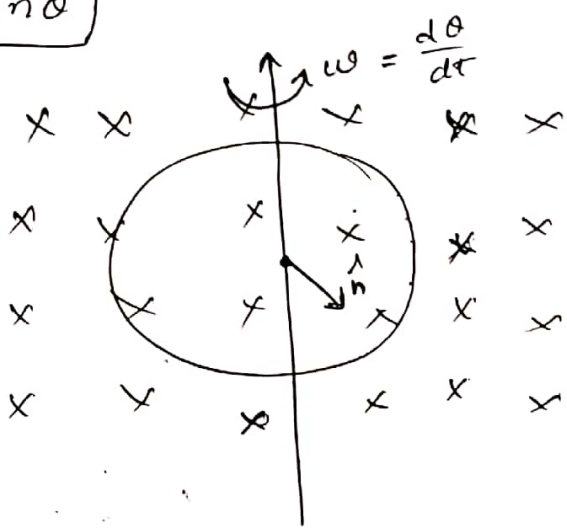
$$e_{inst} = BAN\sin\theta \cdot \frac{d\theta}{dt}$$

$$\boxed{e_{inst} = BAN\omega\sin\theta}$$



when $\theta = 0° / 180°$

$$\boxed{e_{inst} = 0}$$

where as $\phi = N'BA\cos\theta$

$$\boxed{\phi = NBA} \text{ (max)}$$

$\theta = 90°$

$$\boxed{e = NBA\omega} \text{ (max)}$$

but $\boxed{\phi = 0}$

$$\therefore \boxed{e_{inst} = e_{max}\cdot\sin\theta} \quad [\theta = \omega t]$$

↳ only when ω is const

& $$e_{avg} = NBA\frac{(\cos\theta_2 - \cos\theta_1)}{t_2 - t_1}$$

c) when only A is changed,

$$e_{inst} = BN\cos\theta\frac{dA}{dt}$$

$$e_{avg} = BN\cos\theta\frac{(A_2 - A_1)}{t_2 - t_1}$$

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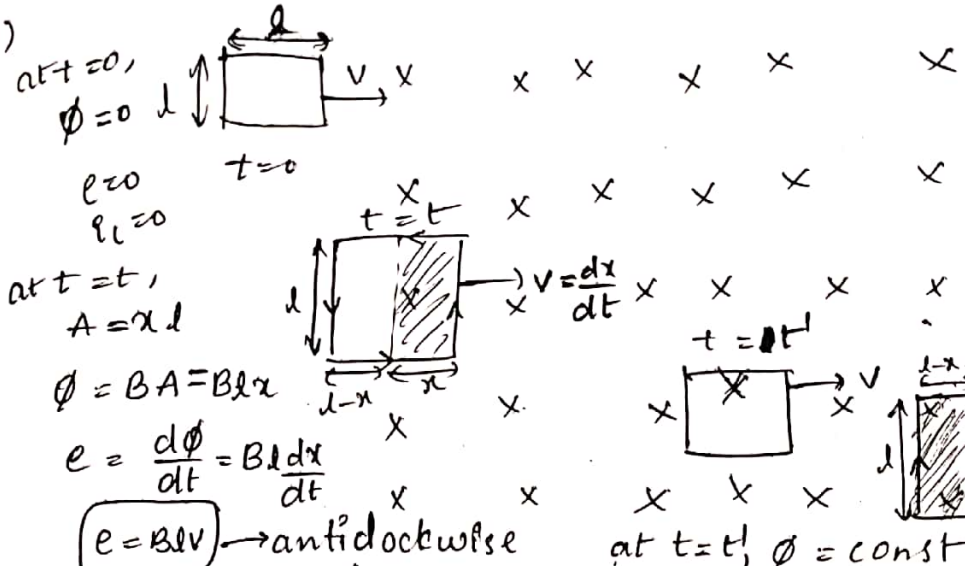
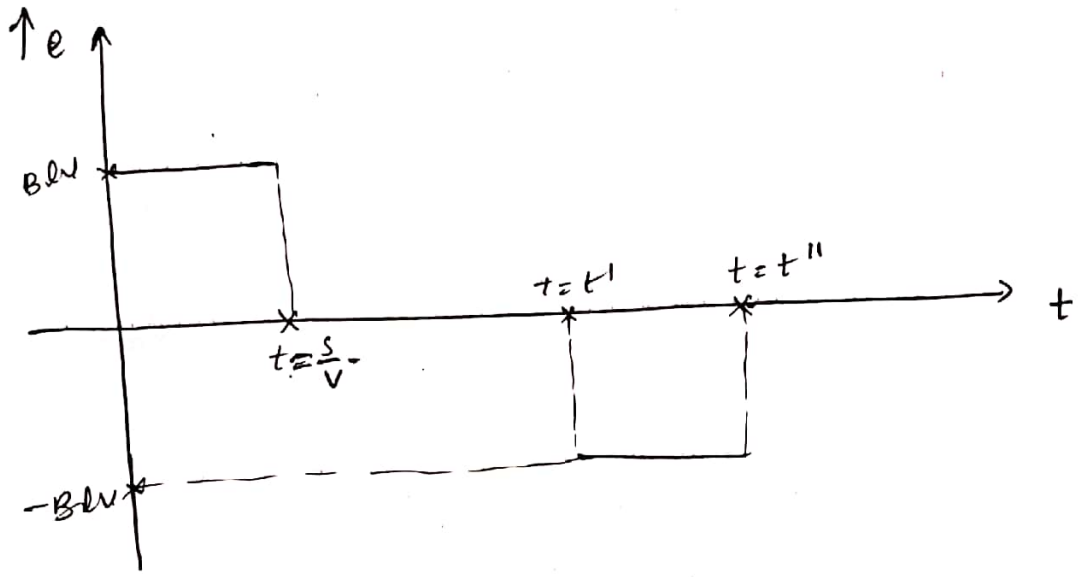
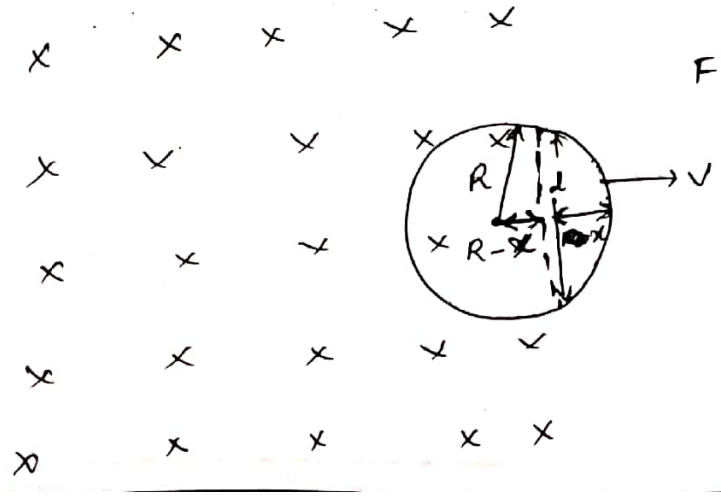
Ex!

1)





2)



Find instantaneous emf.

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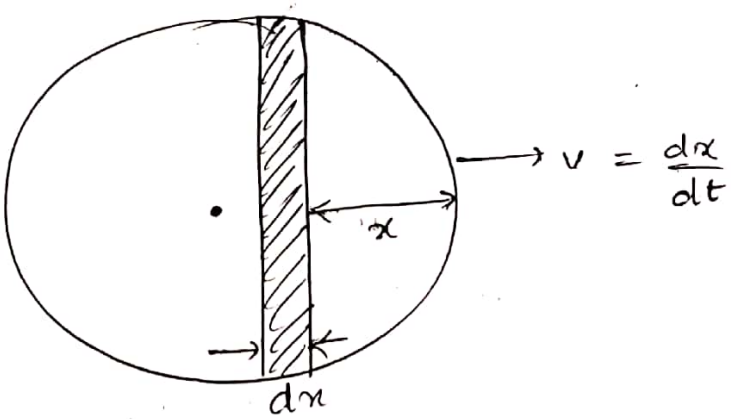
here,

$$l = 2\sqrt{R^2 - (R-x)^2}$$



Change in area,

$$dA = l\,dx$$

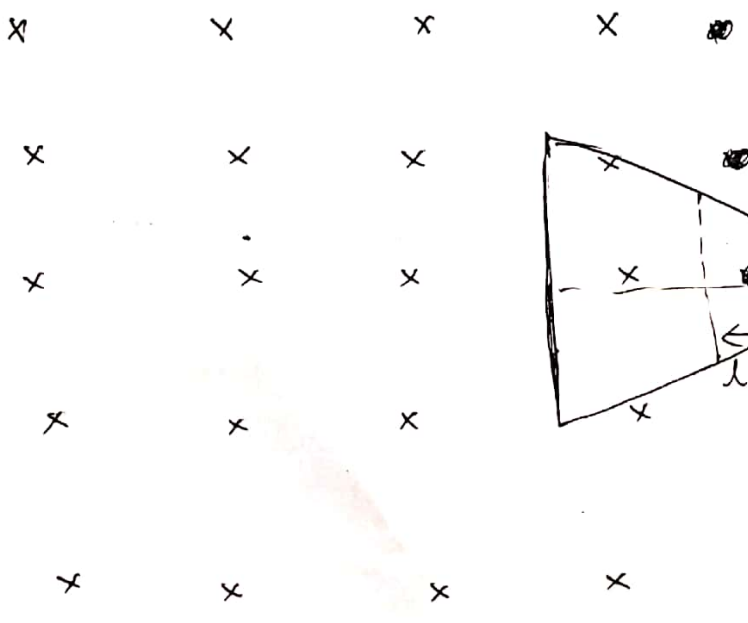
$$dA = \left(2\sqrt{R^2 - (R-x)^2}\right) v \cdot dt$$

$$\frac{dA}{dt} = 2v\sqrt{R^2 - (R-x)^2}$$

now, $e = B\frac{dA}{dt}$

$$\boxed{e = 2BV\sqrt{R^2 - (R-x)^2}}$$

Ex:



Find instantaneous emf.

At any instant

Find e_{inst} & its direction?

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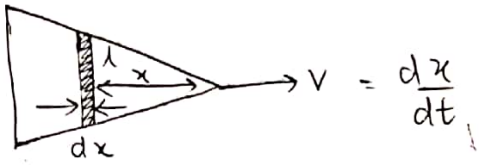
$$A = \frac{\sqrt{3}}{4}l^2 - \frac{1}{2}(2x\tan 30)x = \frac{\sqrt{3}}{4}l^2 - \frac{x^2}{\sqrt{3}}$$

$$\phi = BA$$

$$e = \frac{d\phi}{dt} = B\frac{dA}{dt} = \frac{2xvB}{\sqrt{3}} \text{ clockwise}$$

mll:

$dA = l\,dx$

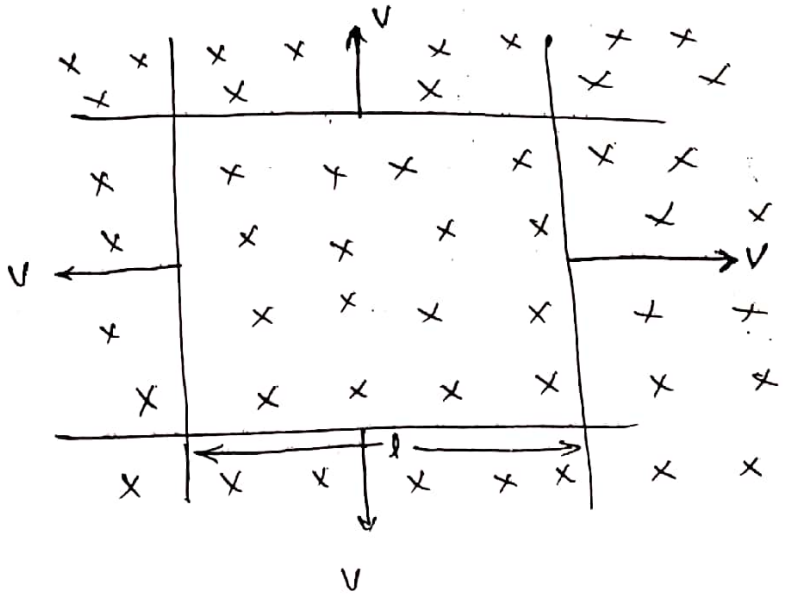


$= (2x\tan 30°)\,v\,dt$

$$\frac{dA}{dt} = \frac{2x}{\sqrt{3}}v$$

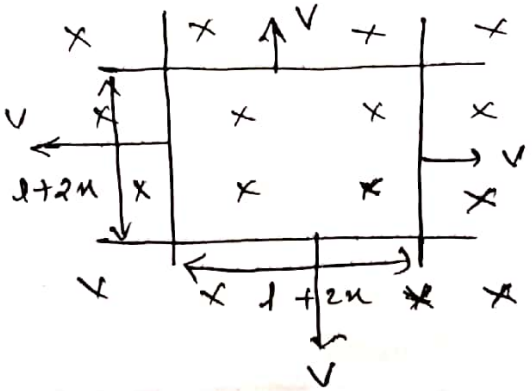
$$e = B\frac{dA}{dt} = \frac{2xvB}{\sqrt{3}}\text{ ; clockwise.}$$

Ex:8



Find induced emf & induced current

Sol:8 At any time 't',



$$A = (l+2x)^2$$

$$\phi = BA$$

$$e = \frac{d\phi}{dt} = B\frac{dA}{dt}$$

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$$e = B(l+2x)(2)(2)\cdot\frac{dx}{dt}$$

$$e = 4B(l+2x)\cdot\frac{dx}{dt}$$

$$e = 4B(l+2x)V = 4BLV$$

* L => length of conductor at that time.

(*) λ – resistance per unit length.

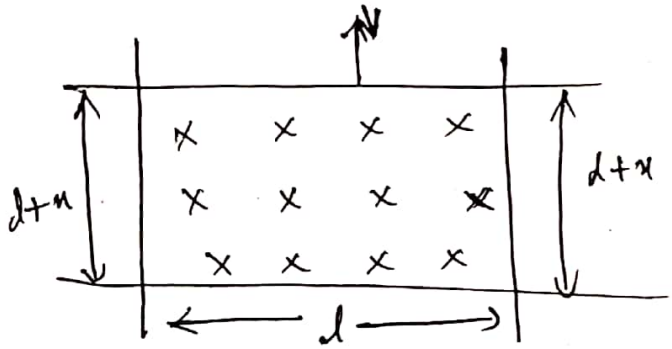
(*) $R_{total} = 4l\lambda$

$$\therefore i = \frac{e}{R_T}$$

$$i = \frac{4BLV}{4L\lambda}$$

$$\boxed{i = \frac{BV}{\lambda}}$$

* If one conductor is in motion,



$$A = l(l+x)$$

$$\phi = BA$$

$$e = \frac{d\phi}{dt} = B\frac{dA}{dt} = Bl\frac{dx}{dt} = Blv$$

* l => length of moving conductor at that time.

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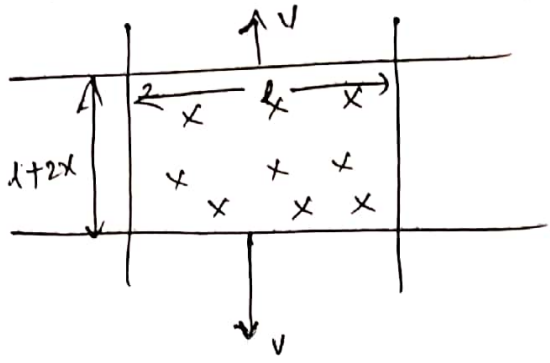
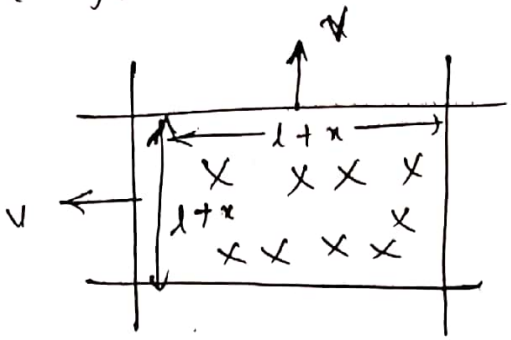
* If two conductors are in motion, (opp)



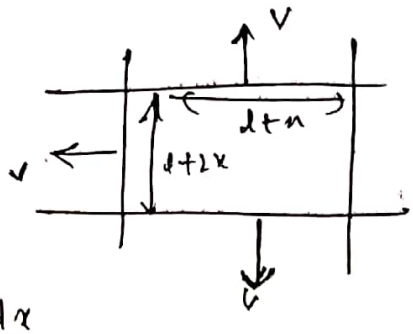
$$e = 2BlV$$

* If two conductors (adj) are in motion,



$$e = 2BLV$$

* If three conductors are in motion,



$$A = (l+2x)(l+x)$$

$$e = B\frac{dA}{dt}$$

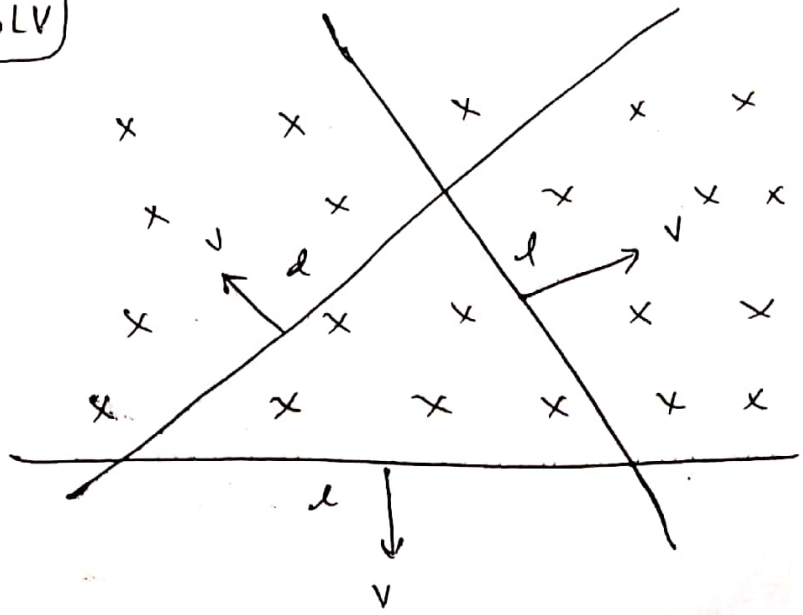
$$= B\left[(l+2x)+2(l+x)\right]\cdot\frac{dx}{dt}$$

$$= B\left[3l+4x\right]\cdot V$$

$$e = BLV$$

Ex:

Find induced emf.



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Sol: At that instant, 'l' is the length of each moving conductor.

$$dA = 3l\,dx$$

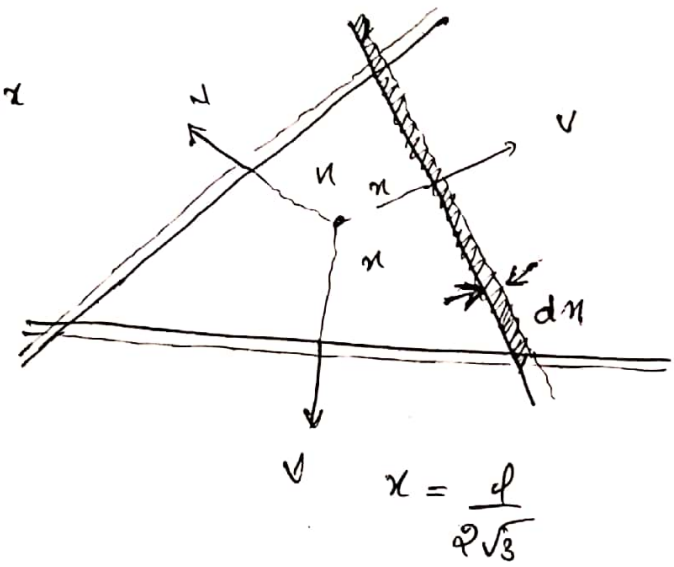
$$dA = 3lv\,dt$$

$$e = \frac{B\,dA}{dt}$$

$$\boxed{e = 3Blv}$$

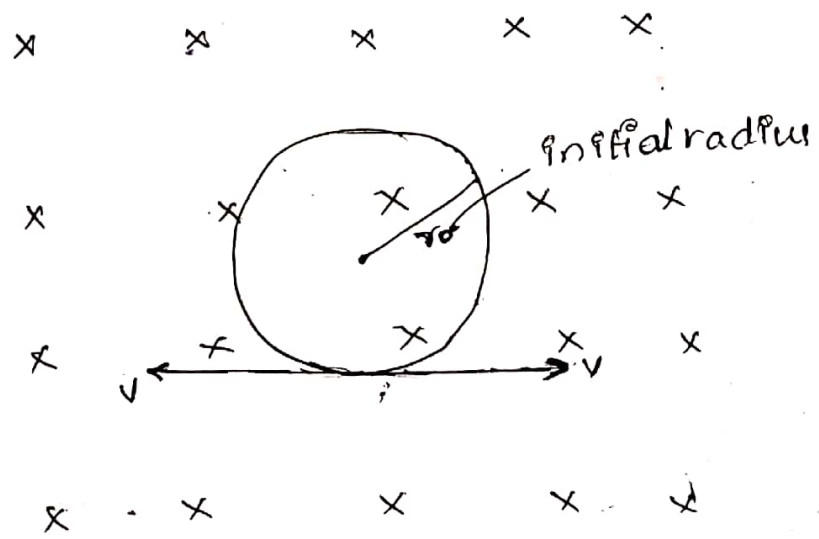
$$i = \frac{3Blv}{\lambda 3l}$$

$$\boxed{i = \frac{Bv}{\lambda}}$$



$$x = \frac{l}{2\sqrt{3}}$$

Ex:



Find 1) radius of loop as a function of time.

2) induced emf of loop as a func^{c} of time.

3) induced current in the loop as f(t) (If λ is resistance per unit length.)

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Soln

1) $l = 2\pi r = 2\pi r_0 - 2vt$

$$r = r_0 - \frac{vt}{\pi}$$

At any time 't'

i.e, radius decreases linearly with time.

2) $A = \pi r^2$

$\phi = BA$

$$e = B\frac{dA}{dt}$$

$$= 2\pi B r \frac{dr}{dt}$$

$$= 2\pi B \left(r_0 - \frac{vt}{\pi}\right)\left(\frac{-v}{\pi}\right)$$

$$e = -2BV\left(-\frac{vt}{\pi} + r_0\right)$$

$$\boxed{e = 2BV\left(r_0 - \frac{vt}{\pi}\right)}$$; clockwise.

i.e, emf decreases linearly with time.

3) $$I = \frac{e}{R} = \frac{2BVr}{\lambda(2\pi r)} = \frac{BV}{\pi\lambda}$$; clockwise.

which is independent of time.

Ex: A square loop of side 12cm with its sides parallel to X & Y-axes is moved with a velocity of 8cm/sec in +ve x direction in magnetic field pointing toward +ve z direction. The field has a gradient of 10^{-3} T/cm. Find magnitude of induced emf

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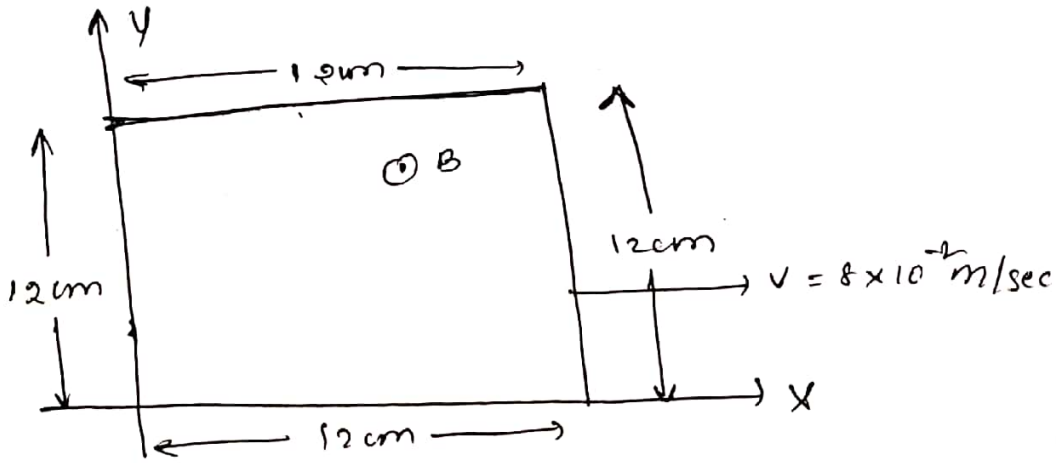
If B changes at a rate of 0.1 T/sec.

Sol:



Given,

$$\frac{\partial B}{\partial x} = 10^{-1}\ T/m$$

$$\frac{\partial B}{\partial t} = 0.1\ T/sec$$

we have,

→ due to change in B with t; $e_1 = \frac{d\phi}{dt} = A\frac{\partial B}{\partial t} = 144 \times 10^{-4} \times 10^{-1}$

$$e_1 = 144 \times 10^{-5}\ ;\ \text{clock wise.}$$

→ due to change in B with x; $e_2 = \frac{d\phi}{dt} = \frac{d\phi}{dx} \times \frac{dx}{dt}$

$$e_2 = A\frac{\partial B}{\partial x} \times \frac{dx}{dt}$$

$$= 144 \times 10^{-4} \times 10^{-1} \times 8 \times 10^{-2}$$

$$e_2 = 1152 \times 10^{-7}\ ;\ \text{clock wise}$$

$$= 11.52 \times 10^{-5}$$

$$e = e_1 + e_2 = 155.52 \times 10^{-5} V = 15.5 \times 10^{-4}$$

$e = 1.55$ mV

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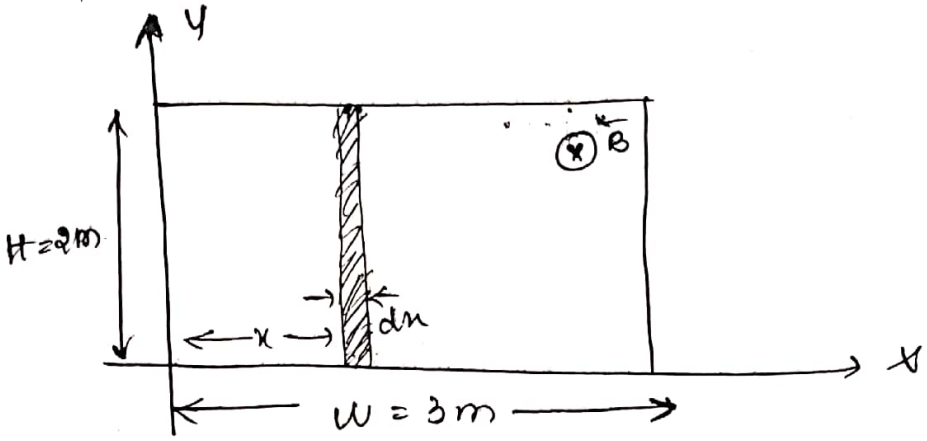
Ex: A rectangular loop of wire immersed in a non-uniform and ~~varying~~ varying $\bar{B}$ that is $\perp$ler to & directed into paper & given by $B = 4t^2x^2$ where B in Tesla, t in sec, x in m, Find magnitude & direction of the induced emf around the loop at $t = 0.1$ sec.



Sol: At a given time 't',

$$d\phi = B\,dA$$

$$d\phi = 4t^2x^2 H\,dx$$

now, total flux,

$$\phi = \int_0^w 4t^2x^2 H\,dx$$

$$= 4t^2H \int_0^w x^2 \cdot dx$$

$$\phi = 8t^2\left(\frac{w^3}{3}\right)$$

$$e = \frac{d\phi}{dt} = \frac{8}{3}w^3 2t = \frac{8}{3}(27)(2)(0.01)$$

$$= 16 \times 9 \times 0.01$$

$$e = 1.44\ V$$

anti clockwise

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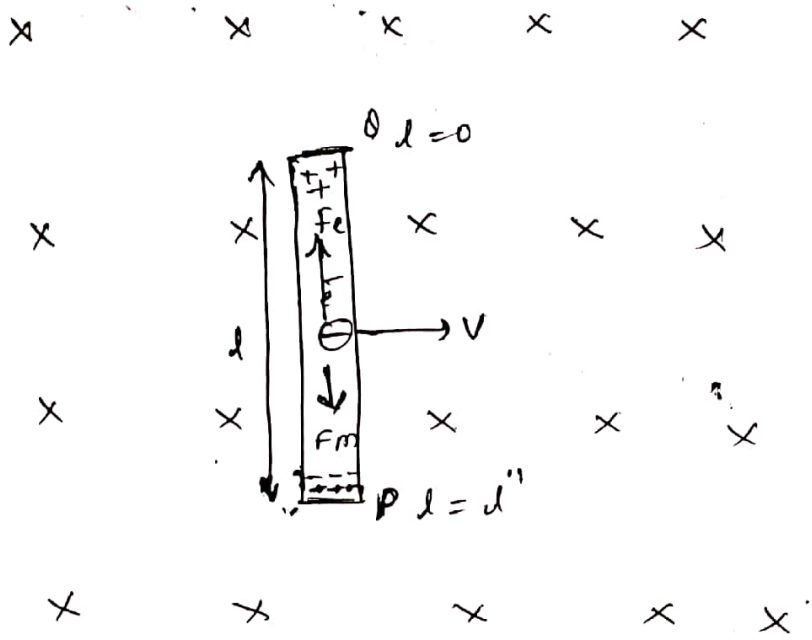
Motional Induced Emf:

1) Due to Translational motion of the conductor:



Magnetic force acting on electron,

$$\bar{F}_m = q(\bar{v} \times \bar{B})$$

$$F_m = -e(\bar{v} \times \bar{B})$$

Electric force on electron,

$$\bar{F}_e = q\bar{E} = -e\bar{E}$$

In eqm,

$$F_{net} = 0$$

$$\bar{F}_e + \bar{F}_m = 0$$

$$-e\bar{E} - e(\bar{v} \times \bar{B}) = 0$$

$$\boxed{\bar{E} = -(\bar{v} \times \bar{B})}$$

& $$\bar{E} = |\bar{v} \times \bar{B}|$$

$$\boxed{E = vB}$$

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now,

$$\int_{V_P}^{V_Q} dV = -\int_{\ell=l}^{\ell=0} \bar{E}\cdot d\bar{l}$$

$$\Rightarrow \int_{V_P}^{V_Q} dV = \int_{l=0}^{l=l} \bar{E}\cdot d\bar{l}$$

$$\boxed{V_Q - V_P = -\int_{\ell=0}^{\ell=l} (\bar{v}\times\bar{B})\cdot d\bar{l}}$$

$V_Q - V_P$: potentials ; $\bar{v}$: velocity of rod

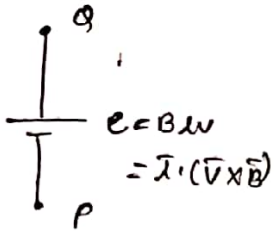
* Here moving conductor behaves as a cell with induced emf

$$\boxed{e = V_Q - V_P = -\int_{l=0}^{\ell=l} (\bar{v}\times\bar{B})\cdot d\bar{l}}$$

* only magnitude,

$$\boxed{e = \int_0^l (\bar{v}\times\bar{B})\cdot d\bar{l}}$$

* If $\bar{v}$ & $\bar{B}$ are const vectors.



then $\boxed{e = (\bar{v}\times\bar{B})\cdot\bar{l}}$

$$\boxed{e = \bar{l}\cdot(\bar{v}\times\bar{B}) = [\,l\ \bar{v}\ \bar{B}\,]}$$

Points!

* If $\bar{l}$, $\bar{v}$, $\bar{B}$ are coplanar vectors.

$$\boxed{e=0}$$

* If $\bar{v}\parallel\bar{B}$ (or) $\bar{v}\parallel\bar{l}$ (or) $\bar{B}\parallel\bar{l}$. $\boxed{e=0}$.

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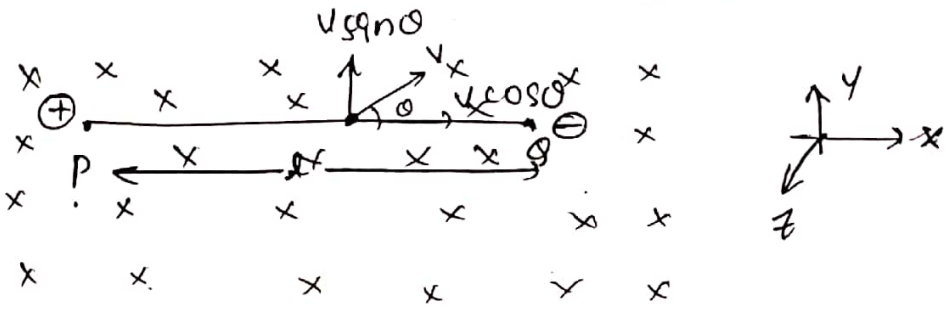
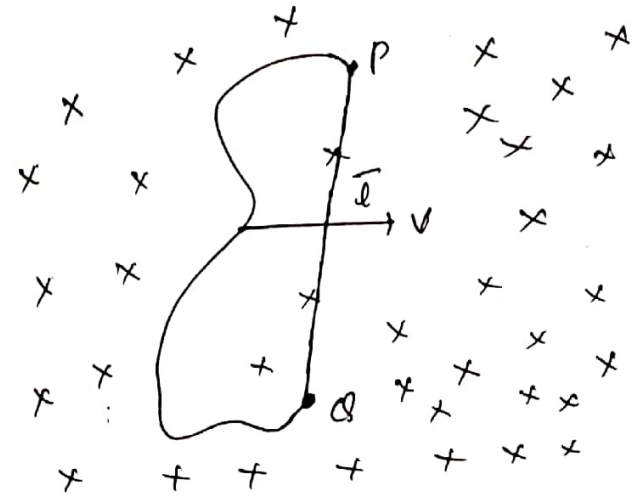
* If $\bar{v} \perp \bar{B} \perp \bar{l}$ then $\boxed{e = Blv}$ (max)

*



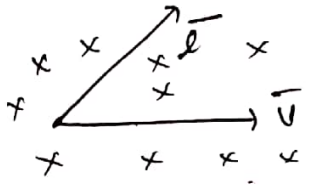
$\boxed{e = Blv\sin\theta}$ $= V_P - V_Q$.

*



*;

=)



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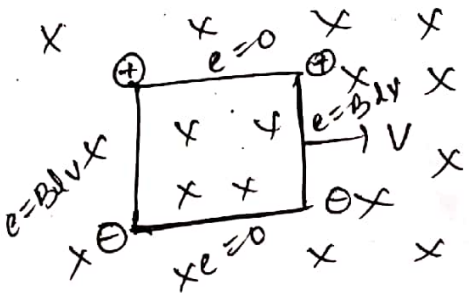

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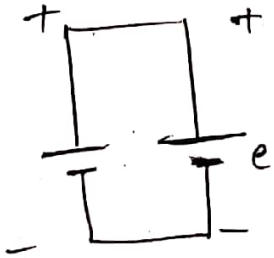
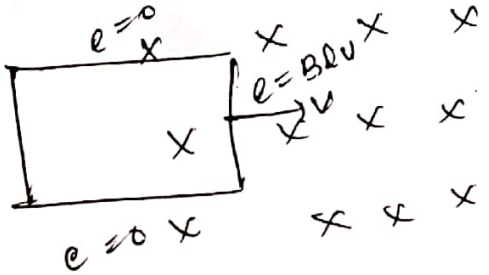


$\boxed{e_{net} = 0}$

*



$\boxed{e_{net} = Blv}$

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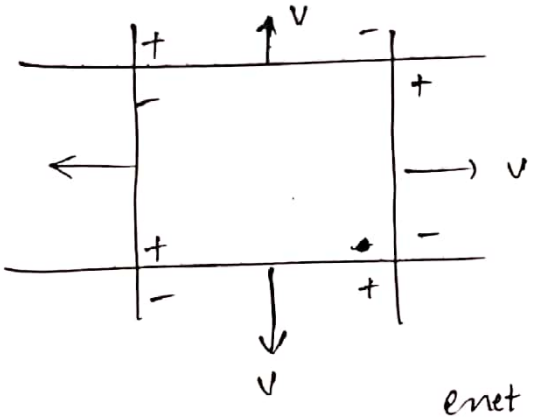
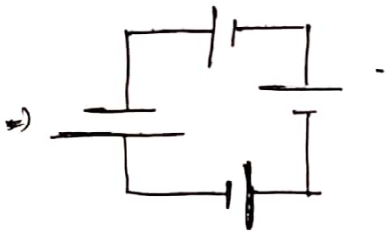
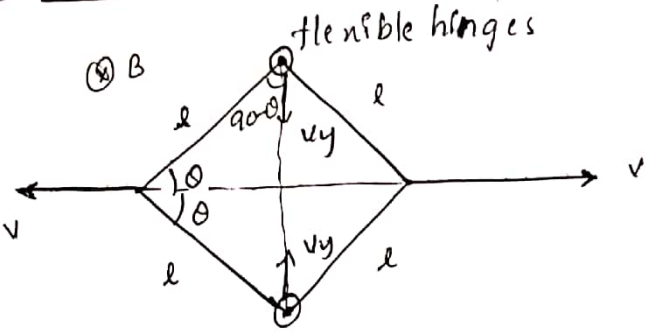




$\mathcal{E}_{net} = 4\mathcal{E} = 4BlV$.

4|06|25| Wednesday

* Constraint Motion :



Constraint equation,

Along the length of rod,

$v\cos\theta = v_y \sin\theta$

$v_y = v\cot\theta \quad -①$

Area of rhombus,

$A = \frac{1}{2} d_1 d_2$

$\frac{dA}{dt} = \frac{1}{2} d_1 \frac{d(d_2)}{dt} + \frac{1}{2} d_2 \frac{d(d_1)}{dt}$

$\frac{d}{dt}(d_1) = 2v$

$\frac{d}{dt}(d_2) = -2v_y$

$\sin\theta = \frac{d_2}{2l}$

$\cos\theta = \frac{d_1}{2l}$

$\frac{dA}{dt} = \frac{1}{2} 2l\cos\theta(-2v_y) + \frac{1}{2} 2l\sin\theta(2v)$

$\frac{dA}{dt} = 2l(v\sin\theta - v_y\cos\theta)$

$= 2l\left(v\sin\theta - v\frac{\cos^2\theta}{\sin\theta}\right)$

$= 2lv\frac{\sin^2\theta - \cos^2\theta}{\sin\theta} = -\frac{2vl\cos 2\theta}{\sin\theta}$

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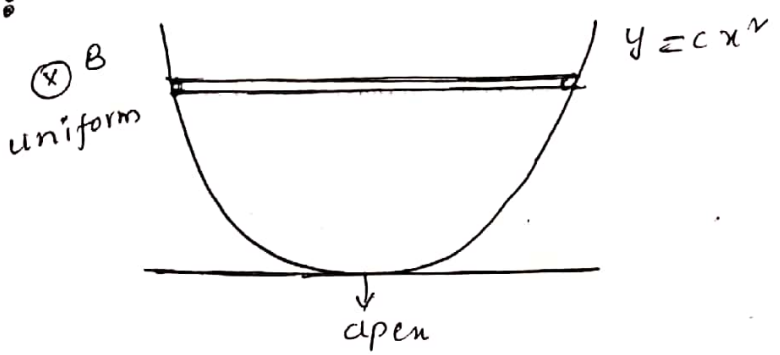
* Induced emf,

$$e = -\frac{B\,dA}{dt}$$

$$e = B(2vl)\frac{\cos 2\theta}{\sin\theta}$$

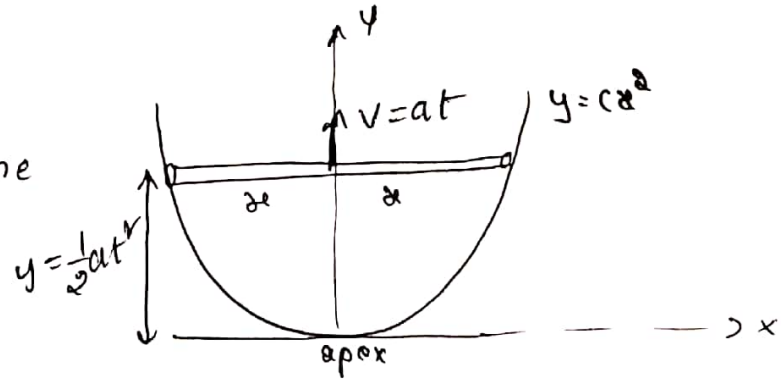
$$\boxed{e = \frac{2Blv\cos 2\theta}{\sin\theta}}$$

Ex:



wire is moving with constant acceleration 'a'. Find induced emf in the loop as a function of time 't'.

Sol:



Induced emf in the rod,

$$e = Blv$$

$$= B(2x)at$$

$$= 2B\sqrt{\frac{y}{c}}\,at$$

$$= 2B\sqrt{\frac{y}{c}} \times a\sqrt{\frac{2y}{a}}$$

$$e = 2By\sqrt{\frac{2a}{c}}$$

$$\boxed{e = By\sqrt{\frac{8a}{c}}}$$

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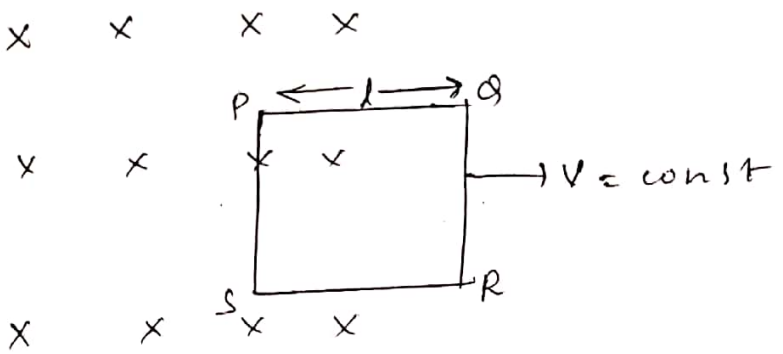
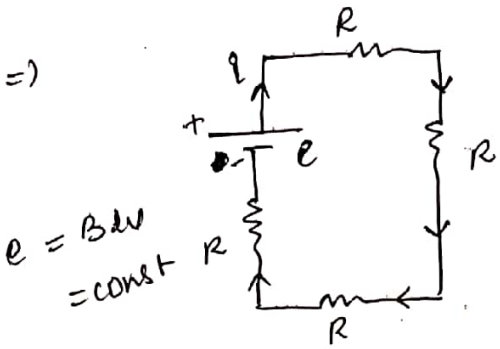
Ex!

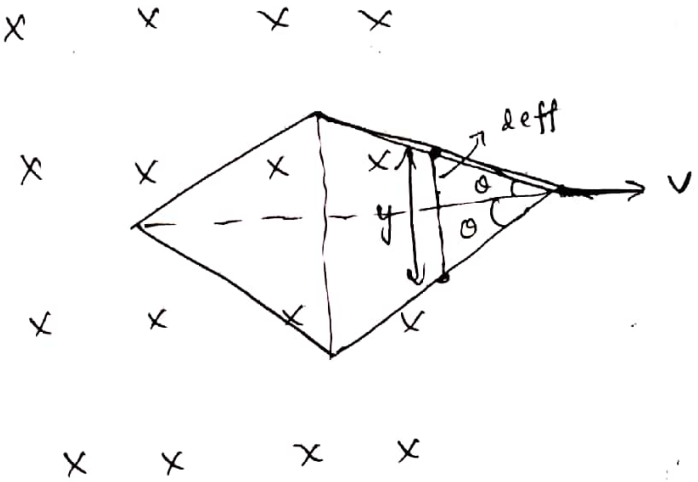




$$i = \frac{e}{4R}$$

$$\therefore i = \frac{BlV}{4R}$$

Ex!



$$e = B\,l_{eff}\,V$$

$$e = B y V$$

As loop is coming out,

'e' first increases from zero to max & then decreases from max to zero.

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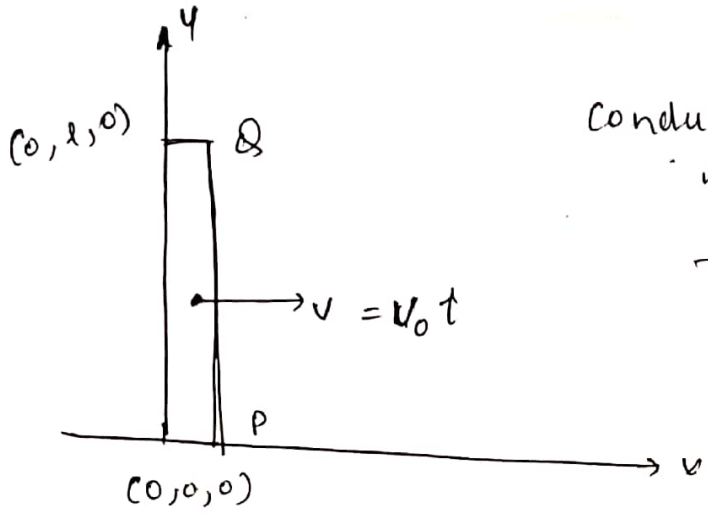
Exe

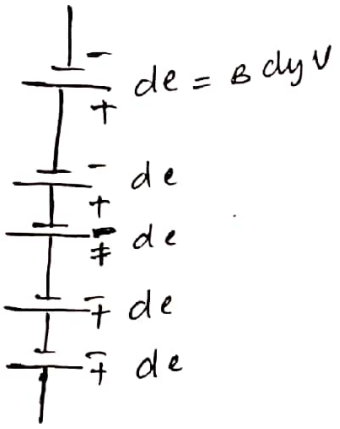


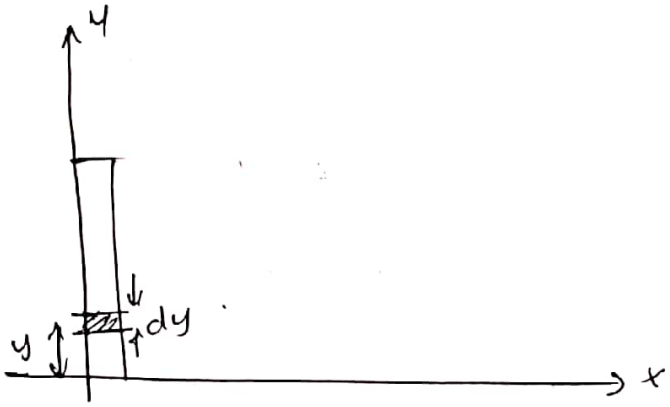
Conductor is placed in "non-uniform magnetic field" $\rightarrow \vec{B} = \frac{B_0 y}{L} \hat{k}$.

Find induced emf across the ends of rod.

Sole

$\Rightarrow$





Induced emf across small element,

$$de = B(dy)v$$

$$= \frac{B_0 y v}{L} dy$$

∴ Total induced emf,

$$e = \int_{y=0}^{y=l} de = \frac{B_0 v}{L} \int_0^L y \, dy$$

$$= \frac{B_0 v}{L} \frac{L^2}{2}$$

$$\boxed{e = \frac{B_0 v L}{2}}$$

∴ end p is +ve.

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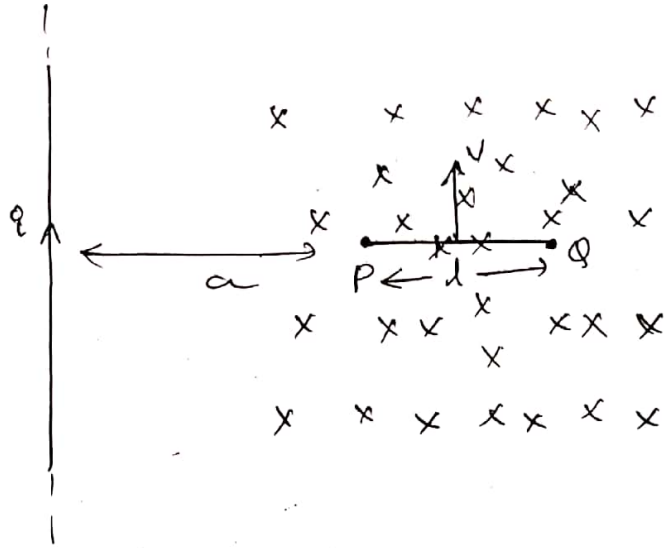
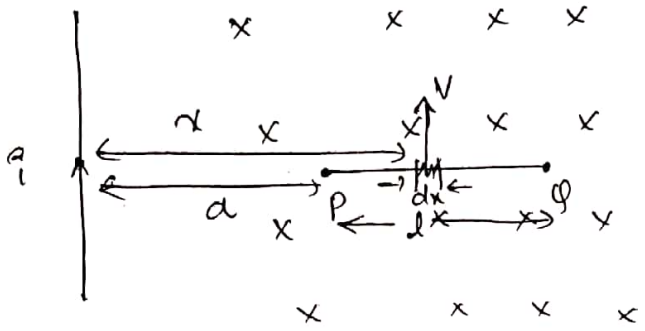
Ex 6



Find Induced emf across P & Q.

Sol:



$$de = B(dx)V$$

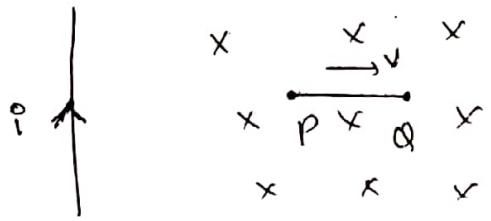
$$= \frac{\mu_0 i}{2\pi x} V\,dx$$

$$\int de = \frac{\mu_0 i V}{2\pi} \int_a^{a+l} \frac{dx}{x}$$

$$e = \frac{\mu_0 i V}{2\pi} (\ln(a+l) - \ln a)$$

$$\boxed{e = \frac{\mu_0 i V}{2\pi} \ln\left(\frac{a+l}{a}\right)}$$

Ex 7



$\boxed{e_{PQ} = 0}$

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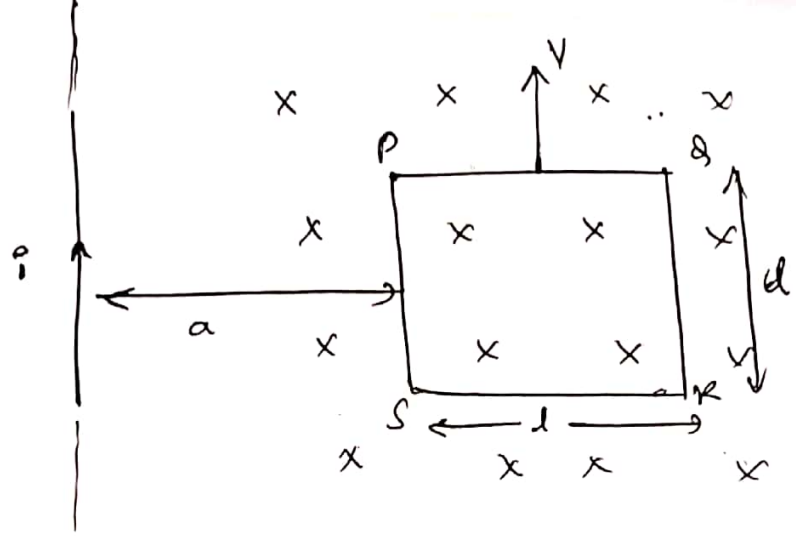
Ex6



loop is moving parallel to wire.

Find induced emf in the loop.

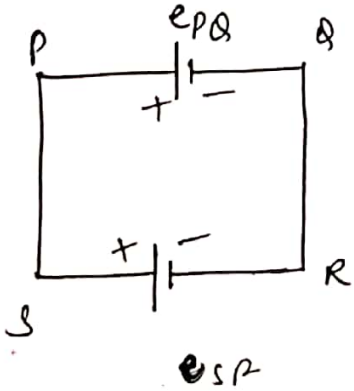
Sol6

$$e_{PQ} = \frac{\mu_0 i v}{2\pi} \ln\left(\frac{a+l}{a}\right)$$

$$e_{QR} = 0$$

$$e_{PS} = 0$$

$$e_{SR} = \frac{\mu_0 i v}{2\pi} \ln\left(\frac{a+l}{a}\right)$$

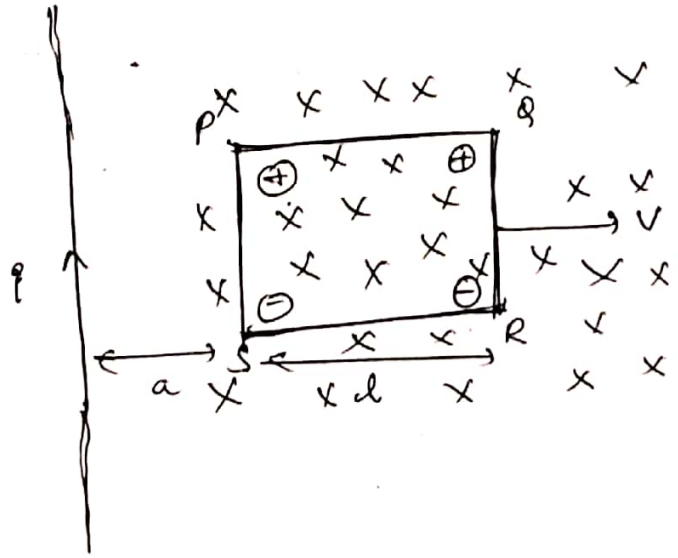


e_{SR}

$$e_{net} = 0$$

It is true for all geometrical shaped loops.

Ex:



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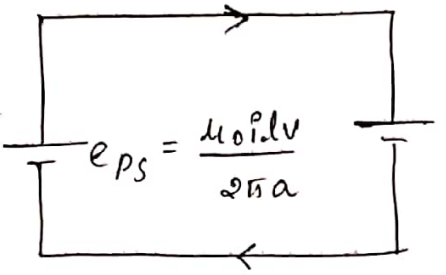
$e_{PQ} = e_{RS} = 0$ $\quad e_{PS} = \frac{\mu_0 i}{2\pi a} dv$

$e_{PS} = B_1 l v$

$e_{QR} = B_2 l v$ $\quad e_{RQ} = \frac{\mu_0 i}{2\pi(a+l)} lv$



$$e_{net} = e_{PS} - e_{QR} = \frac{\mu_0 i l v}{2\pi}\left(\frac{a+l-a}{a(a+l)}\right)$$

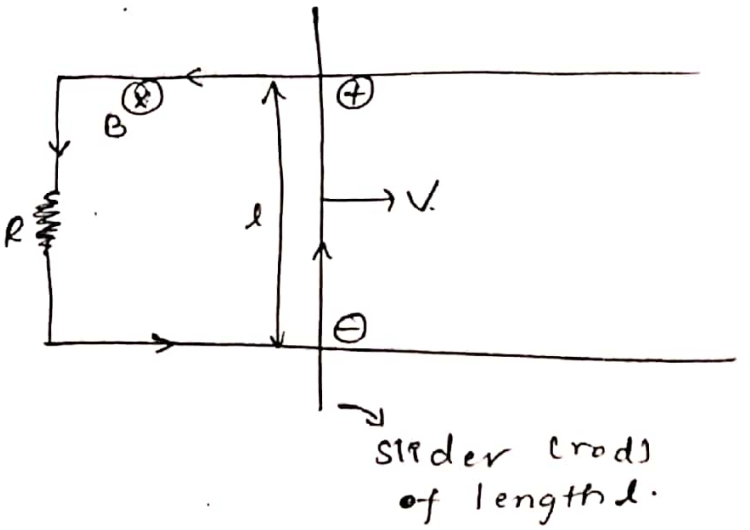
$$\boxed{e = \frac{\mu_0 i l v}{2\pi a(a+l)}} \text{ ; clockwise}$$

* Induced current,

$$i = \frac{e}{R_{Total}}$$

Problems:

Ex: Two smooth conducting rails:-



Let the conductor is projected with initial

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velocity v_0, then

at any time t,

v = speed of rod.

* Induced emf in the rod,

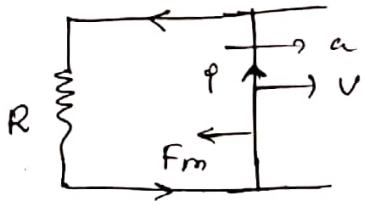
$$e = Blv \; ; \text{anticlockwise}$$

* Induced current,

$$i = \frac{e}{R} = \frac{Blv}{R} \; ; \text{anticlockwise}$$

* Magnetic force acting on the rod,

$$F_m = ilB = \frac{B^2l^2v}{R}$$



From newton's II-Law,

$$ilB = -ma$$

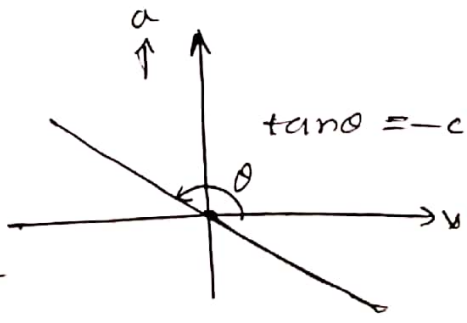
$$\frac{B^2l^2v}{R} = -ma$$

$$\boxed{a = -\frac{B^2l^2v}{mR}}$$

clearly, $\boxed{a \propto v}$

$$\boxed{a = -cv}$$

$$c = \frac{B^2l^2}{mR} = \frac{1}{\tau}$$



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now, $\frac{dv}{dt} = -cv$

$$\int_{v_0}^{v} \frac{dv}{v} = -\int_0^t c\,dt$$

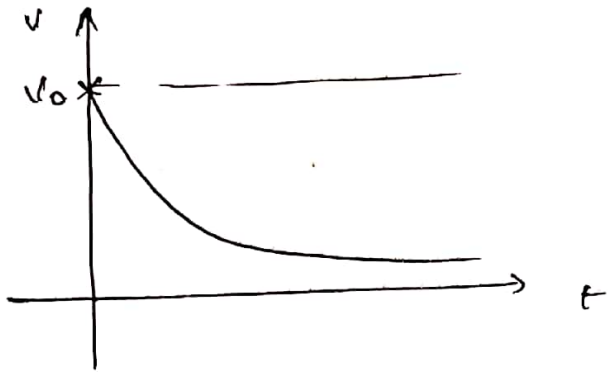
$$\ln(v/v_0) = -ct$$

$$v = v_0 e^{-ct}$$

$$\boxed{v = v_0 e^{-t/\tau}} \quad *$$

* speed decreases exponentially with time.

→ Again,



$a = -cv$

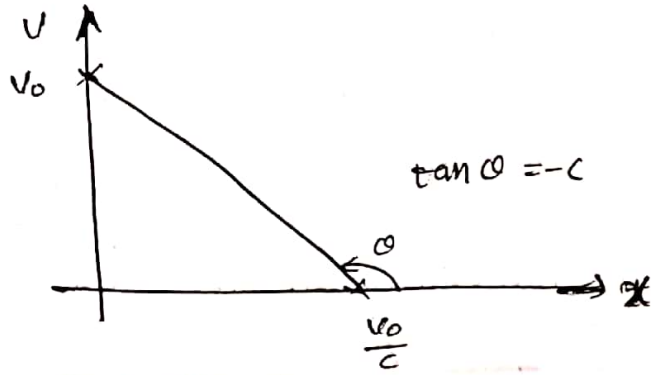
$v\frac{dv}{dx} = -cv$

$$\int_{v_0}^{v} dv = -\int_0^x c\,dx$$

$$v - v_0 = -cx$$

$$\boxed{v = v_0 - cx}$$

* speed decreased linearly with distance travelled.



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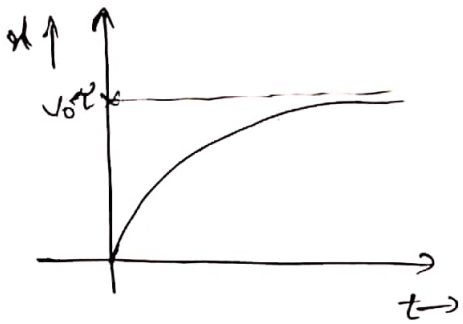
→ Again,

$$v = v_0 e^{-t/\tau}$$

$$\frac{dx}{dt} = v_0 e^{-t/\tau}$$

$$\int_0^x dx = v_0 \int_0^t e^{-t/\tau} \cdot dt$$

$$\boxed{x = +\tau v_0 (1 - e^{-t/\tau})}$$



* distance travelled increases exponentially with time.

→ Induced emf – $e = Blv$

$$\boxed{e = Blv_0 e^{-t/\tau}}$$

* e is decaying exponentially with time.

→ induced current – $i = \frac{e}{R}$

$$\boxed{i = \frac{Blv_0}{R} e^{-t/\tau}}$$

* i_c is decaying exponetially with time.

→ K.E of rod,

$$K.E = \frac{1}{2} mv^2$$

$$\boxed{K.E = \frac{1}{2} m v_0^2 e^{-2t/\tau}}$$

→ Heat produced, $H = \int_0^t i^2 R \, dt$

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$$= \int_0^t \frac{B^2 l^2 v_0^2}{R^2} e^{-2t/\tau} \, (R) \, dt$$

$$= \frac{B^2 l^2 v_0^2}{R} \int_0^t e^{-2t/\tau} \, dt$$

$$= \frac{B^2 l^2 v_0^2}{R} \times \frac{\tau}{2} \left(1 - e^{-2t/\tau}\right)$$

$$= \frac{B^2 l^2 v_0^2 \tau}{2R} \left(1 - e^{-2t/\tau}\right)$$

$$\boxed{H = \frac{1}{2} m v_0^2 \left(1 - e^{-2t/\tau}\right)}$$

* At $t \to \infty$ $\Rightarrow$ 1) $V = 0$

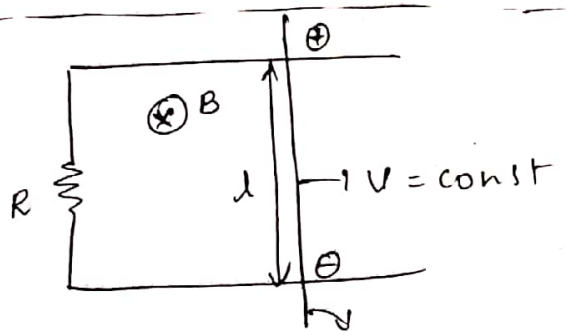
2) $x = v_0 \tau$

3) $e = 0$

4) $i = 0$

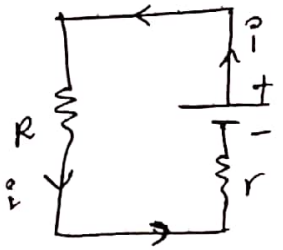
5) $H = \frac{1}{2} m v_0^2$ = initial K.E of rod.

Eg



Resistance = r

$\Rightarrow$



$e = B l v$

$i = \frac{e}{R+r}$

V = const

$\downarrow$

steady state

$\Rightarrow \Delta k = 0$

$\boxed{W = H}$

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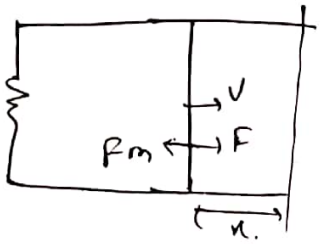

$f = f_m$

$W_F = Fx$

$= f_m x$

$= Bil\, x$

$= B \times \frac{e}{R_{eff}} \times l \times v \times t$

$$W = \frac{e^2}{R_{eff}} t$$

If not in steady state,

$F - F_m = ma$

$F = F_m + ma$

$dW_F = (ilB + ma)dx$

$= ilBdx + madx$

~~$= \frac{e^2}{R_{eff}} dx + \Delta K$~~

~~$= \frac{e^2}{R_{eff}} + \Delta K$~~

$$W = \int dW_F = \int_0^x ilBdx + \int_0^x madx$$

$$= \int_0^t \frac{B^2 l^2 v}{R}(at)\,dt + \Delta K$$

$$= \int_0^t \frac{B^2 l^2 a^2 t^2}{R}\,dt + \Delta K$$

$$W = \frac{B^2 l^2 a^2 t^3}{3R} + \Delta K \Rightarrow W = K + \Delta K$$

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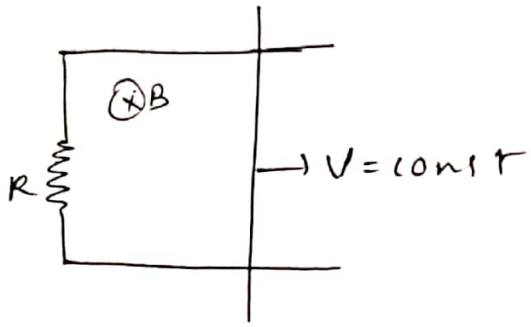
Imp Points

1) $v = \text{const}$

$\Rightarrow$ steady state



$F_{net} = 0$

$F - F_m = 0$

$$F = F_m = IlB = \frac{elB}{reff} = \frac{B^2l^2v}{reff}$$

$$W_F = Fx$$

$$= \frac{B^2l^2v}{Reff} \times vt = \frac{B^2l^2v^2}{ReH}t = \frac{e^2t}{Reff}$$

W_{ext} = heat produced

2)

$F_{net} = F_{nm}$

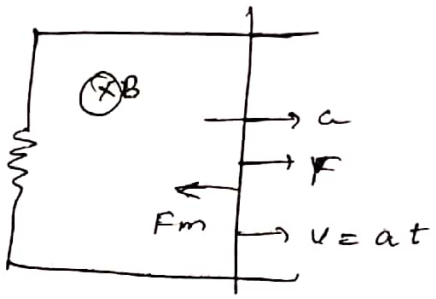
$F - F_m = ma$

$F = F_m + ma$

$= IlB + ma$

$= \frac{Blv}{Reff} + ma$

$$F = \frac{B^2l^2at}{Reff} + ma$$



now,

$$W_F = \int_0^x dW_F = \int_0^x F\,dx$$

$$= \int_0^t Fat \cdot dt$$

$$W_F = \int_0^t \frac{B^2l^2a^2t^2}{Reff} \cdot dt + \frac{ma^2t^2}{2}$$

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$$\omega_F = \frac{B^2 l^2 a^2}{R_{eff}} \times \frac{t^3}{3} + \frac{m a^2 t^2}{2}$$

$$\boxed{\omega_F = H + \Delta K}$$

where,

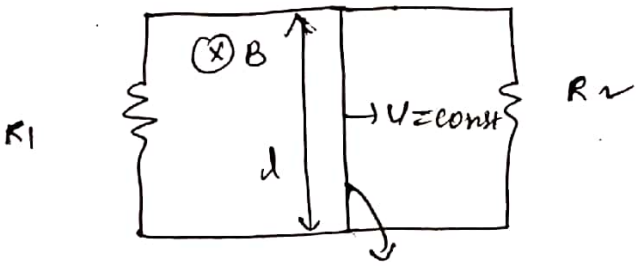
$$H = \int_0^t i^2 R_{eff} \, dt$$

$$= \int_0^t \frac{B^2 l^2 V^2}{R_{eff}} \, dt$$

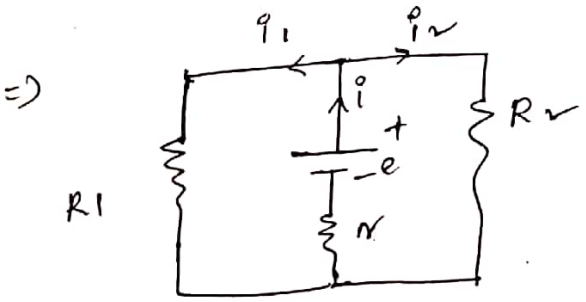
$$= \int_0^t \frac{B^2 l^2 a^2 t^2}{R_{eff}} \, dt$$

$$\boxed{H = \frac{B^2 l^2 a^2 t^3}{3 R_{eff}}}$$

Ex.6



slider of resistance = r



$q = q_1 + q_2$ eq use KVL.

solve for i_1, i_2, i.

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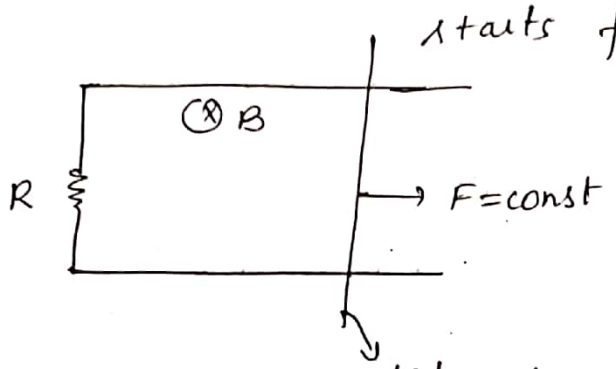
Ex:



Sol: Let v be the speed of rod at any time 't'.

* Induced emf, $e = Blv$

$\Rightarrow$ Induced current, $i = \frac{e}{R} = \frac{Blv}{R}$

* magnetic force on rod,

$$F_m = ilB = \frac{B^2 l^2 v}{R}$$

* net force on the rod,

$$F - F_m = ma$$

$$F - \frac{B^2 l^2 v}{R} = ma$$

$$a = \frac{F}{m} - \frac{B^2 l^2 v}{mR}$$

$$\boxed{a = A - Cv}$$

$A = F/m \qquad C = \frac{B^2 l^2}{mR}$

* acceleration is decreasing linearly with speed.

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Again,

$$\frac{dv}{dt} = A - CV$$

$$\int_0^V \frac{dV}{A-CV} = \int_0^t dt$$

$$\frac{[\ln(A-CV)]_0^V}{-C} = t$$

$$\ln\left(\frac{A-CV}{A}\right) = -Ct$$

$$1 - \frac{CV}{A} = e^{-Ct}$$

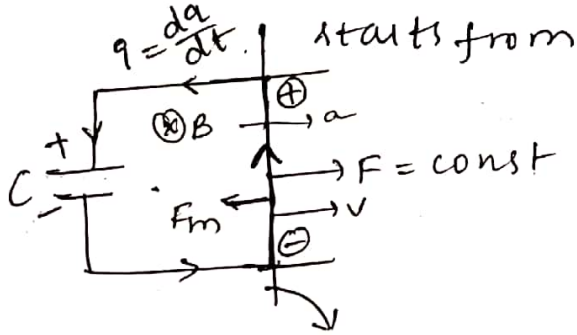
$$1 - e^{-Ct} = \frac{CV}{A}$$

$$\boxed{V = \frac{A}{C}(1-e^{-Ct})}$$

* velocity increased exponentially with time.

$$\boxed{V = \frac{FR}{B^2 l^2}\left(1 - e^{-\frac{B^2 l^2 t}{mR}}\right)}$$

Ex 6



Rod of ⇒ l, m ; R - negligible.

Let 'v' be the speed of rod at any time 't', induced emf, $\boxed{e = Blv}$

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* charge on the plates of capacitor,

$$q = ce$$

$$= c(Blv)$$

now, $$i = \frac{dq}{dt} = \frac{d(Blvc)}{dt}$$

$$i = \frac{dv}{dt}(Blc)$$

$$\boxed{i = Blca}$$

* magnetic force on the conductor,

$$F_m = ilB = (Blca)(lB)$$

$$\boxed{F_m = B^2l^2ca}$$

* net force on the rod,

$$F - F_m = ma$$

$$F - B^2l^2ca = ma$$

$$F = (B^2l^2c + m)a$$

$$\boxed{a = \frac{F}{m + B^2l^2c}} = \text{const.}$$

$\therefore$ $v = at$

$$\boxed{v = \frac{Ft}{m + B^2l^2c}}$$

1) $v = at$

2) $v = \sqrt{2ax}$

3) $x = \frac{1}{2}at^2$

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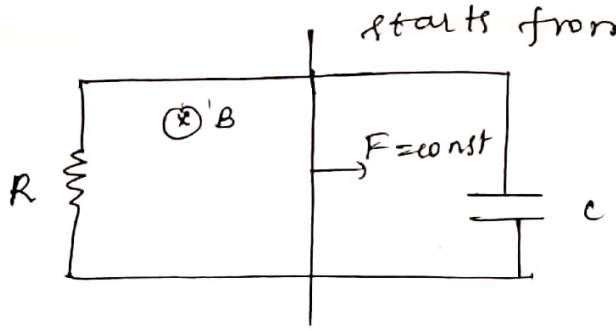
Ex 8

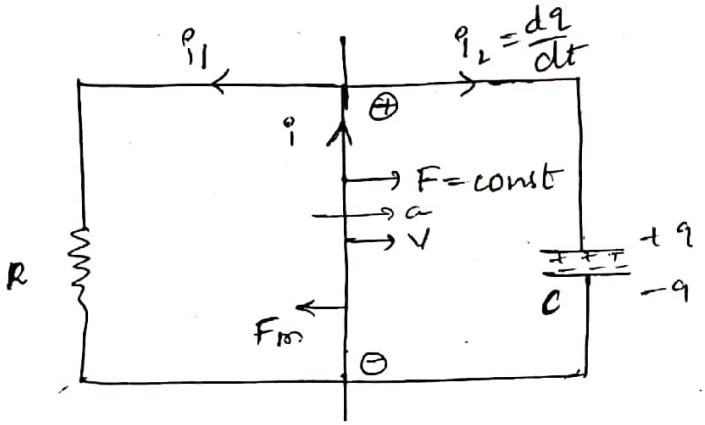


Find speed of rod as a function of time.

Sol 8

Let 'v' be the speed of wire at any instant 't'.

* induced emf, $e = Blv$



we have,

$$e = i, R = \frac{q}{C} = Blv$$

$$i_1 = \frac{Blv}{R} \quad \& \quad q = BlvC$$

$$i_2 = \frac{dq}{dt} = Blca$$

$$\boxed{i = \frac{Blv}{R} + Blca}$$

* magnetic force on the wire,

$$F_m = iLB$$

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* net force,

$$F - F_m = ma$$

$$F - iLB = ma$$

$$F - \left(\frac{BLV}{R} + BLca\right) BL = ma$$

$$F - \frac{B^2L^2V}{R} = (m + B^2L^2c)a$$

$$\boxed{a = k_1 - k_2 V}$$

i.e, acceleration decreases linearly with speed.

$$\frac{dv}{dt} = k_1 - k_2 V$$

$$\int_0^V \frac{dv}{k_1 - k_2 V} = \int_0^t dt$$

$$\frac{\left[\ln(k_1 - k_2 V)\right]_0^V}{-k_2} = t$$

$$\ln(k_1 - k_2 V) - \ln(k_1) = -k_2 t$$

$$\frac{k_1 - k_2 V}{k_1} = e^{-k_2 t}$$

$$1 - \frac{k_2 V}{k_1} = e^{-k_2 t}$$

$$V = \frac{k_1}{k_2}\left(1 - e^{-k_2 t}\right)$$

$$\boxed{V = \frac{FR}{B^2L^2}\left(1 - e^{-\frac{B^2L^2 t}{R(m + B^2L^2c)}}\right)}$$

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i.e, velocity increases exponentially with time.

$$a = k_1 - k_2\left(\frac{k_1}{k_2}\right)(1 - e^{-k_2 t})$$

$$= k_1\left(1 - 1 + e^{-k_2 t}\right)$$

$$a = k_1 e^{-k_2 t}$$

$$a = \frac{F e^{-\frac{B^2 l^2 t}{R(m + B^2 l^2 c)}}}{(m + B^2 l^2 c)}$$

i.e, acceleration decreases exponentially with time.

$$v = \frac{dx}{dt} = \frac{k_1}{k_2}(1 - e^{-k_2 t})$$

$$\int_0^x dx = \frac{k_1}{k_2}\int_0^t (1 - e^{-k_2 t})$$

$$= \frac{k_1}{k_2}\left(t + \frac{e^{-k_2 t}}{k_2}\right)_0^t$$

$$x = \frac{k_1}{k_2}\left(t + \frac{e^{-k_2 t}}{k_2} - \frac{1}{k_2}\right)$$

$$x = \frac{k_1}{k_2}\left(t - \frac{v}{k_1}\right)$$

$$x = \frac{k_1}{k_2} t - \frac{v}{k_2}$$

* x may increase or decrease.

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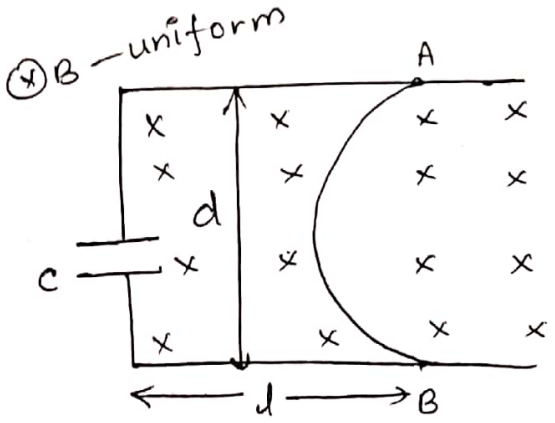
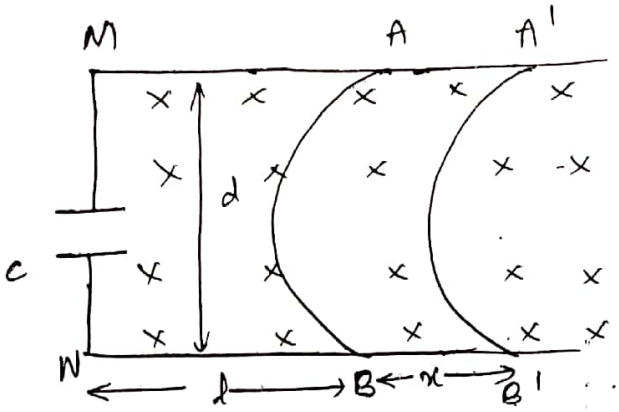
5/06/25 Thursday

Ex 1



If the displacement of semi circular arc wire from initial position is $x = a\sin\omega t$. Find max induced current in the circuit.

Soln



At any time 't',

Area of loop A'MNB'1,

$$A = d(d+x) - \frac{\pi (d/2)^2}{2}$$

$$= d(d+x) - \frac{\pi d^2}{8}$$

$$= dl + xd - \frac{\pi d^2}{8}$$

$\therefore$ flux, $\phi = BA = B\left(dl + dx - \frac{\pi d^2}{8}\right)$

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∴ induced emf,

$$e = \frac{d\phi}{dt} = B\,d\,\frac{dx}{dt}$$

$$= B\,d\,\frac{d}{dt}(a\sin\omega t)$$

$$= B\,d\,a\cos\omega t\,(\omega)$$

$$\boxed{e = Bda\omega\cos\omega t}$$

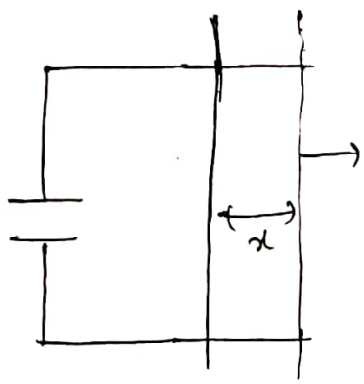
* $q = Ce$

$= Bda\omega C\cos\omega t$

$$i = \frac{+dq}{dt} = -Bdq\omega^{r}C\sin\omega t$$

$$\boxed{q_{max} = Bda\omega^{r}C}$$

Short cut!



$\rightarrow V = \frac{dx}{dt} \Rightarrow e = BdV$

q, i

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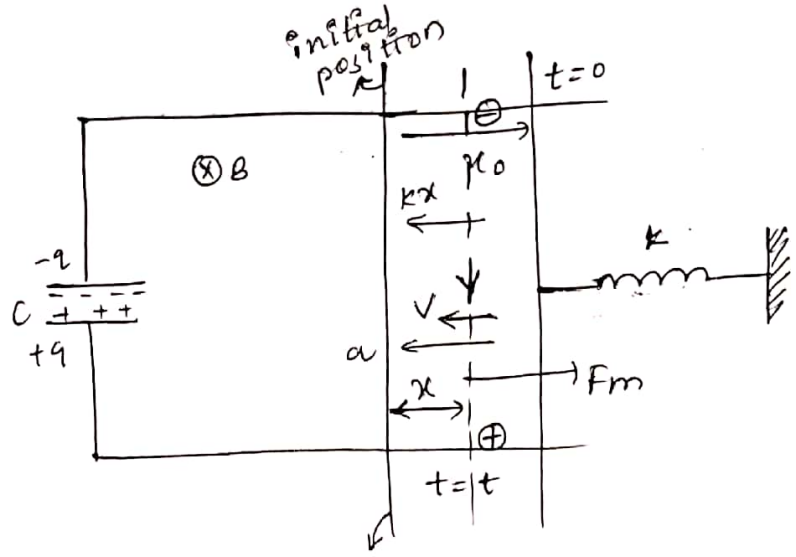
Exe



length ℓ, mass m

[spring is at it's natural length]

Let the spring is compressed by no eq released. Find time taken by wire to reach it's initial position.

Sole At any time 't',

induced emf in wire (e) = $B\ell v$

charge (q) = ce

$$q = B\ell vc$$

$$i = \frac{dq}{dt} = B\ell ca$$

* magnetic force on the wire,

$$F_m = i\ell B = B^2\ell^2 ca$$

net force,

$$kx - F_m = ma$$

$$kx - B^2\ell^2 ca = ma$$

$$kx = (m + B^2\ell^2 c)a$$

$$\boxed{a = \frac{-kx}{m + B^2\ell^2 c}}$$

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clearly, $a \propto -x$

∴ motion of wire is S.H.M.

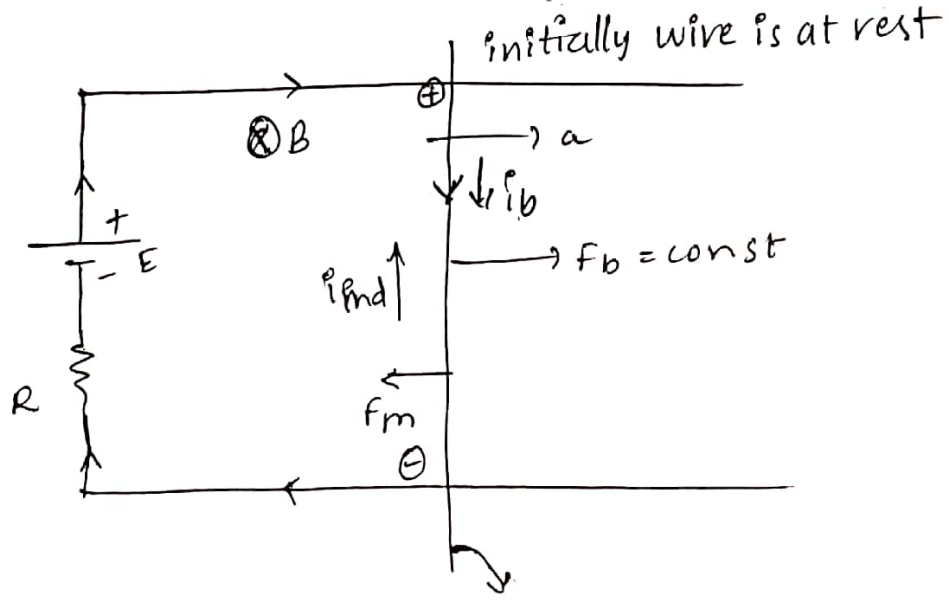
$$\omega = \sqrt{\frac{K}{m + B^2 l^2 C}}$$

$$***T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m + B^2 l^2 C}{K}}$$

Required time,

$$t = \frac{T}{4} = \frac{\pi}{2} \sqrt{\frac{m + B^2 l^2 C}{K}}$$

Ex:



current from battery,

$$i_b = \frac{E}{R}$$

magnetic force on wire,

$$F_b = i_b l B = \frac{E l B}{R}$$

* Let at any time 't' V be the speed of wire,

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induced emf, $e = Blv$

Induced current, $i_{ind} = \frac{e}{R} = \frac{Blv}{R}$

magnetic force on the wire,

$$F_m = i_{in} l_B$$

$$= \frac{B^2 l^2 v}{R}$$

$$F_b - F_m = ma$$

$$\frac{EdB}{R} - \frac{B^2 l^2 v}{R} = ma$$

$$\frac{EdB - B^2 l^2 v}{mR} = a$$

$$k_1 - k_2 v = a$$

where $k_1 = \frac{EdB}{mR}$

$$\frac{dv}{dt} = k_1 - k_2 v$$

$k_2 = \frac{B^2 l^2}{mR}$

$$\int_0^v \frac{dv}{k_1 - k_2 v} = \int_0^t dt$$

$$\frac{[\ln(k_1 - k_2 v)]_0^v}{-k_2} = t$$

$$1 - \frac{k_2}{k_1} v = e^{-k_2 t}$$

$$v = \frac{k_1}{k_2}(1 - e^{-k_2 t})$$

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$$v = \frac{E}{Bl}\left(1 - e^{-\frac{B^2 l^2 t}{mR}}\right)$$

speed increases exponentially with time,

when $t \to \infty$,

$$v = \frac{E}{Bl} = max$$ ← terminal velocity.

At this velocity,

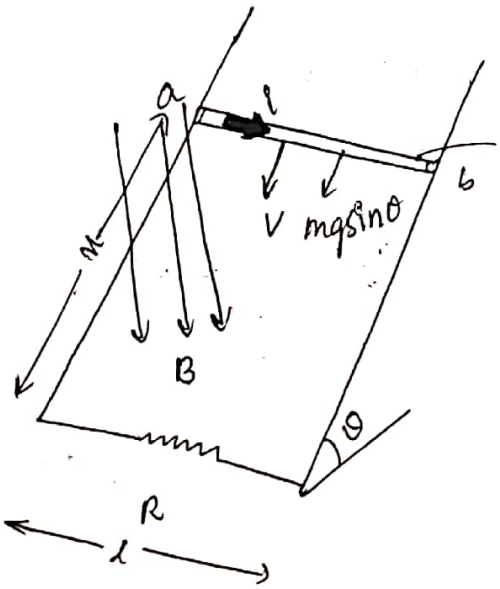
$$i_{ind} = \frac{Blv}{R} = \frac{E}{Bl} \times \frac{Bl}{R}$$

$$i_{ind} = \frac{E}{R}$$

∴ net current through wire $= i_b - i_{ind}$

$= 0.$

Conducting rails on Inclined planes



wire of length 'l' & mass 'm'.

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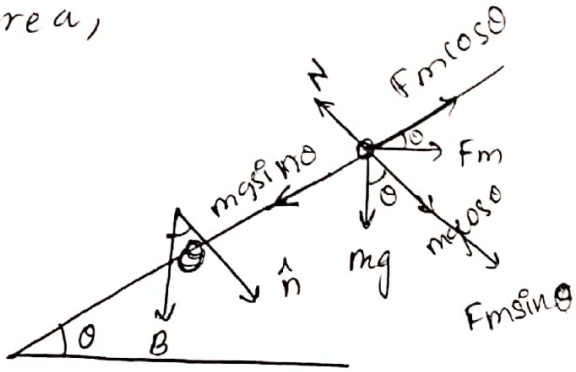
* At any instant,

M-I :
flux linked with area,

$$\phi = \bar{B} \cdot \bar{A}$$



$$\phi = BA\cos\theta$$

$$\phi = B l x \cos\theta$$

now, $e = \frac{d\phi}{dt} = Bl\cos\theta \frac{dx}{dt}$

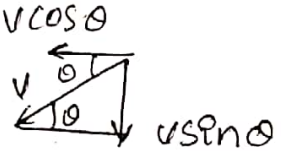
$$\boxed{e = Blv\cos\theta} \quad **$$

M-II :

$$\bar{e} = \bar{l} \cdot (\bar{V} \times \bar{B})$$

$$\bar{l} = l\hat{\uparrow}$$

$$\bar{B} = B(-\hat{\uparrow})$$



$$\bar{V} = v\cos\theta(\hat{k}) + v\sin\theta(-\hat{\uparrow})$$

$$\bar{e} = lBv\,l\,(\hat{\uparrow} \cdot \cos\theta\,\hat{\uparrow})$$

$$\boxed{\bar{e} = Blv\cos\theta}$$

* $i = \frac{e}{R} = \frac{Blv\cos\theta}{R}$

* Magnetic force, $F_m = i(\bar{l} \times \bar{B}) = iJB(-\hat{k})$

$$\therefore F_m = \frac{B^2 l^2 v\cos\theta}{R}$$

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net force on the wire,

$$F_{net} = mg\sin\theta - F_m\cos\theta$$

$$ma = mg\sin\theta - \frac{B^2 l^2 v \cos^2\theta}{R}$$

* at terminal velocity,

$$mg\sin\theta = \frac{B^2 l^2 v_T \cos^2\theta}{R}$$

$$v_T = \frac{mgR\sin\theta\sec^2\theta}{B^2 l^2}$$

$$\boxed{v_T = \frac{mgR\sin\theta}{B^2 l^2 \cos^2\theta}}$$

Here, in steady state.

* rate of workdone by gravity,

$$P_g = (mg\sin\theta)\, v_T$$

$$= (mg\sin\theta)$$

$$P_g = mg\sin\theta \frac{B^2 l^2 v_T^2 \cos^2\theta}{R}$$

* rate of heat produced in circuit,

$$P_R = i^2 R = \frac{B^2 l^2 v_T^2 \cos^2\theta}{R}$$

clearly, $\boxed{P_g = P_R}$

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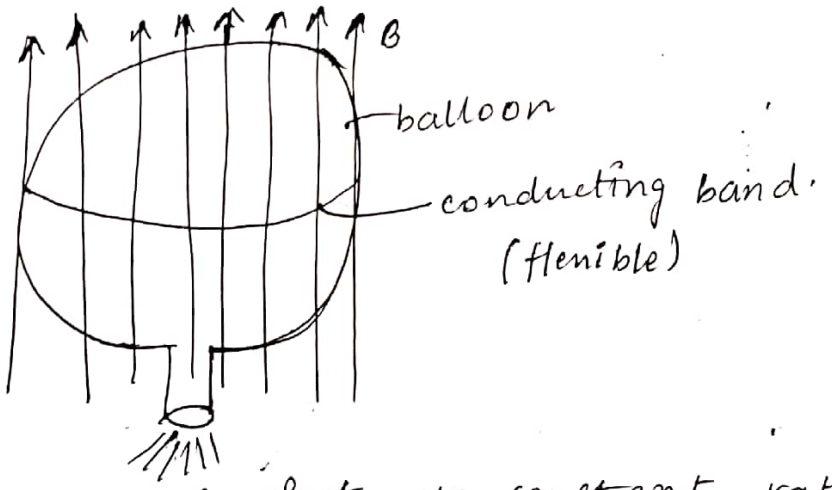
Ex 8



air ejected at constant rate $\frac{dv}{dt}$

At any instant, if r is the radius of balloon,

* volume of balloon,

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dv}{dt} = \frac{4}{3}\pi (3r^2)\cdot \frac{dr}{dt}$$

$$\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt}$$

* magnetic flux in the band,

$$\phi = BA$$

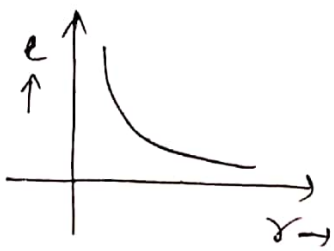
$$\phi = B(\pi r^2)$$

now, $e = \frac{d\phi}{dt} = 2\pi Br\frac{dr}{dt}$

$$e = 2\pi B r \times \frac{1}{4\pi r^2}\cdot \frac{dv}{dt}$$

$$\boxed{e = \frac{B}{2r}\cdot\frac{dv}{dt}}\ *$$

$\Rightarrow (e \propto \frac{1}{r})$



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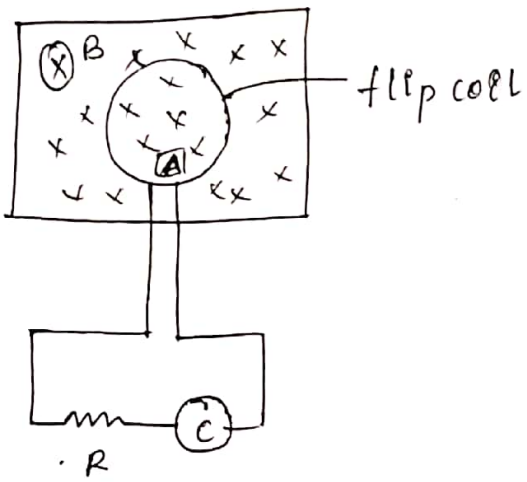
Ex 8



If coil is rotated through an angle of 180° in Δt sees from initial position. Then,

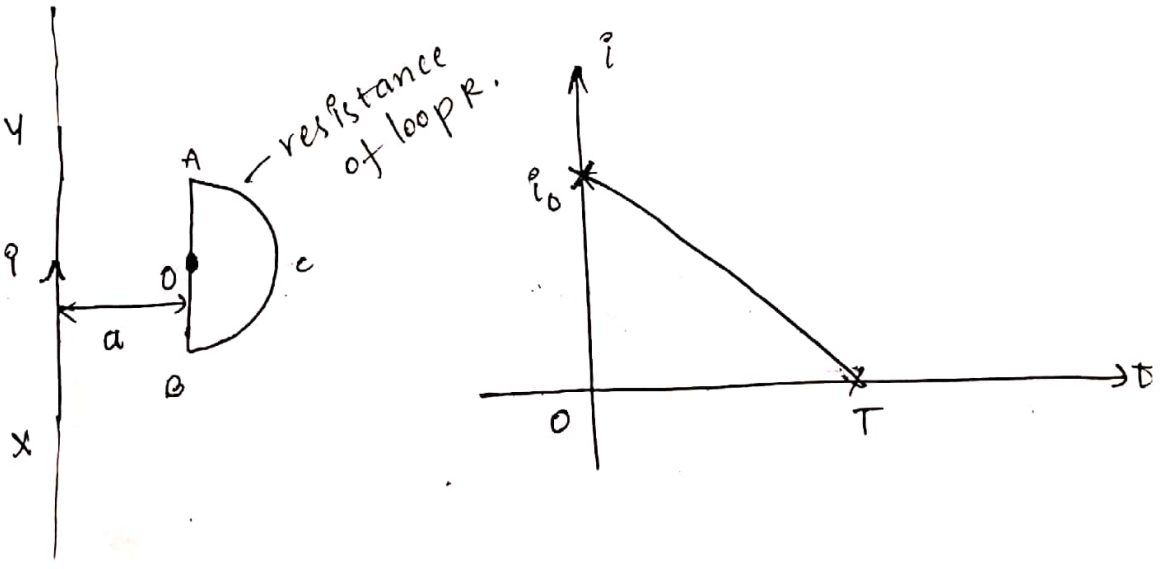
Sol^n $\phi_i = NBA\cos 0° = NBA$, $\phi_f = NBA\cos 180° = -NBA$

1) $e_{avg} = \frac{\Delta\phi}{\Delta t} = \frac{2NBA}{\Delta t}$

2) $i_{avg} = \frac{e_{avg}}{R} = \frac{2NBA}{R\Delta t}$

3) $\Delta q = \frac{\Delta\phi}{R} = \frac{2NBA}{R}$.

Ex 9



$AO = OB = OC = a$

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A total charge Q flows through long wire such that current decreases linearly from initial value to zero in time T. Find

1) current in XY as a function of time

2) Magnetic flux (ϕ) through ACBOA when current in XY is i.

3) Induced current in ACBOA as a function of time. Ans: $i_1 = \frac{\mu_0 a (\pi-2) Q}{\pi R T^2} = \text{const.}$

4) Heat generated in ACBOA in time T.

Ans: $H = \frac{a^2 \mu_0^2 (\pi-2)^2 Q^2}{\pi^2 R T^3}$

5) Force acting on 1st path AOB as a function of time. Ans: $F = \frac{\mu_0^2 Q^2 (\pi-2)}{\pi^2 R T^3}\left(1-\frac{t}{T}\right)$

Sol:

1) Total charge,

Q = Area under $i-t$ graph.

$$Q = \frac{1}{2}(T)(i_0)$$

$$i_0 = \frac{2Q}{T}$$

From fig, $y = mx + c$

$$i = -\frac{i_0 t}{T} + i_0$$

$$i = i_0\left(1-\frac{t}{T}\right)$$

$$\boxed{i = \frac{2Q}{T}\left(1-\frac{t}{T}\right)}$$

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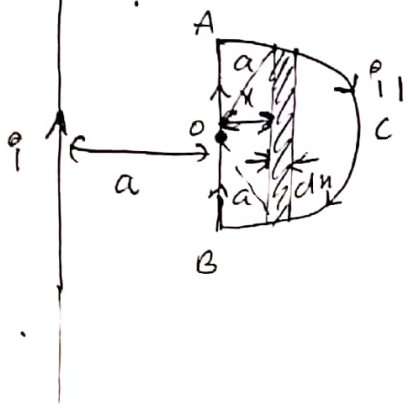
2)



$$dA = 2(\sqrt{a^2-x^2})\cdot dx$$

$$d\phi = B\,dA = 2B\sqrt{a^2-x^2}\cdot dx$$

$$d\phi = 2\frac{\mu_0 i}{2\pi(a+x)}\sqrt{a^2-x^2}\cdot dx$$

$$\phi = \int_{x=0}^{x=a} d\phi = \frac{\mu_0 i}{\pi}\int_{x=0}^{x=a}\frac{\sqrt{a^2-x^2}}{a+x}\cdot dx$$

$$= \frac{\mu_0 i}{\pi}\int_{x=0}^{x=a}\sqrt{\frac{a-x}{a+x}}\cdot dx$$

$x = a\cos 2\alpha$

$dx = -2a\sin 2\alpha\, d\alpha$

$$= \frac{\mu_0 i}{\pi}\int \sqrt{\frac{2\sin^2\alpha}{2\cos^2\alpha}}(-2a\sin 2\alpha)\,d\alpha$$

$$= \frac{\mu_0 i}{\pi}\int \frac{\sin\alpha}{\cos\alpha}(-2a)\,2\sin\alpha\cos\alpha\cdot d\alpha$$

$$= -\frac{\mu_0 i}{\pi}(4a)\int \sin^2\alpha\cdot d\alpha$$

$$= -\frac{\mu_0 i 4a}{\pi}\int \frac{1-\cos 2\alpha}{2}\cdot d\alpha$$

$$= -\frac{\mu_0 i(4a)}{\pi}\left[\frac{\alpha}{2}-\frac{\sin 2\alpha}{4}\right]_{\pi/4}^{0}$$

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$$= \frac{\mu_0 i (ka)}{\pi} \left[\frac{\pi}{\beta_2} - \frac{1}{k} \right]$$

$$\boxed{\phi = \frac{\mu_0 i}{\pi} \frac{a}{2} (\pi - 2)}$$

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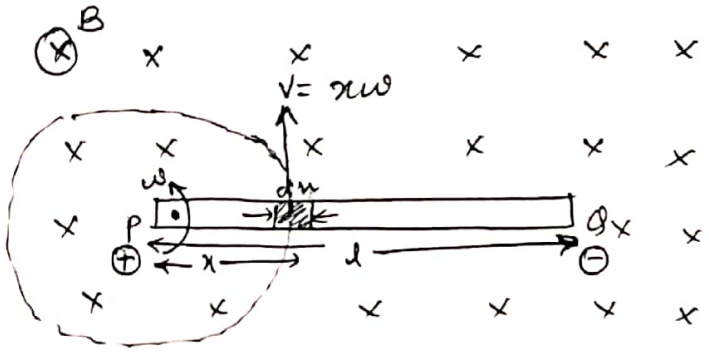
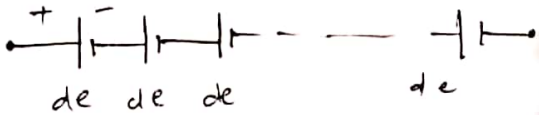
Motional emf due to rotational motion of rod:-

* Rod is conducting rod.



rod is rotating in vertical plane.

* $de = B(dx)V$ ⇒



$de = B x\omega\, dx$

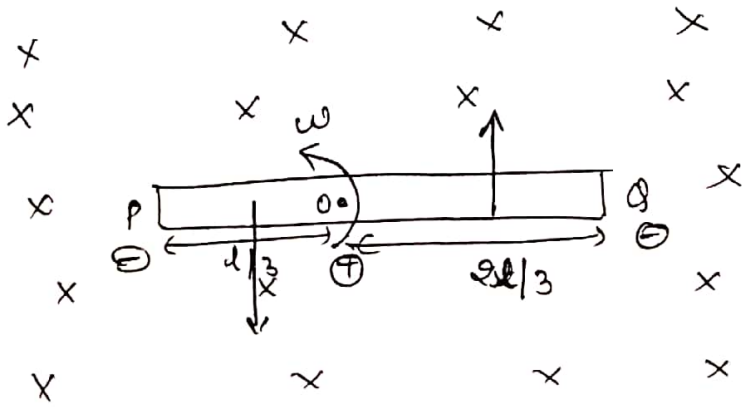
* net induced emf.

$$V_P - V_Q = e = \int_{x=0}^{x=l} de = \int_{x=0}^{x=l} B\omega x \cdot dx = \frac{1}{2} B\omega l^2$$

* If the rod is rotated in horizontal plane,

$$\boxed{e = 0}$$

*



$$V_O - V_Q = \frac{1}{2} B\omega \left(\frac{2l}{3}\right)^2$$

$$V_O - V_P = \frac{1}{2} B\omega \left(\frac{l}{3}\right)^2$$

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now,

$$V_p - V_Q = (V_O - V_Q) - (V_O - V_P)$$

$$= \frac{1}{2} B\omega \left(\frac{4l^2 - l^2}{9} \right)$$

$$V_p - V_Q = \frac{B\omega l^2}{6} = +ve$$

$$\boxed{V_p > V_Q}$$

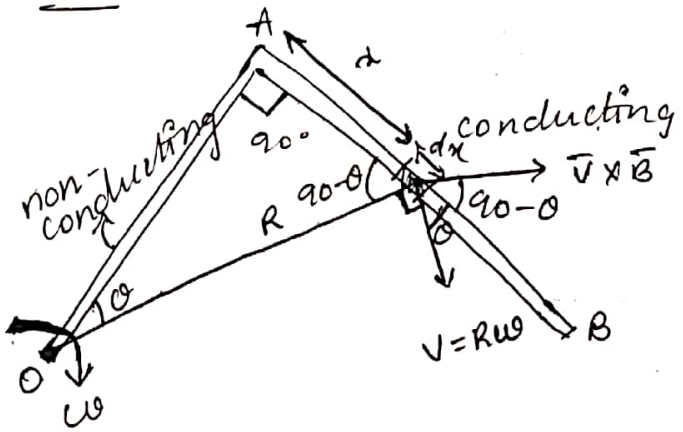
(or)

$$V_p - V_Q = \int_{-l/3}^{2l/3} B\omega x \, dx$$

$$= \frac{1}{2} B\omega \left[\left(\frac{2l}{3} \right)^2 - \left(\frac{l}{3} \right)^2 \right]$$

$$= \frac{B\omega l^2}{6} \quad \therefore$$

L-shaped rod :-



AB = L

from fig,

$$\sin\theta = \frac{x}{R}$$

* all elements are rotating in circular paths with same angular velocities but with different linear speeds.

Induced emf of small element,

$$de = d\vec{l} \cdot (\vec{v} \times \vec{B})$$

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$$de = dx\,(vB\sin 90^\circ)\cos(90-\theta)$$

$$de = vB\,dx\sin\theta$$

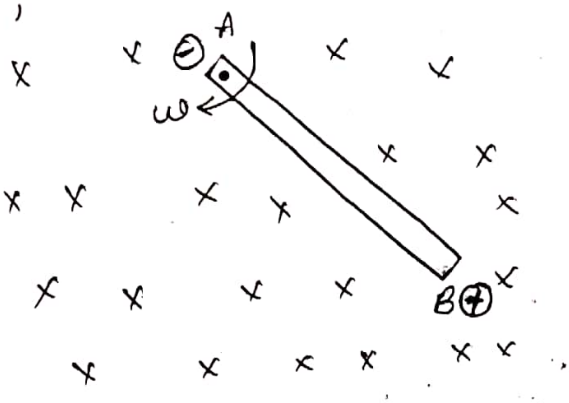
$$de = R\omega B\,dx\sin\theta$$

$$de = B\omega x\cdot dx$$

$$e = \int_{x=0}^{x=l} de$$

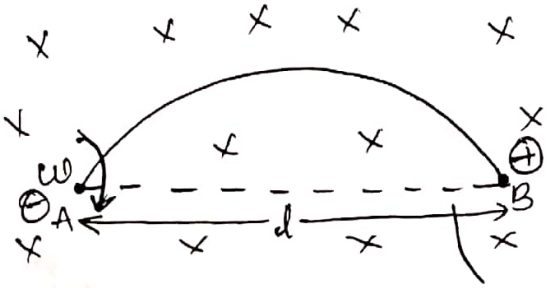
$$\boxed{e = \frac{1}{2} B\omega l^2}$$

Clearly,



$e = \frac{1}{2} B\omega l^2$.

Ex:



$\Rightarrow \phi = \text{const} \Rightarrow e = 0$

considering a line to close the loop.

For st. wire, $e = \frac{1}{2} B\omega l^2$;

B is a high potential

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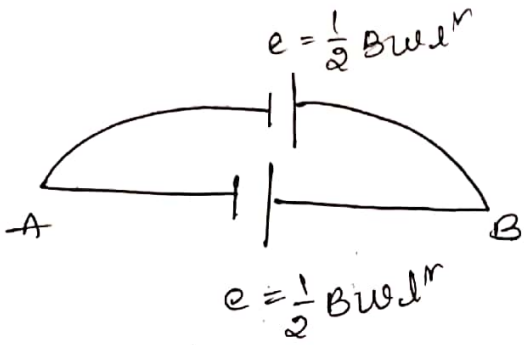
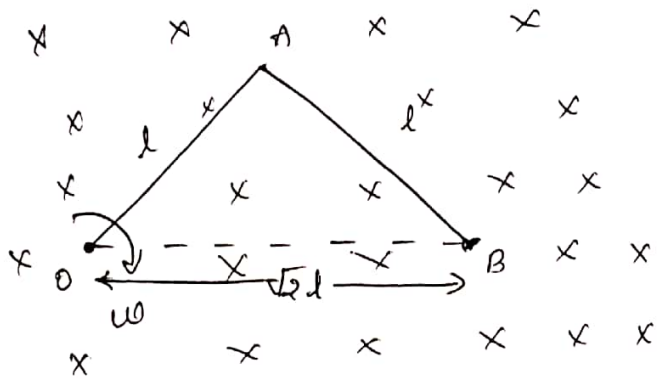


Independent of shape of wire AB.

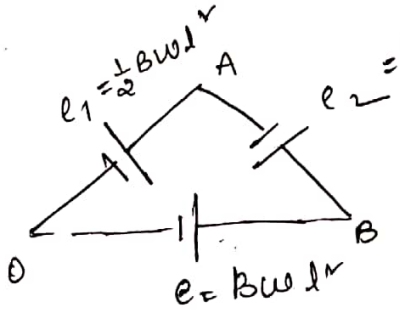
Ex:



Find induced emf across AB

Sol:

$$e_{OB} = \frac{1}{2} B\omega (\sqrt{2}l)^2 = B\omega l^2$$

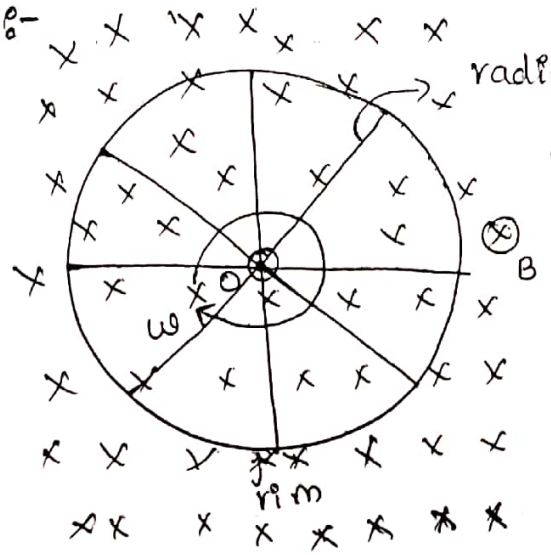


$$e = e_1 + e_2$$

$$B\omega l^2 = \frac{1}{2} B\omega l^2 + e_2$$

$$\boxed{e_2 = \frac{1}{2} B\omega l^2}$$

Rotating Wheel :-



radius, resistance of each spoke is R, R_1 respectively.

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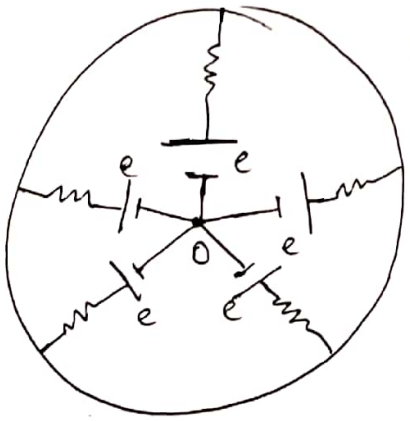
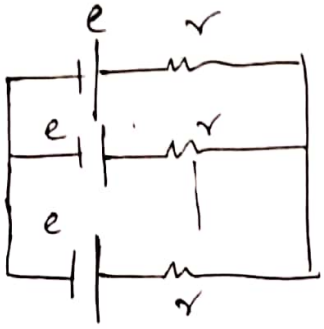


$\Rightarrow$



$\Rightarrow$

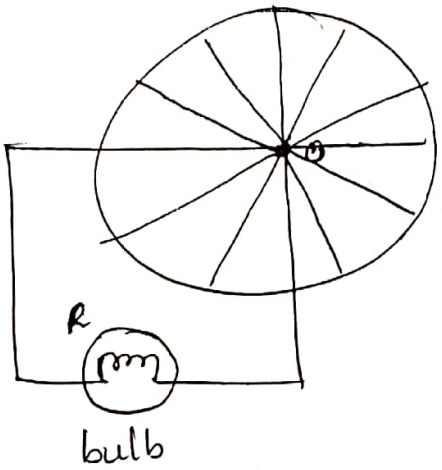
$$e_{net} = \frac{\Sigma e_i/r_i}{\Sigma 1/r_i}$$

$$= \frac{N e/r}{N/r}$$

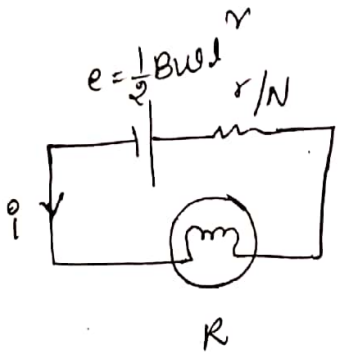
$$r_{net} = r/N$$

$$\boxed{e_{net} = e}$$

If it is connected with a bulb,



$\Rightarrow$



$$\boxed{i = \frac{e}{R + \frac{r}{2N}}}$$

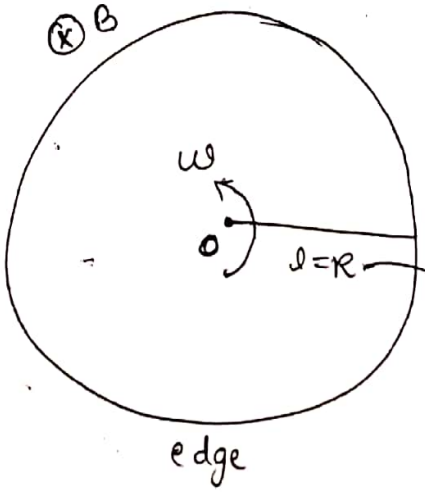
* If $N \to \infty$, wheel becomes disc.

Rotating Conducting Disc :-

$$\boxed{V_{centre} - V_{edge} = \frac{1}{2}B\omega R^2}$$

$\downarrow$

This is dominating



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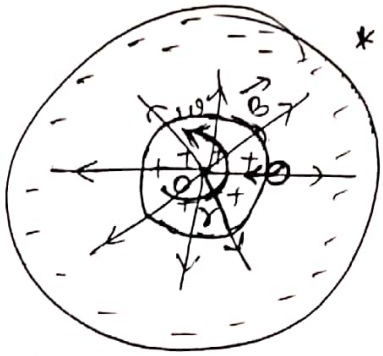
* without magnetic field :-



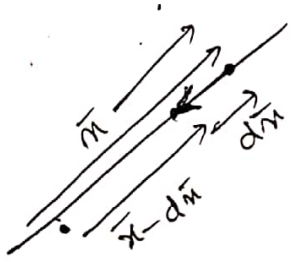
* due to centrifugal force during rotation electrons comes to the edge radially outward

* direction of ω is not necessary.

* after sufficient electric field is created by e^-s, it rotates in circular path with radius r due to neccessary (sufficient centripetal force acting towards centre.

* In eqm,

$$eE = m_e r\omega^2$$



coulombic field $E = \dfrac{m_e r \omega^2}{e}$

$$\int_{V_{edge}}^{V_{centre}} dV = -\int_{r=R}^{r=0} \bar{E}\cdot d\bar{l}$$

$$V_{centre} - V_{edge} = \int_{r=0}^{r=R} \frac{m_e r\omega^2}{e}(\hat{r})\cdot dr(\hat{r})$$

$$\boxed{V_{centre} - V_{edge} = \frac{m_e \omega^2 R^2}{2e}}$$

↓

This is maximum negligible.

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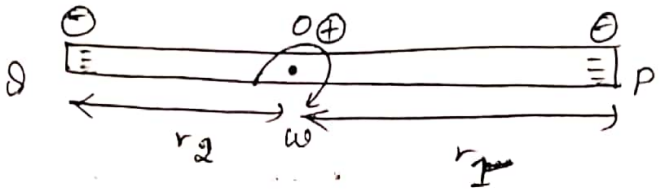


Here, $V_O - V_P = \frac{m_e \omega^2 r_1^2}{2e}$

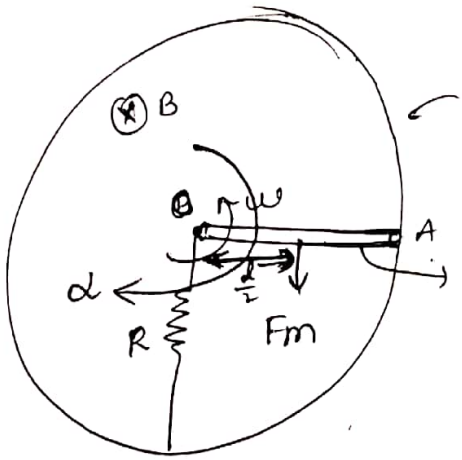
$$V_O - V_Q = \frac{m_e \omega^2 r_2^2}{2e}$$

$$V_Q - V_P = (V_O - V_P) - (V_O - V_Q)$$

$$= \frac{m_e \omega^2 r_1^2}{2e} - \frac{m_e \omega^2 r_2^2}{2e}$$

$$\boxed{V_Q - V_P = \frac{m_e \omega^2}{2e}(r_1^2 - r_2^2)}$$

Ex!



conducting ring of negligible resistance.

rod of l, m

Let ω_0 is the initial angular velocity,

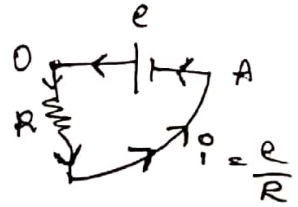
At any time t,

'ω' is the angular velocity of rod.

$\therefore$ induced emf, $e = \frac{1}{2} B\omega l^2$, with O is +ve,

induced current,

$$i = \frac{e}{R} = \frac{\frac{1}{2} B\omega l^2}{R}$$



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* magnetic force on the rod,

$$F_m = iLB = \frac{B^2 l^3 \omega}{2R}$$

* Torque about 'O',

$$\tau = F_m \times \frac{l}{2} = \frac{B^2 \omega l^4}{4R}$$

$\tau = I\omega$

$$\frac{ml^2}{3}\alpha = -\frac{B^2 \omega l^4}{4R}$$

$$\boxed{\alpha = -\frac{3B^2 l^2}{4mR}\omega}$$

$$\Rightarrow \boxed{\alpha = -k_1 \omega}$$ where, $k_1 = \frac{3B^2 l^2}{4mR}$

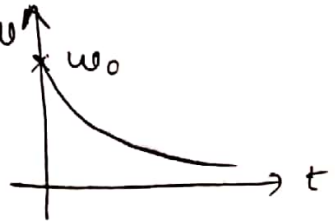
$$\frac{d\omega}{dt} = -\frac{3B^2 l^2}{4mR}\omega$$

$$\int_{\omega_0}^{\omega} \frac{d\omega}{\omega} = \int_0^t k_1 dt$$

$$\ln(\omega/\omega_0) = -k_1 t$$

$$\boxed{\omega = \omega_0 e^{-k_1 t}}$$

i.e, angular velocity decrease exponentially with time



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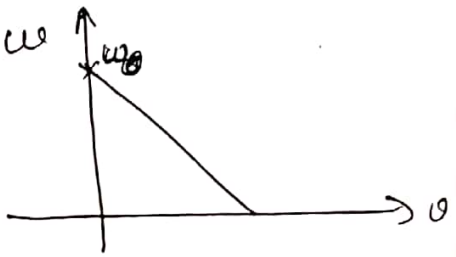
Again,

$$\alpha \frac{d\omega}{d\theta} = -k_1 \alpha$$

$$\int_{\omega_0}^{\omega} d\omega = -\int_0^{\theta} k_1 \, d\theta$$

$$\omega - \omega_0 = -k_1 \theta$$

$$\boxed{\omega = \omega_0 - k_1 \theta}$$



i.e, ω decreases linearly with θ.

and $\omega = \omega_0 e^{-k_1 t}$

$$\frac{d\theta}{dt} = \omega_0 e^{-k_1 t}$$

$$\int_0^{\theta} d\theta = \omega_0 \int_0^{t} e^{-k_1 t} \cdot dt$$

$$\theta = \frac{\omega_0 \left(e^{-k_1 t} - 1\right)}{-k_1}$$

$$\boxed{\theta = \frac{\omega_0}{k_1} \left(1 - e^{-k_1 t}\right)}$$

i.e, θ increases exponentially with time.

* Rotational K.E,

$$K = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} I \omega_0^2 e^{-2k_1 t}$$

At $t = \infty \Rightarrow \boxed{K.E = 0}$

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* Heat produced in 'R',

$$H = \int_0^t i^2 R\,dt = \Delta K = K_f - K_i$$

Here torque due to magnetic field do some work.

$$H = \frac{1}{2} I\omega_0^2 - \frac{1}{2} I\omega_0^2 e^{-2k_1 t}$$

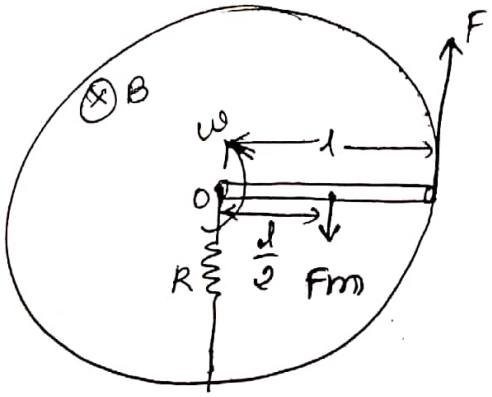
$$= \frac{1}{2} I\omega_0^2 \left(1 - e^{-2k_1 t}\right)$$

* At $t = \infty$, $\boxed{H = \frac{1}{2} I\omega_0^2}$

Here heat is equal to intial K.E. produced

∴ total mechanical energy is converted into heat.

Ex:



To move the rod with constant angular velocity. Find F?

Sol:

net Torque on the rod about O is zero.

$$F(l) = Fm\left(\frac{l}{2}\right)$$

$$F = \frac{Fm}{2}$$

$$= \frac{iLB}{2}$$

$$= \frac{eLB}{2R} = \frac{\frac{1}{2}BL^2\omega \cdot LB}{2R}$$

$$\boxed{F = \frac{B^2\omega l^2 B}{4R}}$$

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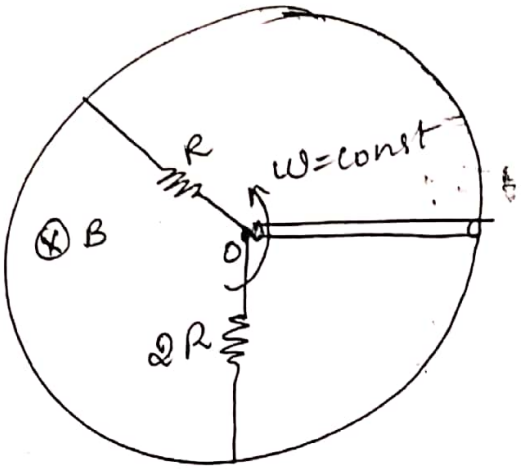
Ex 6

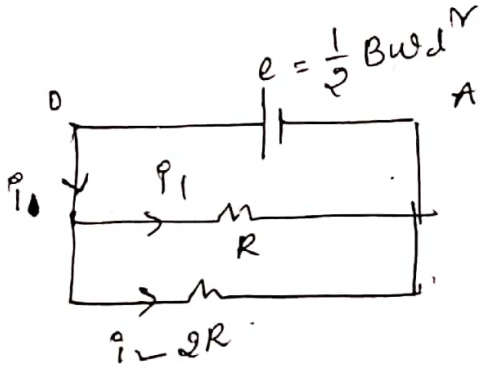


Find

1) i

2) F_m

Sol:



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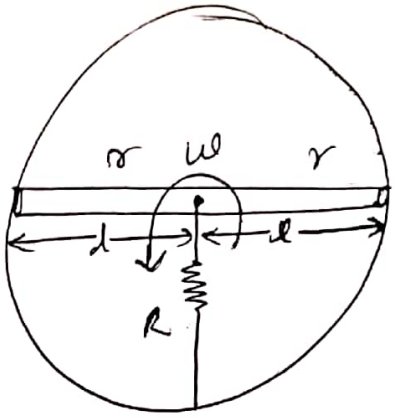
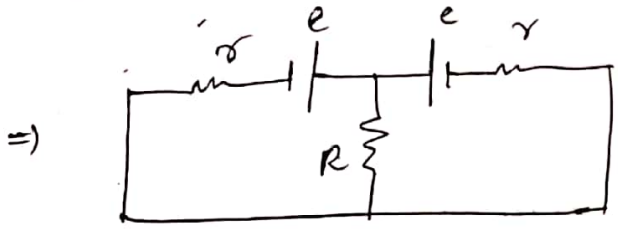
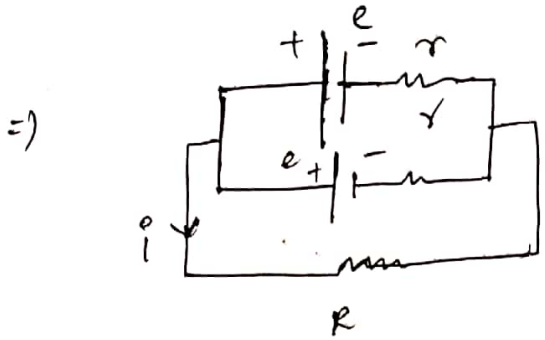
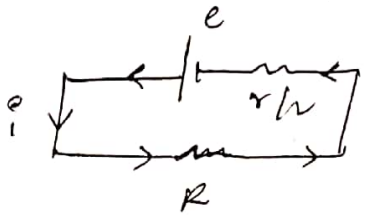
Ex 6



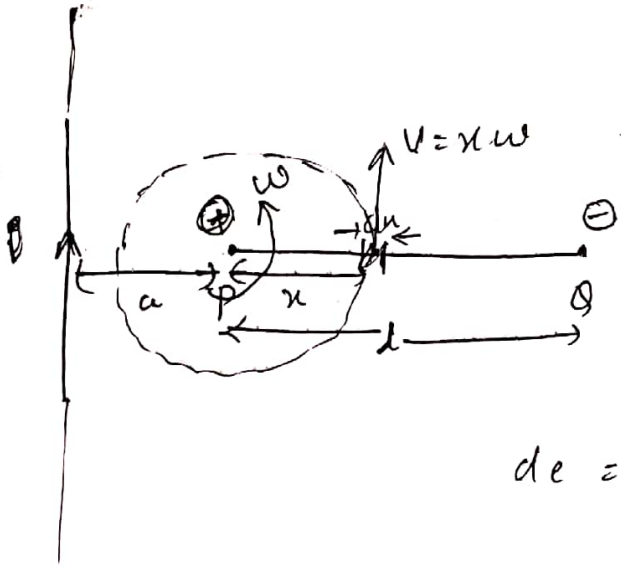






$$i = \frac{e}{R + \frac{r}{n}}$$

Rotation in Non-uniform B (magnetic field) :-



$$de = B l v$$

$$= \frac{\mu_0 i}{2\pi(a+x)} (x\omega) \cdot dx$$

$$e = \int_{x=0}^{x=l} \frac{\mu_0 i \omega}{2\pi} \left(\frac{x}{a+x}\right) \cdot dx$$

=

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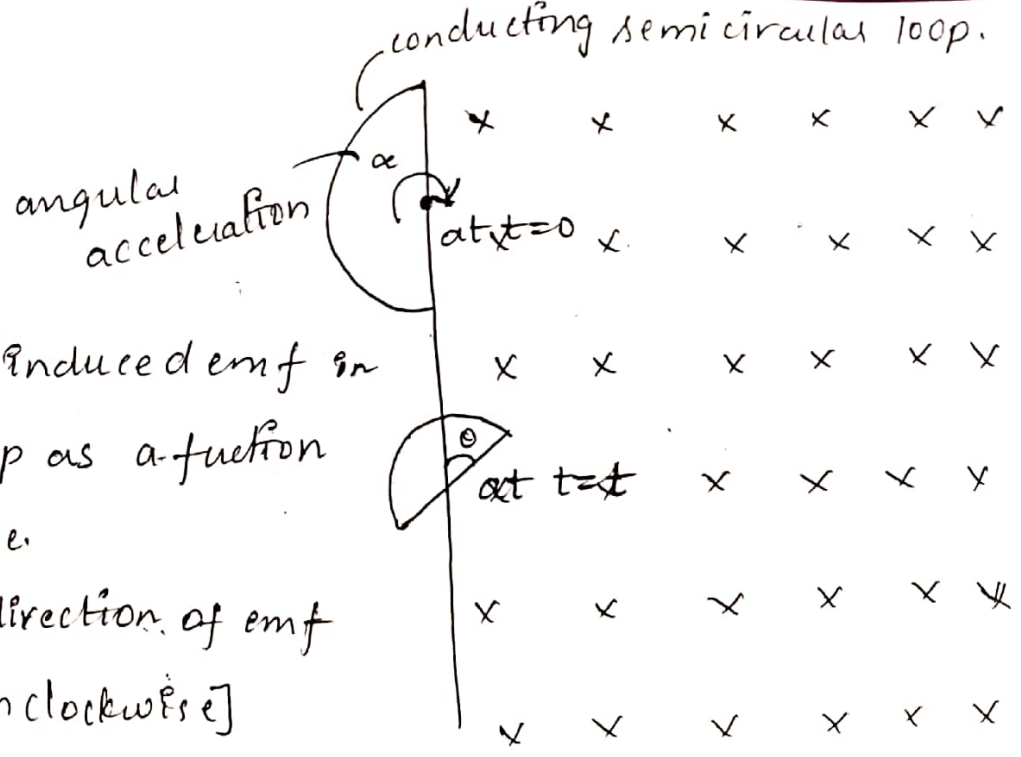
Ex6



Find induced emf in the loop as a function of time.
[take direction of emf +ve in clockwise]

Sol6

at any time 't',

$$\theta = \frac{1}{2}\alpha t^2$$

$$\boxed{t = \sqrt{\frac{2\theta}{\alpha}}}$$

* $\theta \longrightarrow$ 0 to π
  2π to 3π
  4π to 5π
  and so on
  $e \longrightarrow$ -ve

$\theta \longrightarrow$ π to 2π
3π to 4π
5π to 6π
and so on
$e \longrightarrow$ +ve

again,

$$A = \frac{1}{2}R(R\theta)$$

$$= \frac{1}{2}R^2\theta$$

$$= \frac{1}{2}R^2\left(\frac{1}{2}\alpha t^2\right)$$

$$A = \frac{1}{4}R^2\alpha t^2$$

$$\phi = BA$$

$$= \frac{1}{4}BR^2\alpha t^2$$

$$e = \frac{d\phi}{dt} = \frac{1}{2}BR^2\alpha t \quad \Rightarrow \boxed{e \propto t}$$

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complete revolution,

for $\theta = \pi$

$$t_1 = \sqrt{\frac{2\pi}{\alpha}}$$

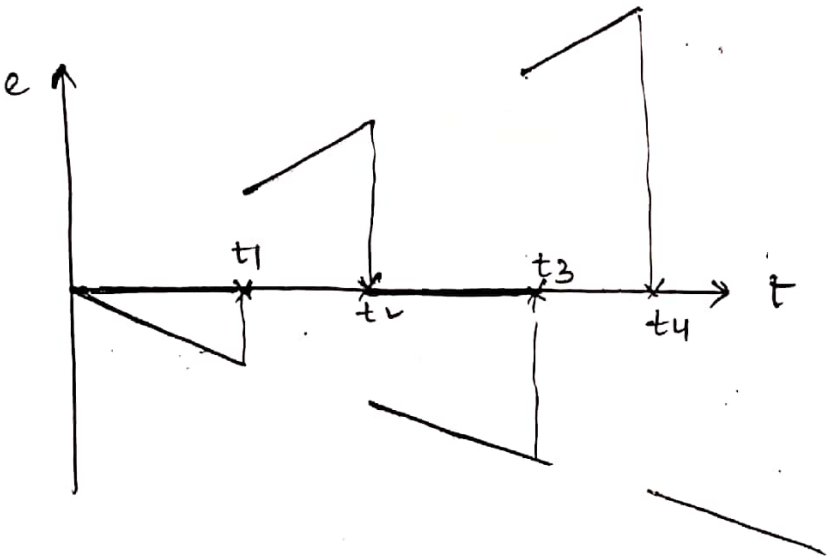
for $\theta = 2\pi$

$$t_2 = \sqrt{\frac{4\pi}{\alpha}}$$

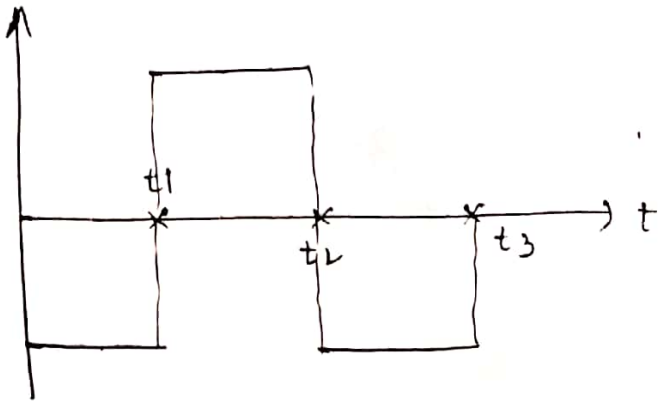
for $\theta = 3\pi$

$$t_3 = \sqrt{\frac{6\pi}{\alpha}}$$

and so on - —



* For $\theta = \omega t$,



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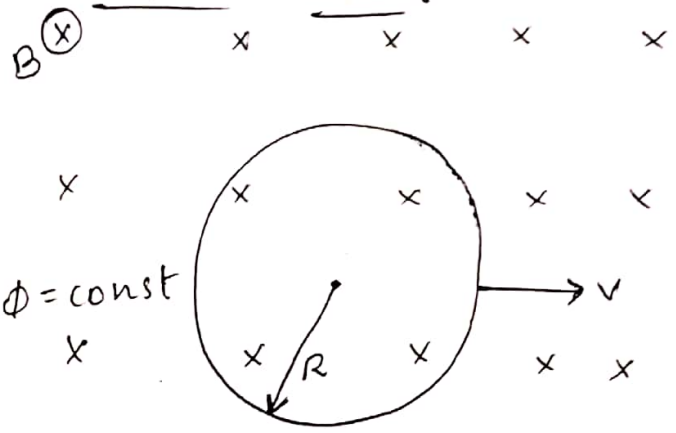
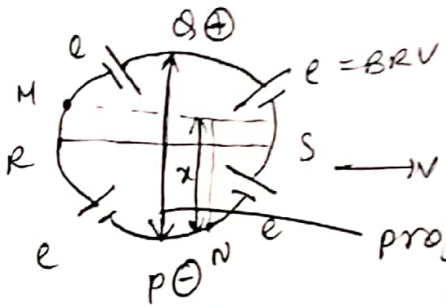
Ex 8 Translation Motion:



$e_{loop} = 0$.

$\Rightarrow$

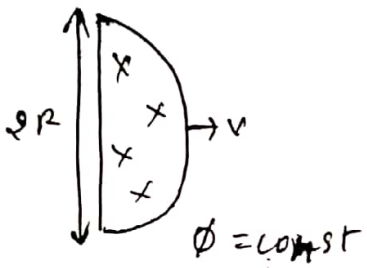


$e_{loop} = 0$

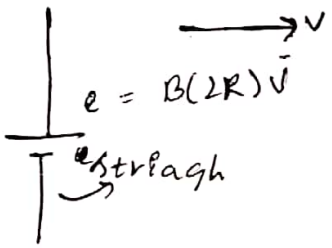
$e_{PQ} = 2BRV$

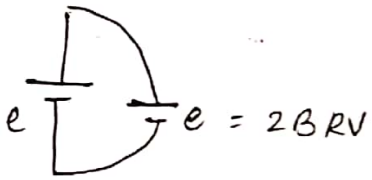
$e_{RS} = 0$

$\boxed{e_{NM} = BxV}$

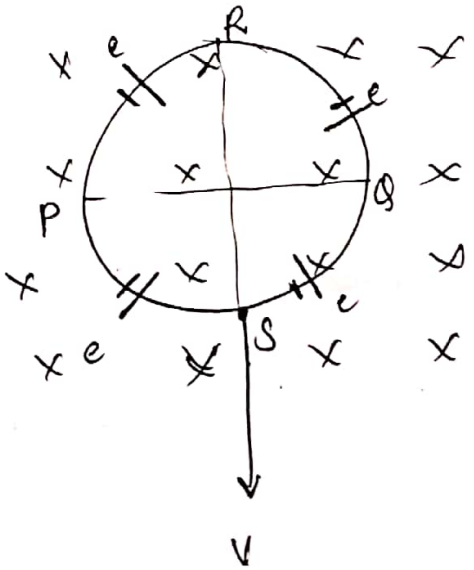


$\Rightarrow e_{loop} = 0$





Ex!



$e_{PQ} = 2BRV$

$e_{SR} = 0$.

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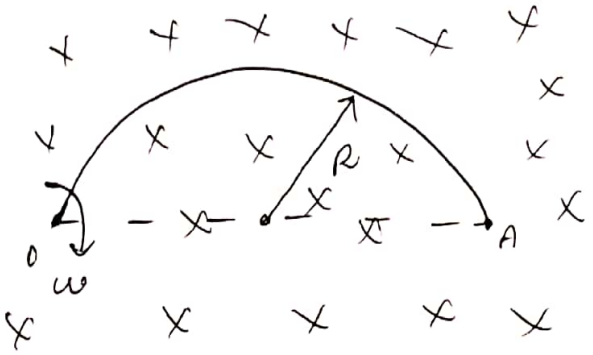
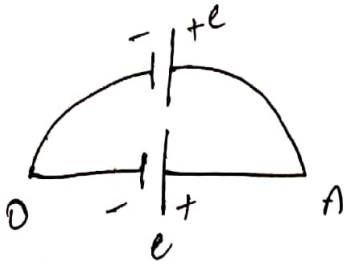
Ex: Rotational Motion

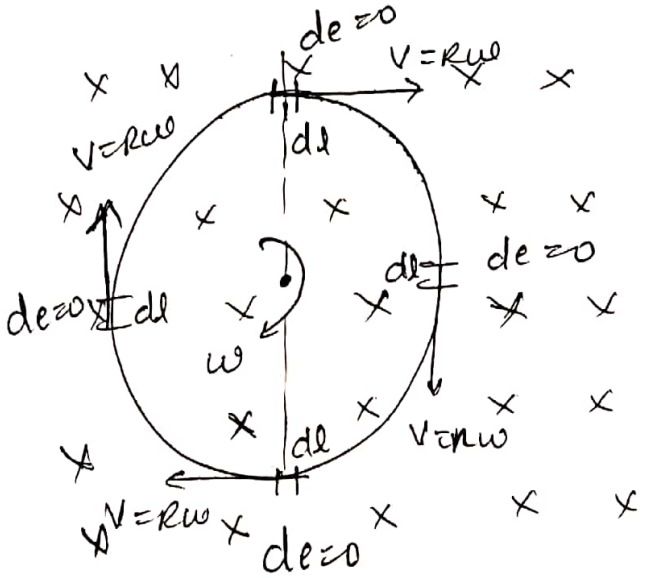


$\phi = \text{const}$

$\Rightarrow e = 0$

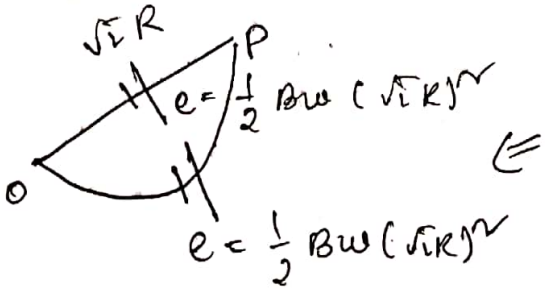
$e_{OA} = \frac{1}{2} B\omega (2R)^2$

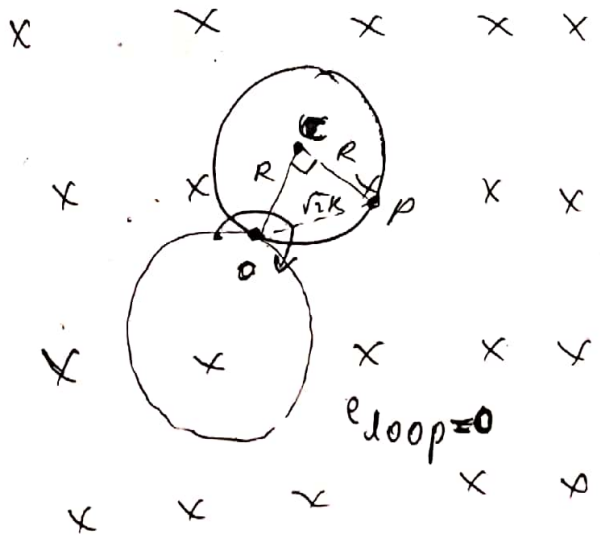




$\Rightarrow e_{loop} = 0$

Ex:





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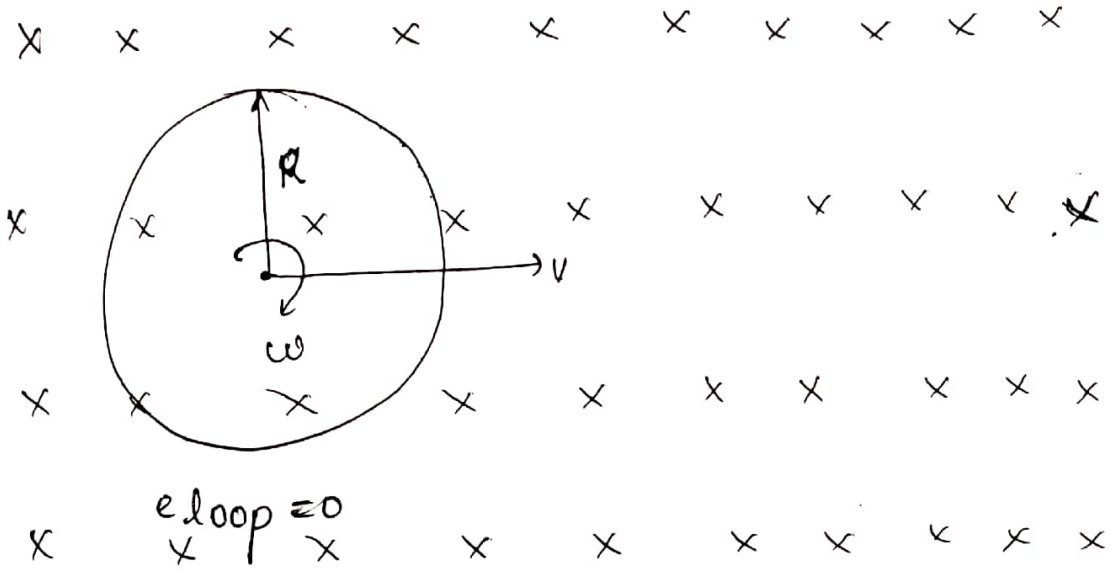
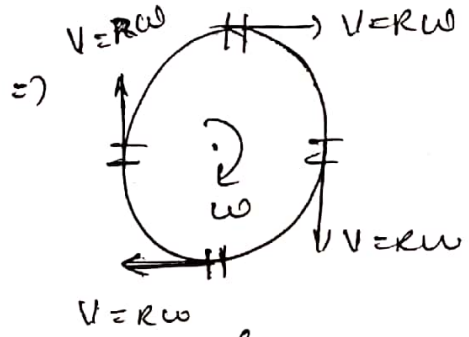
Rolling Motion

↳ Rotation + translation



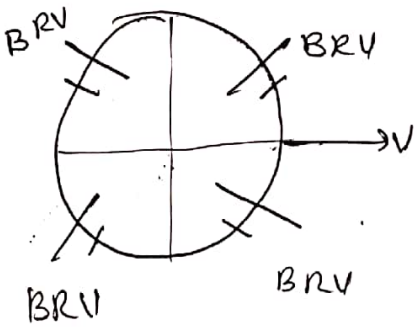
$e_{loop} = 0$



$e_{loop} = 0$

Rotation

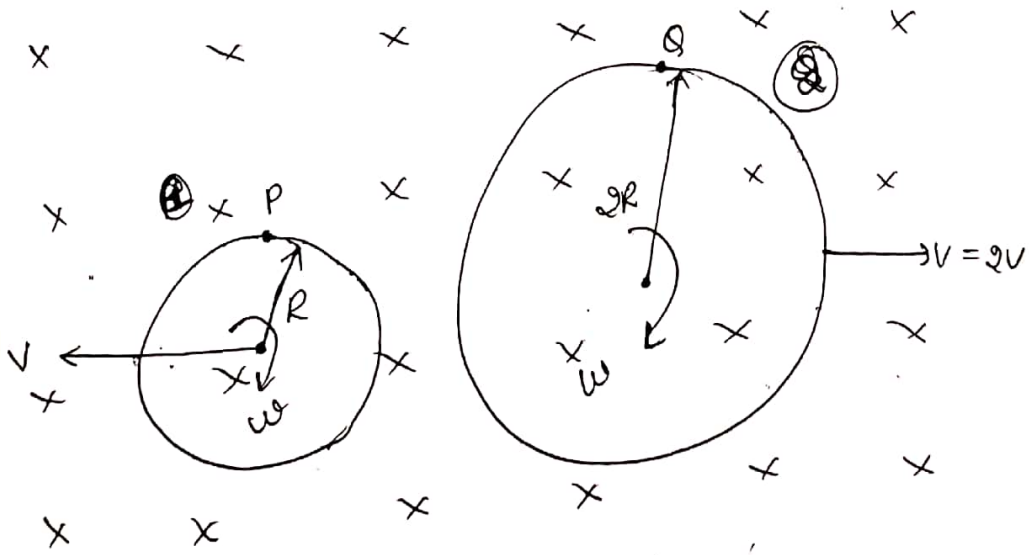
+



$e_{loop} = 0$

Translation.

Ex 8



conducting plates.

Find $V_p - V_q$.

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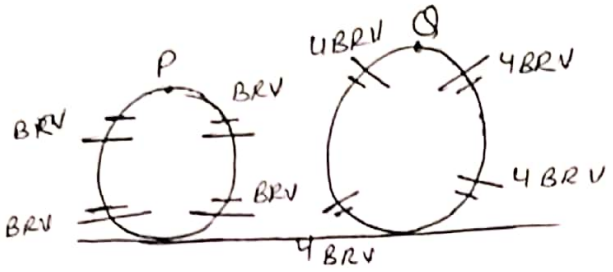
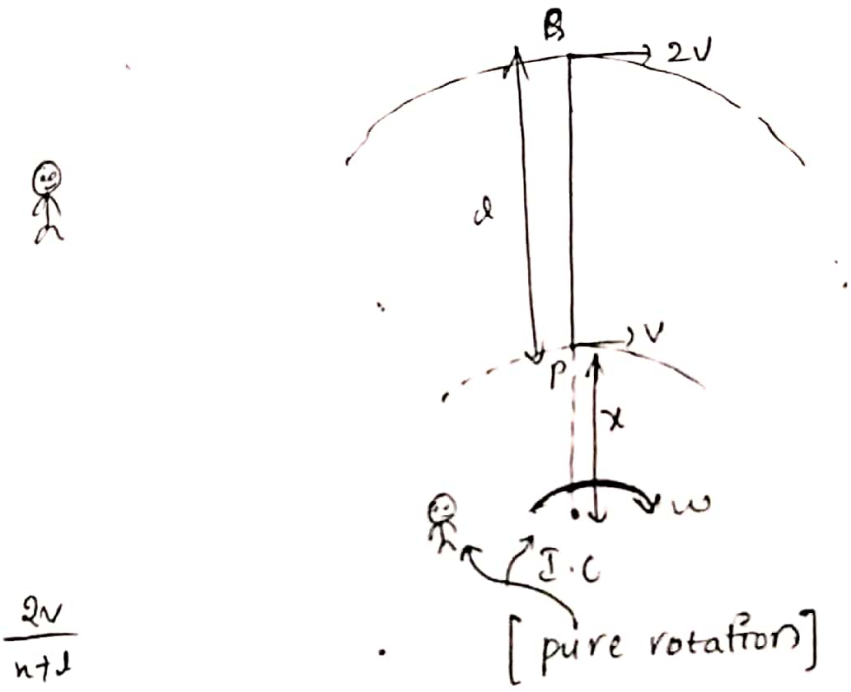


$$V_P + BRV + BRV + 4BRV + 4BRV = V_Q$$

$$\boxed{V_Q - V_P = 10\,BRV}$$

Instantaneous centre (I·C):

The pt at which particle comes to rest for a moment



[pure rotation]

$$\omega = \frac{V}{x} = \frac{2V}{x+d}$$

$$\boxed{x = d}$$

$$\boxed{\omega = \frac{V}{d}}$$

now,

$$V_Q - V_P = e_{PQ} = \int_{x=0}^{x=2d} B x \omega \cdot dx$$

$$= \frac{3B\omega d^2}{2}$$

$$(V_Q > V_P) \quad \Leftarrow \quad \boxed{V_Q - V_P = \frac{3BVd}{2}}$$

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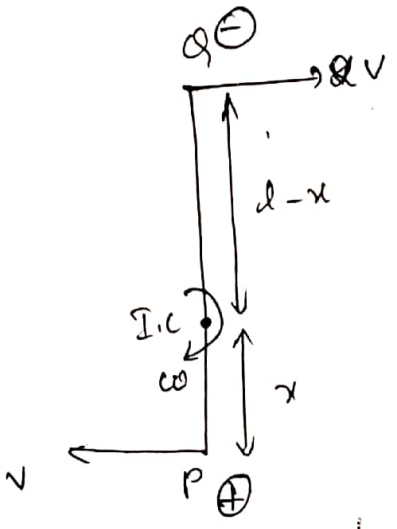
II-Method :

$$V_Q - V_O = \frac{1}{2} B\omega (2l)^2$$

$$V_P - V_O = \frac{1}{2} B\omega l^2$$

$$V_Q - V_P = \frac{3B\omega l^2}{2}$$

Exe :



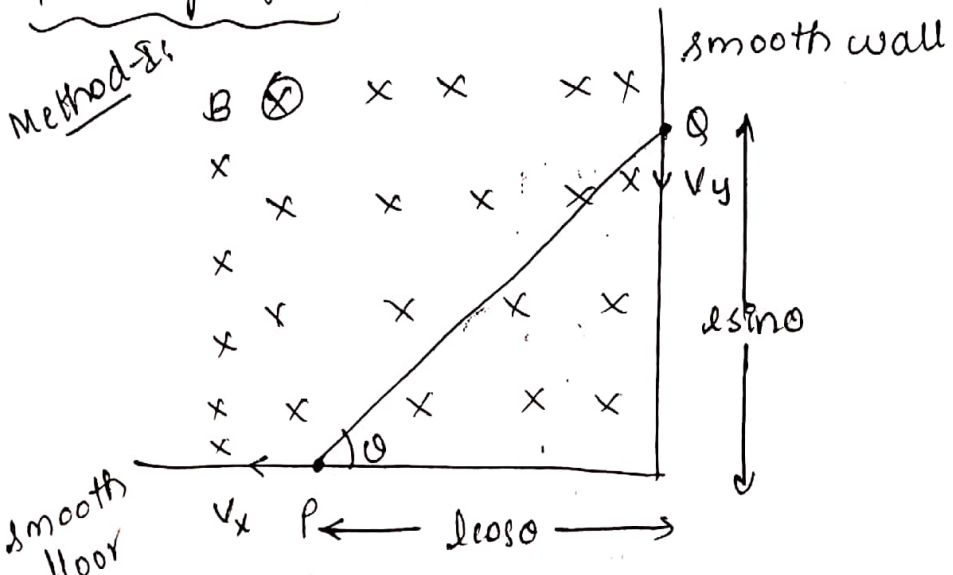
$$\omega = \frac{V}{x} = \frac{2V}{l-x}$$

$$x = l/3$$

$$\omega = \frac{3V}{l}$$

Sliding Rod :

Method-I :



At any time 't',

* constraint eqn,

$$V_x \cos\theta = V_y \sin\theta$$

* Area enclosed by flux,

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$$A = \frac{1}{2} \times l\cos\theta \times l\sin\theta = \frac{l^2}{4}\sin 2\theta .$$

$$\& \quad \phi = BA = \frac{Bl^2 \sin 2\theta}{4}$$

$$e = \frac{d\phi}{dt}$$

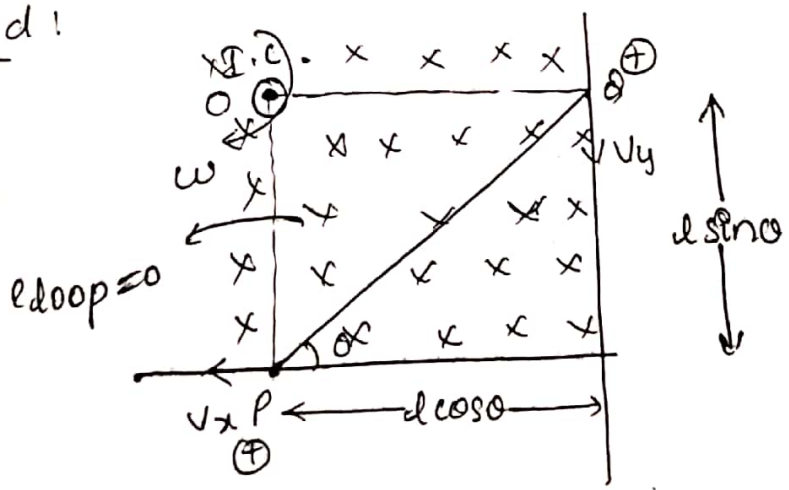
$$e = \frac{Bl^2}{4} \times 2\cos 2\theta \left(\frac{d\theta}{dt}\right)$$

$$\boxed{e = \frac{1}{2} B\omega l^2 \cos 2\theta}$$

$\Rightarrow$ ϕ max when $\theta = 45^\circ$

but $\boxed{e = 0}$

II - method :



$$\therefore \omega = \frac{V_y}{l\cos\theta} = \frac{V_x}{l\sin\theta}$$

$$e_{OP} = \frac{1}{2} B\omega (l\sin\theta)^2$$

$$e_{OQ} = \frac{1}{2} B\omega (l\cos\theta)^2$$

$$\boxed{e_{PQ} = \frac{1}{2} B\omega l^2 \cos 2\theta}$$

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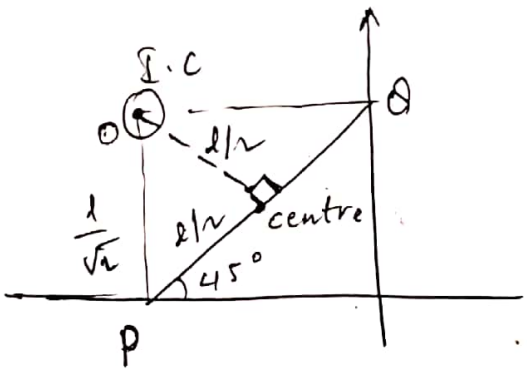
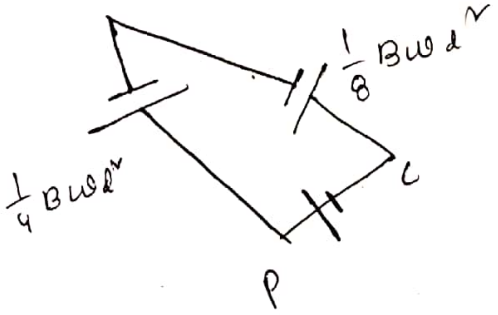
When $\theta = 45°$



$$e_{OP} = \frac{1}{2} B\omega \left(\frac{l}{\sqrt{2}}\right)^2 = \frac{1}{4} B\omega l^2$$

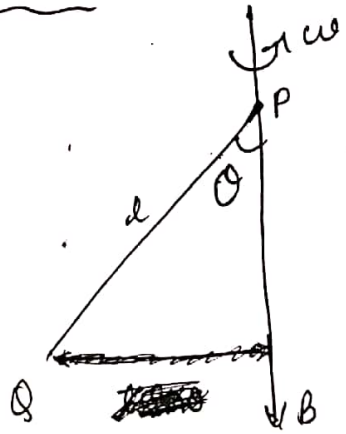
$$e_{OQ} = \frac{1}{2} B\omega \left(\frac{l}{2}\right)^2 = \frac{1}{8} B\omega l^2$$



$$V_P - \frac{1}{4} B\omega l^2 + \frac{1}{8} B\omega l^2 = V_C$$

$$\boxed{V_P - V_C = \frac{1}{8} B\omega l^2}$$

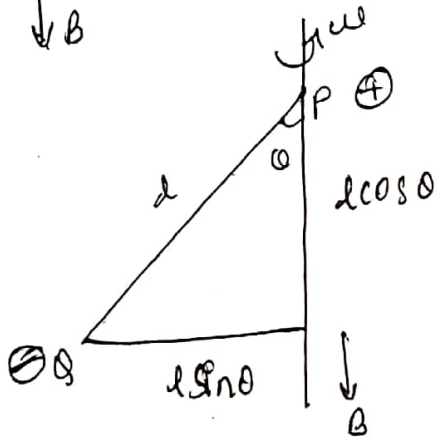
Ex: Conical Pendulum



Find e_{PQ}.

Sol:

$$e = \frac{1}{2} B\omega (l \sin\theta)^2$$



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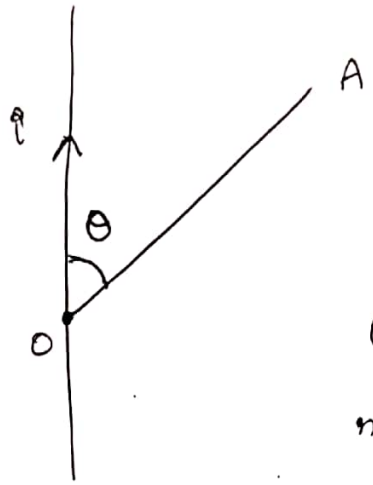
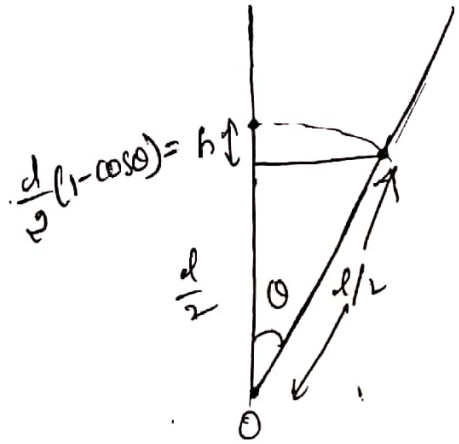
Ex:6



Rotation in Non-Uniform Magnetic Field:-

Conducting rod of length l & mass m is released from vertical position and allowed to move in vertical plane. A very long current wire is placed near to O as shown. Find induced emf in the rod when it is fallen through an angle θ.

Sol:



From L.C.E,

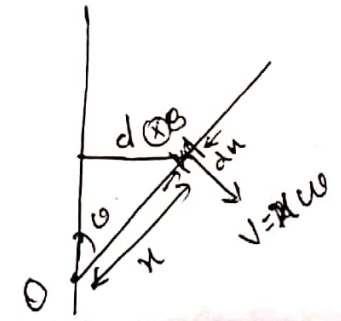
$$\frac{1}{2} I\omega^2 = mg\frac{l}{2}(1-\cos\theta)$$

$$\frac{ml^2}{3}\omega^2 = mgl(1-\cos\theta)$$

$$\boxed{\omega = \sqrt{\frac{3g}{l}(1-\cos\theta)}}$$

* At this position,

'ω' is same for all elements.



* Induced emf is small element,

$$de = B(dx)V$$

$$= \frac{\mu_0 i}{2\pi x}(x\omega)\,dx$$

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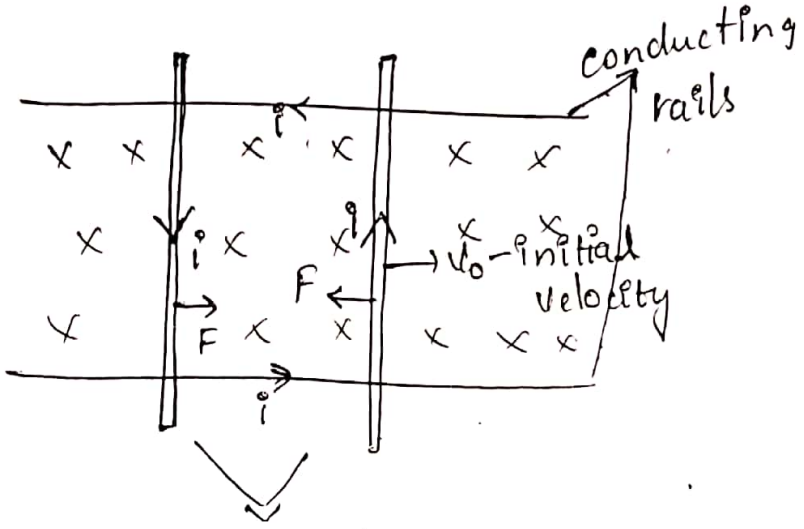
$$de = \frac{\mu_0 i \omega x}{2\pi x \sin\theta} \cdot dx$$

$$e = \int_{x=0}^{x=l} de = \frac{\mu_0 i \omega l}{2\pi \sin\theta}$$

$$\boxed{e = \frac{\mu_0 i \omega l}{2\pi \sin\theta} \sqrt{\frac{3g}{l}(1-\cos\theta)}}$$

Ex:



From L·C·L·M,

$$m v_0 = 2mv$$

$$\boxed{v = v_0/2}$$

In steady state,

$v_1 = v_2 = v$

$A \longrightarrow$ max

$\phi \longrightarrow$ max

$e \longrightarrow 0$

$i \longrightarrow 0$

Heat produced on wire,

$$H = \frac{1}{2} m v_0^2 - \frac{1}{2}(2m)\left(\frac{v_0}{2}\right)^2$$

$$= \frac{1}{2} m v_0^2 - \frac{1}{4} m v_0^2 \Rightarrow \boxed{\frac{1}{4} m v_0^2 = H}$$

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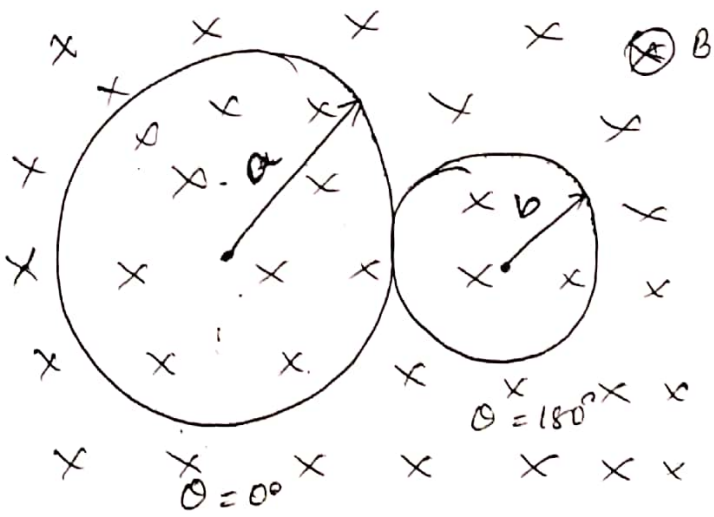
here K.E is not conserved.

Ex:



λ – resistance per unit length

Find i_{max}

$$\phi = \phi_1 + \phi_2$$

$$= BA_1\cos 0^\circ + BA_2\cos 180^\circ$$

$$= \pi Ba^2 - \pi Bb^2$$

$$\phi = \pi B(a^2 - b^2)$$

$$= \pi B_0 (a^2 - b^2)\sin\omega t$$

$$e = \frac{d\phi}{dt} = \pi B_0 \omega (a^2 - b^2)\cos\omega t$$

$$i = \frac{e}{R} = \frac{\pi B_0 \omega (a^2 - b^2)\cos\omega t}{\lambda(2\pi a + 2\pi b)}$$

$$i = \frac{B_0 \omega (a-b)\cos\omega t}{2\lambda}$$

$$\boxed{i_{max} = \frac{B_0 \omega (a-b)}{2\lambda}}$$

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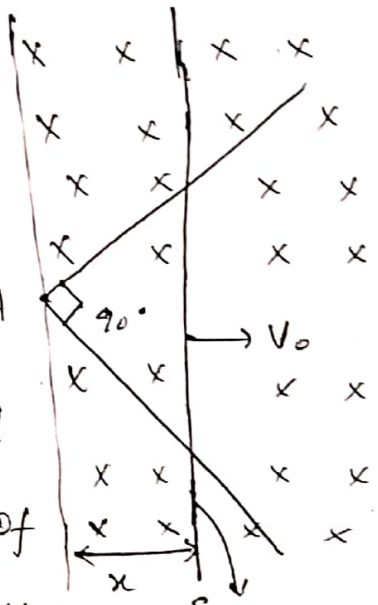
Ex 6

A very long rod of mass 'm' initially starts with a velocity V_0 from the apex A of the bent wire. The resistance of wire and rod is $\lambda\ \Omega/m$. If initial velocity of projection of rod is V_0. Find the [at any time 't'] distance travelled by the rod before coming to rest.

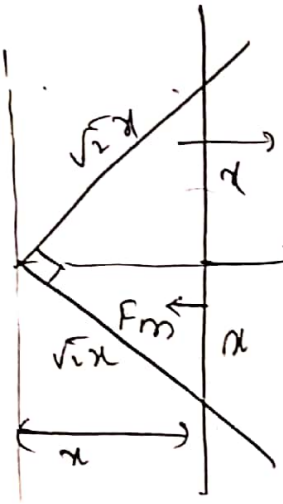


Ans: $x = \sqrt{\frac{mV_0\lambda(1+\sqrt{2})}{B}}$

Sol 6

$F_m = ma$

$Bil = ma$



$l = 2x$.

$\lambda = \frac{R}{2(\sqrt{2}+1)x}$

$B \times \frac{BxV_0}{\lambda(\sqrt{2}+1)}(2x) = m \cdot V_0 \frac{dV}{dx}$

$\frac{2B^2}{\lambda m(\sqrt{2}+1)} \int x \cdot dx = \int dV$

$e = Blv$

$i = \frac{Blv}{R} = \frac{B(2x)V_0}{\lambda(2\sqrt{2}+1)x}$

$\frac{2B^2}{m\lambda(\sqrt{2}+1)} \frac{x^2}{2} = V_0$

$i = \frac{BV_0}{\lambda(\sqrt{2}+1)}$

$x^2 = \frac{mV_0\lambda(\sqrt{2}+1)}{B^2}$

$$\boxed{x = \frac{\sqrt{mV_0\lambda(\sqrt{2}+1)}}{B}}$$

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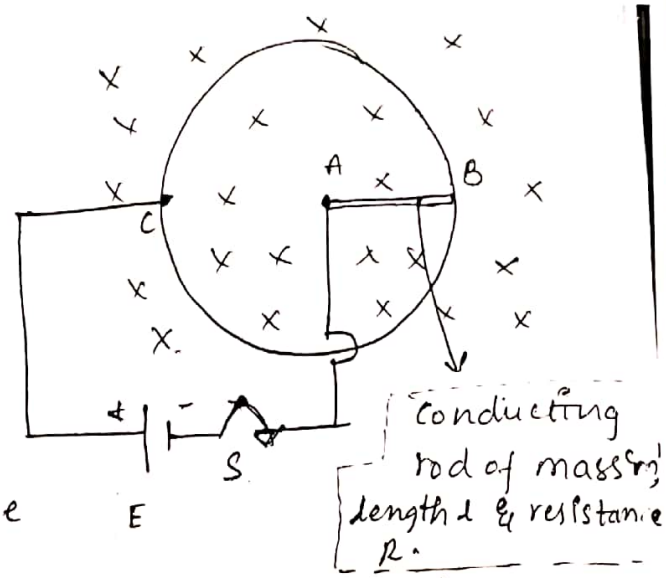
Ex:

1) The rod is free to rotate in a horizontal plane about a vertical axis passing through end A. The end A of the rod & point C on the circumference of the ring are connected to a cell of emf E. Find



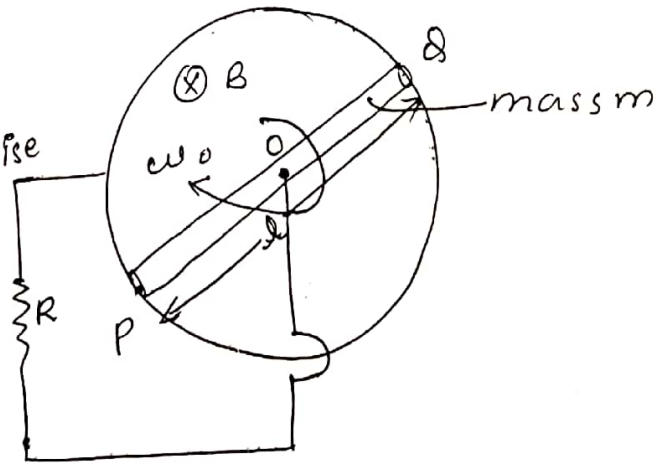
a) Angular speed of rod as a function of time

Ans: $\omega = \frac{2E}{Bl^2}\left(1 - e^{-\frac{3B^2l^2t}{4Rm}}\right)$

b) Heat produced in rod in first 2 sec.

Ans: $H = \frac{2E^2m}{3B^2l^2}\left(1 - e^{-\frac{3B^2l^2t}{2Rm}}\right)$

2) At $t = 0$, rod PQ starts rotating clockwise with angular velocity ω_0. Find



1) magnitude of current as a function of time.

2) Total charge flown through the resistance

3) Heat generated by the circuit at $t = \infty$.

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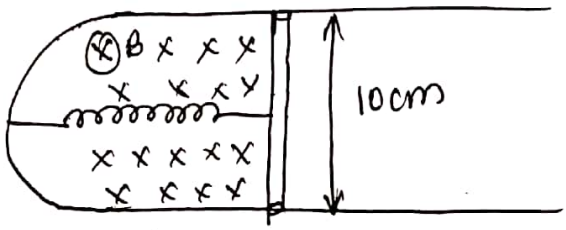
3) A thin strip 10cm long is on a U-shaped wire of negligible resistance and it is connected to a spring of spring const 0.5N/m. The assembly is kept in a uniform electric field of 0.1T. If the spring is pulled from it's eqm position and released, the no. of oscillations it performs before it's amplitude decreases by factor of e is N. If the mass of strip is 50gm, it's resistance 10Ω and air drag is negligible, N will be close to



Sol^{n} 3) $Kx - F_m = ma$

$$kx - Bil = ma$$

$$kx - B\left(\frac{Blv}{R}\right)l = ma$$

$$ma + \frac{B^2l^2v}{R} - kx = 0$$

$$m\frac{d^2x}{dt^2} + \frac{B^2l^2}{R}\frac{dx}{dt} - kx = 0$$

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$$\frac{d^2x}{dt^2} + \frac{B^2l^2}{Rm}\frac{dx}{dt} - \frac{k}{m}x = 0 \qquad b = \frac{B^2l^2}{R}$$

$$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} - \omega_0^2 x = 0 \qquad 2\gamma = \frac{b}{m}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} - \omega_0^2 x = 0$$

Soln, $x = A_0 e^{-\gamma t}\cos(\omega_d t + \phi)$

$$\boxed{\omega_d = \sqrt{\omega_0^2 - \gamma^2}}$$

damped frequency

Amplitude, $A = A_0 e^{-\gamma t}$

$$\frac{A_0}{e} = A_0 e^{-\gamma t}$$

$$\gamma t = 1$$

$$t = 1/\gamma = \frac{2m}{B^2l^2} = \frac{2\times 5\times 10^{-2}\times 10}{10^{-2}\times 10^{-2}}$$

$$\boxed{t = 10^4\ s}$$

$$T = \frac{2\pi}{\omega_d} = \frac{2\pi}{\sqrt{\omega_0^2 - \gamma^2}} \approx \frac{2\pi}{\omega_0} = \frac{2\times\pi}{\sqrt{k/m}} = 2\pi\times\sqrt{\frac{8\times 10^{-4}}{8\times 10^{-4}}}$$

$$= \frac{2\pi}{\sqrt{10}} \approx 2$$

$$N = \frac{t}{T}$$

$$= \frac{10^4}{2}$$

$$N = 5\times 10^3$$

$$\boxed{N = 5000 \text{ oscillations}}$$

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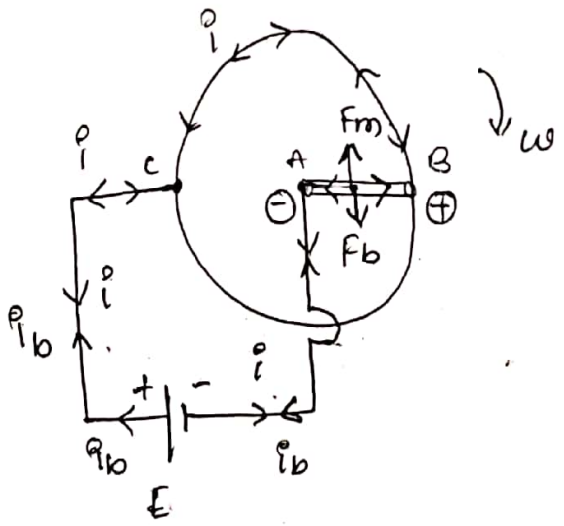
1)



a) $F_b = idB$

$= \frac{EdB}{R}$

$F_m = IdB$

$= \frac{\frac{1}{2}B\omega l^2 \cdot lB}{R} = \frac{B^2\omega l^3}{2R}$

$\tau = I\alpha$

$(F_b - F_m)\frac{l}{2} = \frac{ml^2}{3}\alpha$

$\left(\frac{EdB}{R} - \frac{B^2\omega l^3}{2R}\right)\frac{3}{2ml} = \alpha$

$\alpha = \frac{3EB}{2mR} - \frac{3B^2\omega l^2}{4Rm}$

$\alpha = K_1 - K_2\omega$ where $K_1 = \frac{3EB}{2mR}$

$\frac{d\omega}{dt} = A - B\omega$ $K_2 = \frac{3B^2l^2}{4mR}$

$\int_0^{\omega} \frac{d\omega}{K_1 - K_2\omega} = \int_0^t dt$

0

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$$\frac{[\ln(K_1 - K_2\omega)]_0^{\omega}}{-K_2} = t$$

$$\ln\left(\frac{K_1 - K\omega}{K_1}\right) = -K_2 t$$

$$1 - \frac{K_2\omega}{K_1} = e^{-K_2 t}$$

$$\frac{K_1}{K_2}\left(1 - e^{-K_2 t}\right) = \omega$$

$$\omega = \frac{\frac{BE\ell}{\rho mR}}{\frac{3B^2\ell^2}{2 \cdot 4\rho mR}}\left(1 - e^{-\frac{3B^2\ell^2 t}{4mR}}\right)$$

$$\boxed{\omega = \frac{2E}{B\ell^2}\left(1 - e^{-\frac{3B^2\ell^2 t}{4mR}}\right)}$$

b)

$$i_{net} = i_b - i$$

$$= \frac{E}{R} - \frac{1}{2}\frac{B\omega\ell^2}{R}$$

$$= \frac{E}{R} - \frac{E}{R}\left(1 - e^{-\frac{3B^2\ell^2 t}{4mR}}\right)$$

$$\boxed{i_{net} = \frac{E}{R} e^{-\frac{3B^2\ell^2 t}{4mR}}}$$

$$H = \int_0^t i_{net}^2 R \cdot dt$$

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$$= \int_0^t \frac{E^2}{R^2} \cdot e^{-\frac{3B^2l^2t}{2mR}} \cdot R \cdot dt$$

$$= \frac{E^2}{R} \left[e^{\frac{-3B^2l^2t}{2mR}} \right]_0^t \times \frac{-2mR}{3B^2l^2}$$

$$H = \frac{2mE^2}{3B^2l^2} \left(1 - e^{-\frac{3B^2l^2t}{2mR}} \right)$$

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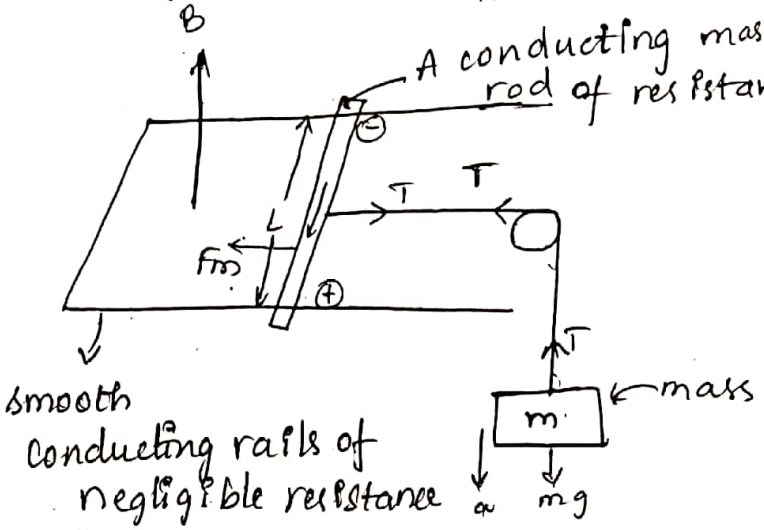
Ex:1) If the system is released from rest. Find

a) terminal velocity attained by the rod.

b) acceleration of the mass at the instant when the velocity of rod is $1/2$ the terminal velocity.

$$v_T = \frac{mgR}{B^2 l^2}$$

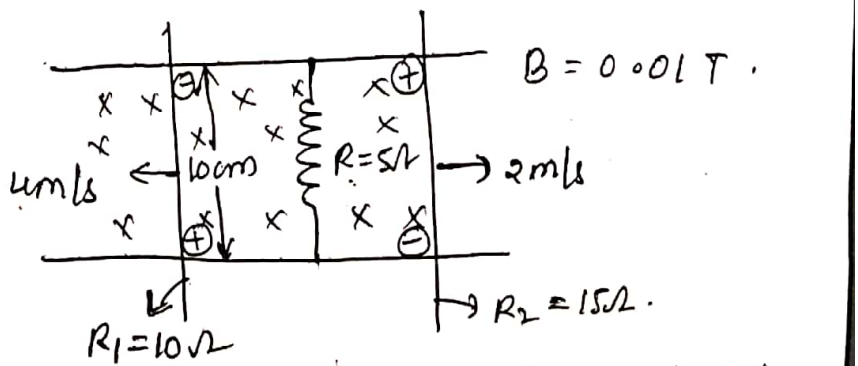
$$a = g/2$$

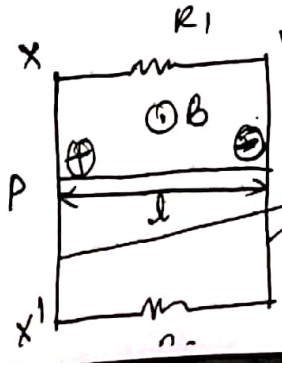


2) A short circuited coil is placed in time valing magnetic field. The electric power is dissipated due to the current induced in the coil if the no. of turns were to be quddrapeled & radius of wire is to be half. Find the electric power consumed. same

3) Find current in 5Ω resistor.

$$i = \frac{\delta}{55}\ mA$$



4) 

The horizontal metal rod PQ of mass m starts falling vertically making

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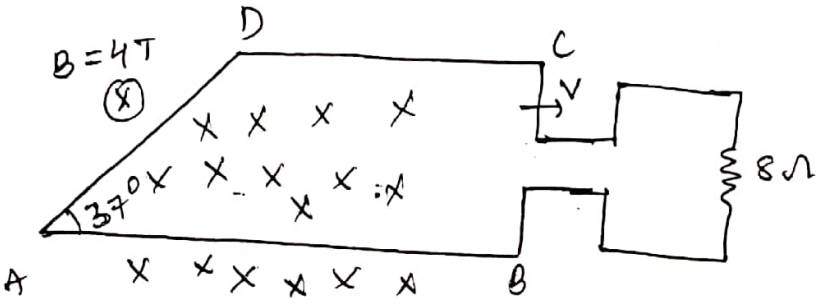
It is observed that in "steady state", the powers dissipated in the resistances are P_1 & P_2 respectively. Find the expressions for R_1, R_2 and terminal velocity attained by the rod.

5)

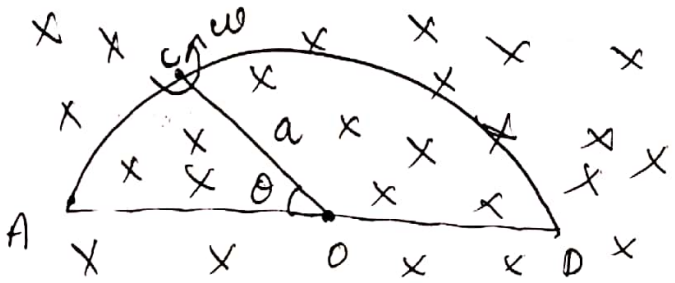


If a steady current of 2A flows in the loop. Then find V.

$V = \frac{100}{3}$ m/s

[If the loop has resistance of 4Ω, AB = 30cm, AD = 30cm

6)



A uniform wire of resistance per unit length λ is bent into a semicircle of radius a. The wire rotates with angular velocity ω in horizontal plane about a vertical axis passing through C.

a) Find p.d b/w A & D.

b) If A & D are connected by a conducting wire of zero resistance, Find p.d b/w A & C.

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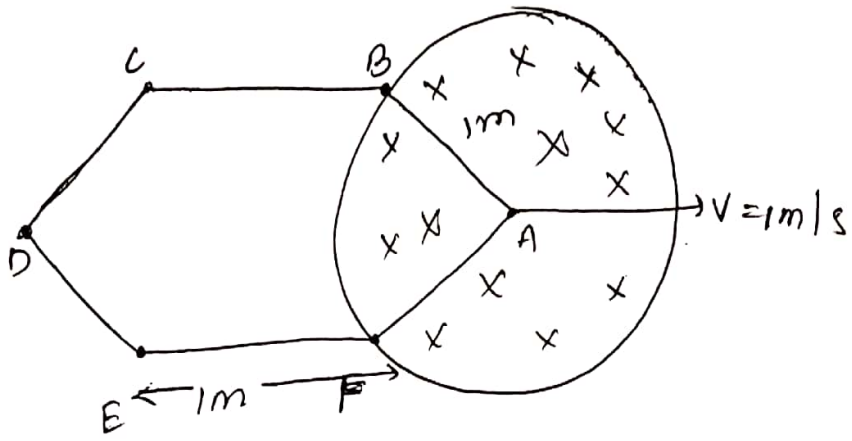
7) A cylindrical region of radius 1m has instantaneous homogeneous magnetic field of 5T and is increasing 2T per sec. A regular hexagonal loop of side 1m is been drawn into a region with constant speed of 1m/s. what is the magnitude of emf developed in the loop just after the shown instant.

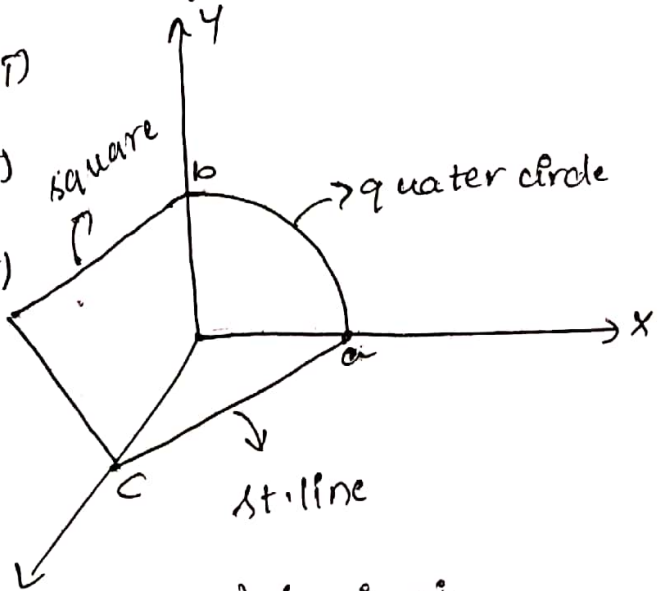


8) In the figure, A wire loop is bent so that it has 3 segments. Here the three choices for the magnetic field through the loop is

1) $\bar{B_1} = 3\hat{i} + 7\hat{j} - 5t\hat{k}$ (mT)

2) $\bar{B_2}$ $5t\hat{i} - 4\hat{j} - 15\hat{k}$ (mT)

3) $\bar{B_3} = 2\hat{i} - 5t\hat{j} - 12\hat{k}$ (mT)



If the induced current due to $\bar{B_1}$, $\bar{B_2}$, $\bar{B_3}$ are i_1, i_2, i_3 are respectively. then

a) $i_1 > i_2 > i_3$

b) $i_2 > i_1 > i_3$

c) $i_3 > i_2 > i_1$

d) $i_1 = i_2 = i_3$

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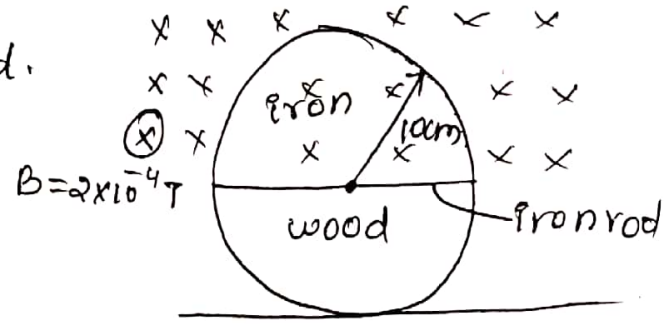
9) The wheel is set in to pure rolling on a horizontal surface. If it takes 2sec for the iron part to come down & wodden part to go up. Find avg emf induced during the period.



Sol^{n}

1)

a) $F_m = T$

$$IlB = mg$$

$$\frac{B^2 l^2 V_T}{R} = mg$$

$$\boxed{V_T = \frac{mgR}{B^2 l^2}}$$

b) $a = g - \frac{F_m}{m}$

$= g - g/2$

$\boxed{a = g/2}$

2)

$$P = \frac{e^2}{R} = \frac{N^2 A^2 \left(\frac{dB}{dt}\right)^2}{\left(\frac{\rho l}{\pi r_1^2}\right)}$$

$$P = \frac{N^2 (\pi r^2)^2 \left(\frac{dB}{dt}\right)^2 \times \pi r_1^2}{\rho (N)(2\pi r)}$$

$\boxed{P \propto N r_1^2}$ ★

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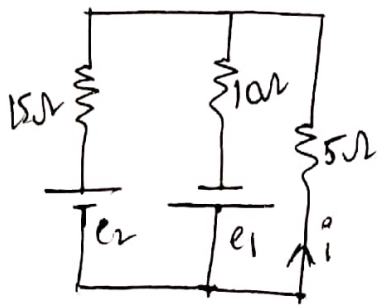
$$\frac{P_1}{P_2} = \frac{N_1 r_1^2}{N_2 r_2^2}$$

$$\frac{P_1}{P_2} = \frac{N r_1^2}{N r_1^2}$$

$$\boxed{P_1 = P_2}$$

3)



$e_1 = Blv = 4mV$

$e_2 = Blv = 2mV$

$$E = \frac{e_1/r_1 - e_2/r_2}{1/r_1 + 1/r_2}$$

$$= \frac{4/10 - 2/15}{1/10 + 1/15} = \frac{40/150}{25/150} = \frac{40}{25}$$

$$r_{eff} = 5 + \frac{150}{25} = \frac{275}{25}$$

$$i = \frac{E}{r_{eff}}$$

$$= \frac{40/25}{275/25}$$

$$= \frac{40}{275} \Rightarrow \boxed{i = 8/55 mA}$$

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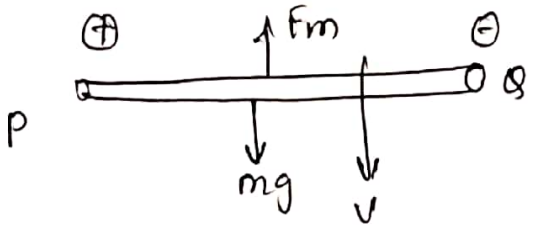
4)



$F_m = mg$

$i\,lB = mg$

$(i_1 + i_2)\,lB = mg$

$\left(\frac{P_1}{e_1} + \frac{P_2}{e_2}\right) lB = mg$

$\frac{(P_1+P_2)}{Blv}\, lB = mg$

$$V = \frac{P_1+P_2}{mg}$$

$P_1 = i_1^2 R_1$

$P_2 = i_2^2 R_2$

$i_1 = e_1/R_1$

$i_2 = e_2/R_2$

$P_1 = i_1^2 R_1$

$\frac{P_1}{i_1^2} = R_1$

$\frac{P_1 R_1^2}{\left(\frac{Bl(P_1+P_2)}{mg}\right)^2} = R/1 \Rightarrow$

$$R_1 = \left[\frac{Bl(P_1+P_2)}{mg}\right]^2 \frac{1}{P_1}$$

Similarly

$$R_2 = \left[\frac{Bl(P_1+P_2)}{mg}\right]^2 \frac{1}{P_2}$$

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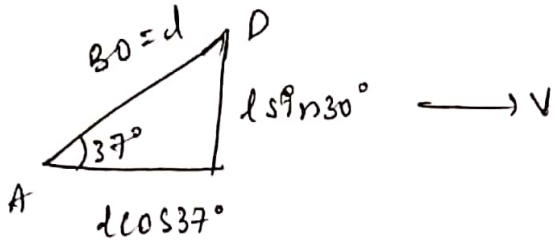
5)



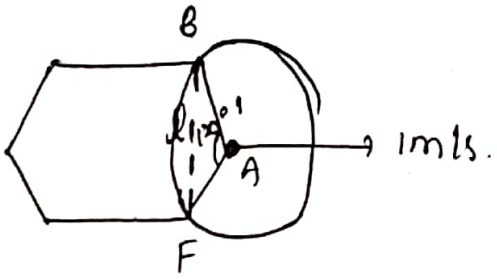
$$\frac{BlV}{R} = 2$$

$$\frac{\cancel{4} \times l \sin 30° \times V}{\cancel{4}_3} = 2$$

$$\cancel{30} \times \frac{3}{\cancel{5}} \times 10^{-\cancel{2}} \times V = \phi$$

$$\boxed{V = \frac{100}{3} \text{ m/s}}$$

7)



$$l = 2 \times 1 \sin 60°$$

$$\boxed{l = \sqrt{3}}$$

$$e_1 = Blv = 5 \times \sqrt{3} \times 1 = 5\sqrt{3}$$

$$e_2 = A\frac{dB}{dt} = \frac{\pi}{3} \times 2 = \frac{2\pi}{3}$$

Area of

$$\text{area} = \frac{\pi r^2}{\cancel{360}_3} \times \cancel{120} = \frac{\pi(1)}{3} = \frac{\pi}{3}$$

$$e = e_1 + e_2 = \left(5\sqrt{3} + \frac{2\pi}{3}\right) V$$

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9)

$$e = \frac{\Delta\phi}{\Delta t}$$

$$= \frac{B \times \frac{\pi r^2}{2} - 0}{2}$$

$$= \frac{2 \times 10^{-4} \times \pi \times 10^{-2}}{2 \times 2}$$

$$= \frac{\pi}{2} \times 10^{-6}$$

$$\boxed{e = 1.57 \times 10^{-6} V}$$

8)

$$\vec{A} = \vec{A_1} + \vec{A_2} + \vec{A_3}$$

$$= l^2 \hat{i} + \frac{l^2}{2} \hat{j} + \frac{\pi l^2}{4} \hat{k}$$

$$\phi_1 = \vec{B_1} \cdot \vec{A} = 3l^2 + \frac{7l^2}{2} - \frac{5\pi}{4} t l^2$$

$$= \left(\frac{13}{2} - \frac{5\pi}{4} t\right) l^2$$

$$\phi_2 = \vec{B_2} \cdot \vec{A} = \left(5t - 2 - \frac{15\pi}{4}\right) l^2$$

$$\phi_3 = \vec{B_3} \cdot \vec{A} = \left(2 - \frac{5t}{2} - 3\pi\right) l^2$$

$$|e_1| = \left|\frac{d\phi_1}{dt}\right| = \frac{5\pi}{4} l^2$$

$$|e_2| = 5l^2$$

$$|e_3| = \frac{5}{2} l^2$$

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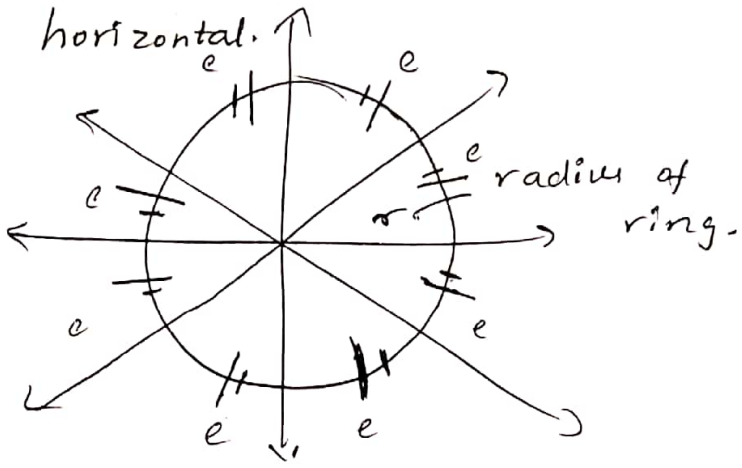
$\Rightarrow$ $e_2 > e_1 > e_3$

~~$e_2 > e_1 > e_3$~~ $\quad \frac{e_2}{R} > \frac{e_1}{R} > \frac{e_3}{R}$

$i_2 > i_1 > i_3$

Ex1 Ring released in radial magnetic field under gravity. plane of ring is horizontal.



* $\ell_{net} = B\ell v$

$= B(2\pi r)v$

* $i = \frac{e}{R}$

* $F_m = i\ell B \ (\uparrow)$

$F_g = mg \ (\downarrow)$

$F_{net} = mg - i\ell B$

$ma = mg - i\ell B$

$a = g - \frac{i\ell B}{m}$

$\frac{dv}{dt} = g - \frac{B^2\ell^2 v}{mR}$

$\frac{dv}{dt} = k_1 - k_2 v$

for terminal velocity,

$a = 0$

$g = \frac{B^2\ell^2 v_T}{mR}$

$\boxed{v_T = \frac{mgR}{B^2\ell^2}}$

$\int \frac{dv}{k_1 - k_2 v} = \int dt$

$\frac{\left[\ln(k_1 - k_2 v)\right]_0^v}{-k_2} = t$

$\ln(k_1 - k_2 v) - \ln(k_1) = -k_2 t$

$1 - \frac{k_2 v}{k_1} = e^{-k_2 t}$

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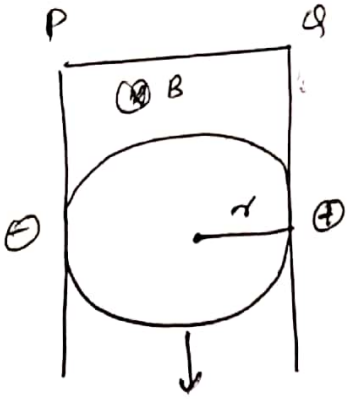
$$v = \frac{k_1}{k_2}\left(1 - e^{-k_2 t}\right)$$

where $k_1 = g$, $k_2 = \frac{B^2 l^2}{mR}$.

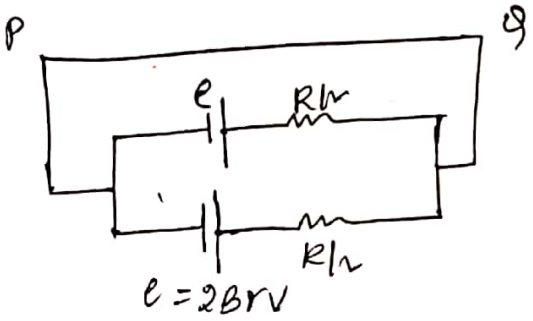
Ex 6



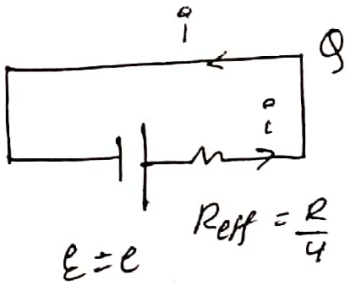
Find induced current in PQ [resistance of ring is R].

Sol:

$\Rightarrow$



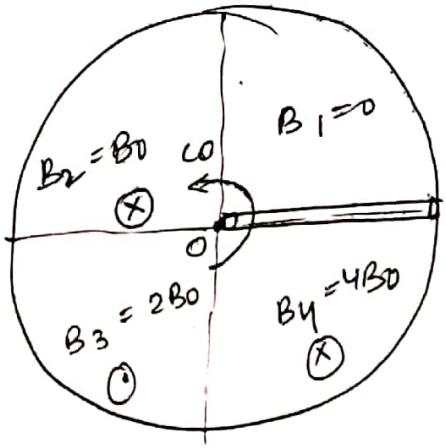
$\Rightarrow$



$$i = \frac{E}{R_{eff}}$$

$$i = \frac{4e}{R} \Rightarrow \boxed{i = \frac{8Brv}{R}}$$

Ex 8



Find average emf in the rod during one complete revolution.

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Sol6

$$T = \frac{2\pi}{\omega}$$

$$t = \frac{T}{4} = \frac{\pi}{2\omega}$$

$$e_{avg} = \frac{0 + \frac{1}{2} B_0 \omega r^2 \cdot \frac{\pi}{2\omega} + \frac{1}{2}(2B_0)\omega r^2 \cdot \frac{\pi}{2\omega} + \frac{1}{2}(4B_0)\omega r^2 \cdot \frac{\pi}{2\omega}}{\frac{2\pi}{4}}$$

$$= \left[\frac{B_0 \omega r^2}{4} - \frac{1}{2} B_0 \omega r^2 + B_0 \omega r^2\right] \times \frac{1}{2}$$

$$\boxed{e_{avg} = \frac{3}{8} B_0 \omega r^2}$$

Ex6

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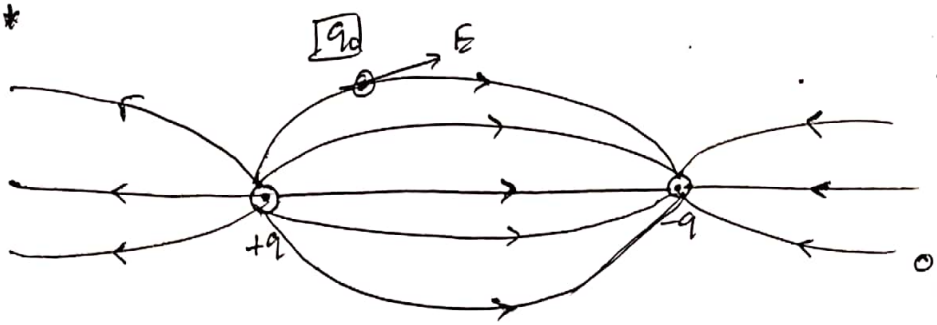
9/06/25 | Monday

Induced Electric Field:

⊛ Sources for electric field:

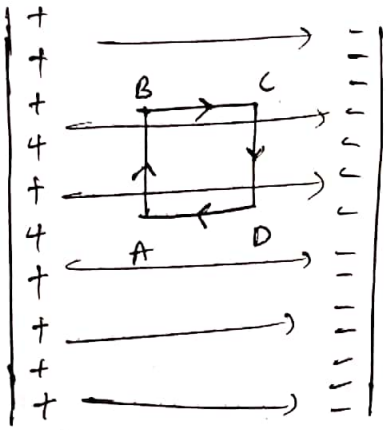
1) Due to static charges (coulomb electrostatic field):



open loop curves

Force on the charge,

$$\bar{F} = q_0 \bar{E}$$



$W_{\text{closed path}} = 0$

∴ Coulomb field is called conservative field.

* Here,

$$F = -\frac{dV}{dr}$$

$$V_f - V_i = -\int_i^f \bar{E} \cdot dr$$

i.e, potential difference can be defined wrt to 'E'.

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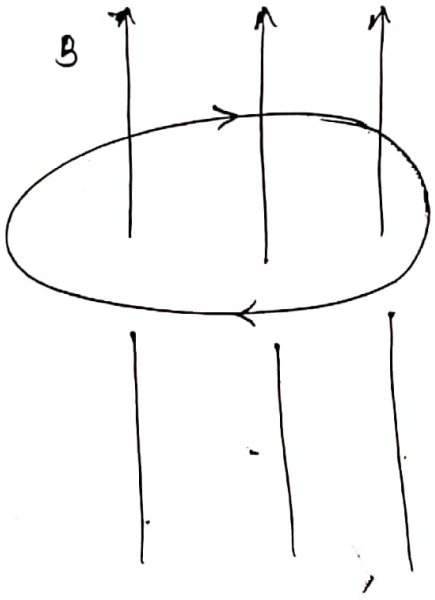
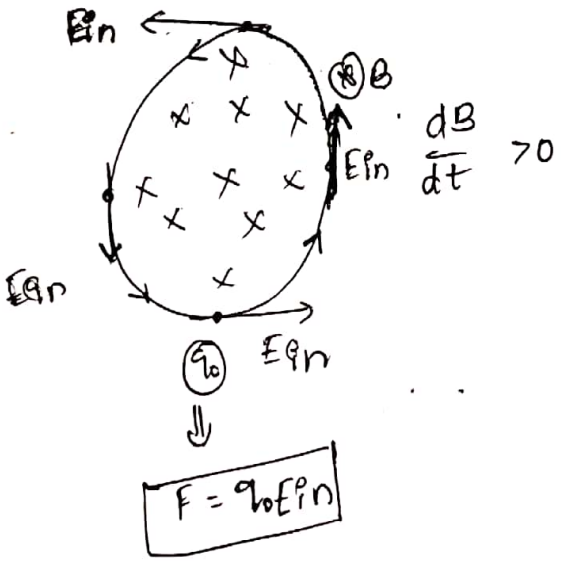
2) Due to time varying magnetic field :-
(Induced electric field)

* 

* B = const
ϕ = const
e = 0, i = 0

* B ↑ with time,
ϕ increases
$e \neq 0$, $i \neq 0$
clockwise

always in the direction of induced current ← electric field lines induced are in circular path.

* 

$F = q_0 E_{in}$

on circle, magnitude of magnetic field is same at every point.

work done by the field in moving the charge around the closed path, $w = \int \bar{F} \cdot d\bar{l} = q_0 \int \bar{E}_{in} \cdot d\bar{l}$

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$$W = q_0 \oint_l \bar{E}_{in} \cdot d\bar{l} \neq 0$$

∴ induced electric field is called non-conservati -ve field.

∴ we cannot define potential difference with fields.

$$V_f - V_i = -\int \bar{E}_{in} \cdot d\bar{l} \quad X$$

* we have,

induced emf,

$$e = \frac{W}{q_0} = \oint_l \bar{E}_{in} \cdot d\bar{l} \quad *$$

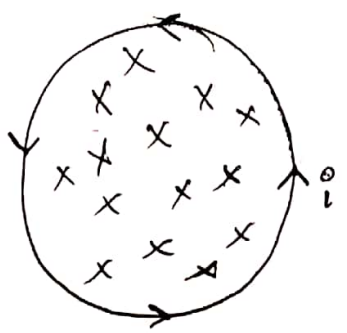
$$e = E_{in}(d)$$

$$E_{in} = e/l$$

*#clearly induced electric field = $\frac{\text{induced emf}}{\text{unit length}}$

* Again,

$$e = -\frac{d\phi}{dt}$$



& $\frac{dB}{dt} > 0$

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$$e = -A\frac{dB}{dt}$$

∴ Induced emf,

$$e = \oint_l \bar{E}_{in} \cdot d\bar{l} = -A\frac{dB}{dt} \quad *$$

Faraday's Law:

$$\oint_l \bar{E}_{in} \cdot d\bar{l} = -\frac{d}{dt}\oint_s \bar{B} \cdot d\bar{s}$$

$$\oint_l \bar{E}_{in} \cdot d\bar{l} = -\oint_s \frac{d\bar{B}}{dt} d\bar{s} \quad *$$

Recap

* $E = \frac{q}{\varepsilon_0 A}$

$q = \varepsilon_0 A E$

displacement current

$i_d = \frac{dq}{dt} = \varepsilon_0 A \frac{dE}{dt}$

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 \varepsilon_0 A \frac{dE}{dt}$$

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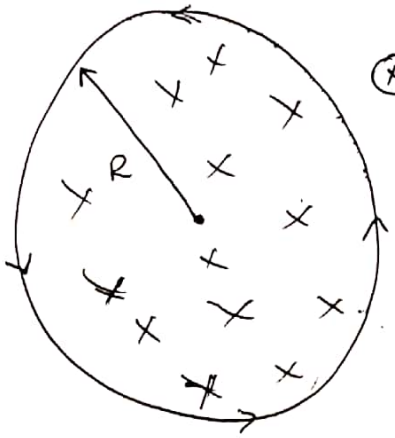
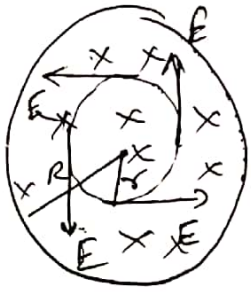
To find induced electric fields



$\frac{dB}{dt} > 0$

λ = resistance per unit length

1) For $r < R$:



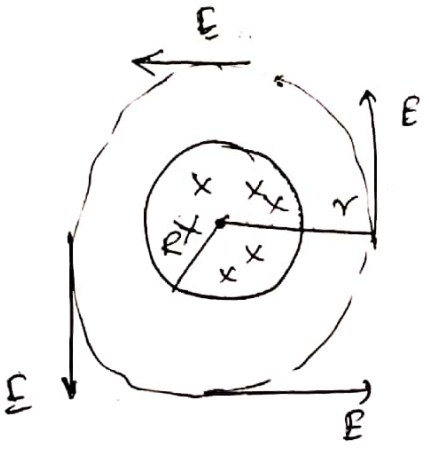
$$\oint \bar{E}_{in} \cdot d\bar{l} = -A \frac{dB}{dt}$$

$$E_{in}(2\pi r) = \pi r^2 \cdot \frac{dB}{dt}$$

$$E_{in} = \frac{r}{2} \frac{dB}{dt}$$

$\boxed{E_{in} \propto r}$ for pts inside the magnetic field.

2) For $r \geq R$:



$$\oint \bar{E} \cdot d\bar{l} = -A \frac{dB}{dt}$$

$$E_{in}(2\pi r) = \pi R^2 \cdot \frac{dB}{dt}$$

$$E_{in} = \frac{R^2}{2r} \frac{dB}{dt}$$

$\boxed{E_{in} \propto \frac{1}{r}}$ till infinity.

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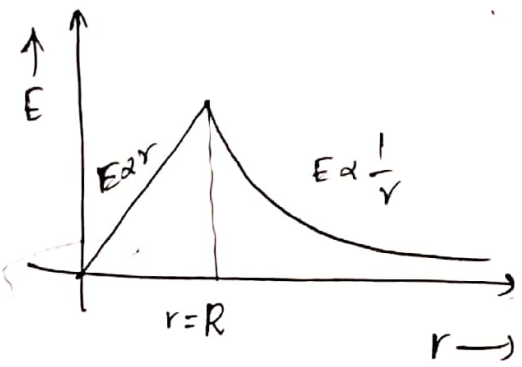
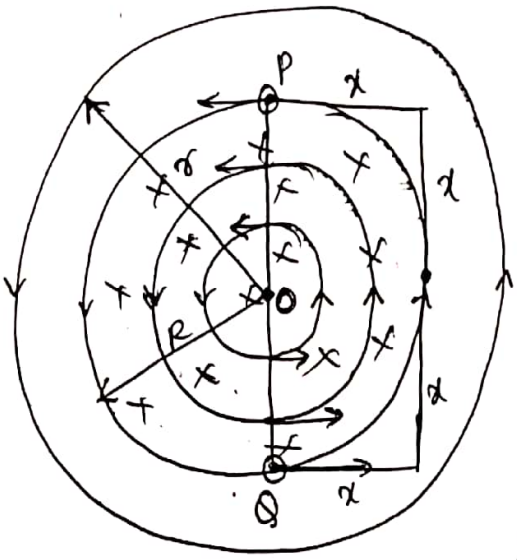
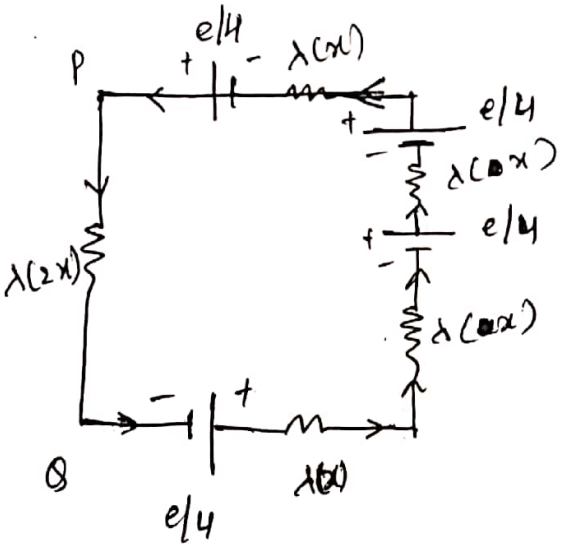
Graph



Ex 1





we have,

$e = A \frac{dB}{dt}$ (A: area enclosed by magnetic field)

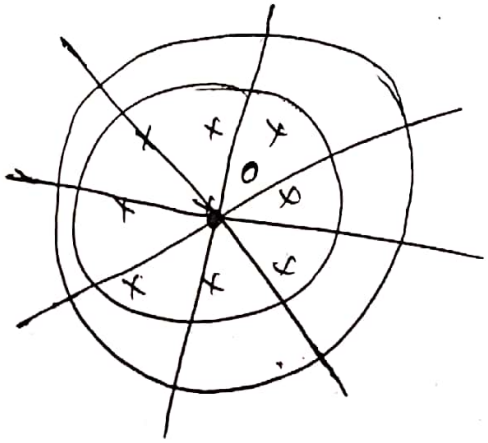
$$\boxed{e = \frac{\pi x^2}{2} \cdot \frac{dB}{dt}}$$

$+\frac{e}{4} - 2\lambda x\hat{i} + \frac{e}{4} - \lambda x\hat{i} - \lambda x\hat{i}$

$+\frac{e}{4} - i\lambda x + \frac{e}{4}$

$-i\lambda x = 0$

Ex 2



Any line passing through centre does not have emf i.e, no pot-difference. These lines are called equipotential lines.

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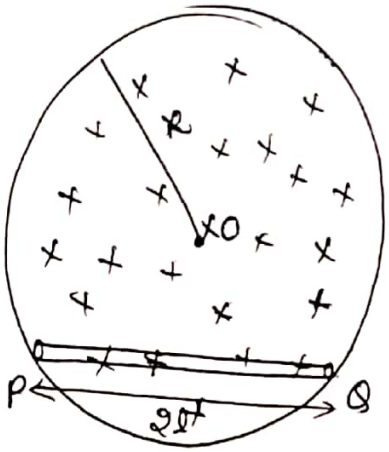
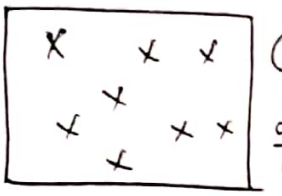
Eg



⊗ B

$\frac{dB}{dt} > 0$



⊗ B

$\frac{dB}{dt} > 0$

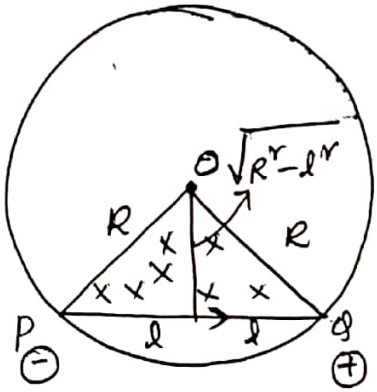
$e = A\frac{dB}{dt}$

Find induced emf across the ends of rod PQ.

Sol

Short cuts

draw two o lines from centre to ends of conductors PQ. (in order to close the loop)



$e = \frac{A\,dB}{dt}$

$= \frac{1}{2} \times 2l \times \sqrt{R^2-l^2} \cdot \frac{dB}{dt}$

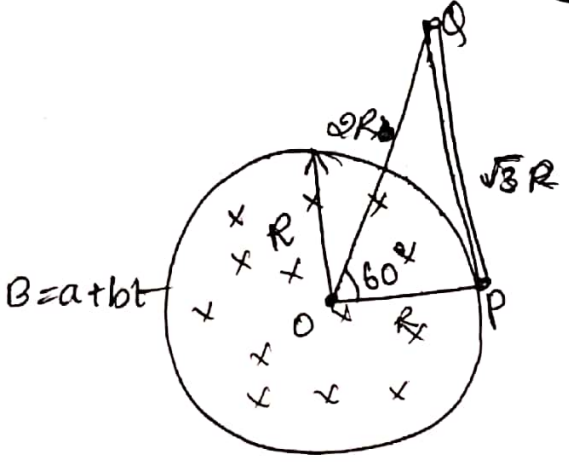
$e_{OPQ} = l\sqrt{R^2-l^2}\,\frac{dB}{dt}$

$e_{OP} + e_{PQ} + e_{QO} = e_{OPQ}$

$e_{PQ} = l\sqrt{R^2-l^2}\,\frac{dB}{dt}$

Q is at high potential.

Eg



If λ is the resistance per unit length, find $V_P - V_Q$.

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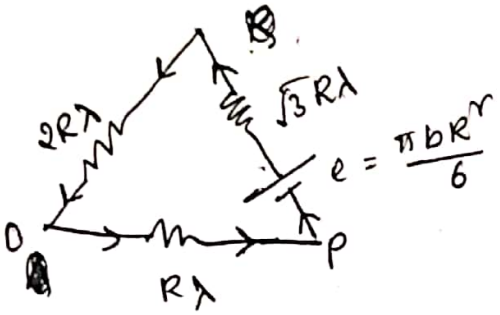
Sol 8

$$e_{OPQ} = A\frac{dB}{dt}$$

$$e_{OPQ} = \frac{\pi R^2}{6}\frac{dB}{dt} = \frac{\pi R^2}{6}(b)$$

$$e_{OP}^{0} + e_{PQ} + e_{QO}^{0} = \frac{\pi R^2 b}{6}$$

$$\boxed{e_{PQ} = \frac{\pi b R^2}{6}}$$



$$e - \sqrt{3}R\lambda i - 3R\lambda i = 0$$

$$\boxed{\frac{e}{(\sqrt{3}+3)R\lambda} = i}$$

~~$V_P + e - \sqrt{3}R\lambda i - 3R\lambda i = V_Q$~~

$$V_P + e - \sqrt{3}R\lambda i = V_Q$$

$$V_P - V_Q = \sqrt{3}R\lambda\left(\frac{e}{(\sqrt{3}+3)R\lambda}\right) - e$$

$$= \frac{e}{\sqrt{3}+1} - e$$

$$= \frac{e - (\sqrt{3}+1)e}{\sqrt{3}+1}$$

$$V_P - V_Q = \frac{-\sqrt{3}e}{\sqrt{3}+1}$$

$$= \frac{-\sqrt{3} \times \frac{\pi b R^2}{6}}{\sqrt{3}+1}$$

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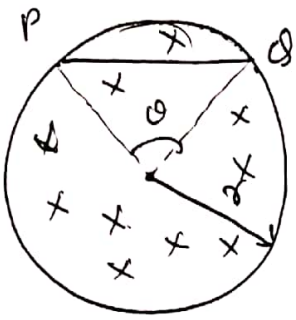
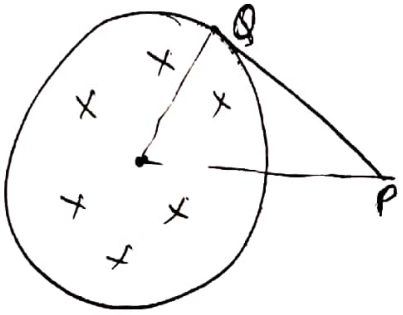
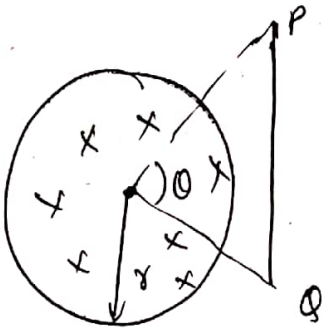
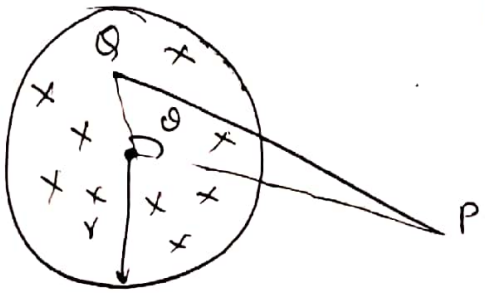
$$V_P - V_Q = \frac{-\pi b R^2}{2(\sqrt{3}+3)}$$

$$\boxed{V_Q - V_P = \frac{\pi b R^2}{2(\sqrt{3}+3)}.}$$

Q8

Column - I	Column - II
a) q, s 	$\frac{dB}{dt}$ = const p) Induced emf in the rod PQ is $\frac{1}{2} r^2 \theta \cdot \frac{dB}{dt}$
b) p, s 	q) Induced emf in rod PQ is less than $\frac{1}{2} r^2 \theta \cdot \frac{dB}{dt}$
c) p, r 	r) end P is positive w.r.to Q.
d) q, s 	s) End Q is positive w.r.to P.

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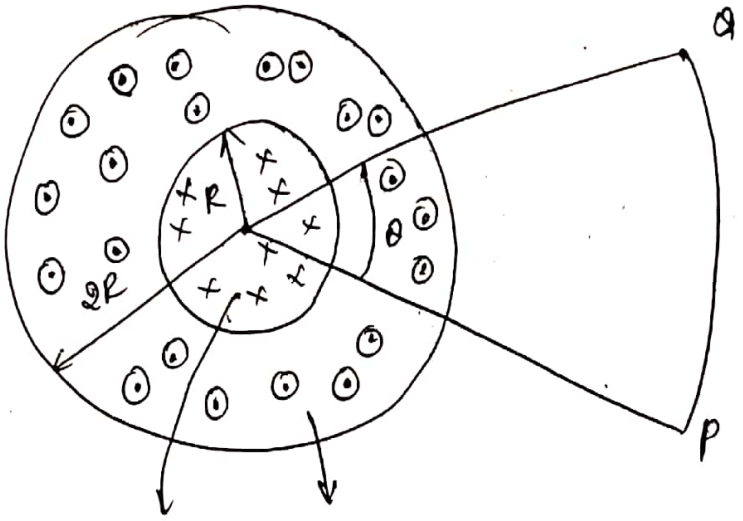
Exle



$\frac{dB}{dt} = 2k_0$ $\frac{dB}{dt} = 4k_0$ $e_{PQ} = ?$

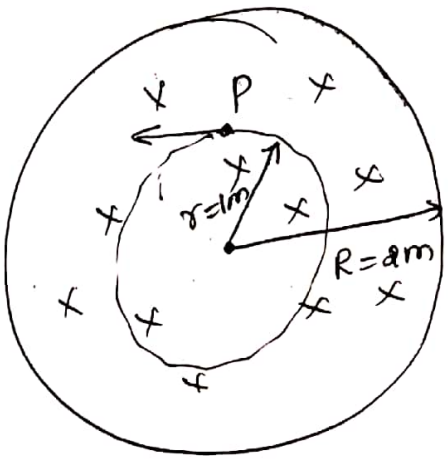
Solu

$$e_{PQ} = \frac{1}{2}R^2\theta\,(-2k_0) + \frac{1}{2}\left[(2R)^2 - R^2\right]\theta\,(2k_0)$$

$$= -R^2\theta k_0 + 6R^2\theta k_0$$

$$\boxed{e_{PQ} = 5R^2\theta k_0}$$

Exle



$B = 2 + 3t^2$

Find induced electric field at $t = 2$ sec.

Solu

$$\text{i.e.,} \quad E \times 2\pi r = \pi r^2 \frac{dB}{dt}$$

$$E = \frac{r}{2}\frac{dB}{dt}.$$

$$= \frac{1}{2} \times 3 \times 2 \times 2$$

$$\boxed{E = 6\ V/m}$$

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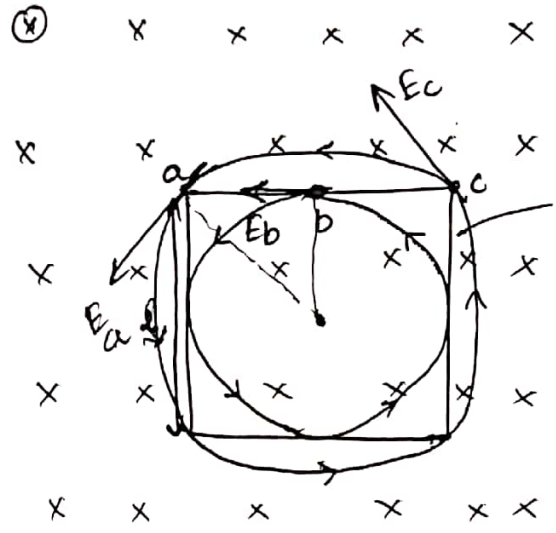
Ex 6



square conducting loop

Resistance of loop = R.

B is increasing with time at a const rate α.

i.e, $\frac{dB}{dt} = \alpha$.

1) $E = \frac{r}{2}\frac{dB}{dt}$

$$E_b = \frac{(l/2)}{2} \cdot \frac{dB}{dt} = \frac{l\alpha}{4}$$

$$E_a = E_c = \frac{(l/\sqrt{2})}{2} \cdot \frac{dB}{dt} = \frac{l\alpha}{2\sqrt{2}}$$

2) Induced emf in the square loop,

$$e = A\frac{dB}{dt}$$

$$\boxed{e = l^2\alpha}$$

3) Induced current,

$$i = \frac{e}{R} \Rightarrow \boxed{i = \frac{l^2\alpha}{R}}$$

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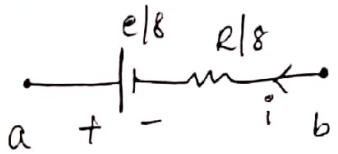
4) $V_a - V_b = ?$

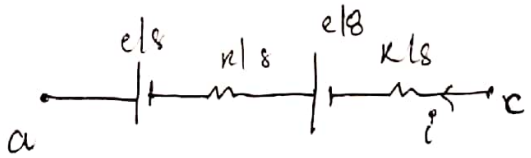


$$V_a - \frac{e}{8} + i\frac{R}{8} = V_b$$

$$V_a - \frac{e}{8} + \frac{e}{8} = V_b$$

$$\boxed{V_a - V_b = 0}$$

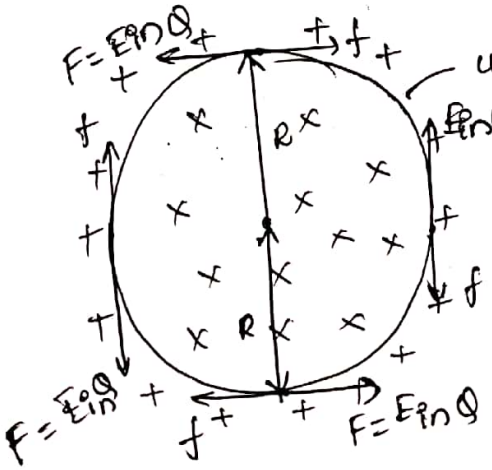
5) $V_a - V_c$



$$V_a - \frac{e}{8} + \frac{e}{8} - \frac{e}{8} + \frac{e}{8} = V_c$$

$$\boxed{V_a - V_c = 0}$$

Ex 6



uniformly charged non-conducting ring of radius r, mass m, charge q is kept on rough horizontal floor.

Magnetic field is $B = B_0 t^r$.

If the ring starts rotating at $t = 2$ sec. Find "μ".

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Sol8

Induced electric field,

$$E = \frac{r}{2} \cdot \frac{dB}{dt}$$

$$= \frac{r}{2} \cdot B_0(2t)$$

$$= 4 B_0 r t$$

* Torque on the small element about AOR,

$$d\tau = (dF)R$$

$$= E\,dq\,R$$

$$d\tau = B_0 R^2 t\, dq$$

∴ net torque on the ring, at time t,

$$\tau = d\tau = B_0 R^2 t \int dq$$

$$\boxed{\tau = B_0 R^2 t Q}, [ACW]$$

* Torque due to friction,

$$\tau = \int d\tau = \int \mu (dm) g R$$

$$\boxed{\tau = \mu g R (m)}, [CW]$$

Just at the time of rotating,

$$\tau_E = \tau_f$$

$$B_0 R^2 t Q = \mu g R (m)$$

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$$\boxed{\mu = \frac{B_0 R t Q}{mg}}$$

$$(\mu)_{\text{at } t=2\text{sec}} = \frac{2B_0RQ}{mg}.$$

* For $t < 2$ sec, ring is at rest.

$$\boxed{\tau_f = B_0 R^2 t Q} \quad [\tau_f \text{ is not } \mu mgR]$$

* For $t > 2$ sec, $\tau_E > \tau_f$

$$\mu < \frac{B_0 R t Q}{mg}$$

net torque, $\tau = \tau_E - \tau_f$

$$= B_0 R^2 t Q - \mu m g R$$

$$= B_0 R^2 t Q - \frac{2B_0RQ}{mg}(mgR)$$

$$\boxed{\tau = B_0 R^2 Q (t-2)}$$

$$\Rightarrow I\alpha = B_0 R^2 Q (t-2)$$

$$mR^2\alpha = B_0 R^2 Q (t-2)$$

$$\boxed{\alpha = \frac{B_0 Q (t-2)}{m}}$$

α increases linearly with time.

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$$\frac{dw}{dt} = \frac{B_0 Q}{m}(t-2)$$

$$\int_0^w dw = \frac{B_0 Q}{m}\int_2^t (t-2)\cdot dt$$

$$= \frac{B_0 Q}{m}\left[\frac{(t-2)^2}{2}\right]_2^t$$

$$w = \frac{B_0 Q}{2m}(t-2)^2$$

$$\boxed{w = \frac{B_0 Q (t-2)^2}{2m}}$$

w increases parabolically with time.

Again,

$$\frac{d\theta}{dt} = \frac{B_0 Q}{m}\frac{(t-2)^2}{2}$$

$$\int_0^\theta d\theta = \frac{B_0 Q}{2m}\int_2^t (t-2)^2 \cdot dt$$

$$\theta = \frac{B_0 Q}{2m}\frac{(t-2)^3}{3}$$

$$\boxed{\theta = \frac{B_0 Q (t-2)^3}{6m}}$$

No. of revolutions, $N = \frac{\theta}{2\pi}$.

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Suppose, if field is switched off at $t = 4$ sec,

$$E = 0 \Rightarrow \tau_E = 0$$

$$\therefore \tau_{net} = -\tau_F$$

$$\Rightarrow I\alpha = -\tau_F$$

$$mR^2\alpha = -\mu mgR$$

$$mR^2\alpha = -\frac{2B_0Rq}{mg}(mgR)$$

$$\boxed{\alpha = -\frac{2B_0q}{m}} \quad \text{const.}$$

for $t > 4$ sec,

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\omega^2 - \omega_0^2 = 2\alpha\theta$$

$$N = \frac{\theta}{2\pi}$$

where $\omega_0 = \omega_{at\ t=4} = \frac{2B_0q}{m}$

time taken by ring to come to rest after field is switched off.

$$\omega = 0, \quad \omega_0 + \alpha t = 0$$

$$t = -\frac{\omega_0}{\alpha} \Rightarrow \boxed{t = 1\ \text{sec}}$$

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for $t > 4$ sec

$$\theta = \frac{B_0 q}{m}$$

$$N = \frac{B_0 q}{2\pi m}$$

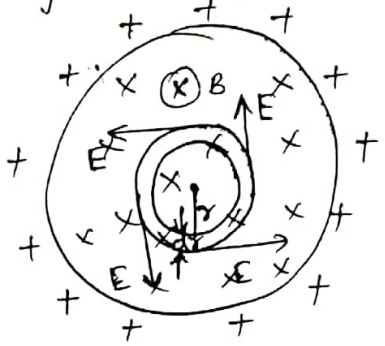
at $t = 4$ sec

$$\theta = \frac{4 B_0 q}{3m}$$

$$N = \frac{4 B_0 q}{3(2\pi m)}$$

total no. of revolutions $= \dfrac{7 B_0 q}{6\pi m}$.

*** Ex 8 Uniformly charged non-conducting disc of radius R, mass m, charge q. $B = B_0 t^2$. If disc starts rotating on rough floor at $t = 2$ sec. Find μ.



Sol:

$$E = \frac{r}{2} \cdot \frac{dB}{dt}$$

$$= \frac{r}{2} \times 2 B_0 t$$

$$E = r B_0 t$$

* Torque on the elementary ring,

$$d\tau = (dF) r$$

$$= r B_0 t \left(\frac{q}{\pi R^2} \cdot 2\pi r \cdot dr \right) r$$

$$\tau_E = \int_0^R d\tau$$

$$= \frac{B_0 t q}{R^2} \cdot \frac{2 R^4}{4} \cdot 2$$

$$\boxed{\tau_E = \frac{B_0 q t R^2}{2}}$$

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* Torque due to friction,

$$d\tau = \mu\, dm\, g\, r$$

$$d\tau = \mu \frac{m}{\pi R^2} g\,(2\pi r\, dr)\cdot r$$

$$\tau_f = \frac{2\mu mg}{R^2}\int_0^R r^2\, dr$$

$$\tau_f = \frac{2}{3}\mu mgR$$

net torque, $\tau = \tau_E - \tau_f$

τ_f ↓ zero at $t = 2$ sec

* At $t = 2$ sec, disc is ready to rotate,

$$\tau_E = \tau_f$$

$$\frac{B_0 \alpha t R^2}{2} = \frac{2}{3}\mu mgR$$

$$\boxed{\mu = \frac{3B_0 \alpha R}{2mg}}$$

At $t > 2$ sec, $\tau_E > \tau_f$

$$\tau = \tau_E - \tau_f$$

$$= \frac{B_0 \alpha R^2 t}{2} - \frac{2}{3}\frac{BB_0 \alpha R}{2mg}(mgR)$$

$$\boxed{\tau = \frac{B_0 \alpha R^2}{2}(t-2)}$$

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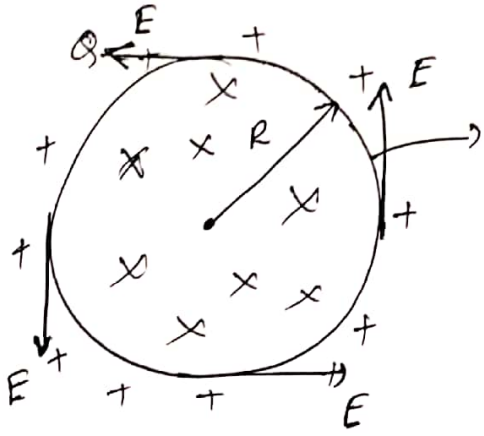
Ex 8



→ non-conducting ring is placed on smooth horizontal surface when field suddenly increased from zero to B. Find angular velocity of ring just after field changing to B.

Sol 8

$$E = \frac{R}{2}\frac{dB}{dt}$$

* Torque on the ring due to field,

$$\tau = (EQ)R = \frac{R^2Q}{2}\cdot\frac{dB}{dt}$$

we have,

Angular impulse,

$$\int \tau \cdot dt = I\omega$$

$$\int \frac{R^2Q}{2}\cdot\frac{dB}{dt}\cdot dt = mR^2\omega$$

$$\frac{R^2Q}{2}(B) = mR^2\omega$$

$$\boxed{\omega = \frac{BQ}{2m}}$$

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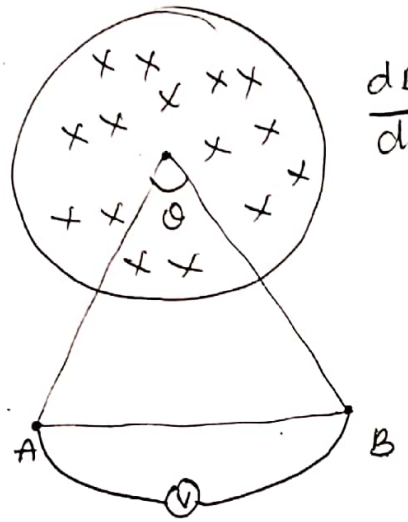
Ex 6

①



$\frac{dB}{dt} > 0$

A & B are two points in space.

1) $e_{AB} = A\frac{dB}{dt} = \frac{1}{2}R^2\theta \cdot \frac{dB}{dt} \neq 0$

2) potential difference b/w points A & B i.e,

$V_A - V_B = 0$

$$\int_{V_A}^{V_B} dV = -\int \bar{E}_c \cdot d\bar{l}$$

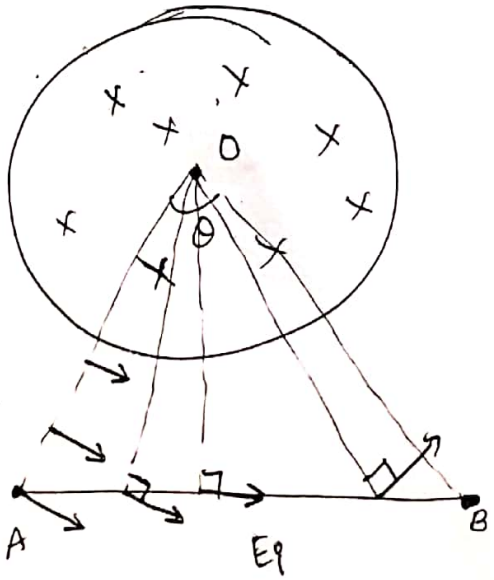
($\bar{E}_c \to 0$)

$$V_B - V_A = 0 \Rightarrow \boxed{V_A = V_B}$$

3) If ideal voltmeter is connected b/w A & B.

It's reading is zero. [no current in AB]

②



AB is rod of negligible resistance.

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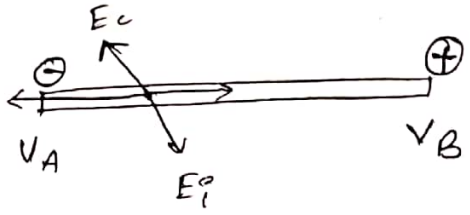
1) $e_{AB} = A \frac{dB}{dt} = \frac{1}{2} R^2 \theta \cdot \frac{dB}{dt}$

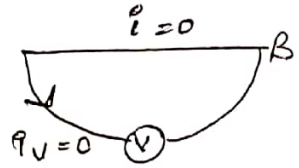
2) $V_A - V_B = -e_{AB}$



3) $i_{induced} = 0$ (∵ there is no closed loop)

4) If ideal voltmeter is connected b/w A & B,

Voltmeter reading = zero.

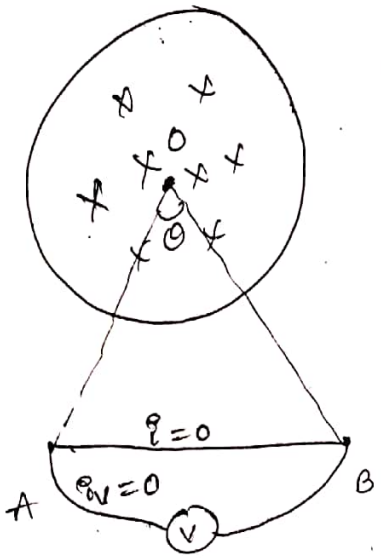


$R_V \longrightarrow \infty$.

$i_V \longrightarrow 0$

but $R_V \cdot i_V$ = finite

(3)



rod AB is having some resistance R

1) $e_A = A \cdot \frac{dB}{dt} = \frac{1}{2} R^2 \theta \cdot \frac{dB}{dt}$

here resistance is considered only when current is flowing through it.

2) $V_A - V_B = -e_{AB}$

3) reading of voltmeter = zero.

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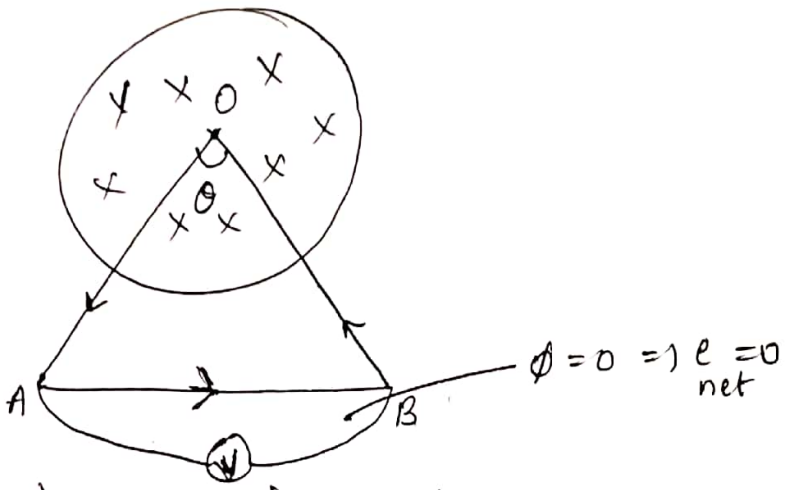
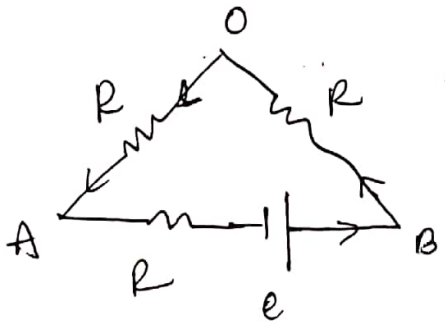
(4)



$\phi = 0 \Rightarrow e_{net} = 0$

1) $e_{AB} = \frac{1}{2} R^2 \theta \cdot \frac{dB}{dt}$

2)



$i = \frac{e}{3R}$

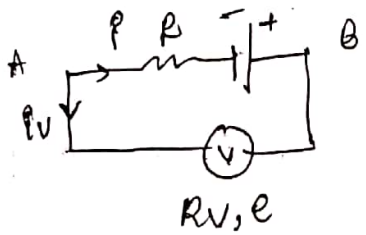
$V_A - iR + e = V_B$

$V_A - V_B = -e + \frac{e}{3}$

$V_A - V_B = -\frac{2e}{3}$

$\boxed{V_B - V_A = \frac{2e}{3}}$

3) If ideal voltmeter is connected b/w A & B,



$-i_V R_V + e - e + iR = 0$

$\boxed{i_V R_V = iR}$

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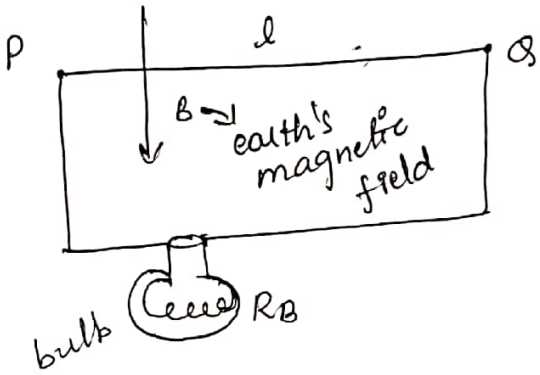
∴ voltmeter reading $= i_V R_V = iR = \frac{e}{3}$.

Exe

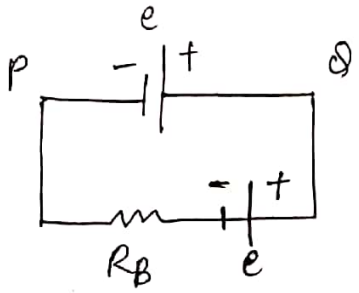


will the Bulb glow?

Sol

$\phi = \text{const} \Rightarrow e = 0 \Rightarrow i_{ind} = 0$ (loop)

bulb will not glow.



$V_P + e = V_Q$

$V_P - V_Q = -e$

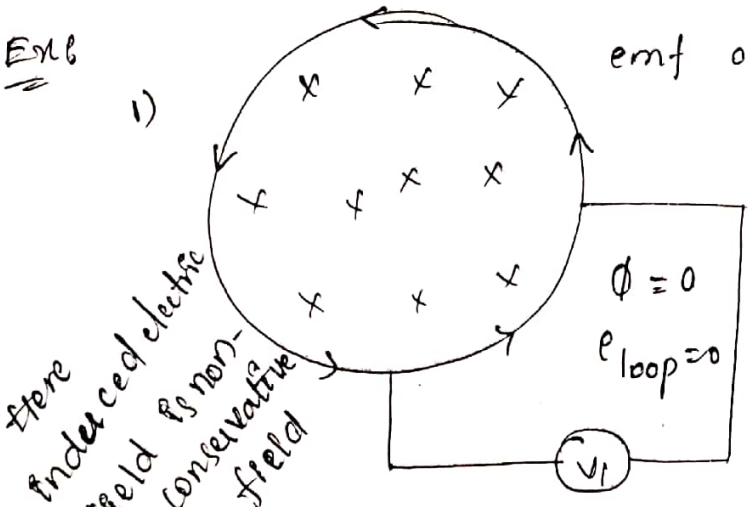
$\boxed{V_Q - V_P = Bdv}$

Exe

emf of loop is 12V

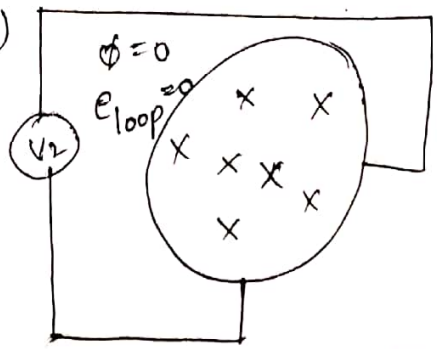
loop resistance $= R$

1)

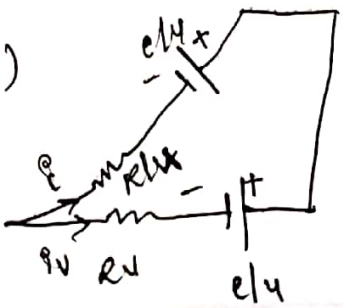


there induced electric field is non-conservative field

2)



1)



$-i_V R_V + \frac{e}{4} - \frac{e}{4} + \frac{iR}{4} = 0$

Voltmeter reading $i_V R_V = \frac{iR}{4} = \frac{12}{4} = 3V$

2) $i_V R_V = \frac{3iR}{4}$

Voltmeter reading $= \frac{3 \times 12}{4} = 9V$

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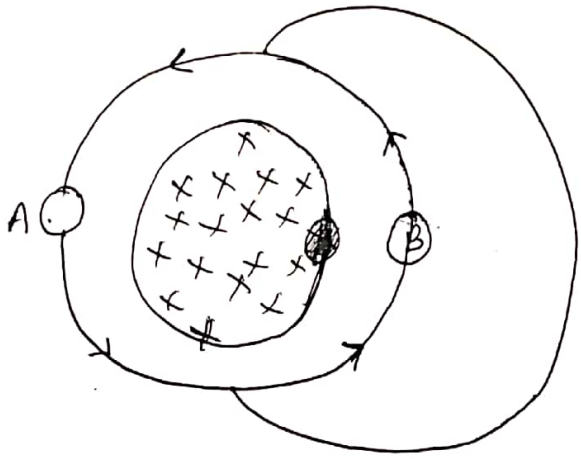
11/06/25 | wednesday

Ex 8



$\frac{dB}{dt} > 0$

Now the two points are shortened by another wire lying to the right of bulb B in the plane of the page. After shortening, wire is inserted which is correct

a) Bulb A goes out

b) Bulb B gets brighter

~~c~~ Bulb B goes out

~~d~~ Bulb A gets brighter.

Sol 8 $e = A \cdot \frac{dB}{dt}$

* current $i = \frac{e}{R_T} = \frac{e}{R_1 + R_2}$

After connecting wire,

no current flows through bulb B.

∴ bulb B goes out.

$i = \frac{e}{R_1}$ increases through bulb A.

∴ bulb A gets brighter.

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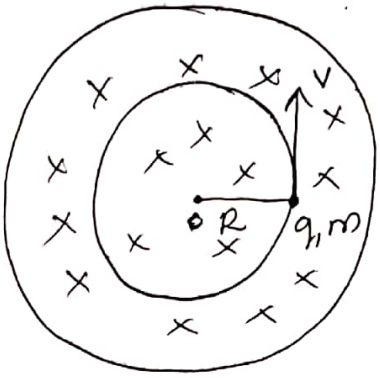
Ex 16



The centre cross-section of a long cylindrical region containing uniform but time varing magnetic field B as shown. A particle of unknow constant mass and variable +ve charge moves in a plane in such a way so that the radius of the circle remains constant. Then Answer the following questions.

1) magnetic field increased by 2% then speed will be
~~a)~~ increase by 1%.
b) decrease by 2%.
c) increase by 2%.
d) decrease by 1%.

2) In the above question, charge will be
a) increase by 2%.
b) increase by 1%.
c) decrease by 2%.
~~d)~~ decrease by 1%.

3) choose the correct option:
a) Both charge and mass can be const
~~b)~~ If mass is const, charge must change.
c) If mass is const, charge must increase

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d) none of these are constants.

Sol:

1) $F = qE$

$$E = \frac{R}{2} \cdot \frac{dB}{dt}$$

$$m\frac{dv}{dt} = \frac{R}{2}\frac{dB}{dt} \cdot q$$

$$dv = \frac{mv}{2Bq \cdot m} \cdot q$$

$$\frac{dv}{v} = \frac{1}{2B} \cdot dB$$

$$\frac{dv}{v} = \frac{1}{2} \cdot \frac{dB}{B}$$

$$\Rightarrow \frac{dv}{v} \times 100\% = \frac{1}{2} \times \frac{dB}{B} \times 100\%$$

$$\frac{dv}{v} \times 100\% = 1\%$$

2) $R = \frac{mv}{Bq}$

$$q = \frac{mv}{BR}$$

$$\frac{dq}{q} = \frac{dm}{m}^{0} + \frac{dv}{v} - \frac{dB}{B} - \frac{dR}{R}^{0}$$

$$\frac{dq}{q} \times 100\% = -1\%$$

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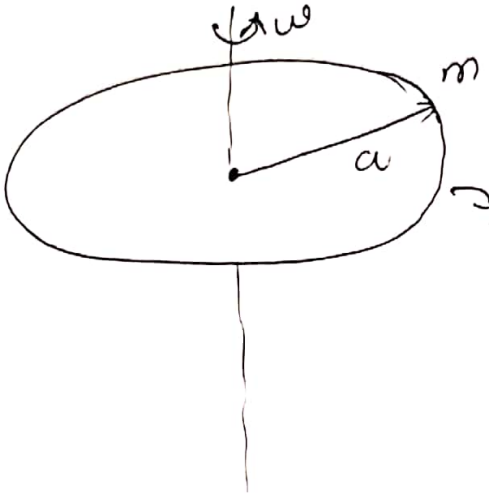
Ex:



Thin non-conducting ring.

At the initial moment, A uniform magnetic field is switched on which is vertically downwards & increases with time according to the law $B = B_0 t$. Now answer the following.

1) The angular acceleration of ring & its direction of rotation as seen from above.

2) The power developed by the forces acting on the ring as a function of time.

Sol:

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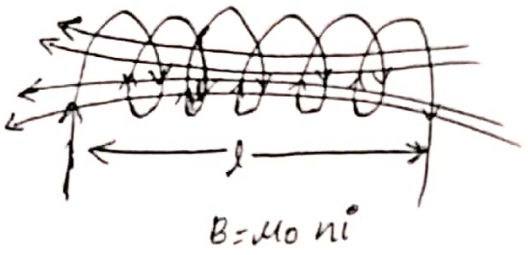
Self induction

* Introduced by Henry.

* Flux through coil,

$$\phi = NBA$$



$B = \mu_0 ni$

$$\boxed{\phi_T = N\mu_0 niA}$$

* when $i(\uparrow) \Rightarrow \phi_T(\uparrow) \Rightarrow$ 'e' is developed.

  when $i(\downarrow) \Rightarrow \phi_T(\downarrow) \Rightarrow$ 'e' is developed.

* The phenomenon in which induced emf is developed in the coil whenever current through the coil is changing is called "Self Induction".

Coefficient of Self Inductance (L) :-

* we have,

$$\phi_T \propto i$$

$$\boxed{\phi_T = Li}$$

$$\boxed{L = \frac{\phi_T}{i}}$$

$\boxed{L = \phi_T}$ when $i = 1A$

* From faraday's Law,

$$e = -\frac{d\phi}{dt} = -\frac{d}{dt}(Li)$$

$$\boxed{e = -L \cdot \frac{di}{dt}}$$

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$L = \frac{|e|}{di/dt} \Rightarrow \boxed{L = e}$ when $\frac{di}{dt} = 1\,A/sec$

Units:

Henry (H)

(or)

Weber/Amp

(or)

$\frac{\text{Volt-sec}}{\text{Amp}}$

(or)

ohm-sec

D.F:

$$L = \frac{V\text{-sec}}{\text{Amp}}$$

$$= \frac{M^1 L^2 T^{-2} A^{-1} T^{-1} \cdot T}{A}$$

$$\boxed{L = [M^1 L^2 T^{-2} A^{-2}]}$$

Inductor — L – henry $= [M^1 L^2 T^{-2} A^{-2}]$

capacitor — C – farad $= [M^{-1} L^{-2} T^4 A^2]$

Resistor — R – ohm

cell / battery — + | − , E – volt

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1) $C = \frac{q}{V} = \frac{i^2 t^2}{W} = \frac{A^2 T^2}{M^1 L^2 T^{-2}} = [M^{-1} L^{-2} T^4 A^2]$

2) $R = \frac{V}{i} = \frac{W}{i^2 t} = \frac{M^1 L^2 T^{-2}}{A^2 T^1} = [M^1 L^2 A^{-2} T^{-3}]$

3) $e = \frac{W}{q} = \frac{M^1 L^2 T^{-2}}{A^1 T^1} = [M^1 L^2 A^{-1} T^{-3}]$

4) $\phi = BA = M^1 A^{-1} T^{-2} \cdot L^2 = [M^1 L^2 T^{-2} A^{-1}]$

5) $B = \frac{F}{il} = \frac{M^1 L^1 T^{-2}}{A^1 L^1} = [M^1 L^0 A^{-1} T^{-2}]$

* we have,

$$\boxed{e = -L \cdot \frac{di}{dt}}$$

* Ideal inductors — only inductance but no resistance.

* practical inductors — both inductance & resistance.

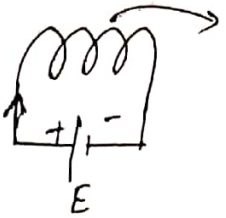
** induced emf opposes the rate of change of current.

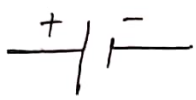
points!

1) const = i

$\frac{di}{dt} = 0 \Rightarrow \boxed{e = 0}$

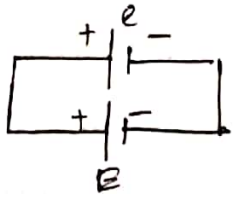
2) $\frac{di}{dt} \neq 0$ i





$e = L \cdot \frac{di}{dt}$

$\Rightarrow$



$\boxed{E_{net} = E - e}$

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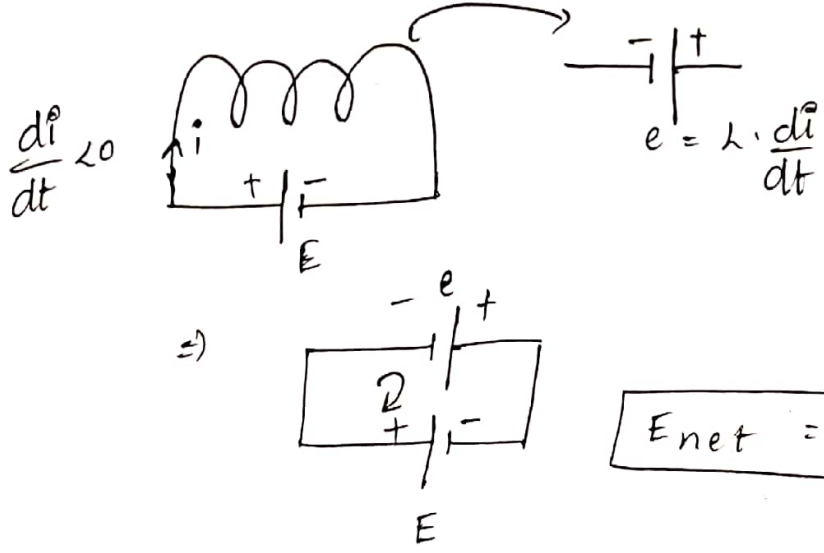
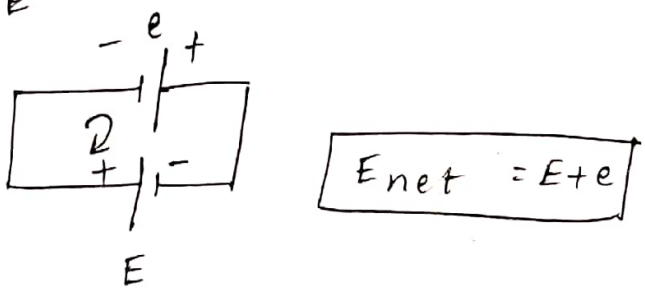
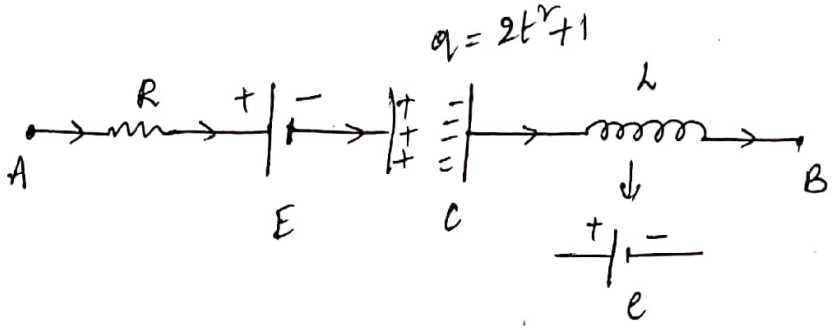
3)



⇒



Ex 8



Sol 8

$$i = \frac{dq}{dt} = 4t$$

$$\frac{di}{dt} = 4$$

$$V_A - iR - E - \frac{q}{C} - L \cdot \frac{di}{dt} = V_B$$

$$V_A - V_B = 4Rt + E + \frac{q}{C} + 4L$$

Calculation of Self Induction :

Steps :-

1) Assume current 'i' in the coil.

2) Find ϕ_T linked with coil.

3) $\phi_T = Li$

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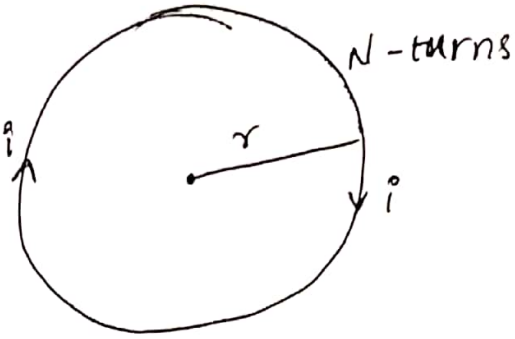
a) Circular Coil:



* magnetic field at the centre of coil,

$$B = \frac{\mu_0 N i}{2r}$$

* ϕ through each turn of coil,

$$\phi_B = BA = \frac{\mu_0 N i}{2r} \times \pi r^2 = \frac{\mu_0 \pi N i r}{2}$$

* Total flux linked with coil,

$$\phi_T = N\phi_B$$

$$\boxed{\phi_T = \frac{\mu_0 \pi N^2 i r}{2}}$$

Finally,

$$\phi_T = Li$$

$$\frac{\mu_0 \pi N^2 i r}{2} = Li$$

$$\boxed{L = \frac{\mu_0 \pi N^2 r}{2}} \quad *$$

If the medium inside the coil is having relative permeability (μ_r) then $\boxed{L = \frac{\mu_0 \mu_r \pi N^2 r}{2}}$

clearly, self inductance of a coil depends on size, shape and total no of turns of coil and

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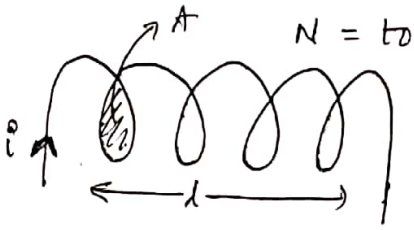
medium inside the coil but doesnot depends on current through the coil.

b) Solenoid :-



$n = \frac{N}{l}$ = no. of turns per unit length

field inside the coil,

$$B = \mu_0 n i$$

& $$\phi = BA = \mu_0 n i A$$

$$\phi_T = N\phi = \mu_0 n N i A$$

$$\phi_T = \mu_0 n^2 i l A$$

$$L i = \mu_0 n^2 i l A$$

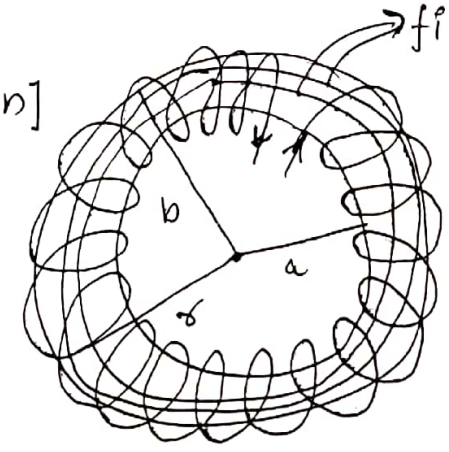
$$\boxed{L = \mu_0 n^2 A l}$$

$$\boxed{L = \mu_0 n^2 A l = \frac{\mu_0 N^2 A}{l}}$$

In general,

$$\boxed{L = \mu_0 \mu_r n^2 A l = \frac{\mu_0 \mu_r N^2 A}{l}}$$

c) Toroid : [circular cross-section]



here, $r = \frac{a+b}{2}$

$$B = \frac{\mu_0 N i}{2\pi r} = \frac{\mu_0 N i}{\pi (a+b)}$$

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b

$$\phi = BA$$

$$\phi_T = N\phi$$

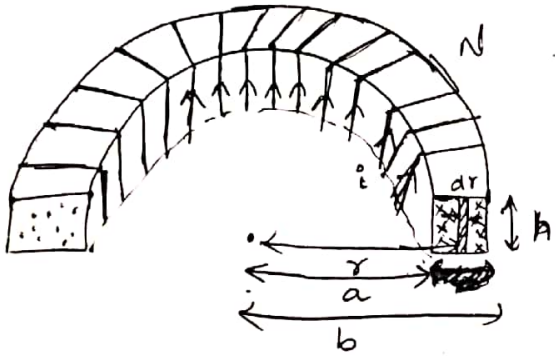
$$\phi_T = \frac{\mu_0 N^2 i A}{\pi(a+b)}$$

$$L I = \frac{\mu_0 N^2 I A}{\pi(a+b)}$$

$$\boxed{L = \frac{\mu_0 N^2 A}{\pi(a+b)}}$$

d) Toroid :

[rectangular cross-section]



Flux through small area,

$$d\phi = BdA$$

$$d\phi = \frac{\mu_0 N i}{2\pi r} \cdot h \cdot dr$$

Total flux through each turn,

$$\phi = \int_{r=a}^{r=b} d\phi = \frac{\mu_0 N i}{2\pi} h \int_a^b \frac{dr}{r}$$

$$\phi = \frac{\mu_0 N i h}{2\pi} \ln(b/a)$$

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Now, $\phi_T = N\phi = \frac{\mu_0 N^2 i h}{2\pi} \ln(b/a)$

* we have,

$\phi_T = Li$

$Li = \frac{\mu_0 N^2 h}{2\pi} \ln(b/a)$

$$\boxed{L = \frac{\mu_0 N^2 h}{2\pi} \ln(b/a)}$$

* If $a >>> b - a$

$$L = \frac{\mu_0 N^2 h}{2\pi} \ln\left(1 + \frac{b-a}{a}\right)$$

$$L = \frac{\mu_0 N^2 h}{2\pi} \left(\frac{b-a}{a}\right)$$

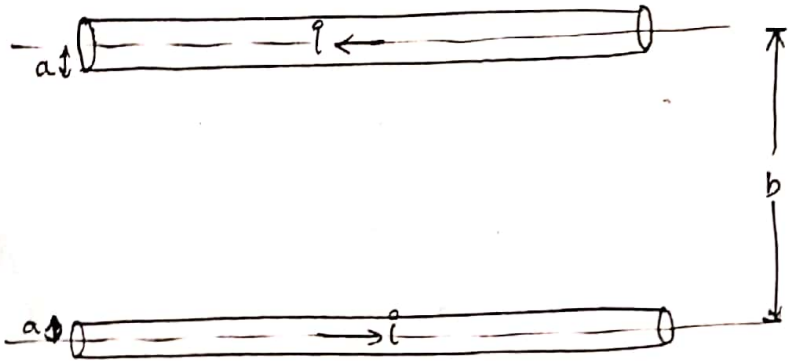
$A = h(b-a)$

$$\boxed{L = \frac{\mu_0 N^2 A}{2\pi a}}$$

↳ similar to toroid with circular cross-section.

Ex:6

Two long current carrying wires.



Find the self inductance per unit length of the system

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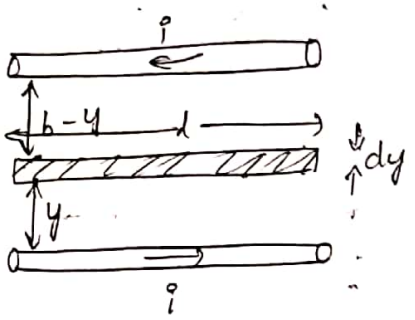
Sol:



1) $B = B_1 + B_2$

$$B = \frac{\mu_0 i}{2\pi y} + \frac{\mu_0 i}{2\pi (b-y)}$$

$$= \frac{\mu_0 i}{2\pi} \left(\frac{b-y+y}{y(b-y)} \right)$$

$$\boxed{B = \frac{\mu_0 i b}{2\pi y(b-y)}}$$

2) Flux through small area,

$$d\phi = B\,dA = lB\,dy$$

$$\phi_T = \int_{y=a}^{y=b} d\phi$$

$$= \frac{\mu_0 i b l}{2\pi} \int_a^b \frac{dy}{y(b-y)}$$

$$= \frac{\mu_0 i l}{2\pi} \int_a^b \frac{b-y+y}{y(b-y)}\,.dy$$

$$= \frac{\mu_0 i l}{2\pi} \int_a^{b-a} \frac{1}{y} + \frac{1}{b-y}\,.dy$$

$$= \frac{\mu_0 i l}{2\pi} \ln\left(\frac{b-a}{a}\right) + \ln\frac{a}{b-a}$$

$$= \frac{\mu_0 i l}{2\pi} 2\ln\left(\frac{b-a}{a}\right)$$

$\phi = Li$ →

$$\boxed{\phi_T = \frac{\mu_0 i l}{\pi} \ln\left(\frac{b-a}{a}\right)}$$

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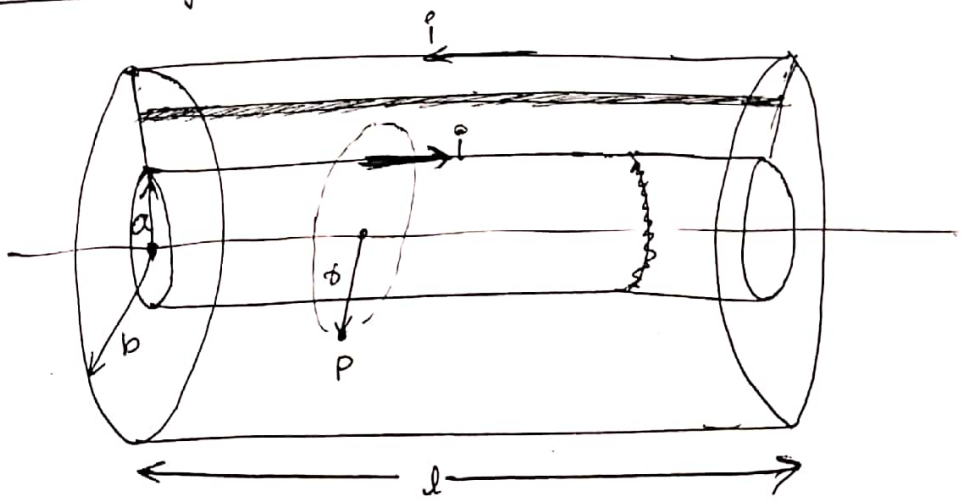
Two co-axial cylindrical conductors   12|06|25| Thursday



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$B \times 2\pi r = \mu_0 i$$

$$\boxed{B = \frac{\mu_0 i}{2\pi r}}$$

* flux through small area,

$$d\phi = B\,dA$$

$$= \frac{\mu_0 i}{2\pi r} \cdot l\,dr$$

$$\phi = \frac{\mu_0 i l}{2\pi} \ln(b/a)$$

$$Li = \frac{\mu_0 i l}{2\pi} \ln(b/a)$$

$$\boxed{L = \frac{\mu_0 l}{2\pi} \ln(b/a)}$$

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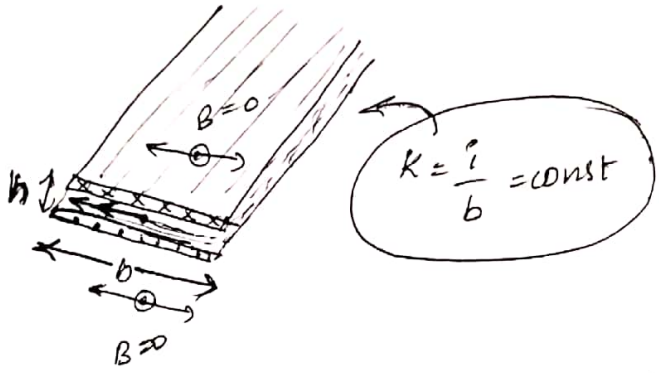
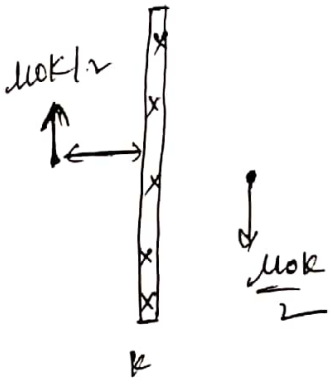
Infinite plane sheets :-





B_{inside} is,

$$B_{net} = B_1 + B_2$$

$$= \frac{\mu_0 K}{2} + \frac{\mu_0 K}{2}$$

$$\boxed{B_{net} = \mu_0 K = \text{uniform}}$$

$$\phi = BA = \mu_0 K l h$$

$$Li = \mu_0 K l h$$

$$\frac{L}{l} = \frac{\mu_0 h i}{b}$$

$$\boxed{\frac{L}{l} = \frac{\mu_0 h}{b}}$$

Energy Stored in an Inductor:

* Solenoid is a good inductor.
  ↓
  best example for inductor.

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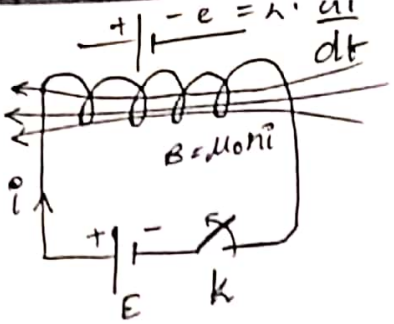

$0 \rightarrow i$

$\Rightarrow \frac{di}{dt} \neq 0$

Power given to inductor by battery.

$$P = \mathcal{E} i$$

rate of energy stored in inductor.

$$\frac{dU}{dt} = \left(L \cdot \frac{di}{dt}\right) i$$

$$\int dU = \int L i \cdot di$$

$$\boxed{U = \frac{1}{2} L i^2} \star$$

Potential energy stored in magnetic field.

* Energy density -

$$u_B = \frac{U}{V} = \frac{\frac{1}{2} L i^2}{V}$$

* For ext.,

In case of solenoid,

$$L = \mu_0 N^2 A l$$

$$u_B = \frac{\frac{1}{2} \mu_0 N^2 A l i^2}{V} = \frac{1}{2} \frac{\mu_0^2 N^2 i^2 A l}{\mu_0 V}$$

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$$u_B = \frac{1}{2}\frac{B^2}{\mu_0} \quad *$$

and $u_B = \frac{dU}{dV}$

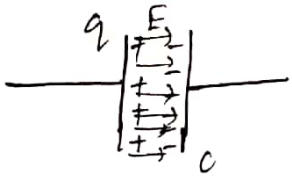
$$dU = u_B \cdot dV$$

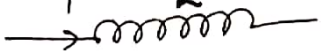
Total energy stored,

$$U = \int dU = \int u_B \cdot dV = \int \frac{B^2}{2\mu_0} \cdot dV$$

* B can be calculated by using Ampere's Law.

* Compaision :-)





1) $V_C = \frac{q}{C}$ $\qquad V = L \cdot \frac{di}{dt}$

2) $E = \frac{q}{\varepsilon_0 A}$ $\qquad B = \mu_0 n i$

3) $U_E = \frac{1}{2} C V^2$ $\qquad U = \frac{1}{2} L i^2$

4) $u_E = \frac{1}{2} \varepsilon_0 E^2$ $\qquad u_B = \frac{1}{2}\frac{B^2}{\mu_0}$

5) $U_E = \int \frac{1}{2} \varepsilon_0 E^2 \cdot dV$ $\qquad U = \int \frac{B^2}{2\mu_0} \cdot dV$

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6) To find E,

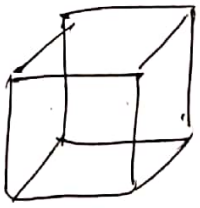
$$\oint \bar{E} \cdot d\bar{s} = \frac{q_{encl}}{\varepsilon_0}$$

To find B,

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i$$

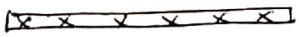
Exp

Find energy stored in cube.



cube is having volume 'V'.

$$U = \int \frac{B^2}{2\mu_0} \cdot dv$$



Infinite plane sheet.

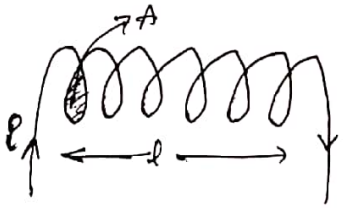
$$= \frac{\left(\frac{\mu_0 K}{2}\right)^2}{2\mu_0} \cdot V$$

$$\boxed{U = \frac{\mu_0 K^2}{8} V}$$

Calculation of Self Induction using energy methods

a) Solenoid:

field inside the solenoid,



$$B = \mu_0 n i = \text{uniform}.$$

$$\text{now, } U = \int \frac{B^2}{2\mu_0} \cdot dv$$

$$U = \frac{B^2}{2\mu_0} V$$

$$= \frac{\mu_0^2 n^2 r_0^2}{2\mu_0} V$$

$$= \frac{\mu_0 n^2 i^2}{2} V$$

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$$U = \frac{\mu_0 n^2 i^2 A.l}{2}$$

$$\frac{1}{2} L i^2 = \frac{\mu_0 n^2 i^2 A l}{2}$$

$$\boxed{L = \mu_0 n^2 A l}$$

If medium inside the solenoid has relative permeability μ_r then $B = \mu_0 \mu_r n i$ (where $\mu = \mu_0 \mu_r$)

$$B = \mu n i$$

$$U = \int \frac{B^2 dv}{2\mu}$$

$$= \frac{B^2}{2\mu} V$$

$$\frac{1}{2} L i^2 = \frac{\mu^2 n^2 i^2}{2\mu} (Al)$$

$$L = \mu n^2 A l$$

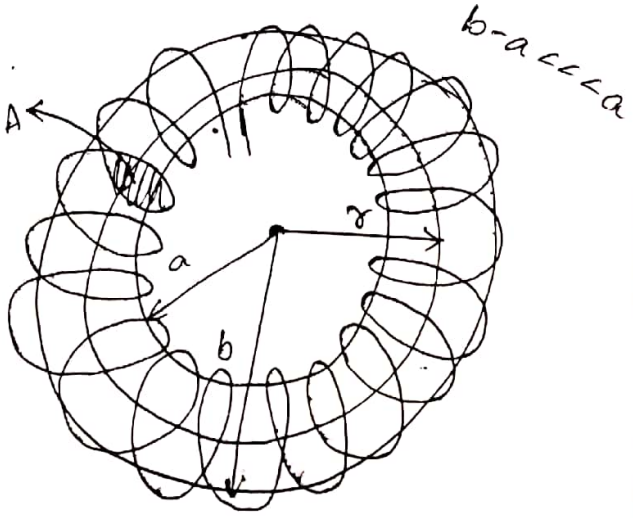
$$\boxed{L = \mu_0 \mu_r n^2 A l}$$

b) Toroid

$$B = \mu_0 n i \qquad V = A \times 2\pi r$$

$$B = \frac{\mu_0 N i}{2\pi r}$$

$$U = \int \frac{B^2}{2\mu_0} \cdot dv$$



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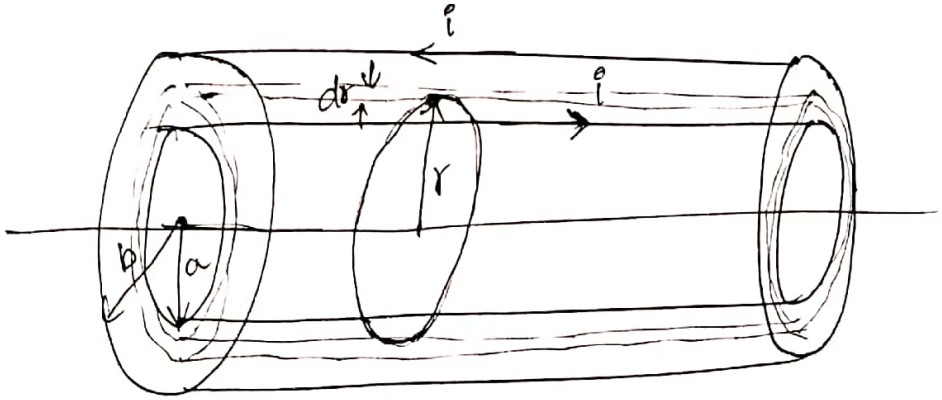
$$= \frac{\mu_0 N^2 i^2}{2\mu_0 (2\pi r)} (A \cdot 2\pi r)$$

$$\frac{1}{2} L i^2 = \frac{\mu_0 N^2 i^2 A}{2(2\pi r)}$$

$$\boxed{L = \frac{\mu_0 N^2 A}{2\pi r}}$$

c) concentric cylindrical shells :



$$B = \frac{\mu_0 i}{2\pi r}$$

$$dv = 2\pi r \cdot l \cdot dr$$

$$U = \int \frac{B^2}{2\mu_0} (2\pi l) \cdot r \cdot dr$$

$$= \frac{2\pi l}{2\mu_0} \times \frac{\mu_0^2 i^2}{(2\pi)^2} \int_a^b \frac{r \cdot dr}{r^2}$$

$$\boxed{U = \frac{\mu_0 i^2 l}{4\pi} \ln(b/a)}$$

$$\frac{1}{2} L i^2 = \frac{\mu_0 i^2 l}{2 \cdot 4\pi} \ln(b/a)$$

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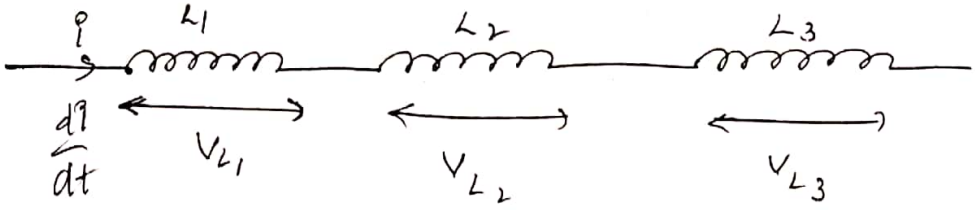
$$L = \frac{\mu_0 l}{2\pi} \ln(b/a)$$

Combination of Inductors:

1) Series Combinations



Here i & $\frac{di}{dt}$ are same for

we have,

$$V_L = V_{L_1} + V_{L_2} + V_{L_3}$$

$$V_{L_{eq}} \cdot \frac{di}{dt} = L_1 \cdot \frac{di}{dt} + L_2 \cdot \frac{di}{dt} + L_3 \cdot \frac{di}{dt}$$

$$L_{eq} = L_1 + L_2 + L_3$$

Energy stored in all 3 inductors,

$$U = U_1 + U_2 + U_3$$

$$U = \frac{1}{2} L_1 i^2 + \frac{1}{2} L_2 i^2 + \frac{1}{2} L_3 i^2$$

$$U = \frac{1}{2} L_{eq} \cdot i^2$$

* we have, $U_1 = \frac{1}{2} \times L_1 i^2$, $U_2 = \frac{1}{2} \times L_2 \times i^2$, $U_3 = \frac{1}{2} L_3 i^2$

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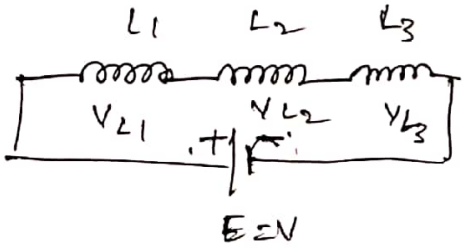
$$V_1 : V_2 : V_3 = L_1 : L_2 : L_3$$

$$V = V_1 + V_2 + V_3$$

$$V_1 = \frac{L_1 V}{L_1 + L_2 + L_3}, \quad V_2 = \frac{L_2 V}{L_1 + L_2 + L_3}, \quad V_3 = \frac{L_3 V}{L_1 + L_2 + L_3}.$$

& $V_{L_1} : V_{L_2} : V_{L_3} = L_1 : L_2 : L_3$

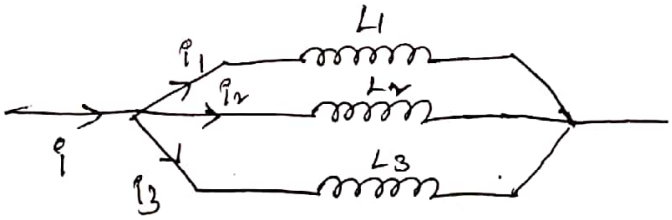
$$V_{L_1} = \frac{L_1 V}{L_1 + L_2 + L_3}$$



$$V_{L_2} = \frac{L_2 V}{L_1 + L_2 + L_3}$$

$$V_{L_3} = \frac{L_3 V}{L_1 + L_2 + L_3}$$

2) Parallel Combination:



Here $e = L\frac{di}{dt}$ is same across all inductors.

we have,

$$i = i_1 + i_2 + i_3$$

$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt} + \frac{di_3}{dt}$$

$$\frac{e}{L_{eq}} = \frac{e}{L_1} + \frac{e}{L_2} + \frac{e}{L_3}$$

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$$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}}$$

* Energy stored in all inductors,

$$U = U_1 + U_2 + U_3$$

$$= \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + \frac{1}{2} L_3 i_3^2$$

$$= \frac{1}{2} L_1 \left(\frac{\phi}{L_1}\right)^2 + \frac{1}{2} L_2 \left(\frac{\phi}{L_2}\right)^2 + \frac{1}{2} L_3 \left(\frac{\phi}{L_3}\right)^2$$

$$= \frac{1}{2} \phi^2 \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}\right)$$

$$\boxed{U = \frac{\phi^2}{2 L_{eq}}}$$

$$U = \frac{(L_{eq} i)^2}{2 L_{eq}}$$

$$\boxed{U = \frac{1}{2} L_{eq} i^2}$$

we have,

$$e = L \cdot \frac{di}{dt}$$

$$L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt} = L_3 \frac{di_3}{dt}$$

same $\Rightarrow L_1 i_1 = L_2 i_2 = L_3 i_3 = L_{eq} i = \phi$

$$U_1 : U_2 : U_3 = \frac{1}{L_1} : \frac{1}{L_2} : \frac{1}{L_3}$$

$$U_1 = \frac{U/L_1}{\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}}, \quad U_2 = \frac{U/L_2}{\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}}, \quad U_3 = \frac{U/L_3}{\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}}$$

$$i_1 : i_2 : i_3 = \frac{1}{L_1} : \frac{1}{L_2} : \frac{1}{L_3}$$

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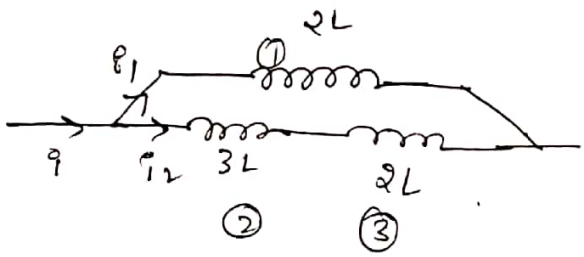
Ex: Find current through each branch and energy stored in all inductors.



$$i_1 : i_2 = \frac{1}{2} : \frac{1}{5}$$

$$i_1 : i_2 = 5 : 2$$

$$i_1 = 5x$$

$$i = i_1 + i_2 \qquad i_2 = 2x$$

$$i = 7x$$

$$x = \frac{i}{7}$$

$$i_1 = \frac{5i}{7}, \quad i_2 = \frac{2i}{7}$$

$$U_1 = \frac{1}{2} L_1 i_1^2 = \frac{1}{2} \times 2L \times \frac{(5i)^2}{7^2} = \frac{25}{49} L i^2$$

$$U_2 = \frac{3L}{2} \times \left(\frac{2i}{7}\right)^2 = \frac{6Li^2}{49}$$

$$U_3 = \frac{2L}{2} \times \left(\frac{2i}{7}\right)^2 = \frac{4i^2 L}{49}$$

$$U_{Total} = \frac{(25+10)\, i^2 L}{49} = \frac{35\, i^2 L}{49}$$

$$\boxed{U_{Total} = \frac{5i^2 L}{7}}$$

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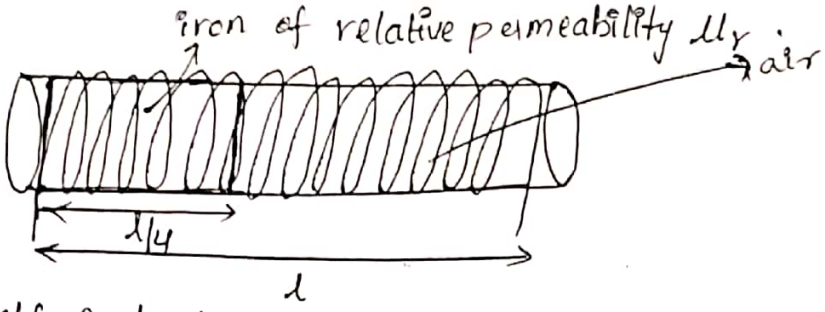
Exe



Find self inductance of solenoid.

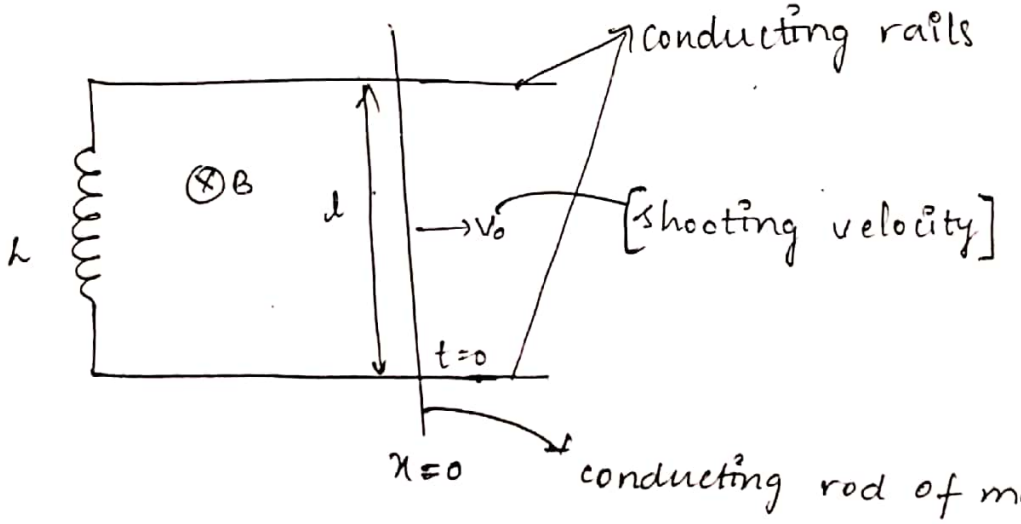
Sol^n

$$L = L_1 + L_2$$

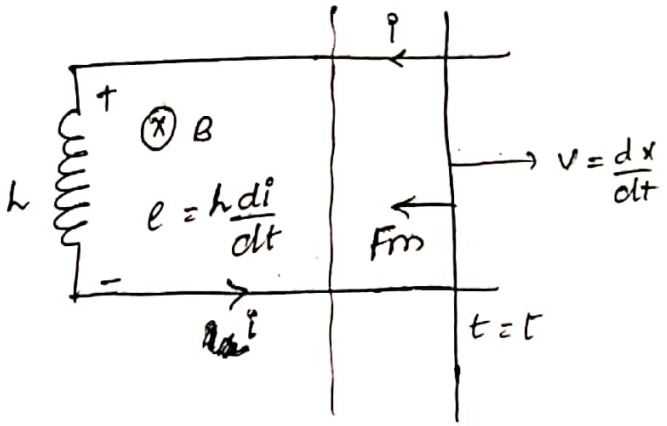
$$= \mu_0 \mu_r n^2 A \frac{l}{4} + \mu_0 n^2 A \left(\frac{3l}{4}\right)$$

$$\boxed{L = \frac{\mu_0 n^2 A l}{4} (\mu_r + 3)}$$

Ex:



Sol^n



Let v be the speed of rod at any time t,

induced emf in the rod, $e = BlV$

* we have, $L \cdot \frac{di}{dt} = B l \frac{dx}{dt}$

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$$Li = Blx$$

$$\boxed{i = \frac{Blx}{L}}$$

magnetic force on the rod,

$$F_m = IlB$$

$$F_m = \frac{B^2 l^2 x}{L}$$

$$F_m = -ma.$$

$$-m\frac{dv}{dt} = \frac{B^2 l^2 x}{L}$$

$$\boxed{\frac{d^2x}{dt^2} = -\frac{B^2 l^2 x}{Lm}}$$

* motion of rod is S.H.M with $\omega = \frac{Bl}{\sqrt{mL}}$ &

$$T = \frac{2\pi}{\omega} = \frac{2\pi\sqrt{mL}}{Bl}$$

* eqn of displacement,

$$x = A\sin(\omega t + \phi)$$

At $t = 0$, $x = 0 \longrightarrow 0 = A\sin\phi$

$$\therefore \boxed{\phi = 0}$$

$$\boxed{x = A\sin\omega t}$$

current $i = \frac{Blx}{L} = \frac{BlA\sin\omega t}{L}$

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* emf of inductor,

$$e = L \cdot \frac{di}{dt}$$

$$e = K \cdot \frac{B l A \omega}{K} \cos \omega t$$

$$\boxed{e = B l A \omega \cos \omega t}$$

* To find Amplitude (A) :-

M-I : (1) speed of rod,

$$v = \frac{dx}{dt} = A\omega \cos \omega t$$

At $t = 0$, $v = A\omega = v_0$ (say)

$$\Rightarrow \boxed{A = \frac{v_0}{\omega} = \frac{v_0 \sqrt{mL}}{Bl}}$$

M-II

L.C.E,

$$\frac{1}{2} m v_0^2 = \frac{1}{2} L i_{max}^2$$

$$m v_0^2 = L \cdot \left(\frac{B l A}{L}\right)^2$$

$$\boxed{A = \frac{v_0 \sqrt{mL}}{Bl}}$$

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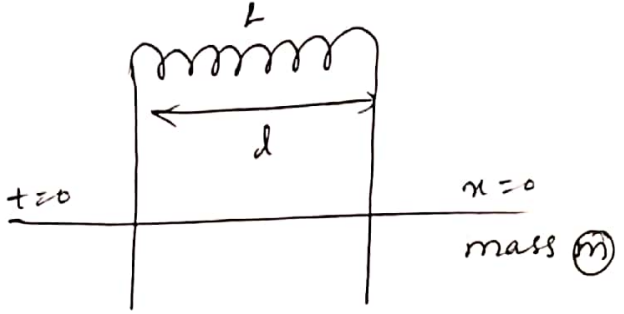
Ex 6



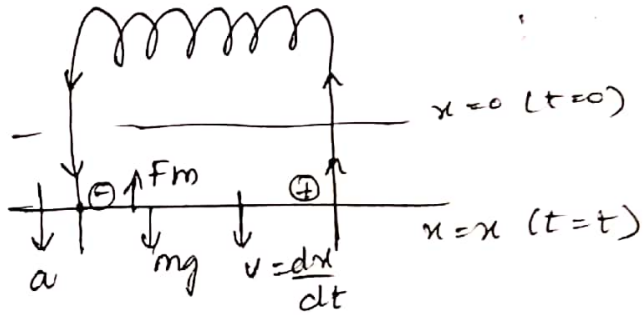
rod is released from rest.

Sol 6 At any time, v be the speed of the rod,

$e = Blv$

we have,

$$e = L\cdot\frac{di}{dt} = Blv$$



$$L\cdot\frac{di}{dt} = Bl\frac{dx}{dt}$$

$$\boxed{i = \frac{Blx}{L}}$$

magnetic force on the rod,

$$F_m = ilB = \frac{B^2l^2x}{L}$$

* net force on the rod,

$$mg - F_m = ma$$

$$m\frac{dv}{dt} = mg - \frac{B^2l^2x}{L}$$

$$m\frac{d^2v}{dt^2} = mg - \frac{B^2l^2v}{L}$$

$$\boxed{\frac{d^2v}{dt^2} = -\frac{B^2l^2v}{mL}}$$

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motion is S.H.M, $\boxed{\omega = \frac{Bl}{\sqrt{mL}}}$

& $T = \frac{2\pi}{\omega} = \frac{2\pi\sqrt{mL}}{Bl}$

$V = V_0 \sin\omega t$

at $t=0$, $v=0 \Rightarrow \phi = 0$

$\boxed{V = V_0 \sin\omega t}$

* At maximum speed (V_0):

$a = 0$, $mg = IlB$

$mg = \frac{B^2 l^2 A}{L}$

$\boxed{A = \frac{mgL}{B^2 l^2}}$

L.C.E)

$0 = -mgA + \frac{1}{2}mV_0^2 + \frac{1}{2}LI^2$

$\frac{1}{2}mV_0^2 = mgA - \frac{1}{2}LI^2$

$V_0^2 = \frac{2gA}{} - \frac{LI^2}{m}$

$= 2g \cdot \frac{mgL}{B^2 l^2} - \frac{B^2 l^2 x^2}{L m}$

$= 2g\frac{mgL}{B^2 l^2} - \frac{B^2 l^2}{Km} \cdot \frac{m^2 g^2 L^2}{B^4 l^4}$

$V_0^2 = \frac{mg^2L}{}$ $\boxed{V_0 = g\sqrt{mL}}$

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* Again,

$$\frac{dx}{dt} = v_0 \sin \omega t$$

$$\int_0^x dx = \int_0^t v_0 \sin \omega t \cdot dt$$

$$\boxed{x = \frac{v_0}{\omega}(1 - \cos \omega t)}$$

* maximum displacement,

$$x_{max} = \frac{2v_0}{\omega} = 2A$$

$$\boxed{A = \frac{v_0}{\omega}}$$

$$A = \frac{g\sqrt{mL}}{Bl} \times \frac{\sqrt{mL}}{Bl}$$

$$\boxed{A = \frac{gmL}{B^2 l^2}}$$

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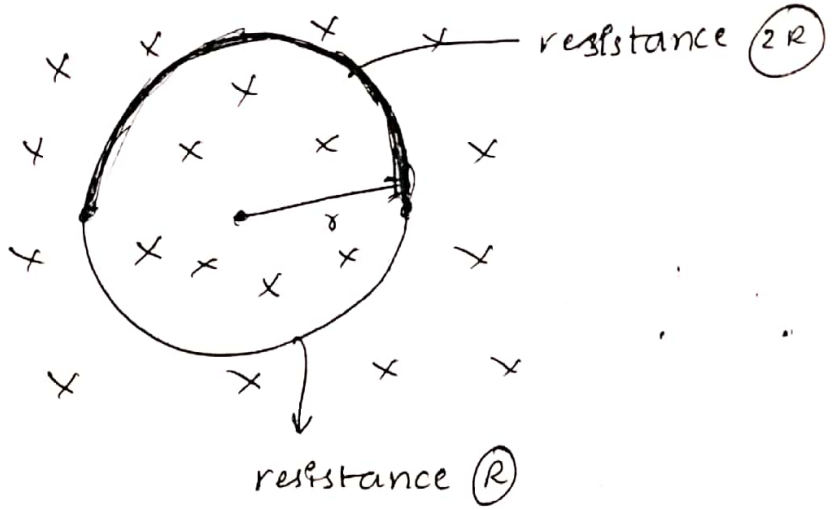
Induced electric field problems

(1)



given: magnetic field, $\boxed{B = B_0 t}$

Find the magnitude of "colomb electric field" at λ is resistance per unit length.

Sole

$$\mathcal{E} = A \cdot \frac{dB}{dt} = \frac{\pi r^2 B_0}{t}$$

induced emf $\Rightarrow \boxed{\mathcal{E} = \pi r^2 B_0}$

$$\& \; i = \frac{\mathcal{E}}{R_{total}} = \frac{\pi r^2 B_0}{3R}$$

we have,

$$\int E_T \cdot dl = iR$$

$$(E_{in} + E_c)\pi r = 2iR$$

$$(E_{in} + -E_c)\pi r = iR$$

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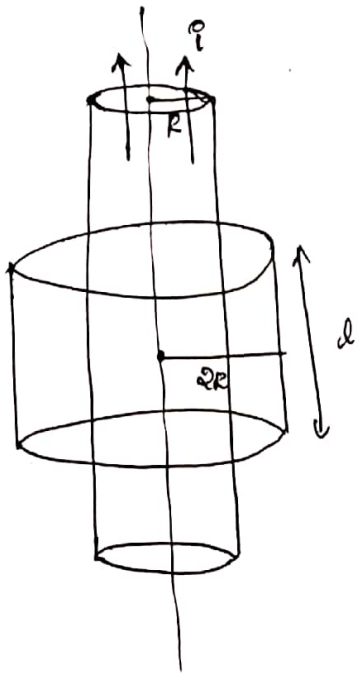
$$E_c = \frac{iR}{2\pi r} = \frac{\pi r^2 B_0}{3t} \times \frac{t}{2\pi r}$$

$$\boxed{E_c = \frac{r B_0}{6}}$$

Exp



Find energy stored in the region ~~[illegible]~~ of cylinder at centre.

$$B_1 = \frac{\mu_0 i r}{2\pi R^2}$$

$$B_2 = \frac{\mu_0 i}{2\pi r}$$

Sol

$$U = \int \frac{B^2}{2\mu_0} \cdot dv$$

$$U = \int_{r=0}^{r=R} \frac{B_1^2}{2\mu_0} (2\pi r l) \cdot dr + \int_{r=R}^{r=2R} \frac{B_2^2}{2\mu_0} (2\pi r l) \cdot dr$$

$$= \int_{r=0}^{r=R} \frac{\mu_0^2 i^2 r^2}{(2\pi R^2)^2} \times \frac{1}{2\mu_0} (2\pi r l) \cdot dr + \int_{r=R}^{r=2R} \frac{\mu_0^2 i^2}{(2\pi r)^2} \cdot \frac{1}{2\mu_0} \cdot 2\pi r l \cdot dr$$

$$= \frac{\mu_0 i^2 l}{2(2\pi R^4)} \int_0^R r^3 \cdot dr + \frac{\mu_0 i^2 l}{4\pi} \int_R^{2R} \frac{1}{r} \cdot dr$$

$$= \frac{\mu_0 i^2 l}{16\pi} + \frac{\mu_0 i^2 l}{4\pi} \ln 2 .$$

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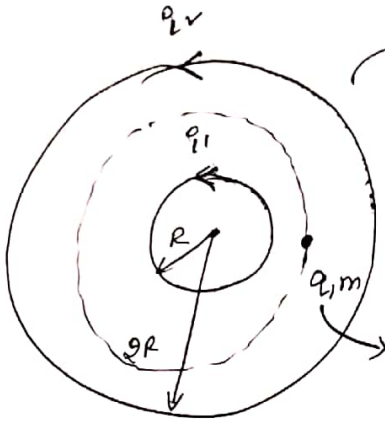
Ex 6



→ Two solenoids of mass 'm'

Given

$i_1 = 2i_2$, $i_2 = kt$

→ Initially charged particle is at rest. If it moves in a circular path of radius 'r'.

Find r?

Sol⁶ Flux through the circular path,

$$\phi = (\mu_0 n i_1 + \mu_0 n i_2)\pi R^2 + (\mu_0 n i_2)\pi (r^2 - R^2)$$

$$= \mu_0 n i_1 \pi R^2 + \mu_0 n i_2 \pi r^2$$

$$= \mu_0 n \pi (i_1 R^2 + i_2 r^2)$$

$$= \mu_0 n \pi (2(kt)R^2 + kt r^2)$$

$$\phi = \mu_0 n \pi k t (2R^2 + r^2)$$

$$e = \frac{d\phi}{dt} = \mu_0 n \pi k (2R^2 + r^2)$$

$$E \times 2\pi r = e$$

$$E = \frac{e}{2\pi r}$$

$$= \frac{\mu_0 n \pi k (2R^2 + r^2)}{2\pi r}$$

$$E = \frac{\mu_0 n k (2R^2 + r^2)}{2r}$$

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$$F_t = Eq$$

$$m a_t = \frac{\mu_0 nk (2R^2 + r^2) q}{2r}$$

$$a_t = \frac{\mu_0 nk (2R^2 + r^2) q}{2rm}$$

$$V = a_t \cdot t$$

$$V_t = \frac{\mu_0 nkq}{2rm} (2R^2 + r^2) t$$

Eq $\quad F_m = BqV$

$$\frac{mv^2}{r} = BqV$$

$$V = \frac{Bqr}{m}$$

$$= \frac{\mu_0 ni_2 qr}{m}$$

$$V = \frac{\mu_0 nqrkt}{m}$$

$$\frac{\mu_0 nkqt}{2rm} (2R^2 + r^2) = \frac{\mu_0 nqrkt}{m}$$

$$2R^2 + r^2 = 2r^2$$

$$\boxed{r = \sqrt{2}R}$$

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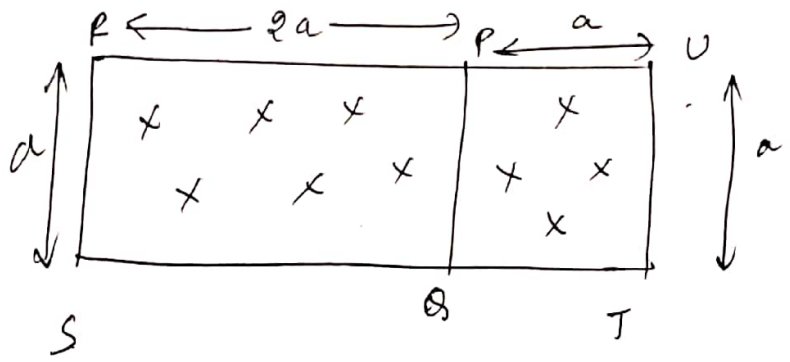
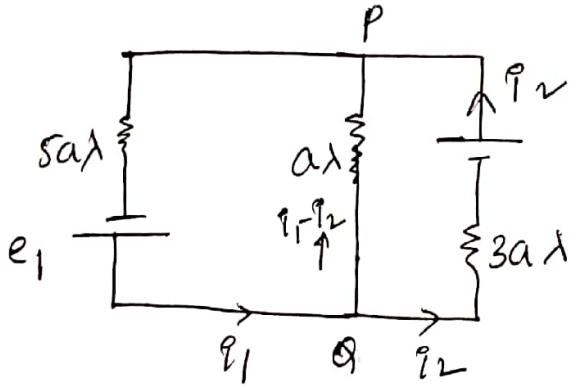
Ex 6



Find current in PQ if $\frac{dB}{dt} = \alpha$ & λ is the resistance per unit length.

Sol 6



$e_1 = 2a^2\alpha$

$e_2 = a^2\alpha$

$$e_1 - (i_1 - i_2)\, a\lambda - 5a\lambda i_1 = 0$$

$$2a^2\alpha - (i_1 - i_2)\, a\lambda - 5a\lambda i_1 = 0$$

$$6i_1 - i_2 = \frac{2a\alpha}{\lambda}$$

$$e_2 + (i_1 - i_2)\, a\lambda - 3a\lambda i_2 = 0$$

$$4i_2 - i_1 = \frac{a\alpha}{\lambda}$$

$$24 i_2 - 6i_1 = \frac{6a\alpha}{\lambda}$$

$$\Rightarrow i_2 = \frac{8a\alpha}{23\lambda}, \quad i_1 = \frac{9a\alpha}{23\lambda}$$

$$\therefore i_{PQ} = \frac{9a\alpha}{23\lambda} \text{ (Q to P)}.$$

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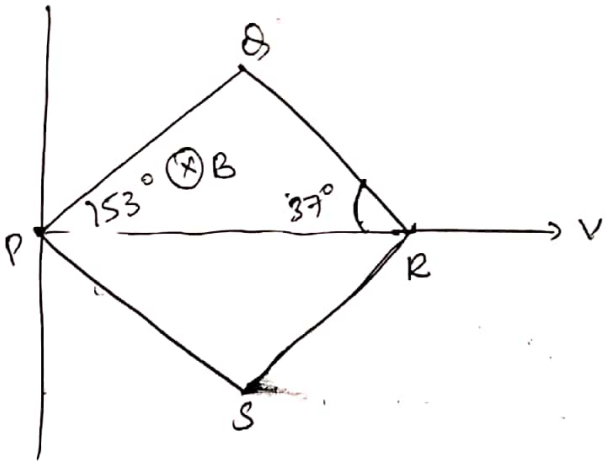
Ex 6



Find induced emf

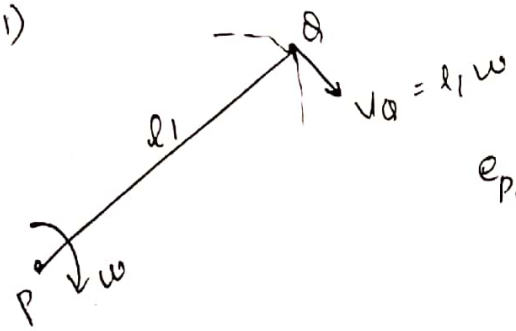
① in rod PQ

② in rod QR

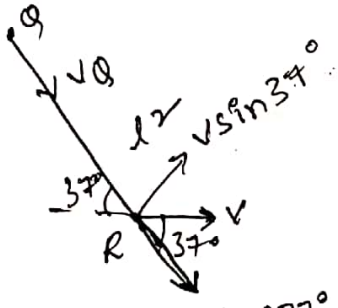
Sol^n Rod PQ ⟶ pure rotation about 'P'

Rod QR ⟶ combined motion about CM of rod QR.

1)



$$e_{PQ} = \frac{1}{2} B\omega l_1^2 \quad -①$$



constraint eq^n, along length of rod,

$$v_Q = v\cos 37° = \frac{4v}{5}$$

$$\omega = \frac{v_Q}{l_1} = \frac{4v}{5l_1}$$

$$e_{PQ} = \frac{1}{2} B\left(\frac{4v}{5l_1}\right)(l_1^2)$$

$$= \frac{2Bvl_1}{5}.$$

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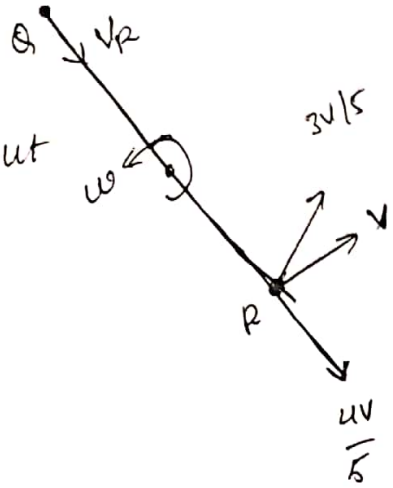
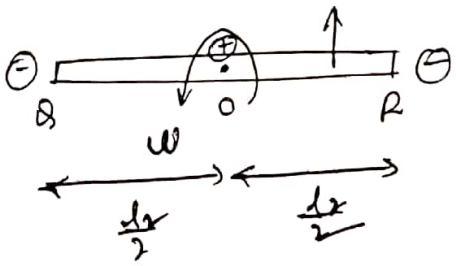
2) For QR,



Due to pure rotation about CM,

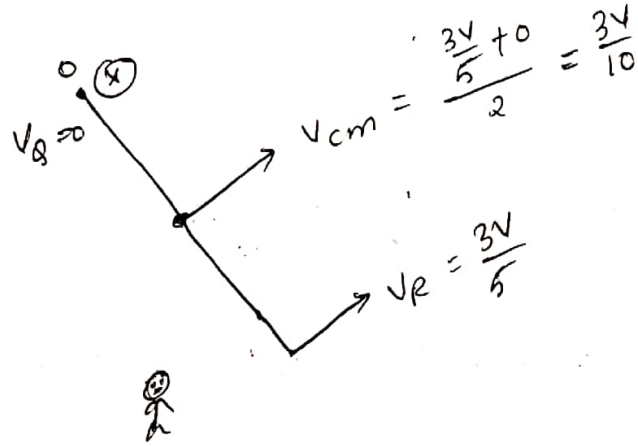


$$V_O - V_R = \frac{1}{2} B\omega \left(\frac{d_2}{2}\right)^2$$

$$V_O - V_Q = \frac{1}{2} B\omega \left(\frac{d_2}{2}\right)^2$$

$$\boxed{V_Q - V_R = 0} \Rightarrow \boxed{e_{QR} = 0}$$

due to translation,



$$\mathcal{E}_{QR} = B d_2 v_{cm}$$

$$= B(d_2)\left(\frac{3v}{10}\right)$$

$$\boxed{\mathcal{E}_{QR} = \frac{3Bd_2v}{10}}$$

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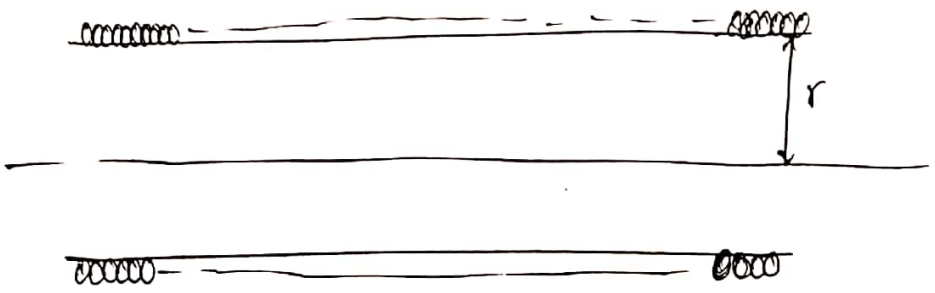
Ex6

A certain vol. of Cu is drawn into a wire of radius 'a' and is rapped in shape of a helix having radius r (r >>> a). The windings are as close as possible without overlapping. Self inductance of inductor so obtained is L_1. Another wire of radius 2a is drawn using same vol of Cu & wound in the fashion as described above. This time the inductance is L_2. Find $\frac{L_1}{L_2}$.



Sol6

$$\text{Length of wire } (l) = \frac{(\text{Volume})_{\text{wire}}}{\text{Area}} = \frac{V}{\pi a^2} \quad \text{--- (1)}$$

$$\text{Total No. of turns } (N) = \frac{l}{2\pi r}$$

$$\text{length of helix} = N(2a) = b = \frac{al}{\pi r}$$

$$\text{No. of turns per unit length} = \frac{N}{b}$$

$$n = \frac{1}{2a}$$

$$L_1 = \mu_0 n^2 A b = \frac{\mu_0 N^2 A}{b}$$

$$= \mu_0 \left(\frac{1}{2a}\right)^2 (\pi r^2)(b) = \frac{\mu_0 \pi^2 r^2 \cancel{4} l / \cancel{\pi} r}{4a^2}$$

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$$= \frac{\mu_0 \gamma (v)/\pi a^2}{4a}$$

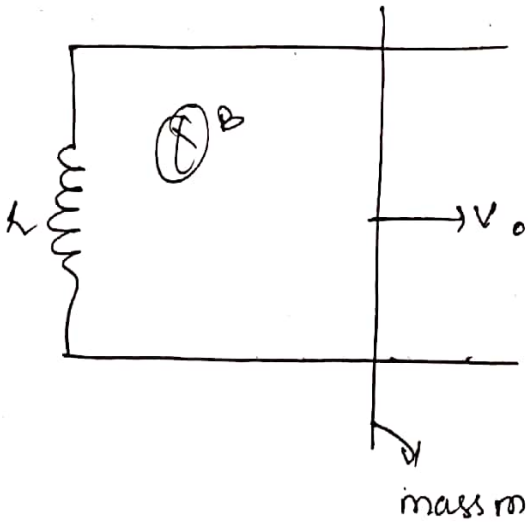
$$\boxed{L = \frac{\mu_0 \gamma}{4\pi a^3}}$$

$$L \propto 1/a^3$$

$$\frac{L_1}{L_2} = \frac{(2)^3}{1^3}$$

$$\boxed{L_1 : L_2 = 8 : 1}$$

Exp



neglect resistance of wire

1) V as function of time

2) i as function of time

3) speed as function of displacement.

4) Is the current in the conductor zero when it stops, If no find it's current.

5) will the conductor move after it stop.

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Sol6

$$F = -ma$$

$$\frac{B^2 l^2 x}{L} = -mv\frac{dv}{dx}$$

$$A \cdot x \cdot dx = -v\,dv$$

$$\frac{Ax^2}{2} = -\frac{v^2}{2} + \frac{v_0^2}{2}$$

$$v^2 = v_0^2 - \frac{B^2 l^2 x^2}{Lm}$$

$$v = \sqrt{v_0^2 - \frac{B^2 l^2 x^2}{Lm}}$$

Velocity when displacement is half of the amplitude.

$$Amp = \frac{v_0\sqrt{mL}}{Bl}$$

$$v_{at\ x = \frac{A}{2}} = \frac{\sqrt{3}}{2} v_0$$

$$v_{at\ x = A} = 0$$

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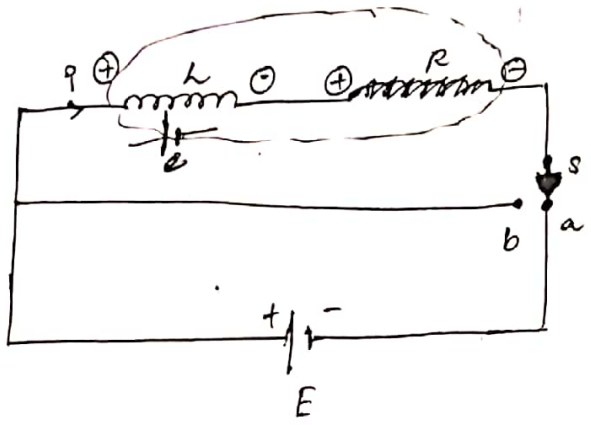
L-R Circuits

1) Growth of current :

* $\frac{di}{dt} > 0$



At $t = 0$, switch 's' is connected to 'a' :

→ current starts flowing in the circuit.

→ Resistance opposes the current.

→ Inductance opposes the change in current.

At $t = t$, let i be the current in the circuit.

\# $E = V_L + V_R$

$$\boxed{E = V_L + iR} \quad -①$$

From KVL,

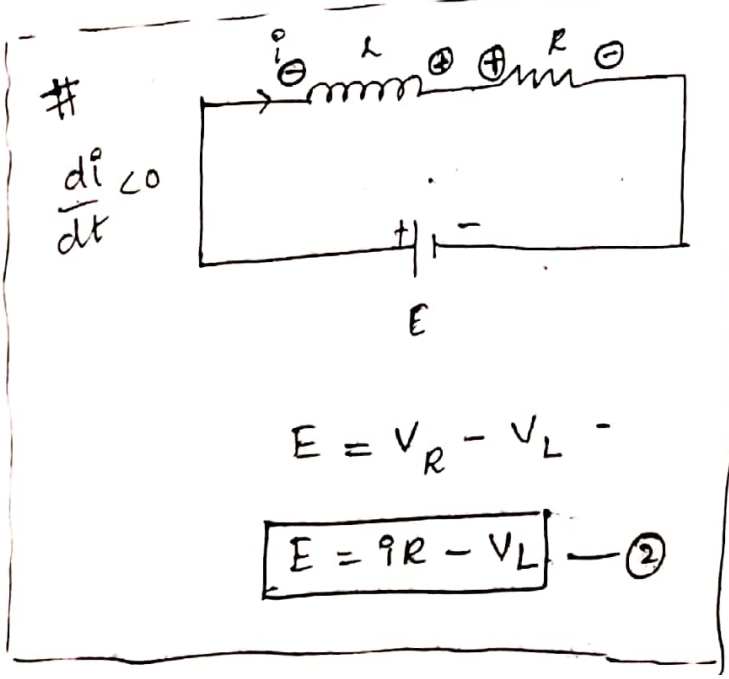
$$E - L\frac{di}{dt} - iR = 0$$

$$E - iR = L\cdot\frac{di}{dt}$$

$$\int_0^t \frac{dt}{L} = \int_0^i \frac{di}{E - iR}$$

$$\frac{[\ln(E - iR)]_0^i}{-R} = \frac{t}{L}$$

$$\ln(E - iR) - \ln E = -\frac{Rt}{L}$$



\# $\frac{di}{dt} < 0$

$$E = V_R - V_L$$

$$\boxed{E = iR - V_L} \quad -②$$

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$$1 - \frac{iR}{E} = e^{-Rt/L}$$

$$\frac{iR}{E} = 1 - e^{-\frac{R}{L}\cdot t}$$

$$i = \frac{E}{R}\left[1 - e^{-R/L\cdot t}\right]$$

$$\boxed{i = i_0\left(1 - e^{-\frac{R}{L}\cdot t}\right)}$$

$\boxed{i_0 = \frac{E}{R}}$ = peak / maximum current.

Exe: Find time taken by the current to reach from $\frac{1}{4}^{th}$ of max value to $\frac{3}{4}^{th}$ of max value.

Sol:

$$\frac{i_0}{4} = i_0\left(1 - e^{-R/L\cdot t}\right)$$

$$e^{-R/L\cdot t} = \frac{3}{4}$$

$$\frac{R}{L}\cdot t = \ln(4/3)$$

$$\boxed{t_1 = \frac{L}{R}\ln(4/3)} \quad (0 \text{ to } i_0/4)$$

$$\frac{3i_0}{4} = i_0\left(1 - e^{-\frac{R}{L}\cdot t}\right)$$

$$\boxed{t_2 = \frac{L}{R}\ln 4} \quad (0 \text{ to } \frac{3i_0}{4})$$

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$$t_2 - t_1 = \frac{L}{R}\left[\ln(4) - \ln(4/3)\right]$$

$$\boxed{t_2 - t_1 = \frac{L}{R}\ln 3}$$

Points:

1. At $t = 0$, $i = i_0(1 - e^0) = 0$ (true for all circuits)

i.e, inductor acts as insulator immediately after switch is closed.

2. At $t = \infty$, $i = i_0(1 - e^{-\infty}) = i_0$ (max)

i.e, inductor behaves as conductor after a long time switch is closed (in steady state).

3. At $t = \frac{L}{R}$, $i = i_0\left(1 - \frac{1}{e}\right) = i_0\left(\frac{e-1}{e}\right) = i_0(0.63)$

the time duration for which current raises from zero to 63% of max value is $\frac{L}{R}$ sec called time constant of L-R circuit.

4. Rate of growth of current in the circuit:

$$\frac{di}{dt} = \frac{d}{dt}\left[i_0(1 - e^{-\frac{R}{L}\cdot t})\right]$$

$$\frac{di}{dt} = i_0\left(\frac{R}{L}\right)\cdot e^{-\frac{R}{L}\cdot t}$$

$$\boxed{\frac{di}{dt} = \frac{E}{L}e^{-R/L\cdot t}}$$

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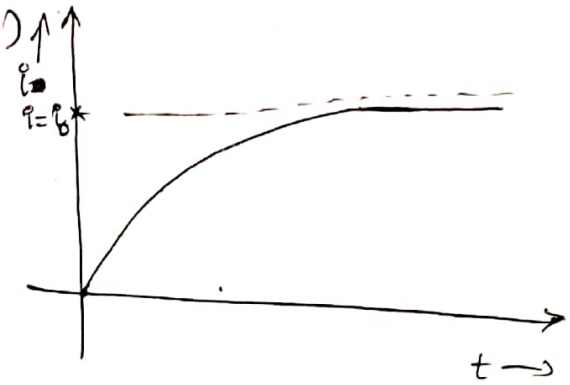
Clearly, $\frac{di}{dt}$ decreases exponentially with time.

* Immediately after closing the switch,

$$\boxed{\frac{di}{dt} = \frac{E}{L}}$$



at $t = \infty$, $\boxed{\frac{di}{dt} = 0}$

5. potential diff across the inductor:

$$V_L = L \cdot \frac{di}{dt} = E \cdot e^{-\frac{R}{L} \cdot t}$$

clearly, V_L decreases exponentially with time.

At $t = 0$, $V_L = E$

At $t = \infty$, $V_L = 0$

6. potential diff across the resistor:

$$V_R = iR = i_0 \left(1 - e^{-\frac{R}{L} \cdot t}\right) R$$

$$V_R = E \left(1 - e^{-R/L \cdot t}\right)$$

clearly, V_R increases exponentially with time.

At $t = 0$, $V_R = 0$

At $t = \infty$, $V_R = E$

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* At any time 't',

$$V_L + V_R = E e^{-R/L \cdot t} + E(1 - e^{-R/L \cdot t})$$

$$\boxed{V_L + V_R = E}$$

$$V_L = V_R$$

$$2 E e^{-R/L \cdot t} = E$$

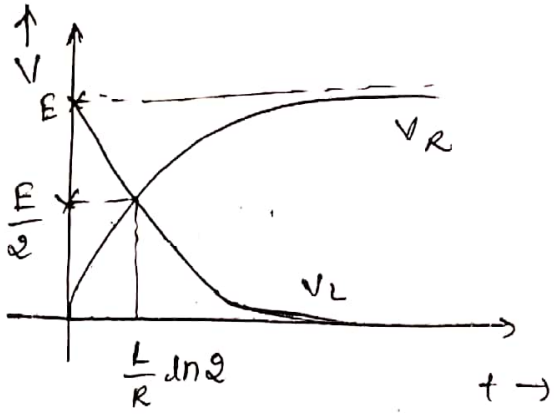
$$\boxed{t = \frac{L}{R} \ln 2}$$

$\tau = \frac{L}{R}$ = time constant.

$$t = \tau (0.693)$$

$$\boxed{t = 0.693\,\tau}$$

Here, $V_L = V_R = \frac{E}{2}$.



Energy stored in Inductor

$$U_L = \frac{1}{2} L i^2$$

$$U_L = \frac{1}{2} L i_0^2 (1 - e^{-\frac{R}{L} \cdot t})^2$$

U_L increases exponentially with time.

At $t=0$, $U_L = 0$; At $t=\infty$, $U_L = \frac{1}{2} L i_0^2$ (max)

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Ex: Find the time taken by the energy stored in inductor to increase from $\frac{1}{4}^{th}$ of max value to $\frac{1}{2}^{th}$ of max value.

Sol:

$$U_L = \frac{1}{2} L i_0^2 \left(1 - e^{-\frac{R}{L} \cdot t}\right)^2$$

$$\frac{1}{4}(U) = U \left(1 - e^{-\frac{R}{L} \cdot t}\right)^2$$

$$1 - e^{\frac{-R}{L} \cdot t} = \frac{1}{2}$$

$$\frac{1}{2} = e^{-R/L \cdot t}$$

$$\boxed{t_1 = \frac{L}{R} \ln 2}$$

$$\frac{U}{2} = U \left(1 - e^{-\frac{R}{L} \cdot t}\right)^2$$

$$1 - e^{-\frac{R}{L} \cdot t} = \frac{1}{\sqrt{2}}$$

$$e^{-\frac{R}{L} \cdot t} = \frac{\sqrt{2} - 1}{\sqrt{2}}$$

$$\boxed{t_2 = \frac{L}{R} \ln \left(\frac{\sqrt{2}}{\sqrt{2} - 1}\right)}$$

$$t_2 - t_1 = \frac{L}{R} \left[\ln \left(\frac{\sqrt{2}}{\sqrt{2} - 1}\right) - \ln 2\right]$$

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$$t_2 - t_1 = \frac{L}{R} \ln\left(\frac{1}{2-\sqrt{1}}\right)$$

Rate of energy stored in the inductor :-

↳ power given to the inductor :

$$P_L = \frac{dU_L}{dt} = \frac{1}{2} L i_0^2 \frac{d}{dt}\left(1 - e^{-\frac{R}{L}\cdot t}\right)^2$$

$$= \frac{1}{2} L i_0^2 \cdot 2\left(1 - e^{-\frac{R}{L}\cdot t}\right)\left(+e^{-\frac{R}{L}t}\left(+\frac{R}{L}\right)\right)$$

$$\boxed{P_L = i_0^2 R \left(1 - e^{-\frac{R}{L}\cdot t}\right)\left(e^{-\frac{R}{L}\cdot t}\right)} \quad \text{—(a)}$$

at $t = 0$, $P_L = 0$

at $t = \infty$, $P_L = 0$

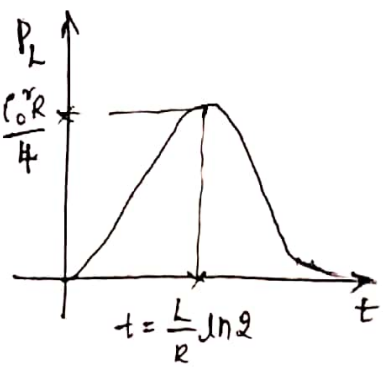
at $t = \ln 2 \cdot \tau$, $P_L = \max$

$P_L - \max$ when $\frac{dP_L}{dt} = 0$ (or) $\frac{d^2 U_L}{dt^2} = 0$

$$\left(1 - e^{-\frac{R}{L}\cdot t}\right)\left(-\frac{R}{L}\cdot e^{-\frac{R}{L}\cdot t}\right) + e^{-\frac{R}{L}\cdot t}\left(+\frac{R}{L}\cdot e^{-\frac{R}{L}\cdot t}\right) = 0$$

$$-1 + 2e^{-\frac{R}{L}\cdot t} = 0$$

$$\boxed{t = \frac{L}{R}\ln 2}$$



$$\boxed{(P_{max})_L = \frac{i_0^2 R}{4}}$$

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Power dissipated in Resistance :

$$P_R = \frac{dH}{dt} = i^2 R$$

$$\boxed{P_R = i_0^2 R \left(1 - e^{-\frac{R}{L} \cdot t}\right)^2} \quad \text{---(6)}$$

at $t=0$, $P_R = 0$

at $t=\infty$; $P_R = i_0^2 R$ (max)

* Heat produced in resistance R,

$$H = \int_0^t dH = \int_0^t i^2 R \cdot dt$$

$$= i_0^2 R \int_0^t \left(1 - e^{-\frac{R}{L} \cdot t}\right)^2 \cdot dt$$

$$= i_0^2 R \int_0^t 1 \cdot dt - 2\int_0^t e^{-\frac{R}{L} \cdot t} \cdot dt + \int_0^t e^{-\frac{2R}{L} t} \cdot dt$$

$$= i_0^2 R \left[(t) - 2\frac{e^{-\frac{R}{L} \cdot t}}{-\frac{R}{L}} + \frac{e^{-\frac{2R}{L} \cdot t}}{-\frac{2R}{L}} - 0 + 2\frac{(1)}{-R/L} - \frac{1}{-\frac{2R}{L}}\right]$$

$$= i_0^2 R \left[t + \frac{2L}{R} e^{-\frac{R}{L} \cdot t} - \frac{L}{2R} e^{-\frac{2R}{L} \cdot t} - \frac{2L}{R} + \frac{L}{2R}\right]$$

$$H = i_0^2 R \left[t + \frac{2L}{R}\left(e^{-\frac{R}{L} \cdot t} - 1\right) - \frac{L}{2R}\left(e^{-\frac{2R}{L} \cdot t} - 1\right)\right]$$

Finally, $P_L + P_R = i_0^2 R \left(1 - e^{-\frac{R}{L} \cdot t}\right)$

$$= (i_0 R)\, i_0 \left(1 - e^{-R/L \cdot t}\right)$$

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$$P_L + P_R = E i$$

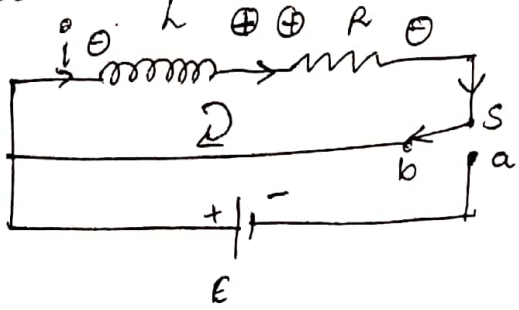
$$\boxed{P_L + P_R = P_{battery}}$$

Decay of current:

when current reaches to max value $i_0 = \frac{E}{R}$, after five time constants (practically).

switch S is now connected to 'b'.

∴ current starts decaying but it is gradual due to inductance.



at time t

$$L\left[\frac{-di}{dt}\right] = iR \quad -(a)$$ [-ve sign indicates currents decreasing with time]

$$-\int_0^t \frac{di}{iR} = \int_0^t \frac{dt}{L}$$

rate of decay $\Rightarrow$ $$\boxed{\frac{iR}{L} = -\frac{di}{dt}}$$

$\frac{L}{R}$

Immediately after switch is connected to b,

$$\frac{-di}{dt} = \frac{i_0 R}{L} = \frac{E}{L} = \max.$$

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we have,

$$-L \cdot \frac{di}{dt} = iR$$

$$\int_{i_0}^{i} \frac{di}{i} = -\frac{R}{L} \int_0^t dt$$

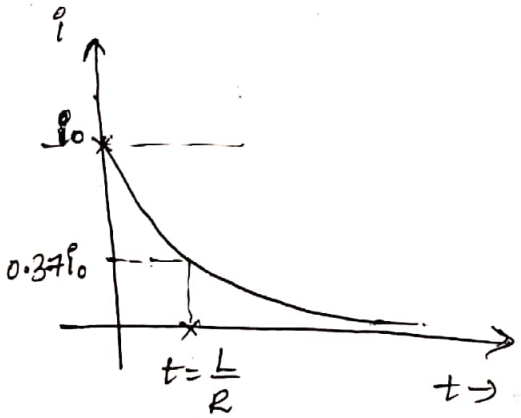
$$\ln(i/i_0) = -\frac{R}{L} \cdot t$$

$$\boxed{i = i_0 e^{-\frac{R}{L} \cdot t}}$$

and rate of decay,

$$-\frac{di}{dt} = \frac{i_0 R e^{-\frac{R}{L} \cdot t}}{L}$$

$$\boxed{-\frac{di}{dt} = \frac{E}{L} e^{-\frac{R}{L} \cdot t}}$$

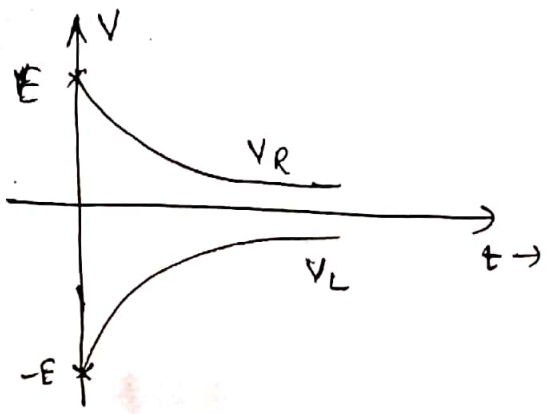


potential diff across inductor;

$$V_L = L \cdot \frac{di}{dt} = -E \cdot e^{-\frac{R}{L} \cdot t}$$

potential diff across R,

$$V_R = iR = i_0 R e^{-\frac{R}{L} \cdot t} = E e^{-\frac{R}{L} \cdot t}$$



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Heat energy produced in R,

$$H = \int_0^t i^2 R \cdot dt$$

$$= \int_0^t i_0^2 e^{-\frac{2R}{L} \cdot t} R \cdot dt$$

$$= i_0^2 R \frac{\left[e^{-\frac{2R}{L} \cdot t} \right]_0^t}{-\frac{2R}{L}}$$

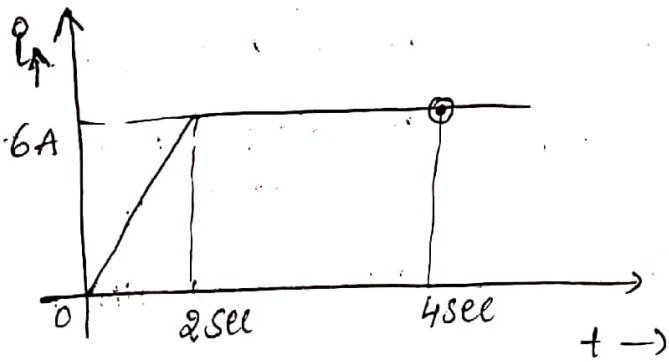
$$= -\frac{1}{2} L i_0^2 \left(e^{-\frac{2R}{L} \cdot t} - 1 \right)$$

$$\boxed{H = \frac{1}{2} L i_0^2 \left(1 - e^{-\frac{2R}{L} \cdot t} \right)}$$

at $t=0$, $\boxed{H = 0}$

at $t = \infty$, $H = \frac{1}{2} L i_0^2$ = initial energy stored in inducto

Ex:



Find the energy absorbe by inductor having L=0 and resistance R=1Ω f the time t=0 to 4

Sol:

$$U = U_1 + U_2$$

here U_1 = energy stored in inductor $= \frac{1}{2} L i^2$

$$= \frac{1}{2} \times 2 \times 36$$

$$= 36$$

U_2 = energy dissipated in the form of heat

$$= \int_0^2 i_1^2 R \cdot dt + \int_2^4 i_2^2 R \cdot dt$$

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$$= \int_0^2 (3t)^2 \cdot dt + \oint (36)(1)(2)$$

$$= \frac{2^3(8)}{3} + 72$$

$$= 24 + 72 = 96$$

$$U = 36 + 96 = 132 \text{ J}$$

Compaision b|w L-R circuit and C-R circuite-

L-R circuit	C-R circuit
* Immediately after closing the switch, inductor behaves as insulator.	* Immediately after closing the switch, capacitor behaves as conductor.
* A long time after closing the switch, inductor behaves as conductor.	* A long time after closing the switch, capacitor behaves as insulator.
* During growth of current, $i = i_0(1 - e^{-\frac{R}{L}t})$, $i_0 = \frac{E}{R}$	* During growth of charge, $q = q_0(1 - e^{-t/RC})$, $q_0 = CE$
* During decay of current, $i = i_0(e^{-\frac{R}{L}t})$	* During decay of charge, $q = q_0(e^{-t/RC})$
* Time constant of L-R circuit, $\tau = \frac{L}{R}$ sec.	* Time const of R-C circuit, $\tau = RC$ sec
D.F = $[T]$	D.F = $[T^1]$

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Calculation of Time constant :

(1) During Growth :

Steps :

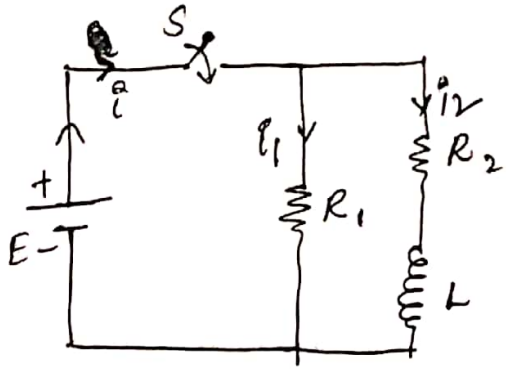
1) short circuit the battery (replace battery with connecting wire)
2) calculate R_{eff} across L.
3) use $\tau = \frac{L}{R_{eff}}$.

(2) During Decay :

steps :

1) remove the ~~bad~~ battery from the circuit.
2) calculate R_{eff} across L.
3) use $\tau = \frac{L}{R_{eff}}$.

Ex6



Find current through battery 1) immediately after closing switch.

2) A long time after closing switch

3) as a function of time.

it,

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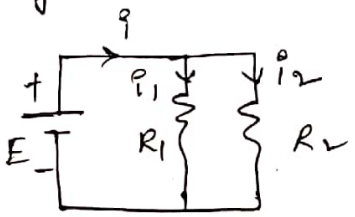
Sol:

1) Immediately after closing switch,

$$i_2 = 0$$

& $\boxed{i = i_1 = \frac{E}{R_1}}$

2) A long time after closing switch,



$$E - i_1 R_1 = 0 \qquad i_1 R_1 = i_2 R_2$$

$$i_1 = \frac{E}{R_1} \qquad i_2 = \frac{E}{R_2}$$

$$i = \frac{E}{R_1} + \frac{E}{R_2}$$

$$\boxed{i = E\left(\frac{R_1 + R_2}{R_1 R_2}\right)}$$

Here energy stored in inductor,

$$U_L = \frac{1}{2} L i_2^2 = \frac{1}{2} \cdot L \cdot \frac{E^2}{R_2^2} = \frac{LE^2}{2R_2^2}$$

3) At any time 't',

$i_1 = \frac{E}{R_1}$, independent of R_2, t.

& $i_2 = \frac{E}{R_2}\left(1 - e^{\frac{-R_2}{L} \cdot t}\right)$

↓

increases exponentially with time.

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$$i = i_1 + i_2$$

$$i = \frac{E}{R_1} + \frac{E}{R_2}\left(1 - e^{\frac{-R_2}{L}\cdot t}\right)$$

$\downarrow$

increases exponentially with time.

Here, potential drop across 'L'.

$$V_L = L \cdot \frac{di_2}{dt}$$

$$= L \cdot \frac{d}{dt}\left(\frac{E}{R_2}\left(1 - e^{\frac{-R_2}{L}\cdot t}\right)\right)$$

$$= \cancel{L} \cdot \frac{E}{\cancel{R_2}} \cdot e^{\frac{-R_2}{L}\cdot t} \cdot \frac{\cancel{R_2}}{\cancel{L}}$$

$$\boxed{V_L = E \cdot e^{\frac{-R_2}{L}\cdot t}}$$

and potential drop across R_2,

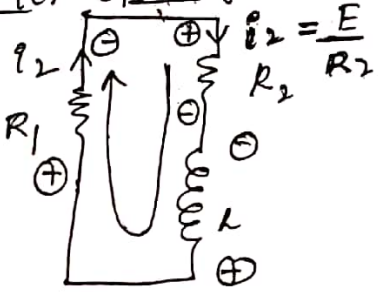
$$V_R = E - V_L$$

$$= E - E \cdot e^{\frac{-R_2}{L}\cdot t}$$

$$\boxed{V_R = E\left(1 - e^{\frac{-R_2}{L}\cdot t}\right)}$$

* Now switch is opened (when steady state is reached)

** Just after opening:



$i_2 = \frac{E}{R_2}$

$i_2 R_1 + i_2 R_2 = L \cdot \frac{di}{dt}$

current through inductor branch doesnot change immediately.

but current changes immediately in resistance branch.

Current through R_1

$= \frac{E}{\cdot}$.

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$$-i_2 R_2 + L\left(\frac{-di_2}{dt}\right) - i_2 R_1 = 0$$

$$L \cdot \frac{di_2}{dt} = i_2 (R_1 + R_2)$$

$$\boxed{\frac{-di_2}{dt} = \frac{i_2 (R_1 + R_2)}{L}} \rightarrow \text{Immediate rate of decay.}$$

$$\star \quad \frac{-di_2}{dt} = \frac{E (R_1 + R_2)}{L R_2}$$

$$-\int_{i_2}^{i} \frac{di_2}{i_2} = \int_0^t \frac{(R_1 + R_2)}{L} dt$$

$$-\ln (i/i_2) = \frac{(R_1 + R_2)}{L} \cdot dt$$

$$i = i_2 \cdot e^{-\frac{(R_1 + R_2)}{L} \cdot t}$$

$$\boxed{i = \frac{E}{R_2} e^{-\frac{(R_1 + R_2)}{L} \cdot t}}$$

* **Heat Energy produced in R_1 & R_2 :-**

$$U = H_1 + H_2 = \frac{1}{2} L i_2^2 = \frac{1}{2} L \cdot \frac{E^2}{R_2^2}$$

$$\boxed{\frac{H_1}{H_2} = \frac{R_1}{R_2}}$$

$$H_1 = \frac{R_1 U}{R_1 + R_2}, \quad H_2 = \frac{R_2 U}{R_1 + R_2}$$

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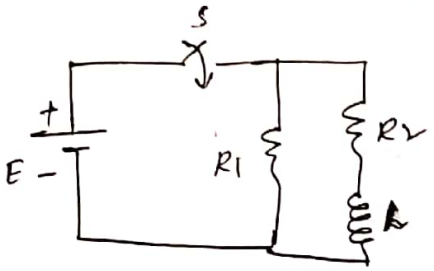
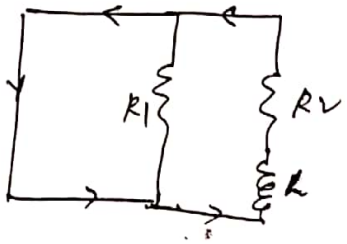
Calculation of Time constant :-



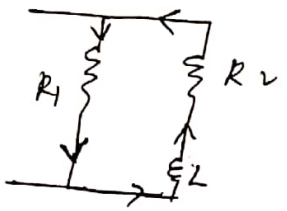
During growth,



$R_{eff} = R_2$

$\tau = \frac{L}{R_2}$

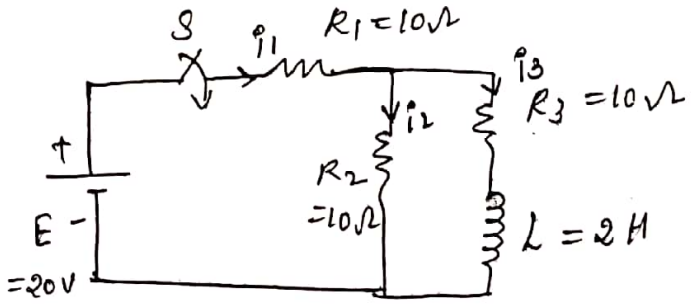
During decay,



$R_{eff} = R_1 + R_2$

$\tau = \frac{L}{R_1 + R_2}$

Ex:



Find current through battery 1) immediately after closing switch.

2) A long time after closing switch.

Sol: Immediately after closing the switch,

$i_3 = 0$, $i_1 = i_2 = \frac{20}{20} = 1\ A$.

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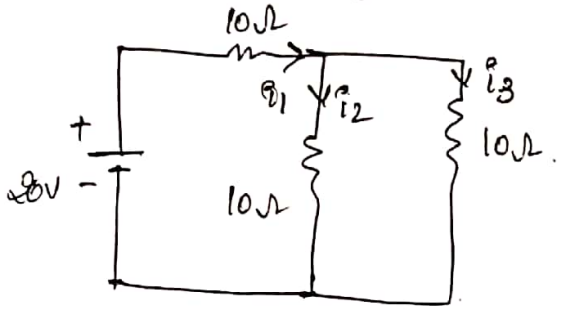
& $V_L = i_2 R_2 = 1 \times 10\Omega = 10$ V

A long time after closing the switch,



$$i_1 = \frac{E}{R_1 + \frac{R_2 R_3}{R_2 + R_3}} = \frac{20}{10+5} = \frac{20}{15} = \frac{4}{3} A$$

and $$i_2 = i_3 = \frac{2}{3} A$$

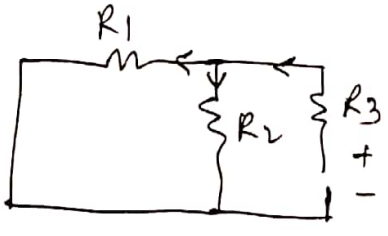
Here $V_L = 0$.

$$U_L = \frac{1}{2} L i_3^2 = \frac{1}{2} \times 2 \times \frac{4}{9}$$

$$\boxed{U_L = \frac{4}{9} J}$$

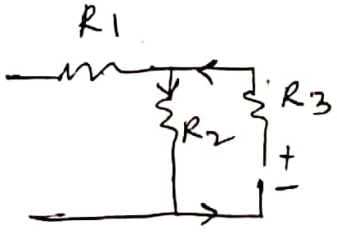
Time const:

during growth,



$$\tau = \frac{L}{R_{eff}} = \frac{2}{15} sec$$

during decay,



$$\tau = \frac{L}{R_{eff}} = \frac{2}{20} = \frac{1}{10} sec$$

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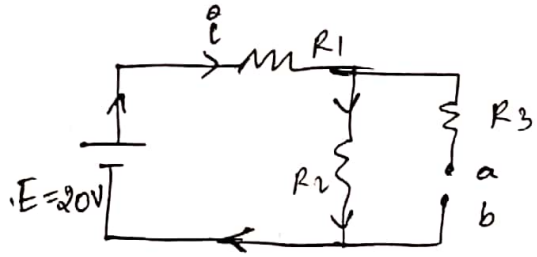
At any time 't',

To find i_3, we use thevinin's theorem,



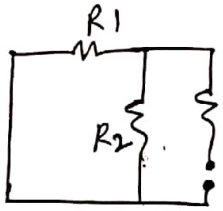
$$i = \frac{E}{R_1+R_2} = \frac{20}{20} = 1A$$

now, $E_{th} = V_a - V_b$

$$\Rightarrow V_a - iR_2 = V_b$$

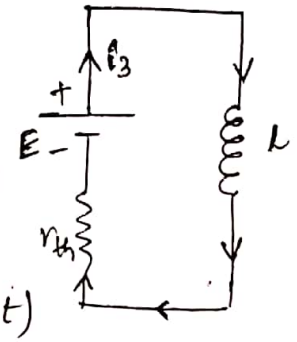
$$V_a - V_b = iR_2$$

$$E_{th} = 1.(10) = 10V$$



$r_{th} = 15\Omega$

Circuit is -



$$i_3 = i_{30}\left(1 - e^{\frac{-R_{th}}{L}\cdot t}\right)$$

$$= \frac{E_{th}}{r_{th}}\left(1 - e^{\frac{-R_{th}}{L}\cdot t}\right)$$

$$i_3 = \frac{2}{3}\left(1 - e^{\frac{-15}{2}\cdot t}\right)$$

potential drop across inductor, $V_L = L\cdot\frac{di_3}{dt}$

$$V_L = 10\, e^{\frac{-15}{2}t}$$

$$= 2 \times \frac{2}{3} \times e^{\frac{-15}{2}\cdot t} \cdot \frac{15}{2}$$

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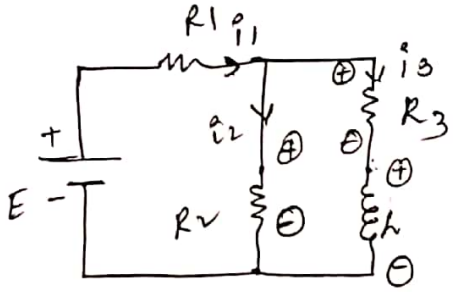
potential drop across R_3 is,

$$i_2 R_2 = i_3 R_3 + V_L$$

$$\boxed{i_3 R_3 = i_2 R_2 - V_L}$$

To find i_2 :-



$$i_1 = i_2 + i_3$$

$$-i_3 R_3 - L \cdot \frac{di_3}{dt} + i_2 R_2 = 0$$

$$i_2 R_2 = 10 e^{-\frac{15}{2} \cdot t} + i_3 R_3$$

$$= 10 e^{-\frac{15}{2} t} + \frac{2}{3}\left(1 - e^{-\frac{15}{2} t}\right)(10)$$

$$= 10 e^{-\frac{15}{2} t} - \frac{20}{3} e^{-\frac{15}{2} t} + \frac{20}{3}$$

$$i_2 (10) = \frac{10}{3} e^{-\frac{15}{2} \cdot t} + \frac{20}{3}$$

$$\boxed{i_2 = \frac{1}{3}\left(e^{-\frac{15}{2} \cdot t} + 2\right)}$$

Finally,

$$i_1 = i_2 + i_3$$

$$= \frac{1}{3}\left(e^{-\frac{15}{2} \cdot t} + 2\right) + \frac{2}{3}\left(1 - e^{-\frac{15}{2} \cdot t}\right)$$

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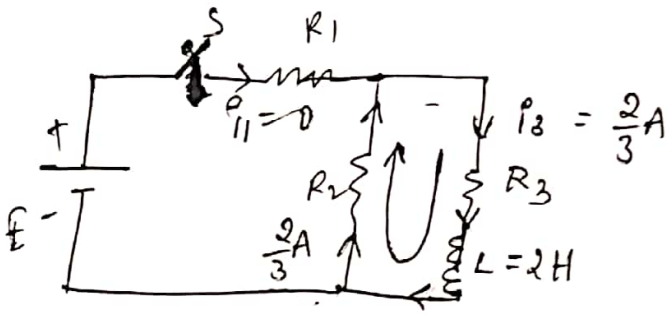
$$\boxed{i_1 = \frac{1}{3}\left(4 - e^{-\frac{15}{2}\cdot t}\right)}$$

* immediately after switch is opened (when steady state is reached)



immediate rate of decay,

$$-L\frac{di_3}{dt} = i_3(R_2 + R_3)$$

$$\Rightarrow -\frac{di_3}{dt} = \frac{i_3(R_2+R_3)}{L}$$

$$= \frac{\frac{2}{3}(20)^{10}}{2}$$

$$= \frac{20}{3}\ A/s\ .$$

o)

* current decay eqn, $i = i_3 e^{-\frac{Rt}{L}}$.

$$\boxed{i = \frac{2}{3}e^{-10t}}$$

* Heat produced in R_2 & R_3.

$$U = H_2 + H_3 = \frac{1}{2}Li_3^2 = \frac{1}{2}\times 2\times\left(\frac{2}{3}\right)^2 = \frac{4}{9}\ J$$

$$\frac{H_2}{H_3} = \frac{R_2}{R_3} \Rightarrow H_2 = H_3 = \frac{2}{9}\ J$$

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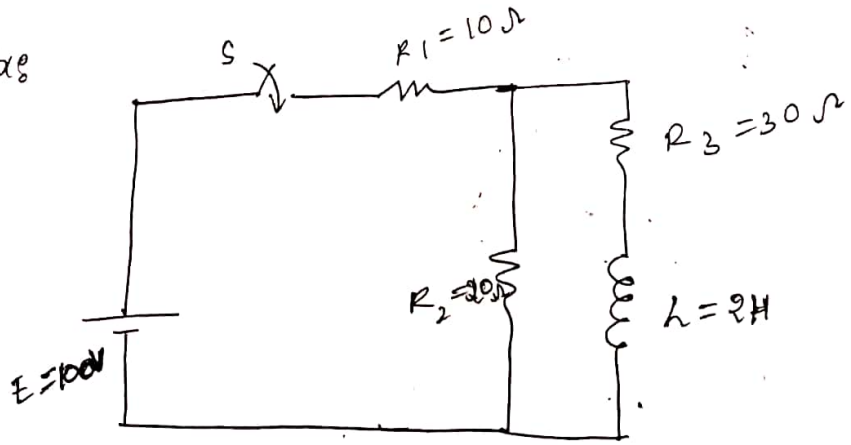
Finally, time constant during decay,

$$\tau = \frac{L}{R_{eff}} = \frac{2}{20} = \frac{1}{10}\ sec.$$

24|06|25 | Tuesday

Ex:



Analyse the circuit.

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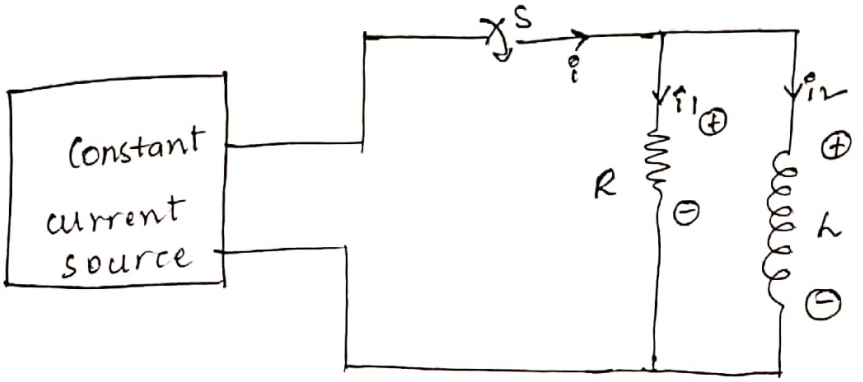
Ex:



At any time t,

$$i = i_1 + i_2 = \text{const}$$

$$\frac{di}{dt} \nearrow 0 = \frac{di_1}{dt} + \frac{di_2}{dt}$$

$$\frac{di_1}{dt} = -\frac{di_2}{dt} \quad -(1)$$

using K.V.L,

$$-L \cdot \frac{di_2}{dt} + i_1 R = 0$$

$$L \cdot \frac{di_1}{dt} = -i_1 R$$

$$\int_{i}^{i_1} \frac{di_1}{i_1} = \int_{0}^{t} -\frac{R}{L} \cdot dt$$

$$\ln(i_1/i) = -\frac{R}{L} \cdot t$$

$$\boxed{i_1 = i \cdot e^{-\frac{R}{L} \cdot t}} \star$$

current through inductor,

$$i_2 = i - i_1$$

$$\boxed{i_2 = i\left(1 - e^{-\frac{R}{L} \cdot t}\right)}$$

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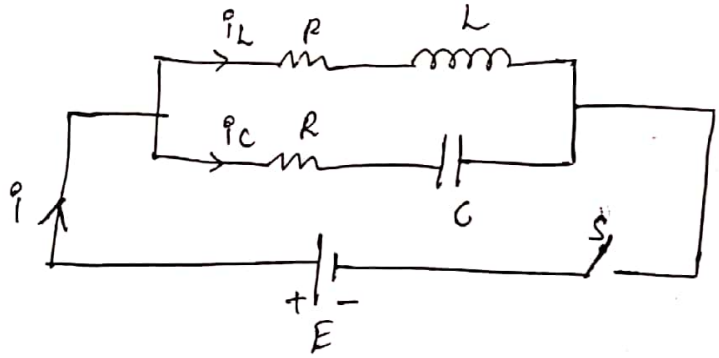
$i_1 = i_L$

$$2e^{-\frac{R}{L}\cdot t} = 1$$

$$t = \frac{L}{R} \ln 2$$

here, $\boxed{i_1 = i_2 = \frac{i}{2}}$

Ex:



Immediately after closing the switch,

$i_L = 0$, $V_L = E$ $\qquad i = i_L + i_C$

$i_C = \frac{E}{R}$, $V_C = 0$ $\qquad \boxed{i = \frac{E}{R}}$

A long time after closing the switch,

$$i_L = \frac{E}{R}, \quad V_L = 0, \quad U_L = \frac{1}{2} L i_L^2$$

$$i_C = 0, \quad V_C = E, \quad U_C = \frac{1}{2} C E^2$$

$$i = i_C + i_L$$

$$\boxed{i = \frac{E}{R}}$$

At any time 't',

$$i = i_C + i_L$$

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when $i_L = \frac{E}{R}\left(1 - e^{-\frac{R}{L}\cdot t}\right)$

$$i_c = \frac{dq}{dt} = \frac{d}{dt}\left(CE\left(1 - e^{-\frac{t}{RC}}\right)\right)$$

$$i_c = \frac{\cancel{C}E}{R\cancel{C}} e^{-t/RC}$$

$$i_c = \frac{E}{R} e^{-t/RC}$$

$$i = \frac{E}{R}\left(1 - e^{-\frac{R}{L}\cdot t} + e^{-t/RC}\right)$$

* current through battery is independent of time when

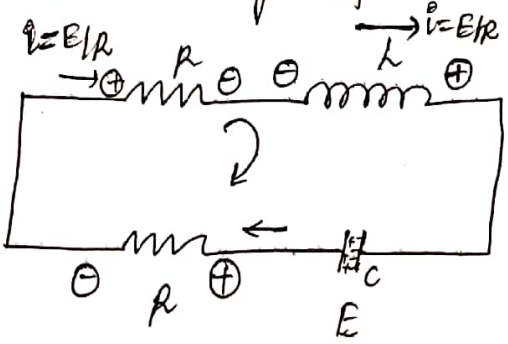
$e^{-\frac{R}{L}\cdot t} = e^{-t/RC}$

$$\frac{R}{L}\cdot\cancel{t} = \frac{\cancel{t}}{RC}$$

$$\boxed{R = \sqrt{\frac{L}{C}}}$$

* when switch is opened

Immediately after this,



$$-\frac{E}{\cancel{R}}\cdot\cancel{R} + L\cdot\frac{di}{dt} + \cancel{E} - \frac{E}{\cancel{R}}\cdot\cancel{R} = 0$$

Immediate rate of decay of current

$$L\cdot\frac{di}{dt} = E \Rightarrow \boxed{\frac{di}{dt} = \frac{E}{L}}$$

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* Heat produced in the circuit,

$$H = U_L + U_C$$

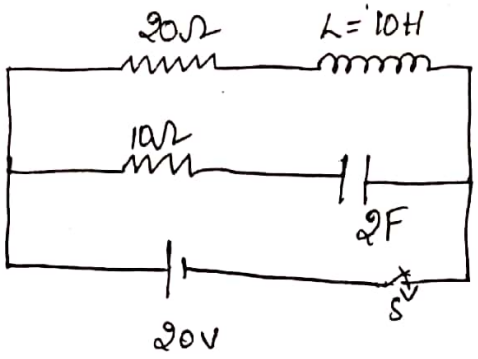
$$= \frac{1}{2} L \left(\frac{E}{R}\right)^2 + \frac{1}{2} C E^2$$

$$H_1 + H_2 = \frac{1}{2} L \left(\frac{E}{R}\right)^2 + \frac{1}{2} C E^2$$

$$\frac{H_1}{H_2} = \frac{R}{R} = 1 \quad \Rightarrow \quad H_1 = H_2$$

$$H_1 = H_2 = \frac{1}{4} L \left(\frac{E}{R}\right)^2 + \frac{1}{4} C E^2$$

Ex 6



The given circuit shown in figure is in steady state when the switch is ~~closed~~ opened at $t = 0$.

Find 1) Immediate rate of current through inductor.

2) Total heat energy produced in 10Ω & 20Ω

Sol 6

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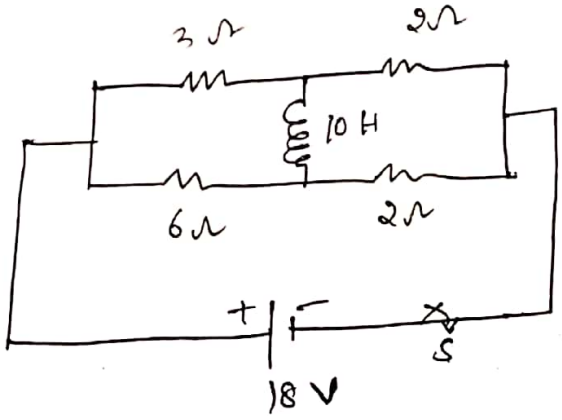

Initially the circuit is in steady state. Find

1) current through inductor

2) now the switch is opened at $t=0$. Find heat energy produced in all resistors.

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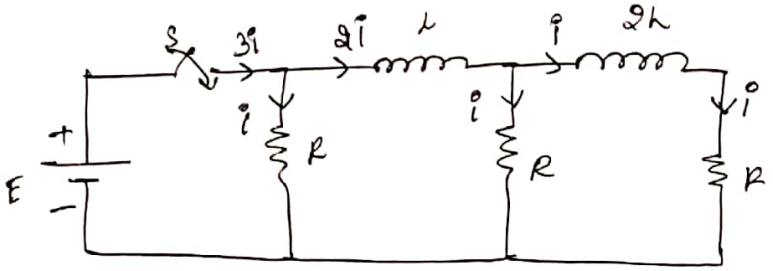
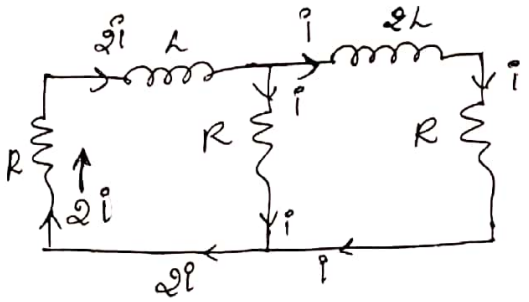
Ex 6



In steady state, the current through each resistor is i. Find currents through the resistors immediately after the switch S is opened. How much heat will be dissipated in the circuit after switch is opened.

Sol:

* Immediately after switch is opened,



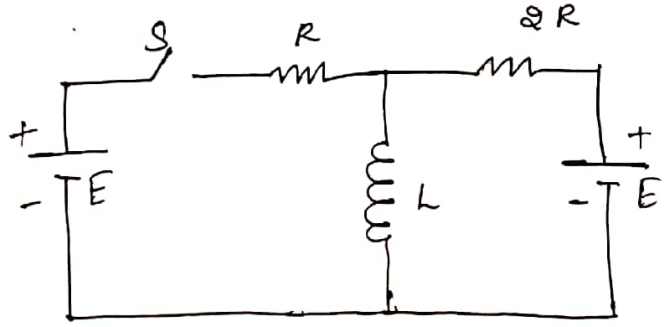
$\therefore$ Total heat produced in circuit $= U_1 + U_2$

$$= \frac{1}{2}L(2i)^2 + \frac{1}{2}(2L)i^2$$

$$= 2Li^2 + Li^2$$

$$\boxed{H_T = 3Li^2}$$

Ex 7



The circuit shown is in steady state. Find the rate of change of current through L immediately

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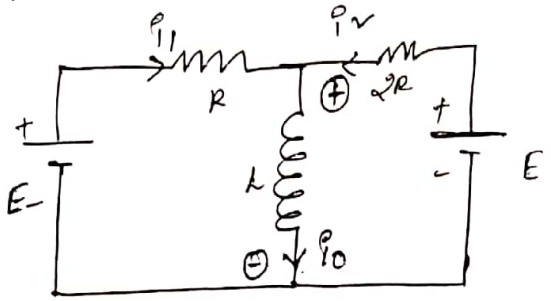
after the switch is closed.

Sol: In steady state,

$$i_0 = \frac{E}{2R} \quad -①$$

Immediately after switch S is closed,



current through L will not change.

$$\therefore \quad i_1 + i_2 = i_0 = \frac{E}{2R}$$

$$E - i_1 R + i_2 (2R) - E = 0$$

$$i_1 = +2i_2$$

$$3i_2 = \frac{E}{2R}$$

$$\boxed{i_2 = \frac{E}{6R}} \Rightarrow \boxed{i_1 = \frac{E}{3R}}$$

$$E - i_1 R - L \cdot \frac{di}{dt} = 0$$

$$L \cdot \frac{di}{dt} = -\frac{E}{3} + E$$

$$\boxed{\frac{di}{dt} = \frac{2E}{3L}}$$ — Immediate rate of growth of current.

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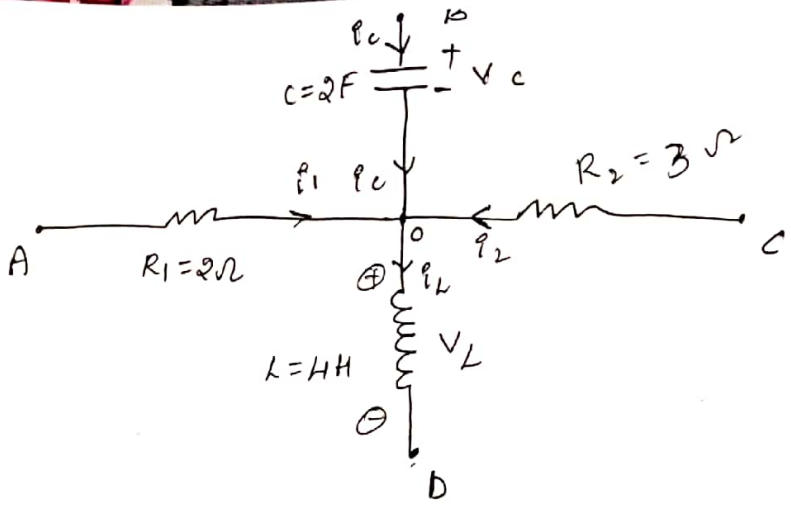
Ex:



Given, $i_1 = 10e^{-2t}$ A, $i_2 = 4A$, $V_c = 3e^{-2t}$ V

Find ① I_L & V_L

② V_{AC}, V_{AB} and V_{CD}.

Soln

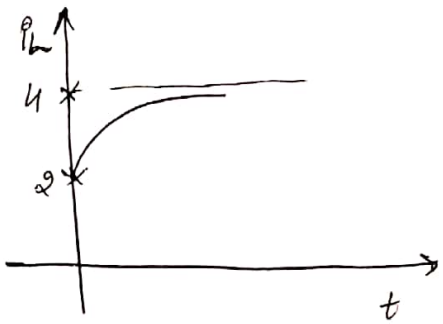
$$q = CV_c = 2 \times 3e^{-2t} = 6e^{-2t}$$

$$i_c = \frac{dq}{dt} = 6(-2)e^{-2t} = -12e^{-2t}$$

from junction law,

$$i_L = i_1 + i_2 + i_c$$

$$\boxed{i_L = 4 - 2e^{-2t}}$$



$$V_L = L \cdot \frac{di_L}{dt}$$

$$= 4 \cdot (+2)(+2)e^{-2t}$$

$$\boxed{V_L = 16e^{-2t}}$$

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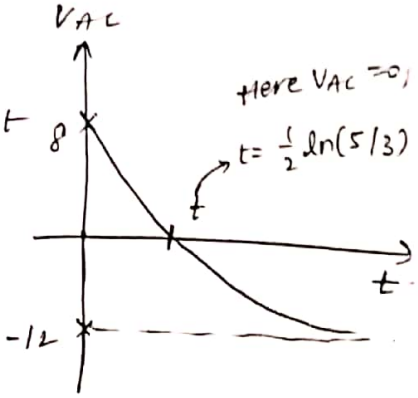
$$V_A - i_1 R_1 + i_2 R_2 = V_C$$

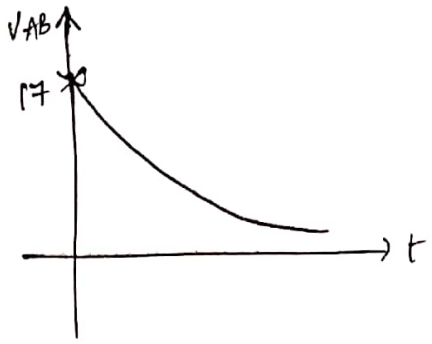
$$V_{AC} = -i_2 R_2 + i_1 R_1$$

$$= -12 + 2(10)e^{-2t}$$

$$\boxed{V_{AC} = -12 + 20e^{-2t}}$$



$$V_A - i_1 R_1 + \frac{q}{C} = V_B$$

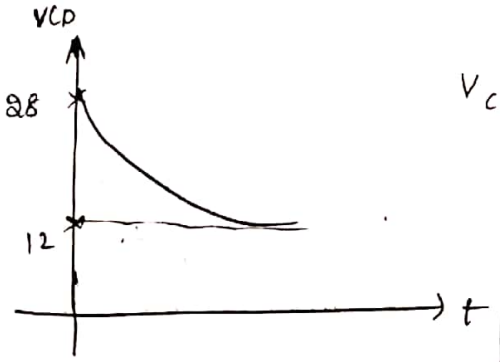


$$V_A - V_B = i_1 R_1 - \frac{q}{C}$$

$$= 20e^{-2t} - 3e^{-2t}$$

$$\boxed{V_{AB} = 17e^{-2t}}$$

$$V_C - i_2 R_2 - V_2 = V_D$$

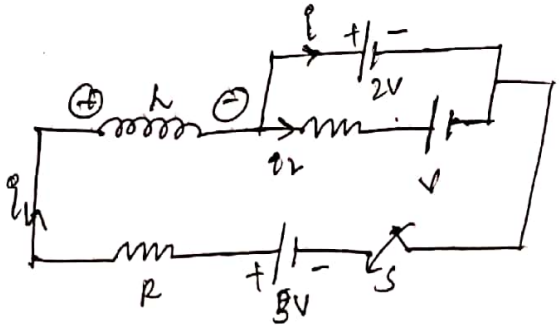


$$V_C - V_D = V_2 + i_2 R_2$$

$$= 16e^{-2t} + 4 \times 3$$

$$\boxed{V_{CD} = 16e^{-2t} + 12}$$

Ex:



The switch is closed at $t=0$, what is the variation of current i as a function t.

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Sol:

At any time t,

$$i_1 = i + i_2 \quad -①$$

$$3V - i_1 R - L \cdot \frac{di_1}{dt} - 2V = 0$$

$$V - i_1 R = L \cdot \frac{di_1}{dt}$$

$$\int_0^t \frac{dt}{L} = \int_0^{i_1} \frac{di_1}{V - i_1 R}$$

$$\frac{t}{L} = \frac{[\ln(V - i_1 R)]_0^{i_1}}{-R}$$

$$-\frac{R}{L} \cdot t = \ln\left(\frac{V - i_1 R}{V}\right)$$

$$e^{-\frac{R}{L} \cdot t} = 1 - \frac{i_1 R}{V}$$

$$\boxed{i_1 = \frac{V}{R}\left(1 - e^{-\frac{R}{L} \cdot t}\right)}$$

and $\quad 2V - i_2 R - V = 0$

$$\boxed{i_2 = \frac{V}{R}}$$

i_2 doesnot depend on time t.

Finally,

$$i = i_1 - i_2$$

$$= \frac{V}{R}\left(1 - e^{-\frac{R}{L} \cdot t}\right) - \frac{V}{R}$$

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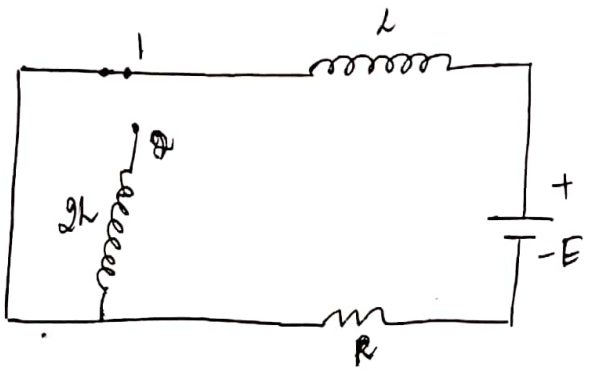
$$i = -\frac{V}{R} e^{-\frac{R}{L} \cdot t}$$

Ex:



In the circuit shown the switch S is shifted to position 2 from position 1 at $t=0$ having been in position 1 for a long time. Find current in the circuit as a function of time.

Sol: when switch is in position 1, $i_1 = \frac{E}{R}$

Immediately after shifting the switch from position one to position 2,

though inductance is changed in circuit there is no immediate change in flux in the circuit.

$$E = \frac{d\phi}{dt \to 0} \times$$

infinity so it's wrong

i.e., $\phi_1 = \phi_2$ ★

$L i_1 = (L + 2L) i_2$

$$i_1 = 3i_2$$

$$\Rightarrow i_2 = \frac{E}{3R}$$

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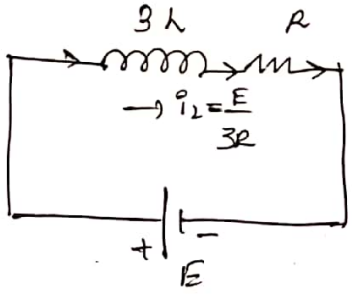
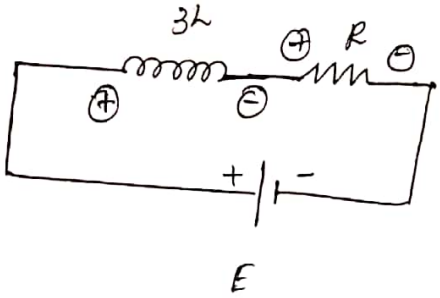
So at $t=0$, circuit is



At any time 't',

i is the current in the circuit.



$$E - 3L\cdot\frac{di}{dt} - iR = 0$$

$$\int_{\frac{E}{3R}}^{q}\frac{di}{E-iR} = \int_{0}^{t}\frac{dt}{3L}$$

$$-\frac{1}{R}\ln\left(\frac{E-iR}{\frac{2E}{3}}\right) = \frac{t}{3L}$$

$$\ln\left(\frac{3}{2} - \frac{3}{2}\frac{iR}{E}\right) = -\frac{Rt}{3L}$$

$$\frac{3}{2} - \frac{3}{2}\frac{iR}{E} = e^{-\frac{Rt}{3L}}$$

$$1 - \frac{2}{3}e^{-\frac{R}{3L}\cdot t} = \frac{iR}{E}$$

$$\boxed{i = \frac{E}{R}\left(1 - \frac{2}{3}e^{-\frac{R}{3L}\cdot t}\right)}$$

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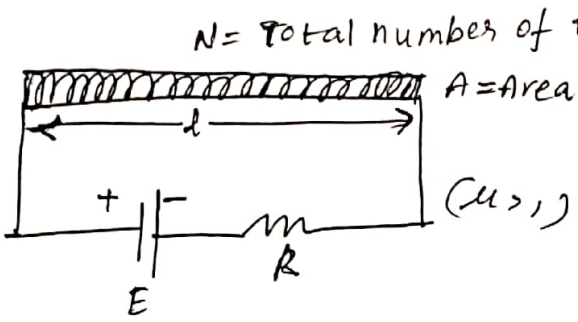
Exe



The iron rod of permeability μ from the core is abruptly (suddenly) removed. Find

1) current as a function of time.

2) workdone to pull out the rod.

Sol:

In steady state,

$$i_1 = \frac{E}{R}$$

Inductance, $L_1 = \frac{\mu_0 \mu N^2 A}{l}$

when the rod is removed abruptly,

immediately after this,

ϕ is not changing.

$$\phi_1 = \phi_2$$

$$L i_1 = L_2 i_2$$

$$\frac{\mu_0 \mu N^2 A}{l} \cdot \frac{E}{R} = \frac{\mu_0 N^2 A}{l} \cdot i_2$$

$$\boxed{i_2 = \frac{\mu E}{R}}$$

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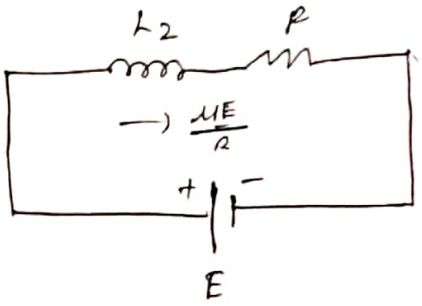
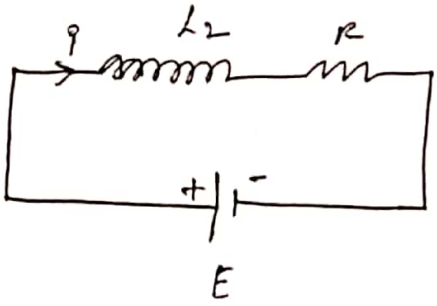
at $t = 0$,



At time 't', current in the circuit is 'i'.



$$E - L_2 \cdot \frac{di}{dt} - iR = 0$$

$$L_2 \cdot \frac{di}{dt} = E - iR$$

$$\int_{\frac{\mu E}{R}}^{i} \frac{di}{E - iR} = \int_{0}^{t} \frac{dt}{L_2}$$

$$\frac{-1}{R} \ln\left(\frac{E - iR}{E - \mu E}\right) = \frac{t}{L_2}$$

$$\frac{E - iR}{E(1-\mu)} = e^{-\frac{Rt}{L_2}}$$

$$\frac{1}{1-\mu} - \frac{iR}{E(1-\mu)} = e^{\frac{-R}{L_2} \cdot t}$$

$$i = \frac{E}{R}\left(1 - (1-\mu)\, e^{-\frac{R}{L_2} \cdot t}\right)$$

$$\boxed{i = \frac{E}{R}\left(1 + (\mu - 1)\, e^{-\frac{R}{L_2} t}\right)}$$

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at $t=0$, $i = \frac{\mu E}{R}$

at $t=\infty$, $i = \frac{E}{R}$

work done in pulling the rod,

$$W = U_f - U_i$$

$$= \frac{1}{2} L_2 i_2^2 - \frac{1}{2} L_1 i_1^2$$

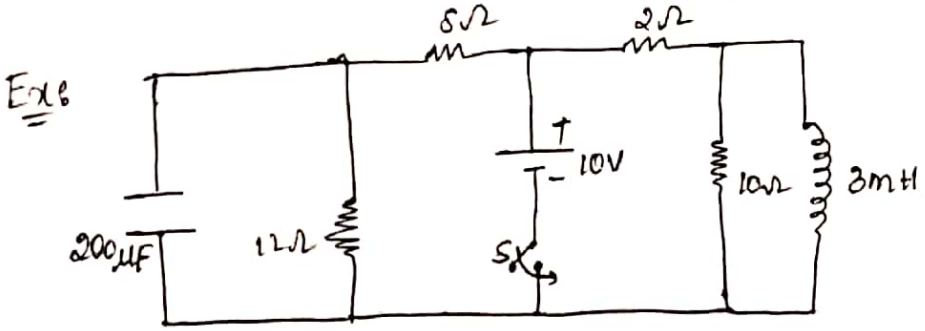
$$= \frac{1}{2}\left(\frac{\mu_0 N^2 A}{\ell}\right)\left(\frac{\mu E}{R}\right)^2 - \frac{1}{2}\left(\frac{\mu_0 \mu N^2 A}{\ell}\right)\left(\frac{E}{R}\right)^2$$

$$= \frac{1}{2}\left(\frac{\mu_0 N^2 A}{\ell}\right)\left(\frac{E}{R}\right)^2 (\mu^2 - \mu)$$

$$W = \frac{\mu(\mu-1)}{2}\left(\frac{\mu_0 N^2 A}{\ell}\right)\left(\frac{E}{R}\right)^2$$

$$\boxed{W = \frac{\mu_0 \mu(\mu-1) N^2 A E^2}{2\ell R^2}}$$

Ex:6



Switch is closed at ~~t=0~~. Find

1) current through battery immediately after this.

2) current through battery a long time after closing

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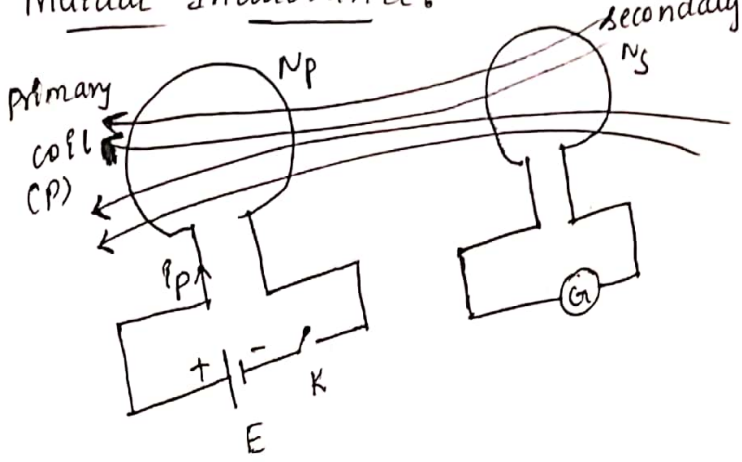
the switch.

3) In steady state, Find energy stored in C & L & p.d across each element.

Sol6

25/06/25 | wednesday

Mutual Inductance



* $N_P \phi_P = L_P i_P$

↳ flux linked with each turn.

$$e_P = -\frac{d}{dt}(N_P \phi_P) = -\frac{d}{dt}(L_P i_P)$$

$$\boxed{e_P = -L_P \frac{di_P}{dt}}$$

Here, $N_S \phi_S \propto i_P$

↳ flux linked with each turn of coil S.

$$\boxed{N_S \phi_S = M_{SP} i_P} \Rightarrow M_{SP} = \frac{N_S \phi_S}{i_P} = \text{Henry (unit)}$$

M_{SP} = mutual inductance of coil 'S' due to coil 'P'

From faraday's law,

$$e_S = -\frac{d}{dt}(N_S \phi_S)$$

$$e_S = -\frac{d}{dt}(M_{SP} i_P) = -M_{SP} \cdot \frac{di_P}{dt}$$

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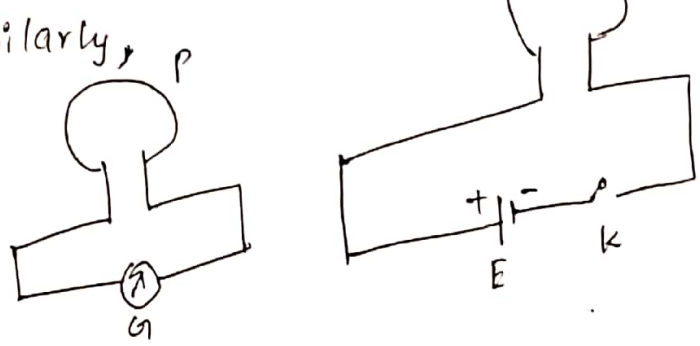
Similarly,



$N_S \phi_S = i_S L_S$

$$e_S = -\frac{d}{dt}(N_S \phi_S)$$

$$e_S = -L_S \frac{di_S}{dt}$$

again,

$$N_P \phi_P = M_{PS} i_S$$

M_{PS} = mutual inductance of coil 'p' due to coil 'S'

Faraday's law,

$$e_P = -\frac{d}{dt}(N_P \phi_P)$$

$$\boxed{e_P = -M_{PS} \frac{di_S}{dt}}$$

By reciprocity theorem,

$$\boxed{M_{PS} = M_{SP}}$$

* Mutual inductance of two coils depends on

1) No. of turns of coil.
2) geometrical shape and size of coils
3) medium inside the coils.

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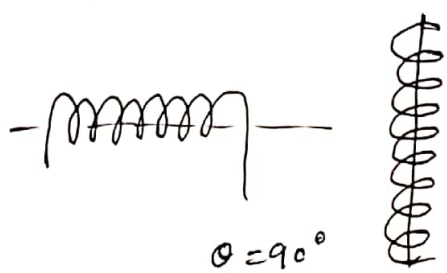
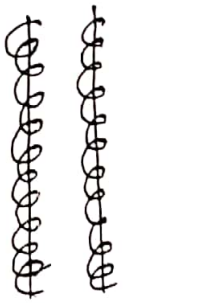
4) distance b/w two coils.

5) Angle b/w axes of coils.



$\theta = 90°$

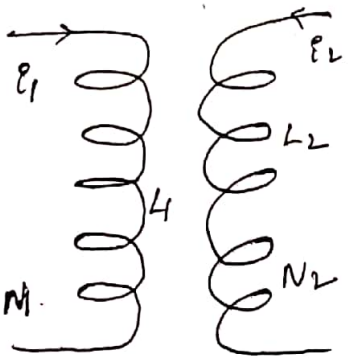
$M \rightarrow min$



$\theta = 0°$

$M \rightarrow max$

Relation b/w L_1, L_2 & M



$N_1 \phi_1 = L_1 i_1$

$$L_1 = \frac{N_1 \phi_1}{i_1}$$

& $N_2 \phi_2 = L_2 i_2$

$$L_2 = \frac{N_2 \phi_2}{i_2}$$

Again,

$N_1 \phi_1 = M_{12} i_2$

$$M_{12} = \frac{N_1 \phi_1}{i_2}$$

$N_2 \phi_2 = M_{21} i_1$

$$M_{21} = \frac{N_2 \phi_2}{i_1}$$

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now,

$$M_{12} \times M_{21} = \frac{N_1 \phi_1}{i_2} \times \frac{N_2 \phi_2}{i_1}$$

$$\therefore M^2 = L_1 \times L_2$$

$$\boxed{M = \sqrt{L_1 L_2}}$$

In general,

$\boxed{M = K\sqrt{L_1 L_2}}$ → This is used when there is some leakage of flux.

K = coeff of coupling. $K = \frac{\text{flux linkage with S}}{\text{flux linkage with P}}$.

generally, $0 < K < 1$

* If $K = 0$ there will be no field line of one coil passing through other coil. So there will be no coupling.

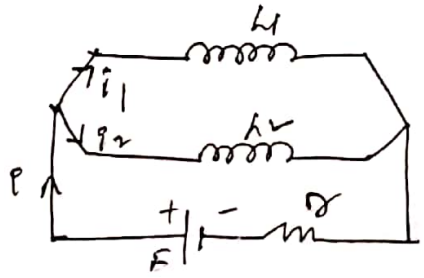
Effective Self Inductance [Leq]

case (1): [without mutual inductance]

a) series combination:

$$L_{eq} = L_1 + L_2 + \cdots$$

b) parallel combination:



$$i = i_1 + i_2$$

$$\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots$$

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$$L_1 \frac{di_1}{dt} = L_2 \cdot \frac{di_2}{dt} = - - -$$

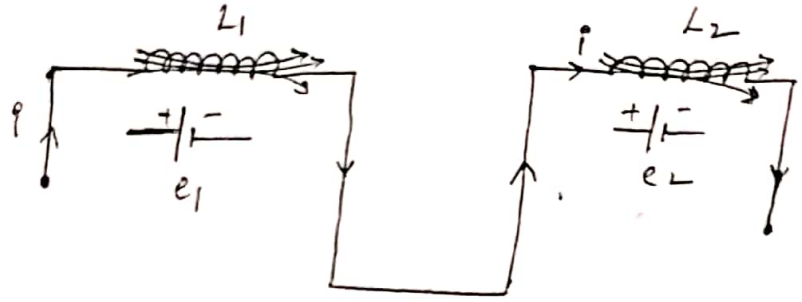
$$L_1 i_1 = L_2 i_2 = - - -$$

$$i_1 = \frac{L_2 i}{L_1 + L_2} \quad , \quad i_2 = \frac{L_1 i}{L_1 + L_2} .$$

$\boxed{i = \frac{E}{R}}$ in steady state

case (2): [with mutual inductance]

a) Flux through the coils supporting each other:



Here, i & $\frac{di}{dt}$ are same through both coils.

Total flux linked with coil 1,

$$\Phi_1 = L_1 i_1 + M_{12} i_2$$

$$e_1 = -\frac{d\Phi_1}{dt} = -L_1 \frac{di_1}{dt} - M_{12} \frac{di_2}{dt}$$

similarly,

Total flux linked with coil 2,

$$\Phi_2 = L_2 i_2 + M_{21} i_1$$

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$$e_2 = \frac{-d\phi_2}{dt} = -L_2 \cdot \frac{di_2}{dt} - M_{21} \frac{di_1}{dt}$$

∴ net Induced emf,

$$e = e_1 + e_2$$

$$= -L_1 \cdot \frac{di_1}{dt} - M_{12} \cdot \frac{di_2}{dt} - L_2 \frac{di_2}{dt} - M_{21} \frac{di_1}{dt}$$

$$\boxed{e = -(L_1 + L_2 + 2M) \frac{di}{dt}}$$

* If two coils are replaced with single coil,

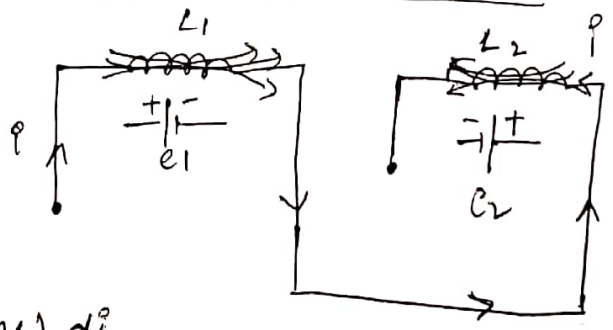
$$e = -L_{eq} \cdot \frac{di}{dt}$$

$$\cancel{+} L_{eq} \left(\frac{\cancel{di}}{\cancel{dt}}\right) = \cancel{+}(L_1 + L_2 + 2M) \cdot \frac{\cancel{di}}{\cancel{dt}}$$

$$\boxed{L_{eq} = L_1 + L_2 + 2M} \star$$

b) Flux through the coils opposing each other:

Here,



$$\phi_1 = L_1 i - Mi$$

$$e_1 = -\frac{d\phi_1}{dt} = -(L_1 - M) \frac{di}{dt}$$

$$\phi_2 = L_2 i - Mi$$

$$e_2 = -\frac{d\phi_2}{dt} = -(L_2 - M) \cdot \frac{di}{dt}$$

∴ net Induced emf,

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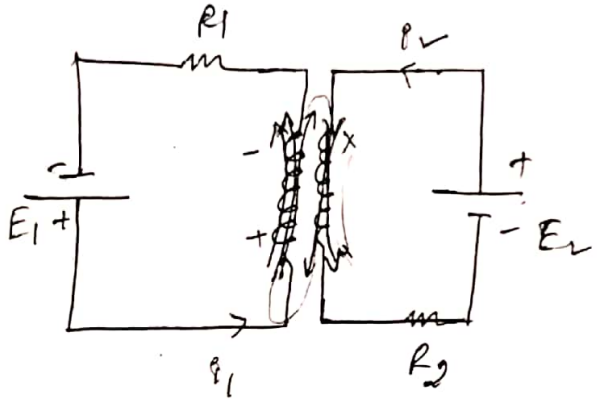
$$e = e_1 + e_2$$

$$\boxed{e = -(L_1 + L_2 \mp 2M)\frac{di}{dt}}$$

$$\mp L_{eq} \cdot \frac{di}{dt} = \mp (L_1 + L_2 - 2M)\frac{di}{dt}$$

$$\boxed{L_{eq} = L_1 + L_2 - 2M} \quad \star$$

Energy stored in both the coils:-



(supporting flux)

If there is no battery, then there will be no raise of current from 0 to max. So battery is continuously doing work against induced emf to rise current from zero to steady state.

* induced emf in coil 1,

$$e_1 = -L_1 \frac{di_1}{dt} - M \cdot \frac{di_2}{dt}$$

similarly,

$$e_2 = -L_1 \frac{di_2}{dt} - M \frac{di_1}{dt}$$

* work done by both the batteries,

$$dw = -e_1 dq_1 - e_2 dq_2$$

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$$\Rightarrow dw = L_1 \frac{di_1}{dt} dq_1 + M \frac{di_2}{dt} \cdot dq_1$$

$$+ L_2 \frac{di_2}{dt} dq_2 + M \cdot \frac{di_1}{dt} \cdot dq_2$$

$$dw = L_1 i_1 \cdot di_1 + M i_1 di_2 + L_2 i_2 di_2 + M i_2 di_1$$

$$dw = L_1 i_1 di_1 + L_2 i_2 di_2 + M d(i_1 i_2)$$

Total work done by battery,

$$w = \int dw = \int_0^{i_1} L_1 i_1 di_1 + \int_0^{i_2} L_2 i_2 di_2 + M \int_0^{i_1} d(i_1 i_2)$$

$$\boxed{w = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M i_1 i_2}$$

work done & also energy stored in this case is work done.

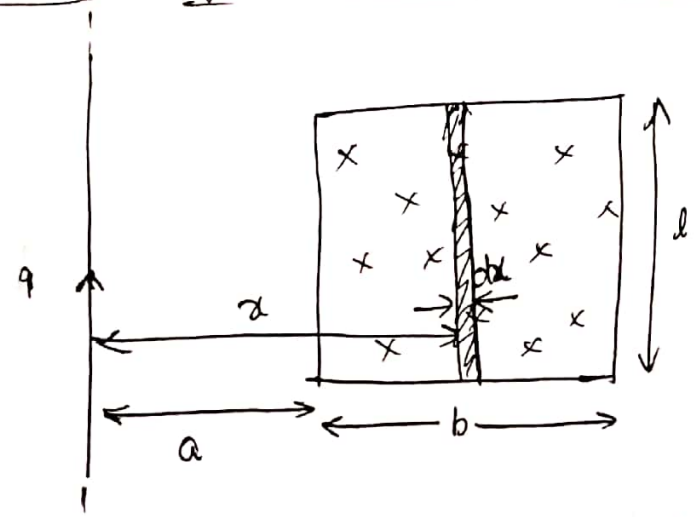
[Supporting flux].

& $\boxed{w = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 - M i_1 i_2}$

(Opposing flux)

Calculation of mutual inductance

①



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$$B = \frac{\mu_0 i}{2\pi x}$$

$$d\phi = B\,dx = Bl\,dx = \frac{\mu_0 i l}{2\pi}\frac{dx}{x}$$

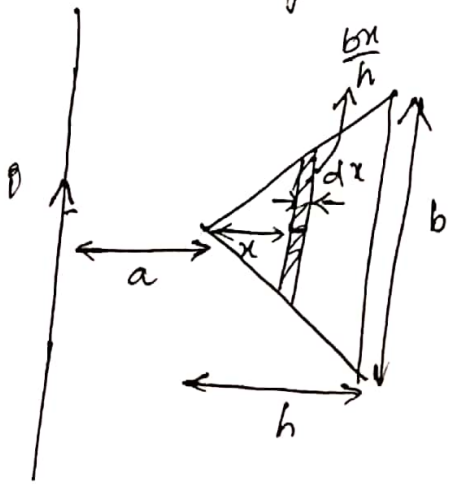
$$\phi_{loop} = \frac{\mu_0 i l}{2\pi} \ln\left(\frac{a+b}{a}\right)$$

$$M\phi_{wire} = \frac{\mu_0 i l}{2\pi} \ln\left(\frac{a+b}{a}\right)$$

$$\boxed{M = \frac{\mu_0 l}{2\pi} \ln\left(1+\frac{b}{a}\right)}$$

* Mutual inductance changes when orientation of both the coils is changed.

(2)



$$\frac{x}{h} = \frac{y}{b}$$

$$\boxed{y = \frac{bx}{h}}$$

$$B = \frac{\mu_0 i}{2\pi(a+x)}$$

$$d\phi = B \cdot dA$$

$$= \frac{\mu_0 i}{2\pi}\frac{bx}{h(a+x)} \cdot dx$$

$$\phi = \int_0^h d\phi = \frac{\mu_0 i b}{2\pi h}\int_0^h \frac{a+x-a}{a+x} \cdot dx$$

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$$= \frac{\mu_0 i b}{2\pi h} \left[x - a \ln (a+x) \right]_0^h$$

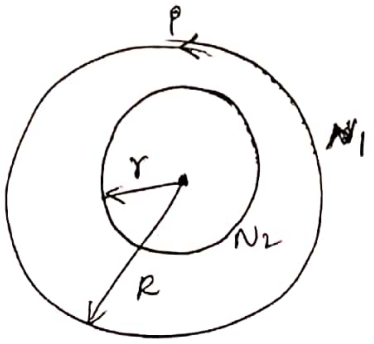
$$= \frac{\mu_0 i b}{2\pi h} \left[h - a \ln (a+h) + a \ln a \right]$$

$$= \frac{\mu_0 i b}{2\pi h} \left[h - a \ln \left(\frac{a+h}{a} \right) \right]$$

$$\boxed{\phi = \frac{\mu_0 i b}{2\pi} - \frac{\mu_0 i a b}{2\pi h} \ln \left(\frac{a+h}{a} \right)} \qquad M = \frac{\phi}{i}$$

③ Concentric circular loops :

$R >>> r$



$$B_1 = \frac{\mu_0 N_1 i}{2R}$$

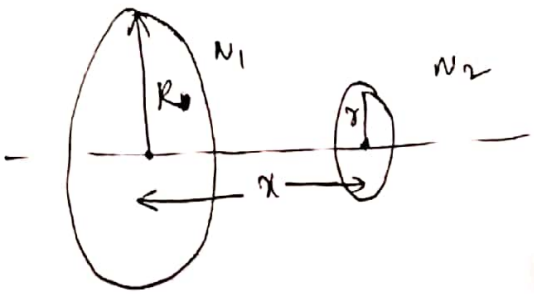
$$\phi_2 = N_2 B_1 A = \frac{N_2 \mu_0 N_1 i}{2R} (\pi r^2)$$

$$\phi_2 = \frac{\mu_0 N_1 N_2 i (\pi r^2)}{2R}$$

$$M i = \frac{\pi \mu_0 N_1 N_2 r^2}{2R} i$$

$$\boxed{M = \frac{\pi \mu_0 N_1 N_2 r^2}{2R}}$$

*



$$B_1 = \frac{\mu_0 N_1 i R^2}{2 [R^2 + x^2)^{3/2}}$$

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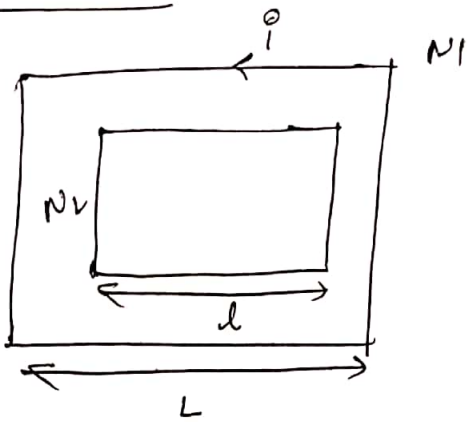
$$\phi_2 = N_2 B_1 A$$

$$M\mathcal{I} = \frac{N_2 \mu_0 N_1 \mathcal{I} R^2}{2(R^2+x^2)^{3/2}} (\pi r^2)$$

$$\boxed{M = \frac{\pi \mu_0 N_1 N_2 R^2 r^2}{2(R^2+x^2)^{3/2}}}$$

(4) Concentric square loops :

$L >> l$



$$B_1 = \frac{\mu_0 N_1 i}{2R} \frac{\tan(\pi/4)}{\pi/4}$$

$$= \frac{\mu_0 N_1 i}{2(L/\sqrt{2})} \times \frac{\tan\pi/4}{\pi/4}$$

$$= \frac{2\sqrt{2}\mu_0 N_1 i}{\pi L}$$

$$\phi_2 = N_2 B_1 A$$

$$M\mathcal{I} = \frac{2\sqrt{2}\mu_0 N_1 N_2 \mathcal{I}}{\pi L} \cdot l^2$$

$$\boxed{M = \frac{2\sqrt{2}\mu_0 N_1 N_2 l^2}{\pi L}}$$

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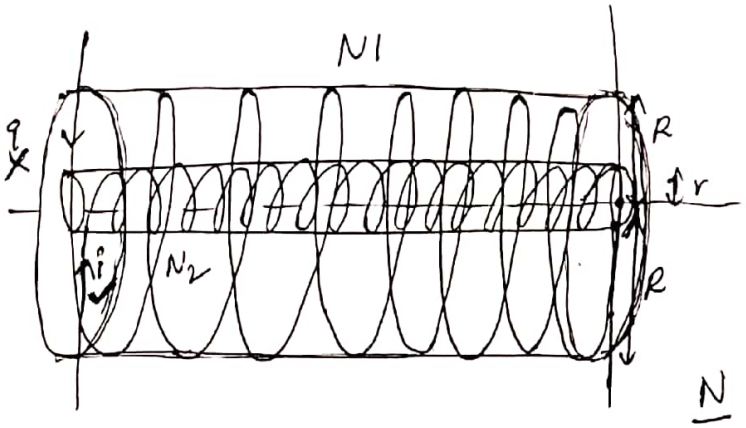
(5) Concentric solenoids



$$\frac{N}{L} = n$$

$$B_1 = \mu_0 n_1 i$$

$$\phi_2 = N_2 B_1 A$$

$$\phi_2 = N_2 \mu_0 n_1 i A$$

$$M i = N_2 \mu_0 n_1 i A$$

$$M = \mu_0 n_1 N_2 A$$

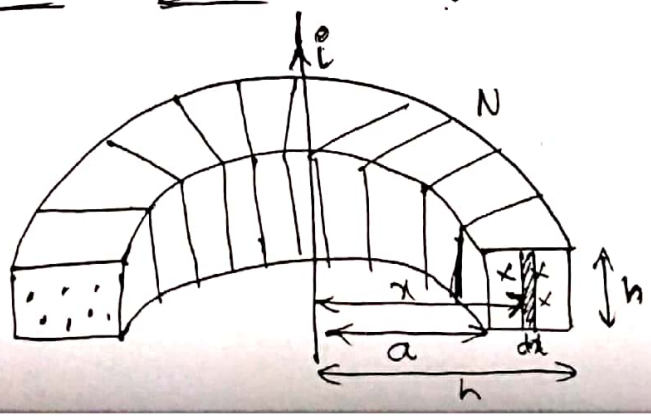
$$\boxed{M = \frac{\mu_0 N_1 N_2 A}{l}}$$

(or) $M = \dfrac{\mu_0 n_1 n_2 A l}{1}$

$$\boxed{M = \mu_0 n_1 n_2 A l}$$

(or) $\boxed{M = \mu_0 \mu_r n_1 n_2 A l}$

(6) Toroid with Rectangular cross-section :



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$$B = \frac{\mu_0 i}{2\pi x}$$

$$d\phi = B\,dA = \frac{\mu_0 i h}{2\pi x} \cdot dx$$

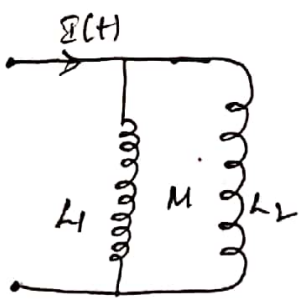
$$\phi_{\text{each turn}} = \frac{\mu_0 i h}{2\pi} \int_a^b \frac{1}{x}\,dx = \frac{\mu_0 i h}{2\pi} \ln(b/a)$$

$$\phi_T = \frac{\mu_0 N i h}{2\pi} \ln(b/a)$$

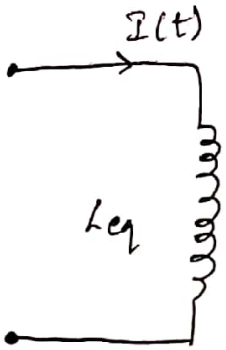
$$M i = \frac{\mu_0 N i h}{2\pi} \ln(b/a)$$

$$\boxed{M = \frac{\mu_0 N h}{2\pi} \ln(b/a)} \quad *$$

Ex: Two Inductors having self inductances L_1 & L_2 are connected in parallel as shown in fig (a). The mutual inductance b/w the 2 inductors is M. Determine the equivalent self inductance L_{eq} for the system as shown in Fig (b).



(a)



(b)

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Sol:

$$I = I_1 + I_2$$

The voltage across the pair,

$$\Delta V = -L_1 \cdot \frac{dI_1}{dt} + M \cdot \frac{dI_2}{dt}$$

$$\Delta V = -L_2 \frac{dI_2}{dt} + M \cdot \frac{dI_1}{dt}$$

$$\Delta V = -L_{eq} \cdot \frac{dI}{dt}$$

[illegible] on solving,

$$L_{eq} = \frac{4L_2 - M^2}{4 + L_2 + 2M}$$

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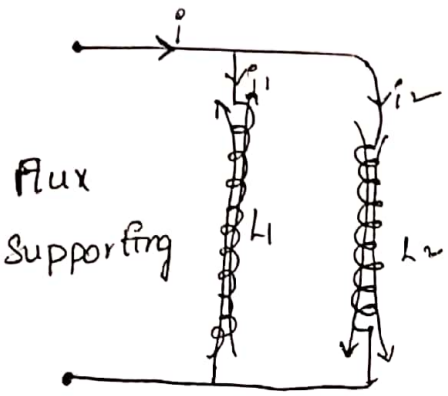
*



In coil 1,

$$e_1 = -L_1 \cdot \frac{di_1}{dt} - M \frac{di_2}{dt}$$

In coil 2,

$$e_2 = -L_2 \cdot \frac{di_2}{dt} - M \frac{di_1}{dt}$$

$$e = e_1 + e_2$$

$$i = i_1 + i_2$$

$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$$

Finally,

$$e = -L_{eq}\left(\frac{di}{dt}\right)$$

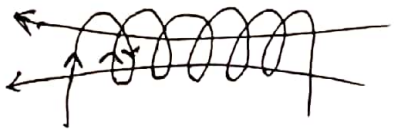
after solving,

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$$

Rheostat!

* Rheostat is a coil having inductance but no resistance.

*



$L_{eq} = L_1 + L_2 - 2M$

& $M = \sqrt{L_1 L_2}$

if $L_1 = L_2$, $L_{eq} = 0$.

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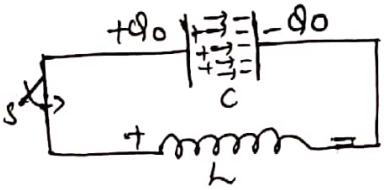
L-C Oscillations

* Circuit has no resistance.

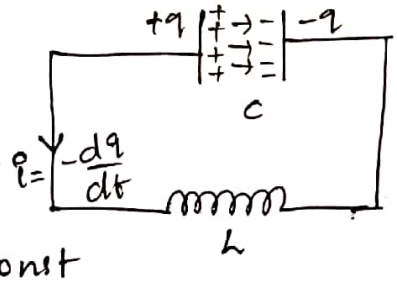


at $t=0$, $q=Q_0$ & $i=0$; $U_i = \frac{Q_0^2}{2C} = U_C$

$$\frac{Q_0}{C} - L\cdot\frac{di}{dt} = 0 \Rightarrow \boxed{\left|\frac{di}{dt}\right| = \frac{Q_0}{LC}}$$

↳ Immediate rate of growth.

At any time 't',



M-I:

L.C.E, $\frac{q^2}{2C} + \frac{1}{2}Li^2 = \frac{Q_0^2}{2C} = \text{const}$

$$\frac{2q}{2C}\cdot\frac{dq}{dt} + \frac{1}{2}L(2i)\frac{di}{dt} = 0$$

$$\frac{q}{C}(-i) + Li\frac{di}{dt} = 0$$

$$-\frac{q}{C} + L\left(\frac{d}{dt}\left(-\frac{dq}{dt}\right)\right) = 0$$

$$\boxed{\frac{d^2q}{dt^2} = -\frac{q}{LC}} \Rightarrow \boxed{\frac{d^2q}{dt^2} = -\omega^2 q}$$

∴ charge is oscillating in circuit (SHM) with angular frequency, $\boxed{\omega = \frac{1}{\sqrt{LC}}}$

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& frequency,

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

* Time period,

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{LC}$$

M-II:

using K.V.L,

$$\frac{q}{C} - L\cdot\frac{di}{dt} = 0$$

$$\frac{q}{C} = L\frac{d}{dt}\left(-\frac{dq}{dt}\right)$$

$$\boxed{\frac{d^2q}{dt^2} = -\frac{q}{LC}} \quad \text{---} \ (a)$$

* Soln of eqn (a) is,

$$q = q_0 \sin(\omega t + \phi)$$

at $t=0$, $q = q_0$

$$q_0 = q_0 \sin(\omega(0) + \phi)$$

$$\sin\phi = 1$$

$$\boxed{\phi = \pi/2}$$

$$\Rightarrow q = q_0 \sin(\omega t + \pi/2)$$

$$\Rightarrow \boxed{q = q_0 \cos\omega t} \quad \text{---} \ (1)$$

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* current in the circuit,

$$i = -\frac{dq}{dt}$$

$$i = -\frac{d}{dt}(Q_0 \cos\omega t)$$

$$\boxed{i = Q_0 \omega \sin\omega t} \quad \text{--- (2)}$$

Here, $i_0 = Q_0 \omega \rightarrow$ max/peak value of current

* ~~The field current i_0~~,

~~use use energy theorem~~

current → max
⇒ charge = zero

charge → max
⇒ current = zero

$$i = Q_0 \omega \sqrt{1 - \cos^2\omega t}$$

$$i = Q_0 \omega \sqrt{1 - \frac{q^2}{Q_0^2}}$$

$$\boxed{i = \omega \sqrt{Q_0^2 - q^2}} \; ; \; \omega = \frac{1}{\sqrt{LC}}$$

* Graphs :

① $q = Q_0 \cos\omega t$

a) $\boxed{q = \pm Q_0}$ when $\cos\omega t = \pm 1$

$\Rightarrow \omega t = 0, \pi, 2\pi, 3\pi, \cdots n\pi$

$\Rightarrow t = 0, \frac{\pi}{\omega}, \frac{2\pi}{\omega}, \frac{3\pi}{\omega}, \cdots \frac{n\pi}{\omega}$

$T = \frac{2\pi}{\omega}$

$t = 0, \frac{T}{2}, T, \frac{3T}{2}, 2T, \cdots \frac{nT}{2}$

b) $q = 0$ when $\cos\omega t = 0$

$\omega t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \cdots$

$t = \frac{\pi}{2\omega}, \frac{3\pi}{2\omega}, \frac{5\pi}{2\omega}, \cdots$

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$$t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \text{-----}$$

② current, $i = q_0 \omega \sin \omega t$

a) $\boxed{i = \pm q_0 \omega}$ (max) $\Rightarrow \sin \omega t = \pm 1$

$$\omega t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \text{--}$$

$$\Rightarrow t = \frac{\pi}{2\omega}, \frac{3\pi}{2\omega}, \frac{5\pi}{2\omega}, \text{---}$$

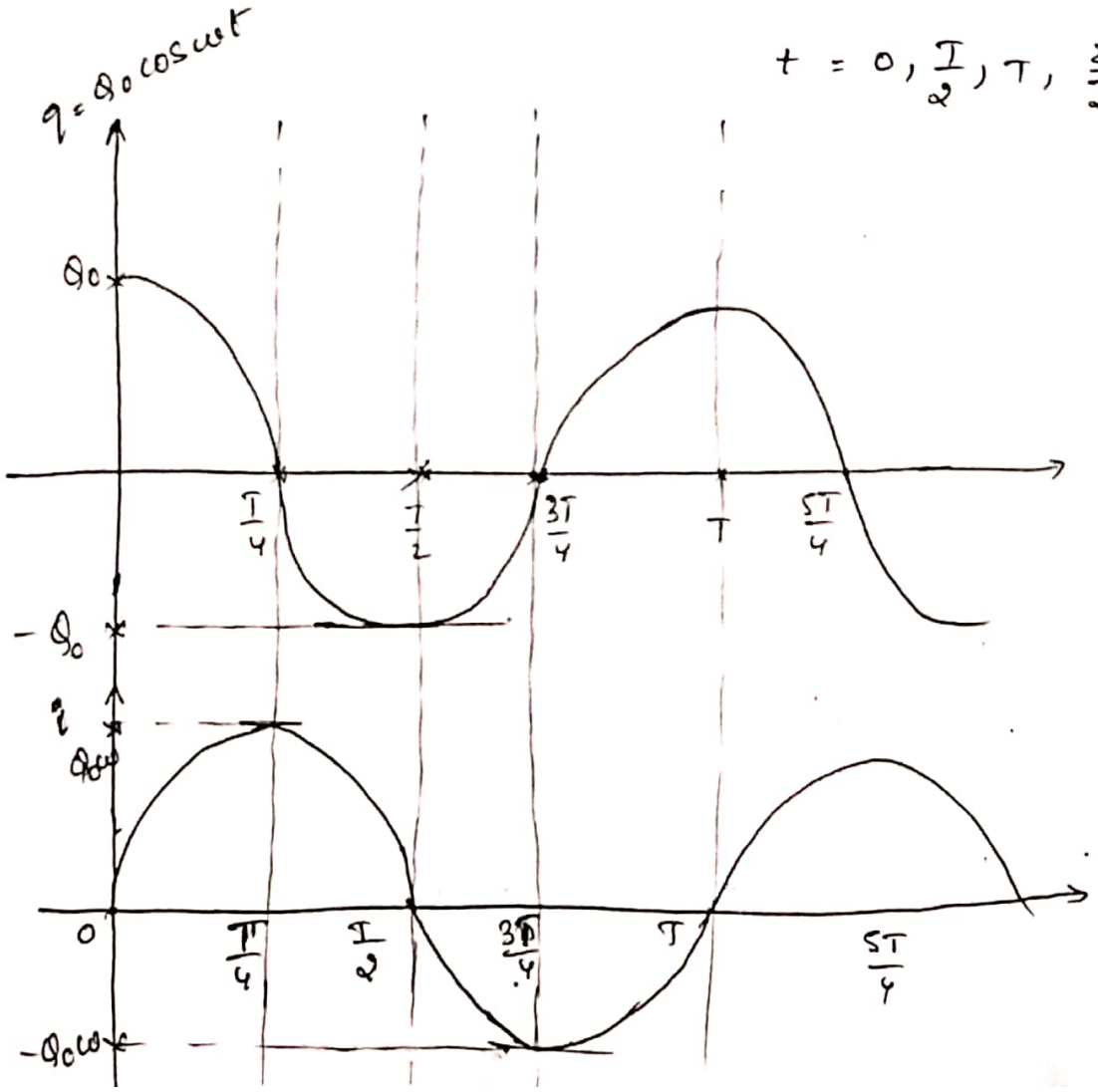
$$\Rightarrow t = \frac{T}{4}, \frac{5T}{4}, \frac{5T}{4}, \text{---}$$

b) $\boxed{i = 0}$ (min) $\Rightarrow \sin \omega t = 0$

$$\omega t = 0, \pi, 2\pi, \text{- ----}$$

$$t = 0, \frac{\pi}{\omega}, \frac{2\pi}{\omega}, \text{- ---}$$

$$t = 0, \frac{T}{2}, T, \frac{3T}{2}, \text{---}$$



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* p.d across capacitor,

$$V_C = \frac{q}{C} = \frac{Q_0 \cos\omega t}{C}$$

* p.d across inductor,

$$V_L = L \frac{di}{dt}$$

$$= L \frac{d}{dt}(Q_0 \omega \sin\omega t)$$

$$= L Q_0 \omega^2 \cos\omega t$$

$$V_L = \frac{Q_0}{C} \cos\omega t$$

clearly, $\boxed{V_L = V_C}$ *

* Energy stored in inductor,

$$U_L = \frac{1}{2} L i^2$$

$$= \frac{1}{2} L (Q_0^2 \omega^2 \sin^2\omega t)$$

$$U_L = \frac{Q_0^2}{2C} \sin^2\omega t$$

$$= \frac{Q_0^2}{2C}(1 - \cos^2\omega t)$$

$$= \frac{Q_0^2}{2C}\left(\frac{1 - \cos 2\omega t}{2}\right)$$

$$\boxed{U_L = \frac{Q_0^2}{4C}(1 - \cos 2\omega t)}$$

* $U_L = \frac{Q_0^2}{2C}$ when $\cos 2\omega t = -1$

$2\omega t = \pi, 3\pi, 5\pi, \ldots$

$t = \frac{\pi}{2\omega}, \frac{3\pi}{2\omega}, \frac{5\pi}{2\omega}, \ldots$

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$$t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \ldots$$

$U_L = 0$ when $\cos 2\omega t = 1$

$$2\omega t = 0, 2\pi, 4\pi, 6\pi \ldots$$

$$t = 0, \frac{\pi}{\omega}, \frac{2\pi}{\omega}, \frac{3\pi}{\omega}, \ldots$$

$$t = 0, \frac{T}{2}, T, \frac{3T}{2}, \ldots$$

* Energy stored in capacitor,

$$U_C = \frac{Q_0^2 \cos^2 \omega t}{2C}$$

$$\boxed{U_C = \frac{Q_0^2}{4C}(1 + \cos 2\omega t)}$$

* $U_C = \frac{Q_0^2}{2C}$ when $\cos 2\omega t = 1$

$$2\omega t = 0, 2\pi, 4\pi, 6\pi, \ldots$$

$$t = 0, \frac{\pi}{\omega}, \frac{2\pi}{\omega}, \ldots$$

$$t = 0, \frac{T}{2}, T, \frac{3T}{2}, \ldots$$

$U_C = \frac{Q_0^2}{2C}$ when $\cos 2\omega t = -1$

$$\Rightarrow t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \ldots$$

$$U_C + U_L = \frac{Q_0^2}{4C}(1 - \cos 2\omega t) + \frac{Q_0^2}{4C}(1 + \cos 2\omega t)$$

$$= \frac{Q_0^2}{4C} + \frac{Q_0^2}{4C}$$

* $$\boxed{U_C + U_L = \frac{Q_0^2}{2C}}$$

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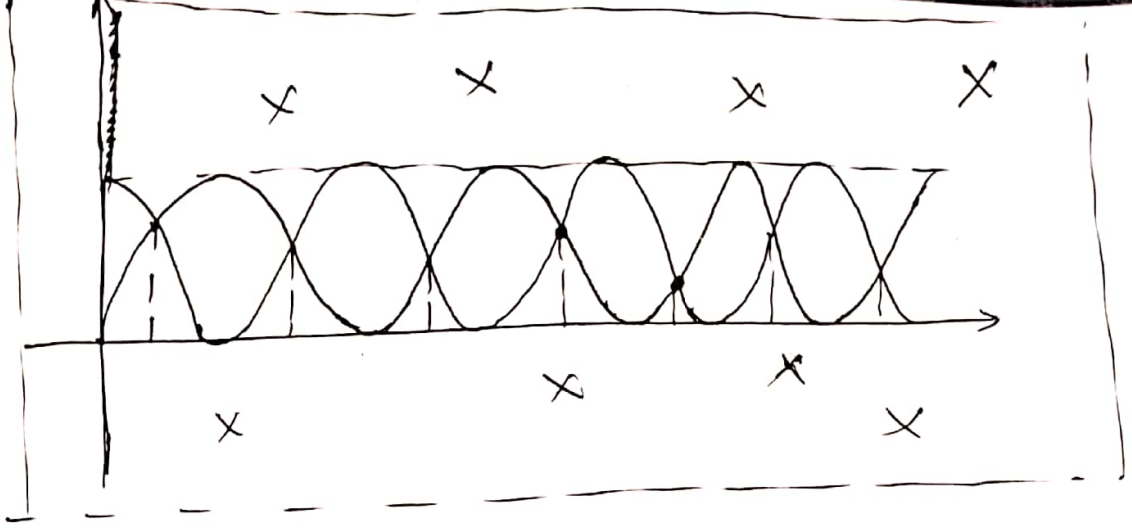

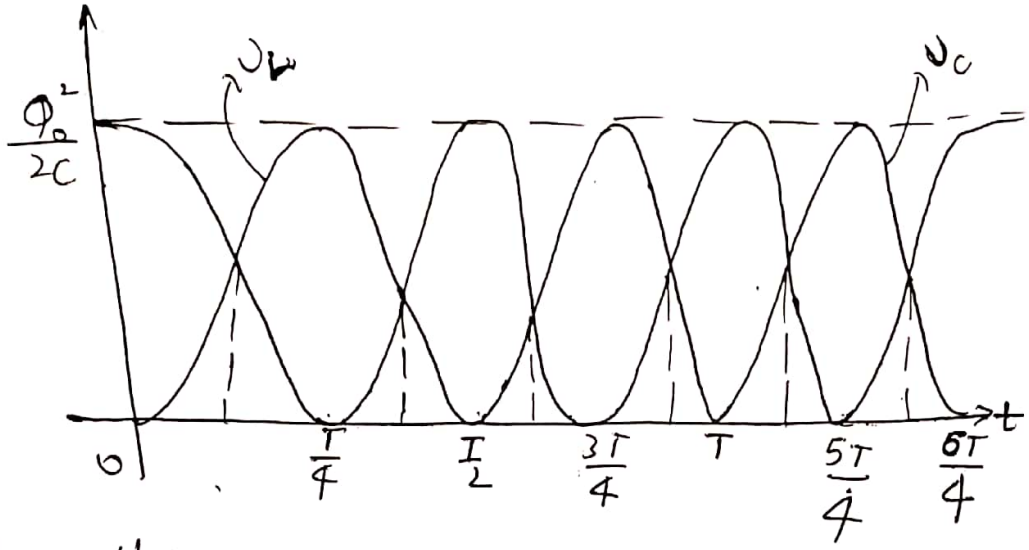

* $U_L = U_C$

$$\frac{q_0^2}{2C}\sin^2\omega t = \frac{q_0^2}{2C}\cos^2\omega t$$

$$\tan^2\omega t = 1$$

$$\tan\omega t = \pm 1$$

$$\omega t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \text{- - - -}$$

$$t = \frac{\pi}{4\omega}, \frac{3\pi}{4\omega}, \frac{5\pi}{4\omega}, \text{- - -}$$

$$t = \frac{T}{8}, \frac{3T}{8}, \frac{5T}{8}, \text{- - -}$$

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Imp Points:

1) charge / current in the circuit oscillates at a frequency 'ω'

$$\omega = \frac{1}{\sqrt{LC}}$$

$$T = 2\pi\sqrt{LC}$$

2) Energy is oscillatory at a frequency $\omega' = 2\omega = \frac{2}{\sqrt{LC}}$ with time period, $T' = \frac{2\pi}{\omega'} = \frac{2\pi}{2\omega} = \frac{T}{2}$.

***

3) To find max current i_0, we use "ω-ε Theorem"

$$U_i + W_b = U_f$$

$$\frac{Q_0^2}{2C} + 0 = \frac{1}{2} L i_0^2$$

$$i_0 = \pm \frac{Q_0}{\sqrt{LC}} = \pm Q_0 \omega$$

4) L-C circuit is similar to spring block system, kept on smooth horizontal surface.

↳ # no friction, no heat & energy is conserved

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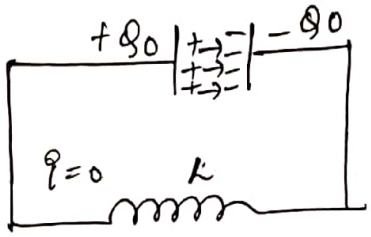
* Diagrams:



at $t=0$

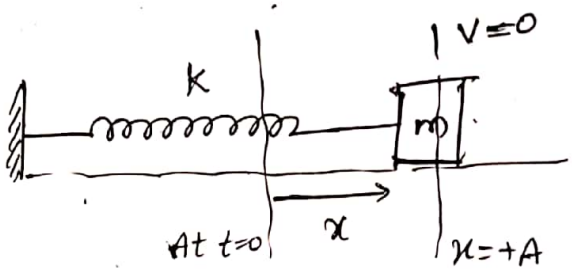
$\omega = \frac{1}{\sqrt{LC}}$

L - electrical inertia

$U_C = \frac{Q_0^2}{2C}$, $U_L = 0$

$q = Q_0 \cos\omega t$

$i = Q_0 \omega \sin\omega t$



$\omega = \sqrt{\frac{K}{m}}$

m - mechanical inertia

#Similarities:

$L \rightarrow m$

$\frac{1}{C} \rightarrow K$

$q \rightarrow x$

$i \rightarrow V$

$U_L \rightarrow K.E$

$U_C \rightarrow P.E$

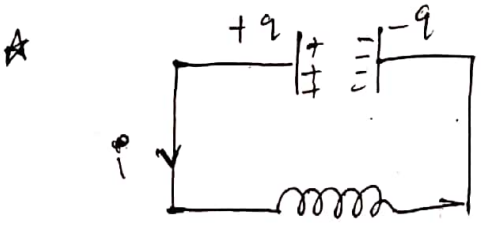
$x = A\cos\omega t$

$V = \frac{-dx}{dt}$

$V = A\omega \sin\omega t$

$PE = \frac{1}{2} KA^2$

$K.E = 0$

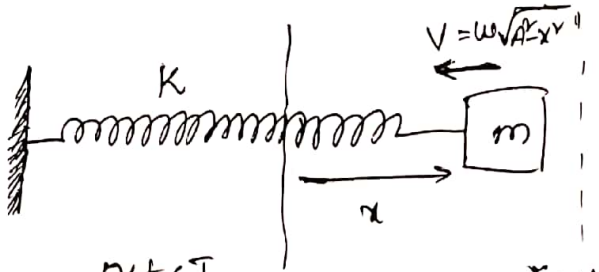


$i = \omega\sqrt{Q_0^2 - q^2}$

$0 < t < \frac{T}{4}$

$U_C = \frac{q^2}{2C}$

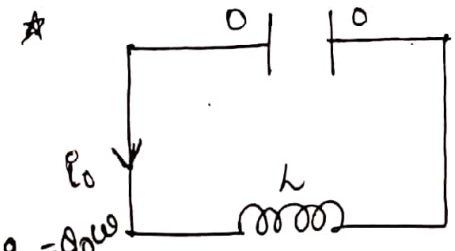
$U_L = \frac{1}{2} L i^2$



$0 < t < \frac{T}{4}$

$P.E = \frac{1}{2} Kx^2$

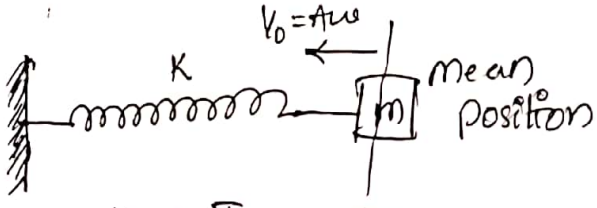
$K.E = \frac{1}{2} mv^2$



$t = \frac{T}{4}$

$U_C = 0$

$U_L = \frac{1}{2} L i_0^2$



$P.E = 0$, $K.E = \frac{1}{2} m v_0^2$

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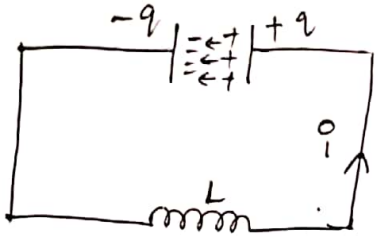
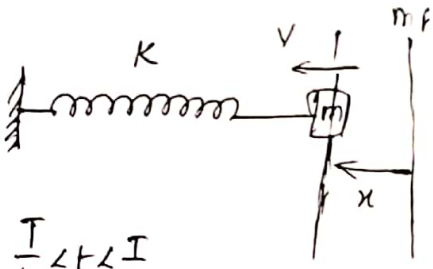


$U_C = \frac{q^2}{2C}$

$U_L = \frac{1}{2} L i^2$

$\frac{I}{4} < t < \frac{I}{2}$

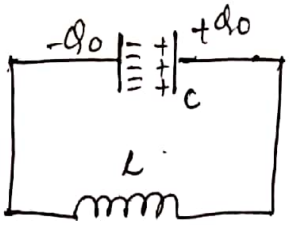


$\frac{T}{4} < t < \frac{I}{2}$

$P.E = \frac{1}{2} K x^2$

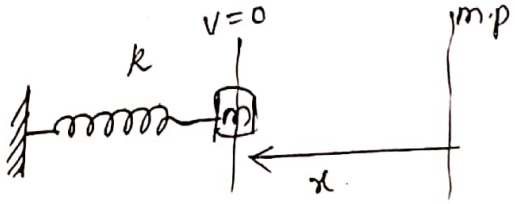
$K.E = \frac{1}{2} m v^2$

★



$U_C = \frac{q_0^2}{2C}$

At $t = \frac{I}{2}$

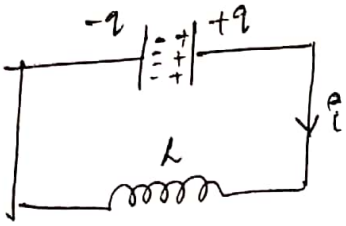


At $t = \frac{I}{2}$ $x = A$

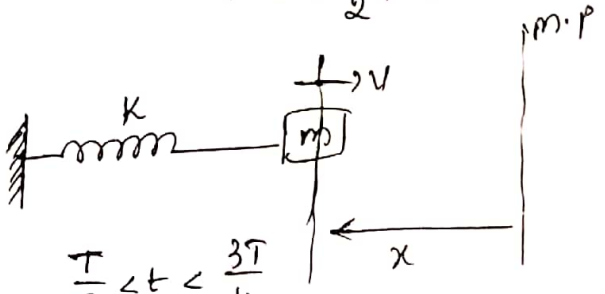
$K.E = 0$

$P.E = \frac{1}{2} K A^2$

★

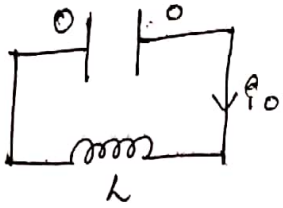


$\frac{I}{2} < t < \frac{3T}{4}$

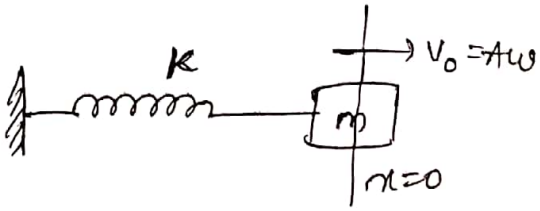


$\frac{T}{2} < t < \frac{3T}{4}$

★

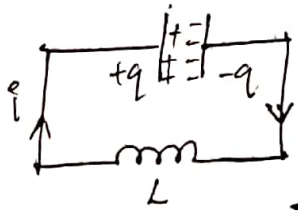


$t = \frac{3T}{4}$

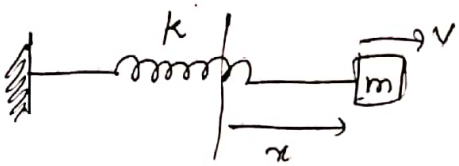


$t = \frac{3T}{4}$ m.p

★



$\frac{3T}{4} < t < T$



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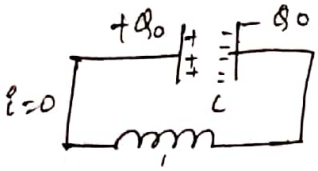
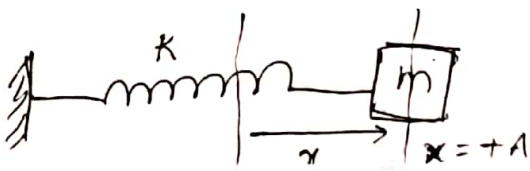


$t = T$



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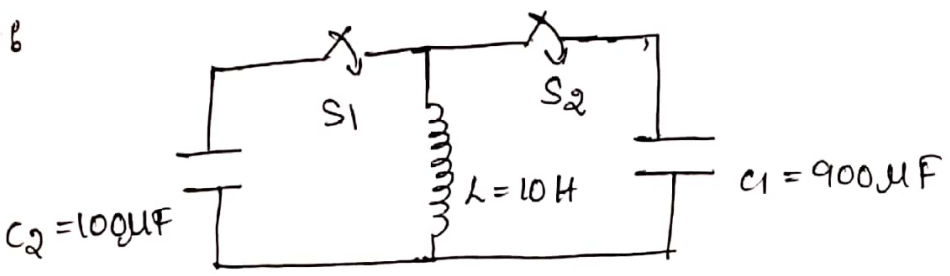
Ex 6



Initially 900 μF capacitor is charged to 100 V and 100 μF capacitor is uncharged. Then the switch S_2 is closed for a time t_1, after which it is opened and at the same instant switch S_1 is closed for a time t_2 and then opened. It is now found that the 100 μF capacitor is charged to 300 V. Find min possible values of the time intervals t_1 & t_2.

Sol: Initially energy stored in C_1 is,

$$U_1 = \frac{1}{2} C_1 V_1^2 = \frac{1}{2} (900 \times 10^{-6}) (100)^2$$

$$= 4.5 \text{ J}$$

Finally, energy stored in C_2 is,

$$U_2 = \frac{1}{2} C_2 V_2^2 = \frac{1}{2} (100 \times 10^{-6}) (300)^2$$

$$= 4.5 \text{ J}$$

Here entire energy transfer from C_1 to C_2 through L.

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we have,

$$T = 2\pi\sqrt{LC}$$

$$t_1 = \frac{2\pi\sqrt{LC_1}}{4}$$

$$= \frac{2\pi}{2}\sqrt{10 \times 900 \times 10^{-6}}$$

$$= \frac{30}{2}\frac{15}{\pi}\sqrt{001}$$

$$\boxed{t_1 = 15\sqrt{001}\,\pi}$$

$$t_2 = \frac{2\pi}{4_2}\sqrt{10 \times 100 \times 10^{-6}}$$

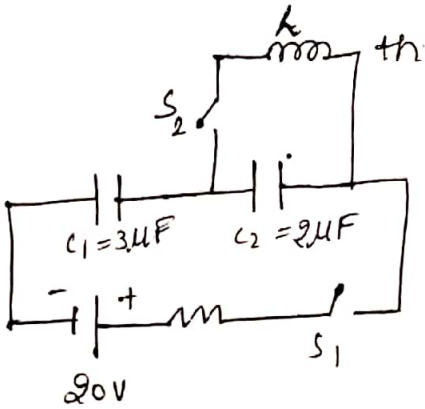
$$\boxed{t_2 = 5\sqrt{001}\,\pi}$$

Ex!

Initially the circuit is in steady state with switch S_1 closed. At $t = 0$, S_1 is opened and S_2 is closed.

1) derive an expression for charge on capacitor C_2 as func of time.

2) determine the 1st instant t when energy in inductor becomes $\frac{1}{3}$rd of that in the capacitor.



$L = 0.2$ mH

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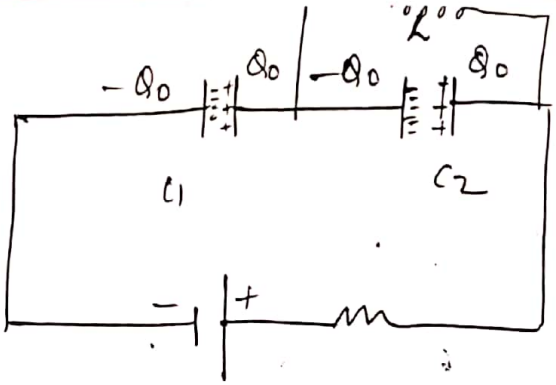
Sol^n



using K.V.L,

$$20 - \frac{Q_0}{C_2} - \frac{Q_0}{C_1} = 0$$

$$Q_0 \left(\frac{5}{6}\right) = 20$$

$$\boxed{Q_0 = 24\,\mu C}$$

1) $q = Q_0 \cos \omega t$

where $Q_0 = 24\mu C$

$$\omega = \frac{1}{\sqrt{LC_2}}$$

$$= \frac{1}{\sqrt{2\times10^{-4}\times2\times10^{-6}}}$$

$$= \frac{10^5}{2}$$

$$= 5 \times 10^4 \text{ rad/sec}$$

$$\boxed{q = (24\times10^{-6}\,C)\cos(5\times10^4 t)}$$

2) we have,

$$U_L + U_C = \frac{Q_0^2}{2C}$$

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Given, $U_L = \frac{U_C}{3}$

$$\frac{U_C}{3} + U_C = \frac{Q_0^2}{2C_2}$$

$$\frac{4U_C}{3} = \frac{Q_0^2}{2C_2}$$

$$\frac{4}{3} \cdot \frac{q^2}{2C_2} = \frac{Q_0^2}{2C_2}$$

$$\boxed{q = \pm \frac{\sqrt{3}}{2} Q_0}$$

$\therefore$ $\boxed{q = + \frac{\sqrt{3}}{2} Q_0}$ (earlier time)

$\frac{\sqrt{3}}{2} q = Q_0 \cos \omega t$

$$\omega t = \frac{\pi}{6}$$

$$t = \frac{\pi}{6\omega}$$

$$= \frac{\pi}{6 \times 5 \times 10^4}$$

$$t = \frac{\pi}{30 \times 10^4}$$

$$\boxed{t = \frac{\pi}{3} \times 10^{-4} \text{ sec}}$$

Ex: Let Q be the instantaneous charge on the capacitor and i be the current in the circuit. It is found that max value of Q is

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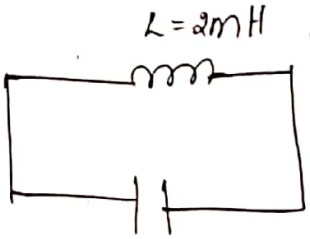
200μc



1) when $Q = 100\mu c$. Find the value of $\left|\frac{di}{dt}\right|$ & value of i.

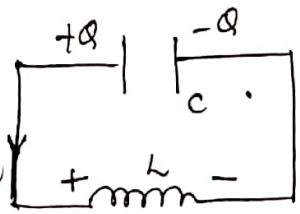
2) when $Q = 200\mu c$. Find value of i.

3) Find max value of i.

4) when i is equal to $\frac{1}{2}$ it's max value, what is ~~that~~ the ~~is~~ value of $|Q|$.

Sol:

1) From KVL,



$$\frac{+Q}{C} - L\cdot\frac{di}{dt} = 0$$

$$\frac{di}{dt} = \frac{Q}{LC} = \frac{10 \times 10^{-6}}{2\times10^{-3}\times5\times10^{-6}} = 10^4 \text{ A/s}$$

& we have,

$$\frac{q^2}{2C} + \frac{1}{2}Li^2 = \frac{Q_0^2}{2C}$$

2) $Q = 200\mu c$ (charge is max)

$\Rightarrow$ current $i = 0$

3) $I_{max} = q_0\omega = 2A$

4) given, $\frac{q_0\omega}{2} = q_0\omega\sin\omega t$

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$\Rightarrow \omega t = \pi/6$

$$|Q| = 200 \times 10^{-6} \times \frac{\sqrt{3}}{2}$$

$$\boxed{q = \sqrt{3} \times 10^{-4}\ C}$$

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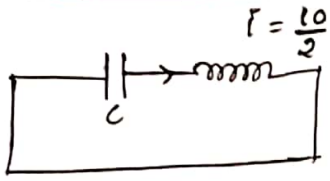
04/05/25 | Friday

Ex:8



At $t=0$; current in inductor is $i = \frac{I_0}{2}$ & magnetic energy stored in inductor is increasing at this instant. Find

1) charge on the plates at this instant

2) current as a function of time.

Sol:8

From L.C.E,

$$\frac{q^2}{2C} + \frac{1}{2}Li^2 = \frac{1}{2}LI_0^2$$

$$\frac{q^2}{2C} + \frac{1}{2}L\left(\frac{I_0^2}{2}\right) = \frac{1}{2}LI_0^2$$

$$\frac{q^2}{2C} = \frac{1}{2}LI_0^2 - \frac{LI_0^2}{8}$$

$$\frac{q^2}{2C} = \frac{3LI_0^2}{8}$$

$$\boxed{q = \frac{\sqrt{3LC}}{2} I_0}$$

2) current,

$$i = I_0 \sin(\omega t + \phi)$$

At $t=0$, $i = \frac{I_0}{2}$

$$\frac{I_0}{2} = I_0 \sin\phi$$

$$\sin\phi = \frac{1}{2}$$

$$\phi = \frac{\pi}{6}, \frac{5\pi}{6}$$

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$$\therefore \boxed{q = q_0 \sin\left(\omega t + \frac{\pi}{6}\right)}$$

* current reaches to maximum for the first time (charge becomes zero)

$$\omega t + \frac{\pi}{6} = \frac{\pi}{2}$$

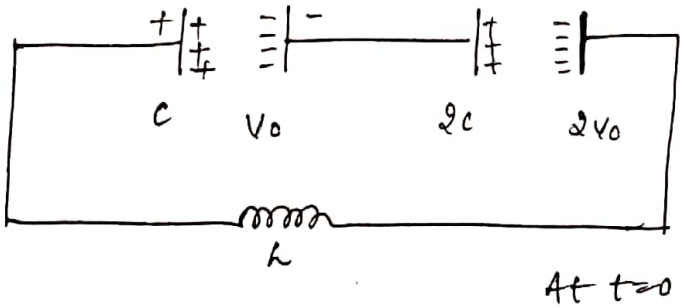
$$\omega t = \frac{\pi}{3}$$

$$\boxed{t = \frac{\pi}{3\omega}}, \quad \omega = \frac{1}{\sqrt{LC}}$$

* current becomes zero (charge becomes max) for the first time.

$$t = \frac{\pi}{3\omega} + \frac{\pi}{2\omega} = \frac{5\pi}{6\omega} = \frac{5\pi}{6}\sqrt{LC}$$

Eg



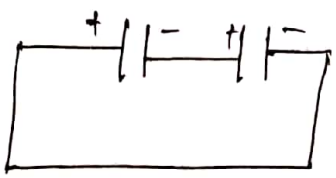
Find max current in the circuit.

Sol

At max current,

$$\frac{di}{dt} = 0$$

$$V_L = L \cdot \frac{di}{dt} = 0$$



i.e, inductor can be replaced with connecting wire.

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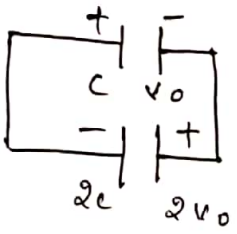
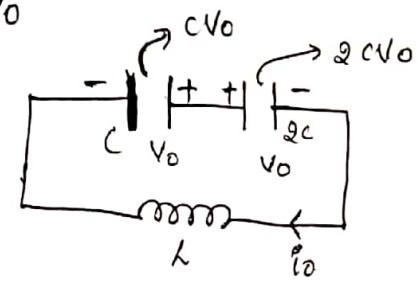
* Common potential,

$$V_c = \frac{CV_0 - (2C)(2V_0)}{C+2C}$$



$$= \frac{-3CV_0}{3C} = -V_0$$



* To find i_0,

use ω-ε Theorem,

$$U_i + W_b = U_f$$

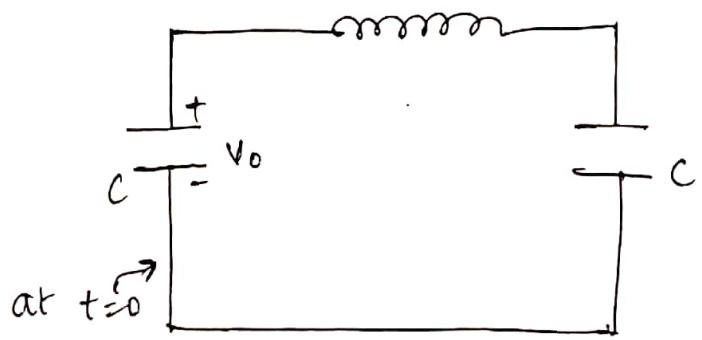
$$\frac{1}{2}C\cancel{V_0^2} + \frac{1}{2}(\cancel{2}C)(2V_0)^2 = \frac{1}{2}C\cancel{V_0^2} + \frac{1}{\cancel{2}}(\cancel{2}C)(V_0)^2 + \frac{1}{2}Li_0^2$$

$$3CV_0^2 = \frac{1}{2}Li_0^2$$

$$6CV_0^2 = Li_0^2$$

$$\boxed{i_0 = \sqrt{\frac{6C}{L}}\,V_0}$$

Ex:



Find

1) charge on capacitor as a function of time.

2) Max current in the circuit.

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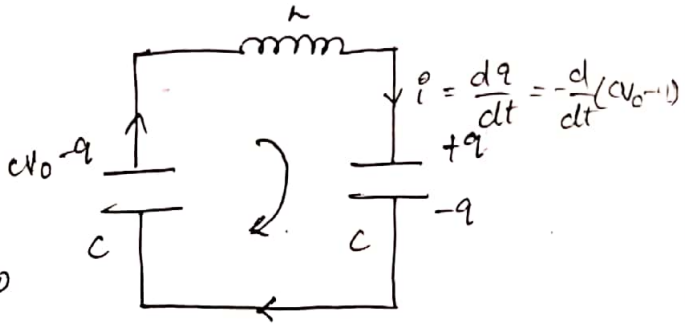
Sol^{n}

At any time t,



From KVL,

$$\frac{CV_0 - q}{C} - L\cdot\frac{di}{dt} - \frac{q}{C} = 0$$

$$V_0 - \frac{2q}{C} = L\cdot\frac{di}{dt}$$

on diff,

$$-\frac{2}{C}\left(\frac{dq}{dt}\right) = L\cdot\frac{d^2i}{dt^2}$$

$$\boxed{\frac{d^2i}{dt^2} = -\frac{2i}{LC}} \text{ — SHM}$$

$$\boxed{\omega = \sqrt{2/LC}} \rightarrow \text{angular frequency of charge/current}$$

* Genelal soln,

$$i = i_0(\sin(\omega t + \phi))$$

At $t=0$, $i=0$

$$0 = i_0 \sin\phi$$

$$\sin\phi = 0$$

$$\boxed{\phi = 0}$$

* when i_0 = max current,

we use K.V.L-Theorem to find i_0.

Common potential $\Rightarrow V_C = \frac{CV_0}{C+C} = \frac{V_0}{2}$

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$$U_i + W_b = U_f$$

$$\frac{1}{2}CV_0^2 + 0 = \frac{1}{2}C\left(\frac{V_0}{2}\right)^2 + \frac{1}{2}C\left(\frac{V_0}{2}\right)^2 + \frac{1}{2}Li_0^2$$

$$\frac{1}{2}CV_0^2 = \frac{1}{4}CV_0^2 + \frac{1}{2}Li_0^2$$

$$\frac{CV_0^2}{2} = Li_0^2$$

$$\boxed{i_0 = \sqrt{\frac{C}{2L}}\,V_0}$$

* Again,

$$\frac{dq}{dt} = i_0 \sin\omega t$$

$$dq = i_0 \sin\omega t \cdot dt$$

$$\int_0^q dq = \int_0^t i_0 \sin\omega t \cdot dt$$

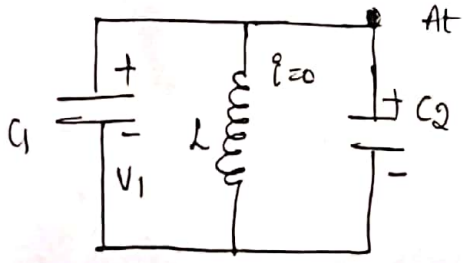
$$\boxed{q = \frac{i_0}{\omega}(1-\cos\omega t)} \Rightarrow$$ on right capacitor.

on left capacitor-

$$q' = CV_0' - q$$

$$= CV_0 - \frac{i_0}{\omega}(1-\cos\omega t)$$

Eg



Find max current through inductor.

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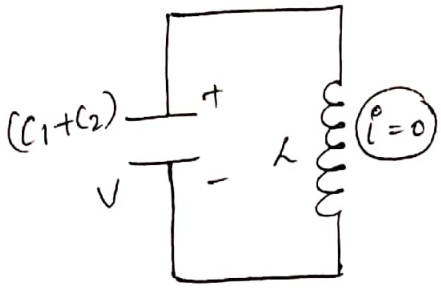
Sol⁸ * immediately capacitors aquire common potential

$$V_c = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

* During this period,

energy lost in the circuit converts into heat in the connecting wire.

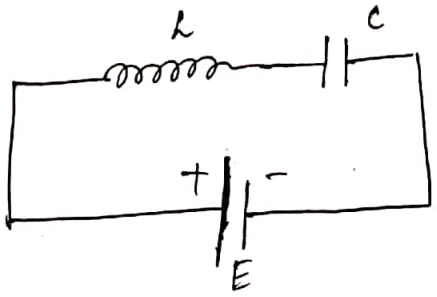


we have,

$$\frac{1}{2}(C_1+C_2)V_i^2 = \frac{1}{2}Li_0^2$$

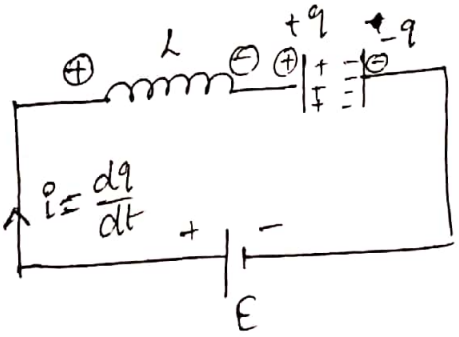
$$(C_1+C_2)V^2 = Li_0^2$$

Ex⁸



At $t=0$, $q=0$ & $i=0$.

At $t=t$,



using KVL,

$$E - L\frac{di}{dt} - \frac{q}{C} = 0$$

on differentiating,

$$0 - L\cdot\frac{d^2i}{dt^2} - \frac{i}{C} = 0$$

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$$\frac{d^2i}{dt^2} = \frac{-i}{LC} \rightarrow SHM$$

$$\omega = \frac{1}{\sqrt{LC}}$$

Soln,

$$i = i_0 \sin(\omega t + \phi)$$

At $t=0$, $i=0$

$$\Rightarrow \boxed{\phi = 0}$$

$$\boxed{i = i_0 \sin \omega t} \quad -①$$

i_0 = max current.

use ω-E theorem to find i_0

At $i = i_{max}$, $V_L = 0$ & $V_C = E$

$$q = CE$$

$$U_i + W_b = U_f$$

$$0 + (CE - 0)E = \frac{1}{2} CE^2 + \frac{1}{2} L i_0^2$$

$$\frac{1}{2} L i_0^2 = \frac{1}{2} CE^2$$

$$i_0^2 = \frac{CE^2}{L}$$

$$\boxed{i_0 = E\sqrt{\frac{C}{L}}}$$

* ① $\Rightarrow$ $i = \frac{dq}{dt}$

$$\frac{dq}{dt} = i_0 \sin \omega t$$

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$$\int_0^q dq = \ell_0 \int_0^t \sin\omega t \cdot dt$$

$$\boxed{q = \frac{\ell_0}{\omega}(1-\cos\omega t)}$$

$$q = E\sqrt{\frac{c}{\lambda}} \cdot \sqrt{\lambda c}\,(1-\cos\omega t)$$

$$\boxed{q = cE(1-\cos\omega t)}$$

Here, $q = 2cE$ (max)

when $\cos\omega t = -1$

$$\omega t = \pi, 3\pi, 5\pi, \ldots$$

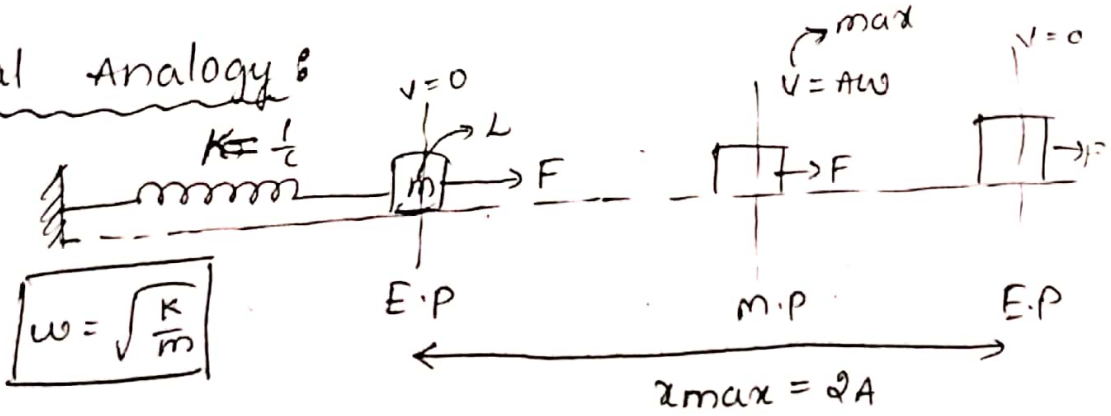
$$t = \frac{\pi}{\omega}, \frac{3\pi}{\omega}, \frac{5\pi}{\omega}, \ldots$$

$q = 0$, $\cos\omega t = 1$.

$$\omega t = 0, 2\pi, 4\pi, \ldots$$

$$t = 0, \frac{2\pi}{\omega}, \frac{4\pi}{\omega}, \ldots$$

**Mechanical Analogy:



$$\boxed{\omega = \sqrt{\frac{K}{m}}}$$

* To find max elongation,

W.E-Theorem,

$$F x_{max} - \frac{1}{2} K x_{max}^2 = 0$$

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$$x_{max} = \frac{2F}{K}$$

$$\boxed{q_{max} = 2EC}$$

To find max speed,

$$\frac{dv}{dt} = 0 \quad , \quad a = 0 \implies F = 0$$

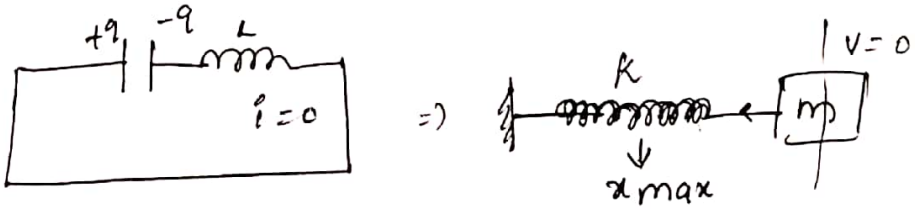
i.e, at mean position,

$$V_{max} = A\omega$$

$$= \frac{x_{max}}{2}\sqrt{\frac{K}{m}}$$

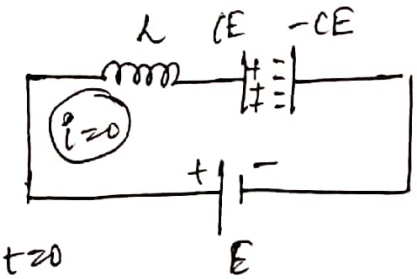
$$i_{max} = \frac{2CE}{2}\sqrt{\frac{1}{LC}}$$

$$\boxed{q_{max} = E\sqrt{\frac{C}{L}}}$$



$$\boxed{\frac{1}{2}m v_{max}^2 = \frac{1}{2}k x_{max}^2}$$

Ex:



At t=0

Find 1) max charge on plates of capacitor.

2) max current in the circuit.

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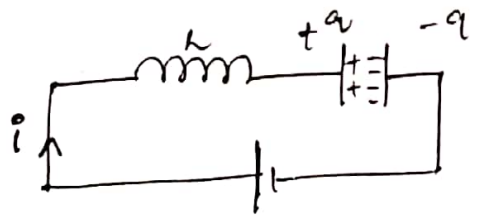
Sol^{n}

$$U_i + W_b = U_f$$

At $t = t$,



use KVL,

$$E - L\cdot\frac{di}{dt} - \frac{q}{C} = 0$$

$$E = L\frac{di}{dt} + \frac{q}{C}$$

diff $\Rightarrow$ $\quad 0 = L\cdot\frac{d^2i}{dt^2} + \frac{i}{C}$

$$\boxed{\frac{d^2i}{dt^2} = -\frac{i}{LC}} \rightarrow \text{SHM.}$$

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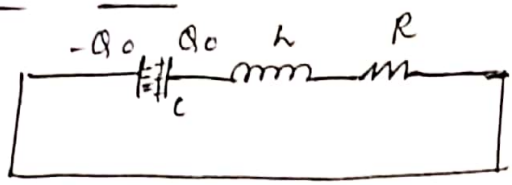
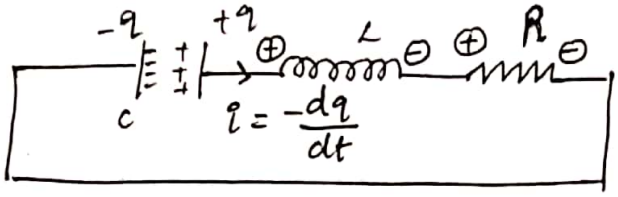
Damped L·C oscillations :



At t=0

At t=t,



use K·V·L,

$$\frac{q}{C} - L\cdot\frac{di}{dt} - iR = 0$$

$$\frac{q}{C} - L\frac{d}{dt}\left(-\frac{dq}{dt}\right) - R\left(-\frac{dq}{dt}\right) = 0$$

$$\frac{q}{C} + L\frac{d^2q}{dt^2} + R\cdot\frac{dq}{dt} = 0$$

$$\frac{d^2q}{dt^2} + \frac{R}{L}\cdot\frac{dq}{dt} + \frac{q}{LC} = 0$$

$$\Rightarrow \boxed{\frac{d^2q}{dt^2} + 2\gamma\frac{dq}{dt} + \omega^2 q = 0} \quad -(1)$$

where $2\gamma = \frac{R}{L}$, $\omega^2 = \frac{1}{LC} \Rightarrow \omega = \frac{1}{\sqrt{LC}}$

which is similar to,

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega^2 x = 0 \quad \text{(damped S.H.O)}$$

here $2\gamma = \frac{b}{m}$ & $\omega = \sqrt{\frac{K}{m}}$.

general Soln is (1) $\Rightarrow$ $q = Ae^{\alpha t}$ — (2) $\quad \frac{dq}{dt} = A\alpha e^{\alpha t} \quad \frac{d^2q}{dt^2} = A\alpha^2 e^{\alpha t}$

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eqn ① becomes,

$$\alpha^2 e^{\alpha t} + 2\vartheta \alpha e^{\alpha t} + \omega^2 e^{\alpha t} = 0$$

$$\alpha^2 + 2\vartheta\alpha + \omega^2 = 0$$

roots, $$\alpha = \frac{-2\vartheta \pm \sqrt{4\vartheta^2 - 4\omega^2}}{2}$$

$$\boxed{\alpha = -\vartheta \pm \sqrt{\vartheta^2 - \omega^2}}$$

eqn ② becomes,

$$q = A_1 e^{(-\vartheta + \sqrt{\vartheta^2 - \omega^2})t} + A_2 e^{(-\vartheta - \sqrt{\vartheta^2 - \omega^2})t}$$

To find A_1 & A_2, use boundary condns.

at $t=0$, $q = Q_0$

$$q = A_1 e^0 + A_2 e^0 = A_1 + A_2$$

$$\boxed{Q_0 = A_1 + A_2} \quad -③$$

Again, $$i = -\frac{dq}{dt}$$

$$= -\left[A_1(-\vartheta + \sqrt{\vartheta^2 - \omega^2}) e^{(-\vartheta + \sqrt{\vartheta^2 - \omega^2})t} + A_2(-\vartheta - \sqrt{\vartheta^2 - \omega^2}) e^{(-\vartheta - \sqrt{\vartheta^2 - \omega^2})t}\right]$$

at $t=0$, $i=0$,

$$i = +A_1\vartheta - A_1\sqrt{\vartheta^2 - \omega^2} + A_2\vartheta + A_2\sqrt{\vartheta^2 - \omega^2}$$

$$(A_1 + A_2)\vartheta - (A_1 - A_2)\sqrt{\vartheta^2 - \omega^2} = 0$$

$$Q_0\vartheta = (A_1 - A_2)\sqrt{\vartheta^2 - \omega^2}$$

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$$A_1 - A_2 = \frac{q_0 \vartheta}{\sqrt{\vartheta^2 - \omega^2}} \quad -(4)$$

solving eqns (3) & (4)

$$\boxed{A_1 = \frac{q_0}{2}\left(1 + \frac{\vartheta}{\sqrt{\vartheta^2 - \omega^2}}\right)}$$

$$\boxed{A_2 = \frac{q_0}{2}\left(1 - \frac{\vartheta}{\sqrt{\vartheta^2 - \omega^2}}\right)}$$

cases:

1) over damped / dead beat :-

$\vartheta^2 > \omega^2$ → damping is more than storage.

$$\frac{R^2}{4L^2} > \frac{1}{LC}$$

$$\boxed{R^2 > \frac{4L}{C}}$$

* roots are real.

charge decays slowly and exponentially with time (there is no oscillation of charges)

2) critical damped :

$$\vartheta^2 = \omega^2$$

$$\frac{R^2}{4L^2} = \frac{1}{LC}$$

$$\boxed{R^2 = \frac{4L}{C}}$$

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Here, $q = A_1 e^{-\vartheta t} + A_2 e^{-\vartheta t}$

$= (A_1 + A_2) e^{-\vartheta t}$

$$\boxed{q = q_0 e^{-\vartheta t}}$$

charge decays rapidly and exponentially with time.

3) under damped

$$\vartheta^2 < \omega^2$$

$$\frac{R^2}{4L^2} < \frac{1}{LC}$$

$$\boxed{R^2 < \frac{4L}{C}}^{*}$$

$$\sqrt{\vartheta^2 - \omega^2} = i\sqrt{\omega^2 - \vartheta^2}$$

* Finally we get,

$$\boxed{q = q_0 e^{-\vartheta t} \cdot \sin(\omega_d t + \phi)}^{*}$$

$q_0 e^{-\vartheta t}$ → Amplitude of charge that decays exponentially with time.

whereas charge q oscillates with frequency,

$$\omega_d = \sqrt{\omega^2 - \vartheta^2}$$

$$= \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

Time period $\Rightarrow T_d = \frac{2\pi}{\omega_d}$

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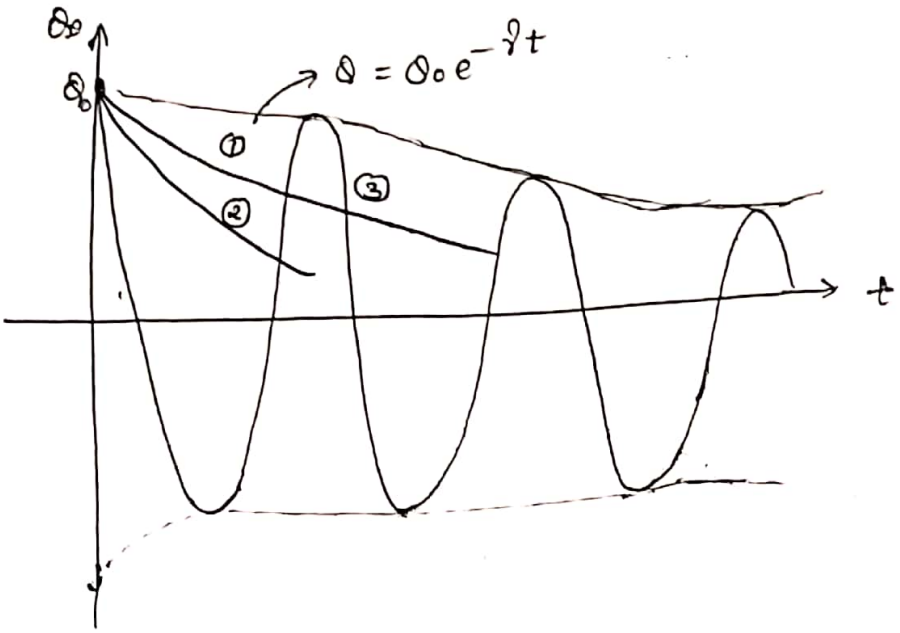

Compaision :

Electrical Oscillation	Mechanical Oscillation
charge (q)	displacement (x)
current (i)	velocity (v)
inductance (L)	mass (m)
capacitance (C)	$\frac{1}{k}$
$U_C = \frac{q^2}{2C}$	$P.E = \frac{1}{2} k x^2$
$U_L = \frac{1}{2} L i^2$	$K.E = \frac{1}{2} m v^2$
resistance (R)	daming coeff (b)
Battery (E)	external force (F)
damped eqn, $q = Q_0 e^{-\gamma t}$	$A = A_0 e^{-\gamma t}$
$\gamma = \frac{R}{2L}$	$\gamma = \frac{b}{2m}$

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damped frequency,

$$\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

damped frequency,

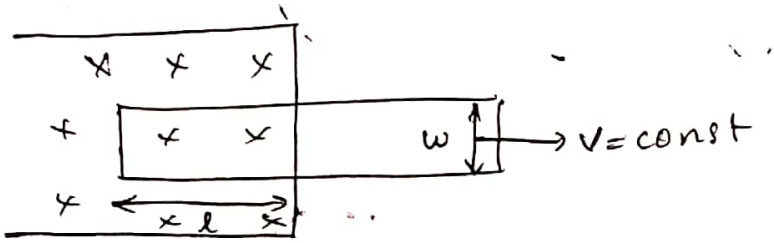
$$\omega_d = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

28/06/25 Saturday

Super Conducting Loop:

↳ resitance of loop = 0.

Ex6



At t=0,

induced current, $i = \frac{e \to 0}{R \to 0}$

we have, $e = \frac{d\phi \to 0}{dt}$

$\downarrow$

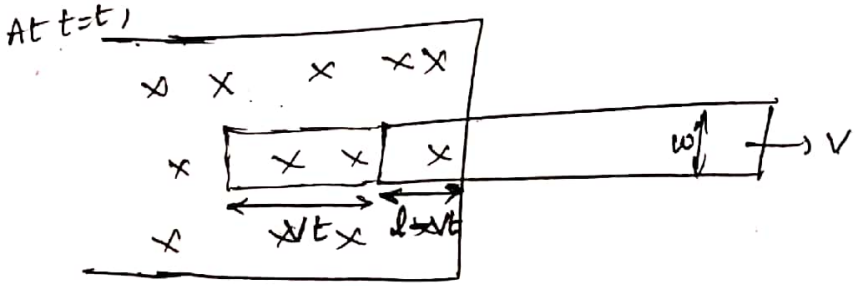
$e \to 0$.

ϕ = const (key point)

we have,

$$\phi_i = Blw + Li_0 \quad -①$$

At t=t1



$$\phi_f = B(l-Vt)w + Li$$

we have, $\phi_i = \phi_f$

$$Blw + Li_0 = Blw - BVtw + Li$$

$$i = \frac{Li_0 + BVtw}{L}$$

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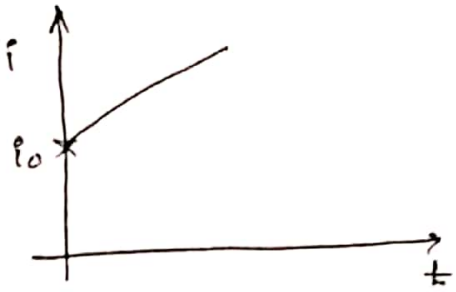
$$i = i_0 + \frac{BVw}{L}(t)$$

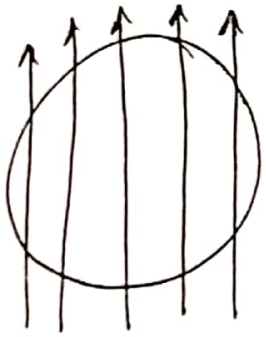


Ex:1 Initially plane of superconducting ring is paallel to magnetic field 'B' and current in it is zero. Now, the ring is rotated through an angle 90°. Find

1) current in the ring at this positon.

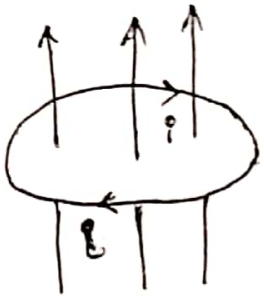
2) workdone in rotating the ring slowly during this process.

Soln At $t=0$,



$i = 0$

$\phi_i = 0$



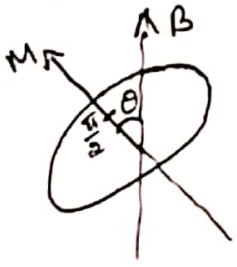
$\phi_f = B(\pi a^2) - L i_f$

we have, $\phi_i = \phi_f$

$$B(\pi a^2) = L i_f$$

$$\boxed{q_f = \frac{B\pi a^2}{L}}$$

2) at any intermediate state,



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flux, $\phi = B\pi a^2 \cos\left(\frac{\pi}{2} - \theta\right) \overline{\cdot} Li$

we have, $\phi_i = \phi$

$$Li = B\pi a^2 \sin\theta$$

$$\boxed{i = \frac{B\pi a^2 \sin\theta}{L}}$$

Torque acting on the ring,

$$\bar{\tau} = \bar{M} \times \bar{B}$$

$$\bar{\tau} = MB \sin\left(\frac{\pi}{2} - \theta\right)$$

$$= iAB\cos\theta$$

$$= \frac{B^2(\pi a^2)^2 \sin\theta\cos\theta}{L}$$

$$\tau = \frac{B^2\pi^2 a^4 \sin 2\theta}{2L}$$

$$\omega = \int d\omega = \int_{\theta=0}^{\theta=90^\circ} \tau \cdot d\theta$$

$$\omega = \frac{B^2\pi^2 a^4}{2L} \int_{\theta=0}^{\theta=90^\circ} \sin 2\theta \cdot d\theta$$

$$= \frac{B^2\pi^2 a^4}{2L} \frac{[\cos 2\theta]_0^{\pi/2}}{2}$$

$$= \frac{B^2\pi^2 a^4}{2L} \cdot \frac{[-1-1]}{2}$$

$$\boxed{\omega = \frac{B^2\pi^2 a^4}{2L}}$$

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